

5 August 2002

TO: Pre-Installation Distribution Tab

SUBJECT: **DIRECTION 2142460, SIGNA LX UPGRADE PRE-INSTALLATION, REV 9**

Enclosed is *Direction 2142460, Signa LX Upgrade Pre-Installation*, Rev 9 update. Table 1 below summarizes the major information incorporated in this revision. Changed content on individual pages is indicated by revision bar in the left hand margin.

TABLE 1
DIRECTION 2142460 REVISION 9 MAJOR CHANGES INCORPORATED

TAB	MAJOR INFORMATION CHANGES
1 SYSTEM CONFIGURATION	●Sec 1-3 flowcharts updated for M3033KH, M3033KJ, & M3033KK ●Corrected 1.0T CX Magnet listing in flowchart Illustration 1-20
4 SITE ENVIRONMENT	●Revised system acoustic noise levels
10 TOOLS & TEST EQUIPMENT	●Added Resistance Meter to table of Rigger/Customer Supplied Equipment

UPDATE INSTRUCTIONS FOR YOUR DIRECTION 2142460 REV 8

Refer to Table 2 for instructions to properly update your Direction 2142460 Rev 8 with the enclosed Revision 9 pages.

TABLE 2
DIRECTION 2142460 REVISION 8 UPDATE INSTRUCTIONS

TAB	REMOVE PAGE(S)	INSERT PAGE(S)
Front Matter	Title Page	Title Page
	A/B	A/B
	i to iv	i to iv
1 SYSTEM CONFIGURATION	1-1/2 1-27 to 1-50	1-1/2 1-27 to 1-51
4 SITE ENVIRONMENT	4-7/8	4-7/8
10 TOOLS & TEST EQUIPMENT	10-1 to 10-10 (entire section)	10-1 to 10-11 (entire section)

Debra C. Balis
MR SERVICE ENGINEERING



GE Medical Systems

Technical Publications

Direction 2142460

Revision 9

Signa® LX Upgrade Pre-Installation

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Operating Documentation



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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
Rev 0	... Oct. 31, 95	... Initial release of document.
Rev 1	 Dec. 6, 96	... Added Signa Horizon LX information throughout manual. Incorporated miscellaneous corrections.
Rev 2	 Jan. 9, 98	... Added Signa Horizon LX SGD information throughout manual. Incorporated miscellaneous corrections. All pages revised due to text font change required for CD-ROM and web page compatibility of this document.
Rev 3	... Sep. 18, 98	... Added Signa CVMR upgrades information throughout manual. Revised Cx magnet ground wires connect to single ground stud. Vibration specification revised for clarity and Appendix A revised to enhance vibration guidelines information. Incorporated miscellaneous corrections.
Rev 4	... Apr. 30, 99	... Revised Recon Processor catalogs to new Reconstruction Module catalogs. Updated Signa CV/i upgrade catalogs. RF/Pen Cabinet front mounting hole height dimension changed. Operator Workspace Table illustration added note for integral part of system. Added dimension illustrations for equipment position on top of Operator Workspace Table. Revised Heat Output values. Revised Facility Input Power Requirements. Updated tools lists, added new items and deleted obsolete items.
Rev 5	 Aug. 8, 00	... Incorporated Release 8.3 software/hardware catalogs (i.e. UCERD). Revised Signa CVMR information for Release 8.4 (ADW to AW). Deleted discontinued Horizon Upgrade catalogs information. Revised Spectroscopy Upgrade/Options information & added MNS Cabinet throughout the manual. Revised Heat Output values. Revised Power Distribution System information. Updated Interconnect Data new catalogs & discontinued catalogs. Clarified RF Penetration Panel upgrade information for RF Shielded Room. Updated references to Upgrade Ordering Checklist documents. Section 10 Table Of Contents corrections.
Rev 6	 Feb. 6, 01	... Changed document title to Signa LX Upgrade Pre-Installation. Incorporated Release 8.X ASP1 Release hardware (i.e. ACGD/PDU Cabinet & SRF Cabinet) information throughout the manual. Revised Signa CVMR information for Release ASP1. Deleted discontinued Horizon Upgrade catalogs information. Revised Spectroscopy Upgrade/Options information. Changed vibration values to G units of measure at Field's request. Updated Interconnect Data new catalogs & discontinued catalogs.
Rev 7	 Mar. 7, 01	... Sec 1 added WARNING for system upgrade to ACGD/PDU to check facility power, MDC, and full system UPS (if present) compatibility with ACGD system config power requirements. Sec 5 updated Critical Power Requirements max. momentary demand for revised design specifications; expanded WARNING to check full system UPS (if present); added 5-2-2 Power Demands Duration & Regulation to clarify power requirements for Signa system with ACGD.
Rev 8	 June 3, 02	... Replaced Direction 2128126 Rev 0 with Rev 1. Added 1.5T Signa MR/i with EXCITE Technology system upgrade information, Added Upgrade to CXK4 Magnet information/references, and deleted Signa CVMR Upgrade information throughout the document. Other major changes listed by section include the following Sec 1: Throughout section added note for systems with Oxford & S-I magnet must upgrade to LCC Magnet; Added 1.5T Signa MR/i with EXCITE Technology catalogs; Deleted OW Color Monitor information; Updated Signa LX Upgrade Reconstruction Module & BAM Memory catalogs; Deleted Oxford & S-1 Magnet related upgrade catalogs which are no longer PriceBook listed; Deleted Signa CVMR related upgrade catalogs which are no longer PriceBook listed. Sec 2: Updated Gradient Coil Assembly, cradle & cart dimensions; Added Network connection and telephone lines option to system connectivity requirements. Sec 3: Changed Signa Horizon to Signa LX. Sec 4: Revised Magnet Room temperature range per System Specifications; Added upgrade to LCC Magnet information references; Deleted OW Color Monitor heat output; Updated Magnet Room Temperature & Humidity Specifications to match system specifications; Updated System Acoustic Noise Level Operating Conditions & wall construction examples. Sec 5: Revised Voltage Transient specification; Added NEC 2002 Article references; Update Critical Power Requirements; Added Ground Fault Protection information. Sec 6: Added 1.5T Signa MR/i with EXCITE Technology information; Updated miscellaneous system interconnects. Sec 7: Added Upgrade to CXK4 Magnet references. Sec 9: Updated SOI Request Form web address; Added SAFETY subsection; Added Upgrade to CXK4 Magnet subsections. Sec 10: Added new & deleted obsolete test equipment & part numbers.
Rev 9	.. Aug. 5, 2002	... Sec 1: Sec 1-3 flowcharts updated for M3033KH, M3033KJ, & M3033KK; Corrected 1.0T CX Magnet listing in flowchart Illustration 1-20. Sec 4: Revised system acoustic noise levels. Sec 10: Added Resistance Meter to table of Rigger/Customer Supplied Equipment.

REVISION HISTORY

<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>
Title Page	9	2-1	6	4-3	8	6-1 to 6-5	8	PREINSTALL.	
GE Logo	—	2-2	blank	4-4	6	6-6	6	CHECKLIST	
Direction 2128126 ...	1*	2-3 to 2-7	8	4-5	8	6-7 to 6-58	8	Tab MR259	—
A to B	8	2-8 to 2-9	6	4-6 to 4-7	6			9-1	8
i	9	2-10	8	4-8	9	RF SHIELDED ROOM		9-2	blank
ii to iii	8	2-11 to 2-12	6	4-9 to 4-15	8	Tab MR257	—	9-3 to 9-7	8
iv	9	2-13 to 2-26	8			7-1	8		
v	7	2-27 to 2-51	9	POWER		7-2	blank	TOOLS & TEST	
SYSTEM CONFIG.		MAGNETIC FIELD		REQUIREMENTS		7-3 to 7-5	8	EQUIPMENT	
Tab MR251	—	CONSID.		Tab MR255	—			Tab MR260	—
1-1	9	Tab MR253	—	5-1	8	SHIPPING &		10-1	9
1-2	blank	3-1	8	5-2	blank	DELIVERY		10-2	blank
1-3 to 1-26	8			5-3 to 5-9	8	Tab MR258	—	10-3 to 10-11	9
1-27 to 1-51	9	SITE ENVIRONMENT		5-10 to 5-13	7	8-1	6		
		Tab MR254	—	5-14	8	8-2	blank	A	
ROOM LAYOUTS		4-1	8	INTERCONNECT DATA		8-3	8	Tab MR301	—
Tab MR252	—	4-2	blank	Tab MR256	—	8-4 to 8-26	6	A-1 to A-7	6

* This revision number/letter corresponds to the indicated document's revision control system.

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
TABLE OF CONTENTS	
INTRODUCTION	iv
SECTION 1 – SYSTEM CONFIGURATION	
1-1 INTRODUCTION	1-3
1-2 UPGRADES TO 1.5T SIGNA LX WITH EXCITE TECHNOLOGY	1-4
1-2-1 Signa Horizon LX	1-6
1-2-2 Signa Horizon 5.X System	1-9
1-2-3 Signa Advantage 1.5T 5.X & 4.X	1-14
1-3 UPGRADES TO SIGNA HORIZON LX WITH ACGD	1-22
1-3-1 Signa Horizon LX with SGD & Signa MR/i System	1-24
1-3-2 Signa Horizon LX With 8645 Gradients System	1-27
1-3-3 Signa Horizon 5.X System	1-29
1-3-4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System	1-35
1-3-5 Signa Advantage 4.X System	1-41
1-4 SIGNA HORIZON LX SYSTEM OPTIONS	1-46
1-5 UPGRADES TO SIGNA HORIZON LX WITH SGD SYSTEM	1-49
1-6 SIGNA HORIZON 5.5 SYSTEM OPTIONS	1-49
1-7 TRANSPORTABLE/RELOCATABLE UPGRADE	1-50
1-8 TRUE MOBILE UPGRADE	1-50
1-9 VAN MANUFACTURERS CONTACT INFORMATION	1-51
SECTION 2 – ROOM LAYOUTS	
2-1 INTRODUCTION	2-3
2-2 TWO MAGNET SITE LAYOUT	2-4
2-3 MINIMUM DOOR/HALLWAY SIZES	2-4
2-3-1 Gradient Coil Assembly, Cradle, & Cart Requirements	2-6
2-4 CABLING CONSIDERATIONS	2-8
2-5 SPECIAL SITING CONSIDERATIONS	2-12
2-5-1 Power Distribution Unit (PDU)	2-12
2-5-2 Blower Box (MG6)	2-13
2-5-3 Water Chiller (WC1) For Gradient Coil Cooling	2-13
2-5-4 SPT Phantom Set Shipping/Storage Cabinet	2-13
2-5-5 Spectroscopy for 1.5T Magnet Only (Optional)	2-14
2-5-6 Magnet Monitor	2-14
2-5-7 System Monitoring & Support Connectivity Requirements	2-15
2-6 ARCHITECTURAL REMINDERS	2-15
2-7 FLOOR LOADING AND WEIGHTS	2-16
2-7-1 Operator Workspace Mounting Requirements For Signa LX	2-17
2-8 COMPONENT DIMENSIONS	2-18
SECTION 3 – MAGNETIC FIELD CONSIDERATIONS	3-1

<u>SECTION</u>	<u>PAGE</u>
SECTION 4 – SITE ENVIRONMENT	
4-1 INTRODUCTION	4-3
4-2 TEMPERATURE AND HUMIDITY SPECIFICATIONS	4-3
4-3 COOLING REQUIREMENTS	4-4
4-3-1 Air Cooling	4-4
4-3-2 Water Cooling	4-6
4-4 ALTITUDE	4-6
4-5 LIGHTING	4-6
4-6 ACOUSTICS	4-7
4-7 ROOM VENTILATION	4-11
4-8 CRYOGENIC VENTING	4-11
4-9 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS	4-12
4-10 CONSTRUCTION MATERIALS	4-12
4-11 VIBRATION	4-12
4-11-1 Definition	4-12
4-11-2 Types Of Vibration Image Quality Issues	4-12
4-11-3 Specifications For 1.5T S-II, & S-III (Unshielded) Magnets and 1.5T S-III Magnet With Magnishield	4-13
4-11-4 Specifications For 1.5T S-IV, S-V, & CX And 1.0T S-X & CX Magnets ..	4-14
4-11-5 Specifications For 1.5T And 1.0T LCC Magnets Without Gradient Isolation	4-14
4-11-6 Specifications For 1.5T And 1.0T LCC Magnets With Gradient Isolation ..	4-15
SECTION 5 – POWER REQUIREMENTS	
5-1 INTRODUCTION	5-3
5-2 CRITICAL POWER REQUIREMENTS	5-4
5-3 POWER DISTRIBUTION SYSTEM	5-6
5-3-1 Main Disconnect Control (MDC)	5-6
5-3-2 System Power Distribution Unit	5-9
5-4 GROUNDING	5-10
5-4-1 Facility Ground	5-10
5-4-2 System Ground	5-12
5-5 GROUND FAULT PROTECTION	5-14
SECTION 6 – INTERCONNECT DATA	
6-1 INTRODUCTION	6-3
6-1-1 Component Designators	6-4
6-1-2 Definition of Terms	6-5
6-2 SYSTEM INTERCONNECTS ADDED BY UPGRADES	6-6
6-2-1 Introduction	6-6
6-2-2 M1033L – Indigo to Octane II with EXCITE Upgrade	6-7
6-2-3 M1060JW – Shield/Cryo Cooler Compressor	6-7
6-2-4 M1060LC – 1.5T CXK4 Magnet Upgrade with Wide Open Enclosure	6-8
6-2-5 M1090JP & M1090JR – Enclosure Upgrade For Fixed Site	6-8
6-2-6 M1090JW – Laser to Halogen Alignment Lights Upgrade	6-9
6-2-7 M1090PC – 5.5 Phased Array Compatibility Kits for 1.5T System	6-9
6-2-8 M1090PH – Pre LX to LX Without Phase Array To 4 channel Phase Array Adder Kit For 1.5T System	6-9
6-2-9 M1090PW – 1.5T Phased Array Upgrade Kit (UCERD) For LX Systems Without Phase Array	6-10
6-2-10 M1090SJ – 5.5 To LX Upgrade For Fixed Site	6-10
6-2-11 M1191JF – Enclosure Upgrade For Transportable/Relocatable	6-10

SECTION**PAGE****SECTION 6 – INTERCONNECT DATA (Continued)**

6-2-12	M1191JM – 5.5 To LX Cable Upgrade for True Mobile with Rear Mounted Penetration Panel	6-10
6-2-13	M1191JN – 5.5 To LX Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel	6-10
6-2-14	M1191SN – Collector Kit for Upgrade to 1.5T LCC Magnet	6-11
6-2-15	M1191SP – Collector Kit for Upgrade to 1.5T LCC Magnet	6-13
6-2-16	M1860LC – 1.0T CXK4 Magnet Upgrade with Wide Open Enclosure	6-13
6-2-17	M1890PA – 5.5 Phased Array Compatibility Kits For 1.0T System	6-13
6-2-18	M1890PH – Pre LX to LX Without Phase Array to 4 channel Phase Array Adder Kit For 1.0T System	6-13
6-2-19	M1890PW – 1.0T Phased Array Upgrade Kit (UCERD) for LX Systems without Phased Array	6-13
6-2-20	M1891SN – Collector Kit for Upgrade to 1.0T LCC Magnet	6-14
6-2-21	M1891SP – Collector Kit for Upgrade to 1.0T LCC Magnet	6-15
6-2-22	M3090BA – 1.0T ACGD Upgrade for system with SGD Gradient Cabinet & RF/Penetration Cabinet	6-16
6-2-23	M3090BB – ACGD Upgrade for system with SGD Gradient Cabinet & RF/PDU Cabinet	6-16
6-2-24	M3090BC – 1.5T ACGD Upgrade for system with SGD Gradient Cabinet & RF/Penetration Cabinet	6-16
6-2-25	M3090CA – 4.X To LX With ACGD Cable Upgrade For Fixed Site	6-17
6-2-26	M3090CB – 5.1–5.4 To LX With ACGD Cable Upgrade For Fixed Site ...	6-21
6-2-27	M3090CC – 5.5 Software Only Site To LX With ACGD Cable Upgrade For Fixed Site	6-23
6-2-28	M3090CD – 5.5 To LX With ACGD Cable Upgrade For Fixed Site	6-24
6-2-29	M3090CH – 3.X–5.4 to LX with ACGD Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel	6-25
6-2-30	M3090CJ – 5.5 Software Site to LX with ACGD Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel ..	6-25
6-2-32	M3090CM – 5.X To LX with ACGD Cable Upgrade for Fixed Site	6-28
6-2-33	M3090CN – 5.5 To LX Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel	6-30
6-2-34	M3090EB – 1.5T EXCITE Hardware Kit	6-30
6-3	ADVANTAGE WORKSTATION INTERCONNECTS	6-31
6-4	INTERCONNECTS RE-ROUTED BY HORIZON LX WITH ACGD UPGRADES	6-31
6-5	INTERCONNECTS REMOVED BY LX WITH EXCITE TECHNOLOGY UPGRADES	6-32
6-6	INTERCONNECTS REMOVED BY HORIZON LX WITH ACGD UPGRADES	6-34
6-6-1	Signa Horizon LX System With SGD & RF/PDU Cabinet	6-34
6-6-2	Signa Horizon LX System With SGD & RF/Penetration Cabinet	6-35
6-6-3	Signa Horizon LX System With 8645 Gradients & RF/Penetration Cabinet	6-37
6-6-4	Signa Horizon 5.X System	6-39
6-6-5	Signa Horizon 5.X Software & Signa Advantage 5.X System	6-42
6-7	INTERCONNECTS REMOVED BY SIGNA HORIZON 5.X TO SIGNA LX COMPUTER ONLY UPGRADE	6-52
6-8	REAR PEDESTAL AND ENCLOSURE CABLE ACCESS	6-54
6-9	SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH ACGD ..	6-55
6-9-1	M3090DA – 4.X–5.4 MNS Upgrade Kit For ACGD Site	6-55
6-9-2	M3090DB – 5.5 MNS Upgrade Kit For ACGD Site	6-56
6-9-3	M3090DB – LX MNS Upgrade Kit For ACGD Site	6-56
6-10	SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH SGD ..	6-57

<u>SECTION</u>	<u>PAGE</u>
SECTION 7 – RF SHIELDED ROOM	
7-1 RF SHIELDED ROOM SPECIFICATION	7-3
7-2 VENTS	7-3
7-3 PLUMBING	7-3
7-4 ELECTRICAL	7-3
7-5 RF PENETRATION PANEL	7-3
7-6 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM	7-5
7-6-1 Magnet Mounting	7-5
7-6-2 Blower Box	7-5
7-7 MAGNET ROOM EQUIPMENT MOUNTING	7-5
7-7-1 Magnet Rundown Unit (MS4)	7-5
7-7-2 Emergency Off Buttons	7-5
7-8 ROOM VENTILATION SWITCH	7-5
SECTION 8 – SHIPPING AND DELIVERY DATA	
8-1 INTRODUCTION	8-3
8-2 UPGRADE CATALOGS SHIPPING INFORMATION	8-6
SECTION 9 – PREINSTALLATION CHECKLIST	
9-1 INTRODUCTION	9-3
9-2 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE DELIVERY	9-3
9-3 GENERAL PREINSTALLATION REMINDERS	9-5
9-4 SAFETY	9-5
9-5 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE LCC MAGNET DELIVERY AND MOVING INTO MAGNET ROOM	9-6
9-6 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE LCC MAGNET CRYOGEN TRANSFER/MAGNET FILL	9-7
9-7 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE LCC MAGNET RAMP-UP	9-7
SECTION 10 – TOOLS AND TEST EQUIPMENT	
10-1 INTRODUCTION	10-3
10-5 INSTALLATION EQUIPMENT	10-6
10-7 TEST EQUIPMENT	10-8
10-8 CALIBRATION TOOLS/FIXTURES	10-9
10-9 TOOL KIT	10-10
APPENDIX A – MR SITE VIBRATION TEST GUIDELINES	
A-1 TEST MEASUREMENTS	A-1
A-2 EQUIPMENT (SPECTRAL ANALYZER) SET-UP	A-1
A-3 TEST MEASUREMENTS	A-2
A-3-1 Ambient Baseline Condition	A-2
A-3-2 Normal (Operating) Condition	A-2
A-4 PRESENTATION/INTERPRETATION OF RESULTS	A-3

INTRODUCTION

This document contains the physical, magnetic, cryogenic, plumbing and electrical data necessary for planning and preparing a site for a Signa Horizon system upgrade. “Preinstallation work” is done to prepare the customer’s premises for the installation of the upgrade sold. It is the responsibility of the purchaser to arrange for performance of this work. Such work includes:

- Installation of the electrical conduit, junction boxes, ducts, surface raceways, outlets and line safety switches.
- Installation of wires not supplied by GE Medical System such as facility input power line to the Main Disconnect Control and Power Distribution Unit. If new wires are installed then the electrical contractor shall ring identify and tag all wires at both ends. Color-coded wires are recommended for easier identification. Wires shall be continuous without splices. Insulation on all equipment ground wires must be green with a yellow stripe.
- Installation of non-electrical lines such as: water plumbing, helium venting systems and air conditioning equipment. Also, installation of recommended air, vacuum and oxygen lines into the Magnet Room. All lines must be clearly labeled.
- Site renovation.
- Installation of structural reinforcements as required.

All work **MUST** be in compliance with national and local building and safety codes.

All site plans and preliminary concepts should be reviewed by GE Medical Systems MR Siting group prior to construction.

Unless specifically mentioned, GE Medical Systems does not provide or install: the facility input power lines to the Main Disconnect Control and Power Distribution Unit or the power lines required in the Magnet Room, the Main Disconnect Control option, nor raised flooring, conduit, junction boxes, ducts, plumbing, water treatment equipment, cryogenic venting outside the Magnet Room, non standard cryogenic venting inside the Magnet Room, or the RF shielded room specified in this document.

All electrical installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations, and testing shall be performed by qualified GE Medical Systems personnel or third-party service companies with equivalent training. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE’s electrical work on these products will comply with the requirements of the applicable electrical codes. The purchaser of GE equipment shall only utilize qualified personnel (i.e. GE’s field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

Pre-installation information is continually changing due to the evolution of the product. GE will make every reasonable effort to maintain accuracy of the pre-installation information.

SECTION 1 – SYSTEM CONFIGURATION

TABLE OF CONTENTS

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
1-1	INTRODUCTION	1-3
1-2	UPGRADES TO 1.5T SIGNA LX WITH EXCITE TECHNOLOGY	1-4
1-2-1	Signa Horizon LX	1-6
1-2-2	Signa Horizon 5.X System	1-9
1-2-3	Signa Advantage 1.5T 5.X & 4.X	1-14
1-3	UPGRADES TO SIGNA HORIZON LX WITH ACGD	1-22
1-3-1	Signa Horizon LX with SGD & Signa MR/i System	1-24
1-3-2	Signa Horizon LX With 8645 Gradients System	1-27
1-3-3	Signa Horizon 5.X System	1-29
1-3-4	Signa Horizon 5.X Software Only & Signa Advantage 5.X System	1-35
1-3-5	Signa Advantage 4.X System	1-41
1-4	SIGNA HORIZON LX SYSTEM OPTIONS	1-46
1-5	UPGRADES TO SIGNA HORIZON LX WITH SGD SYSTEM	1-49
1-6	SIGNA HORIZON 5.5 SYSTEM OPTIONS	1-49
1-7	TRANSPORTABLE/RELOCATABLE UPGRADE	1-50
1-8	TRUE MOBILE UPGRADE	1-50
1-9	VAN MANUFACTURERS CONTACT INFORMATION	1-51

1-1 INTRODUCTION

Magnetic Resonance (MR) system upgrades to Signa LX With EXCITE Technology 1.5T, Signa LX with ACGD, Signa Horizon, and Signa CVMR are documented throughout this publication. Refer to the appropriate section listed below for overview of the upgrade content, refer to the remainder of this publication for upgrade requirements. System cable impacts are provided in SECTION 6 INTERCONNECT DATA.

Note

Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

- SECTION 1-2 UPGRADES TO 1.5T SIGNA LX WITH EXCITE TECHNOLOGY
 - Signa Horizon LX SmartSpeed with EXCITE Technology for 1.5T
 - Signa Horizon LX HiSpeed Plus with EXCITE Technology for 1.5T
 - Signa Horizon LX EchoSpeed Plus with EXCITE Technology for 1.5T
- SECTION 1-3 UPGRADES TO SIGNA LX WITH ACGD
 - Signa Horizon LX SmartSpeed with ACGD for 1.5T or 1.0T
 - Signa Horizon LX HiSpeed Plus with ACGD for 1.5T or 1.0T
 - Signa Horizon LX EchoSpeed Plus with ACGD for 1.5T
- SECTION 1-4 SIGNA LX SYSTEM OPTIONS
- SECTION 1-5 UPGRADES TO SIGNA LX WITH SGD SYSTEM
- SECTION 1-6 SIGNA HORIZON 5.5 SYSTEM OPTIONS
- Refer to Direction 2246012 Pre-Sale/Site Survey For Upgrades to CXK4 Magnet With WideOpen Enclosure which must be completed and signed-off by Local Customer Team to qualify a site for magnet upgrade if such an upgrade is being considered. Direction 2246012 requires input supplied by the customer and by GE. Electronic copy of Direction 2246012 is available to GEMS personnel via the GE Intranet at <http://3.87.118.28/optec3/common/upgradeinfo/upgrade.htm>.

1-2 UPGRADES TO 1.5T SIGNA LX WITH EXCITE TECHNOLOGY

Illustration 1-17 shows the system's major equipment after a 1.5T Signa LX with EXCITE Technology upgrade is complete. The upgrade content is dependent on the configuration of the system being upgraded and the final Signa LX with EXCITE Technology product configuration. The following sub-sections provide an overview of the various Signa LX with EXCITE Technology upgrade hardware.

Note

At this time upgrades to EXCITE Technology are not available for systems with 1.0T Magnets or systems with CRM Body Coils.

WARNING!

ALL SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE FACILITY TRANSFORMER AND FEEDER WIRES ARE CORRECTLY SIZED. ALSO THE MAIN DISCONNECT CONTROL NEEDS TO BE INVESTIGATED FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS. REFER SECTION 5 – POWER REQUIREMENTS.

SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS. REFER SECTION 5 – POWER REQUIREMENTS.

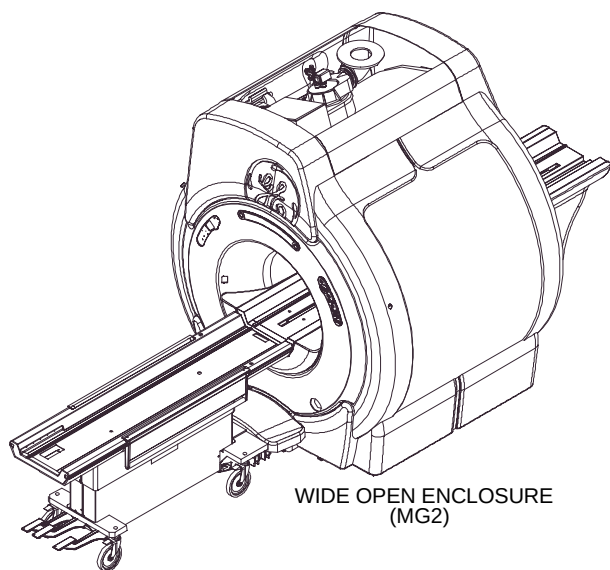
Note

Only catalogs which include hardware are documented in this publication. Software only catalogs are not included, existing system site parameters apply for software only upgrades.

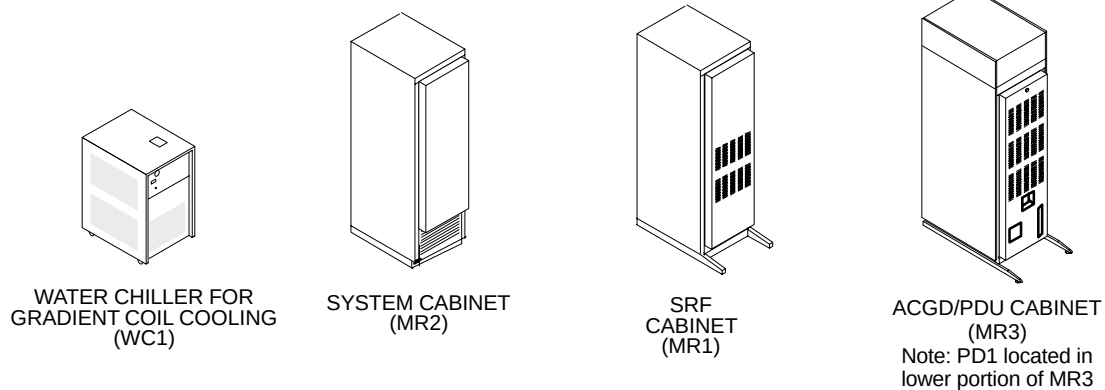
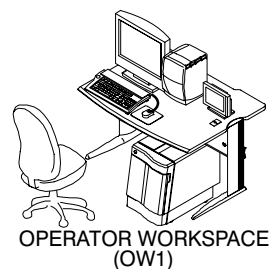
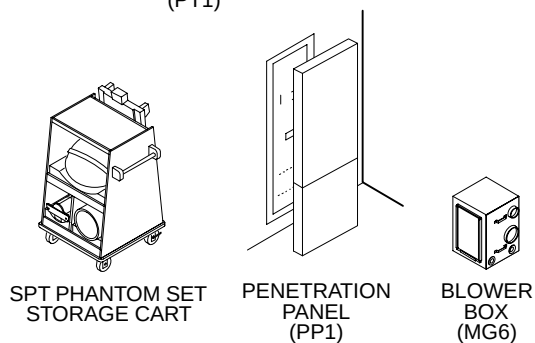
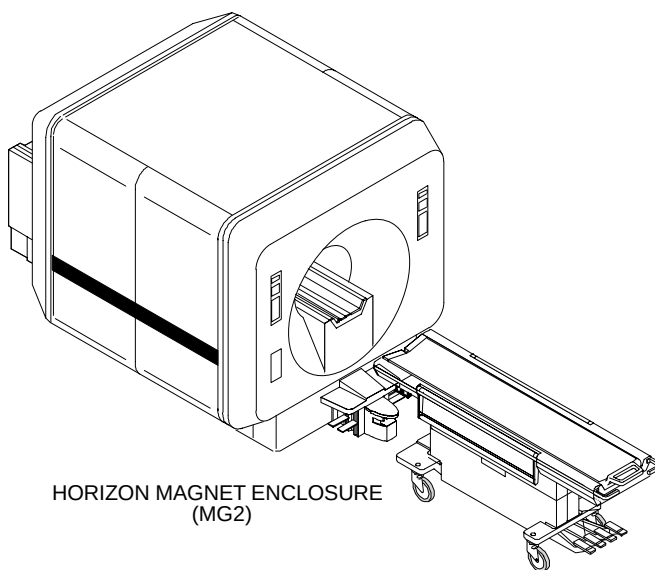
Note

FOR USA, FIXED SITE ONLY CONSIDERING MAGNET UPGRADE: Refer to *Direction 22461012 Pre-Sales/Site Survey For Upgrade To CXK4 Magnet With WideOpen Enclosure* for the many environmental factors and logistics that need to be understood to qualify a site for a magnet upgrade. The Magnet Upgrade Option requires the MR system to be operating at Release 8.X prior or coincident with the Magnet Upgrade option.

1-2 UPGRADES TO 1.5T SIGNA LX WITH EXCITE TECHNOLOGY (Continued)



OR



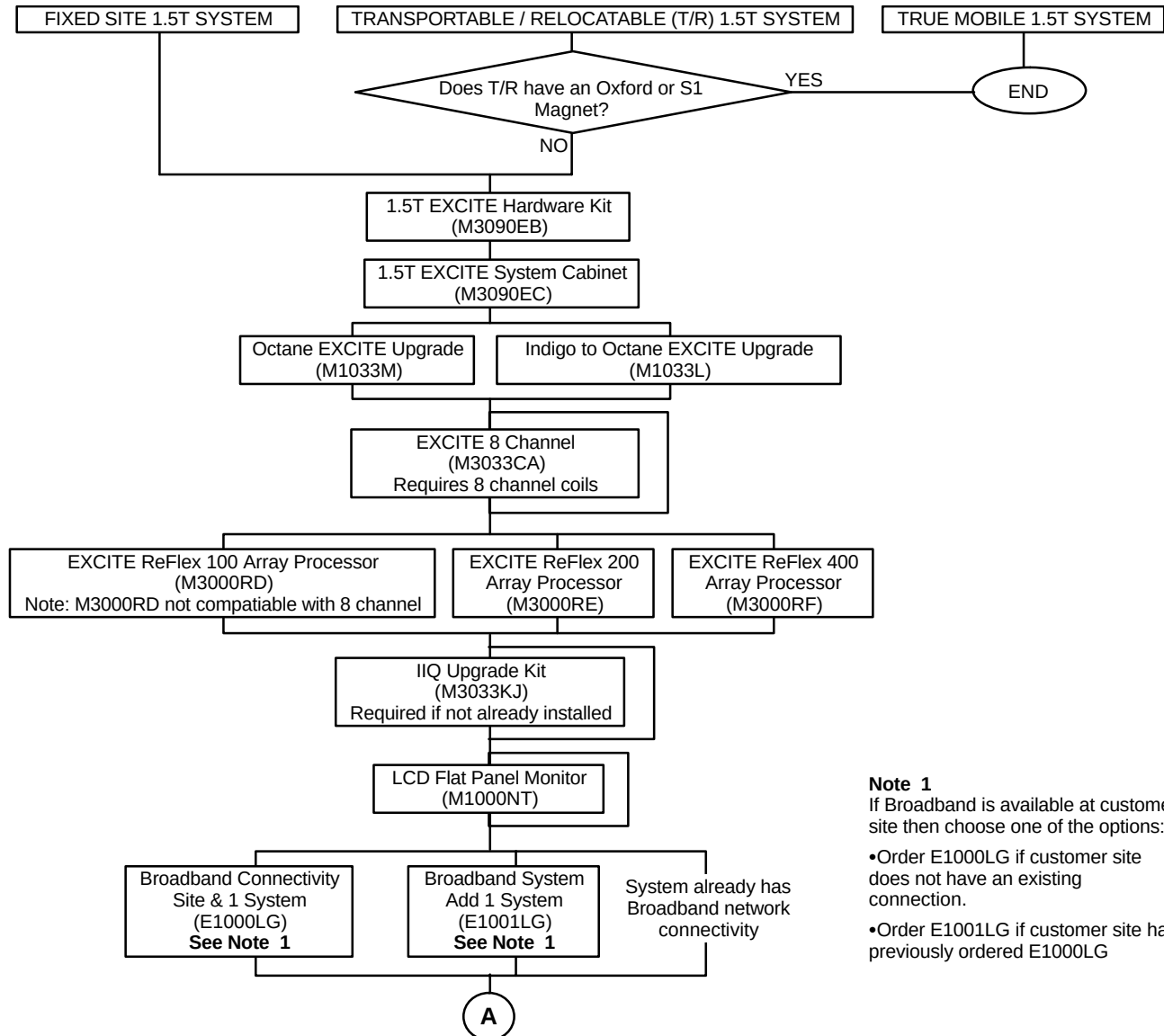
SIGNA LX WITH EXCITE TECHNOLOGY SYSTEM MAJOR EQUIPMENT
ILLUSTRATION 1-1

1-2-1 Signa Horizon LX System

Illustrations 1-2 and 1-3 lists the upgrade catalogs for Signa LX with ACGD or SGD and Signa MR/i system to Signa LX With EXCITE Technology. Illustration 1-4 shows the cabinets replaced and modified by the listed catalogs.

Note

Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



Note 1

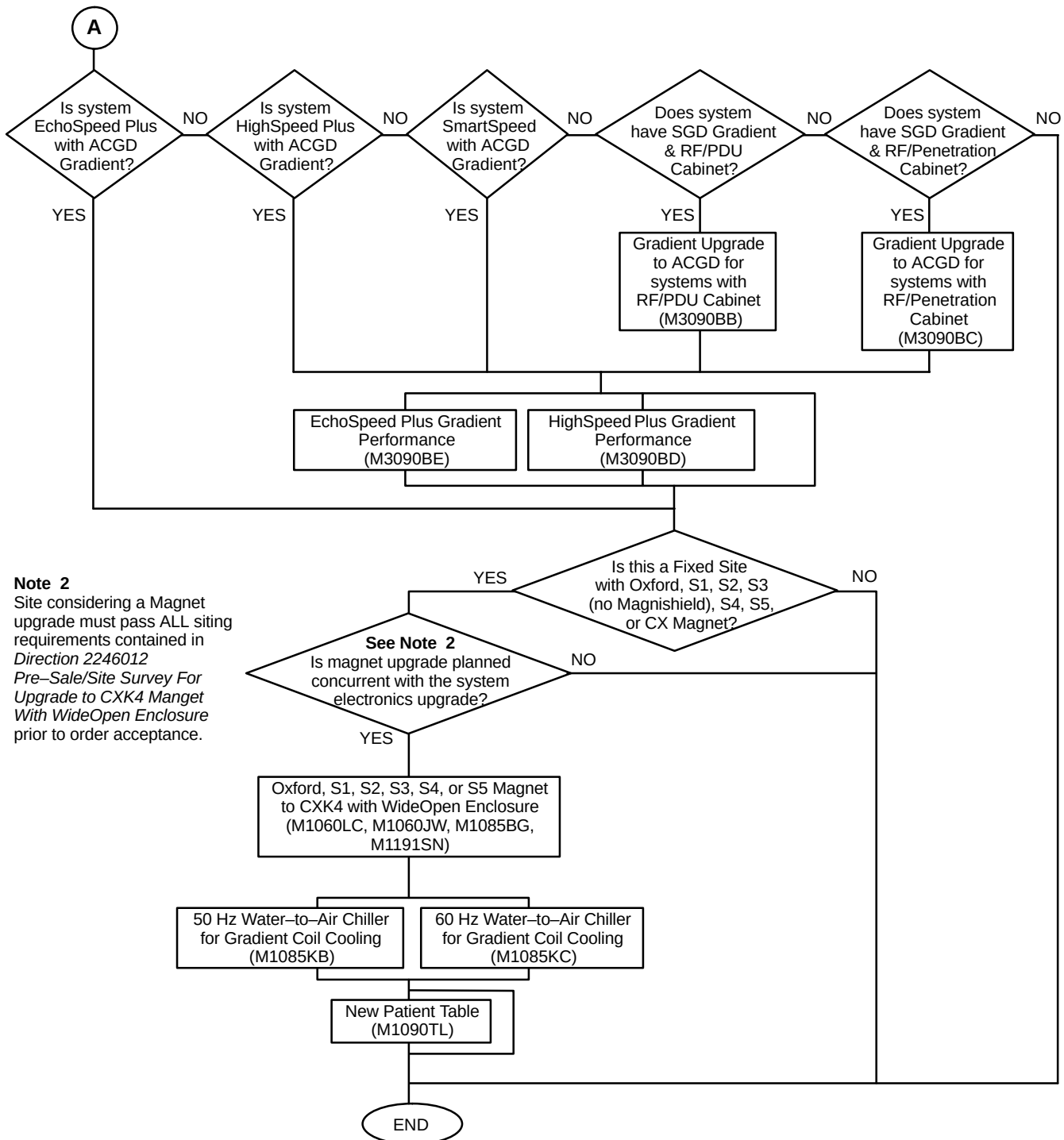
If Broadband is available at customer site then choose one of the options:

- Order E1000LG if customer site does not have an existing connection.
- Order E1001LG if customer site has previously ordered E1000LG

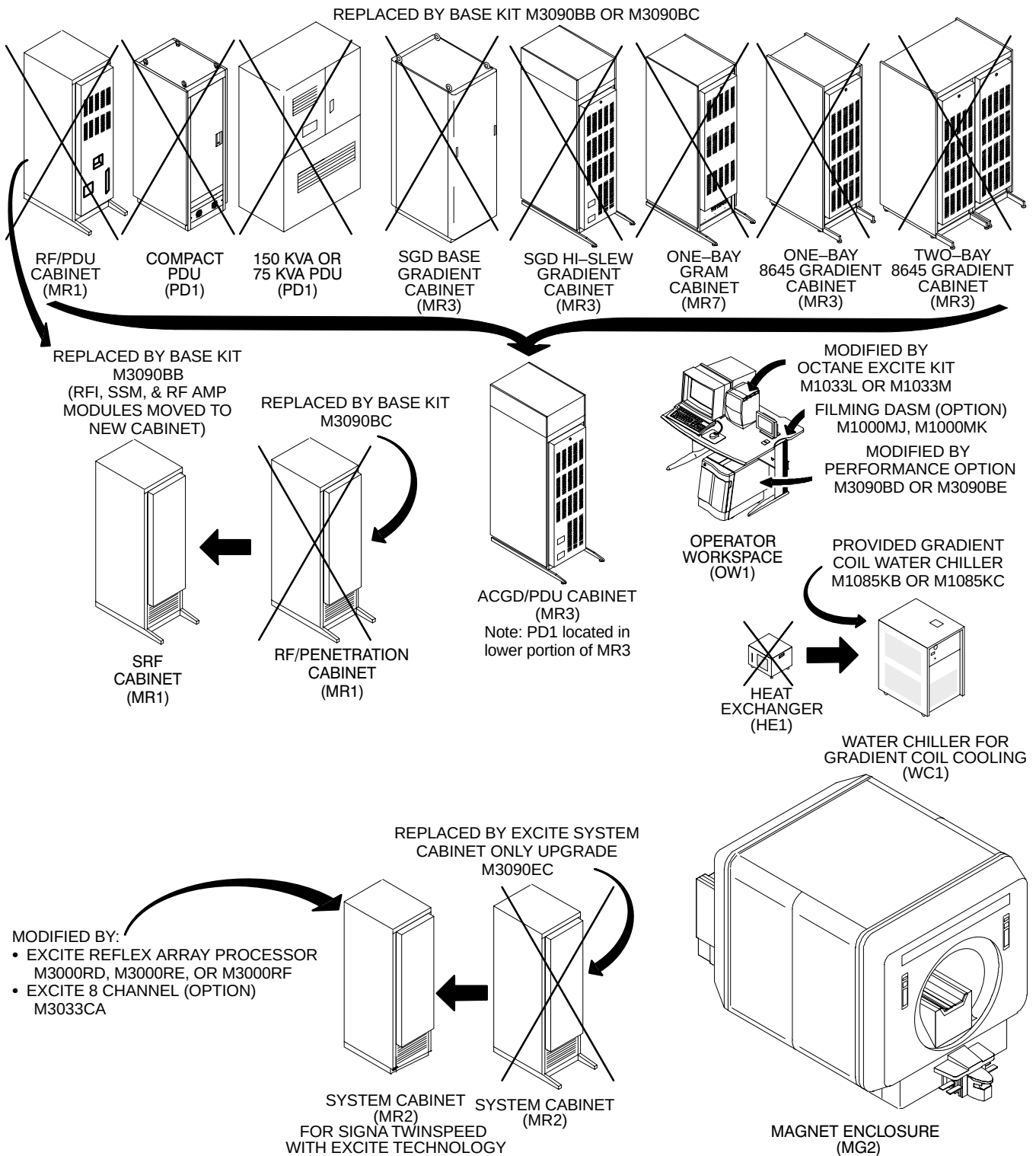
1.5T SIGNA LX TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS

ILLUSTRATION 1-2

1-2-1 Signa Horizon LX System(Continued)



1-2-1 Signa Horizon LX System(Continued)



SIGNA HORIZON LX SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-4

1-2-2 Signa Horizon 5.X System

Illustrations 1-5 through 1-7 lists the upgrade catalogs for Signa Horizon 5.X system to Signa Horizon LX With EXCITE Technology. Illustrations 1-8 and 1-9 how the cabinets replaced or modified and new equipment added by the listed catalogs.

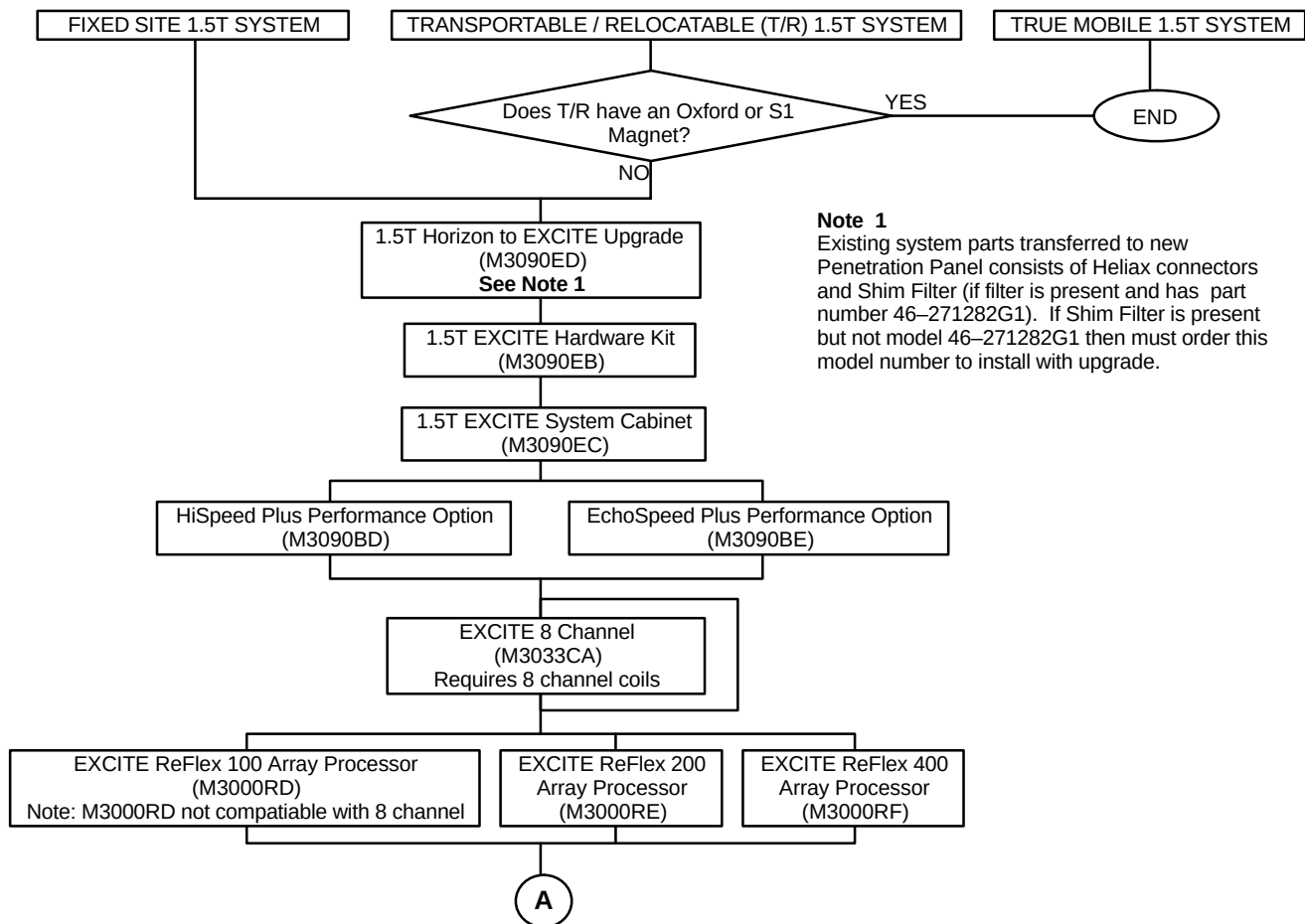
Refer to Section 1-4 SIGNA HORIZON LX SYSTEM OPTIONS for additional option catalogs.

Note

Signa Horizon 5.5 Software only systems have a 55 cm Body Coil and require upgrade catalogs defined in Section 1-2-3, Signa Horizon 5.X Software Only & Signa Advantage 5.X System.

Note

Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

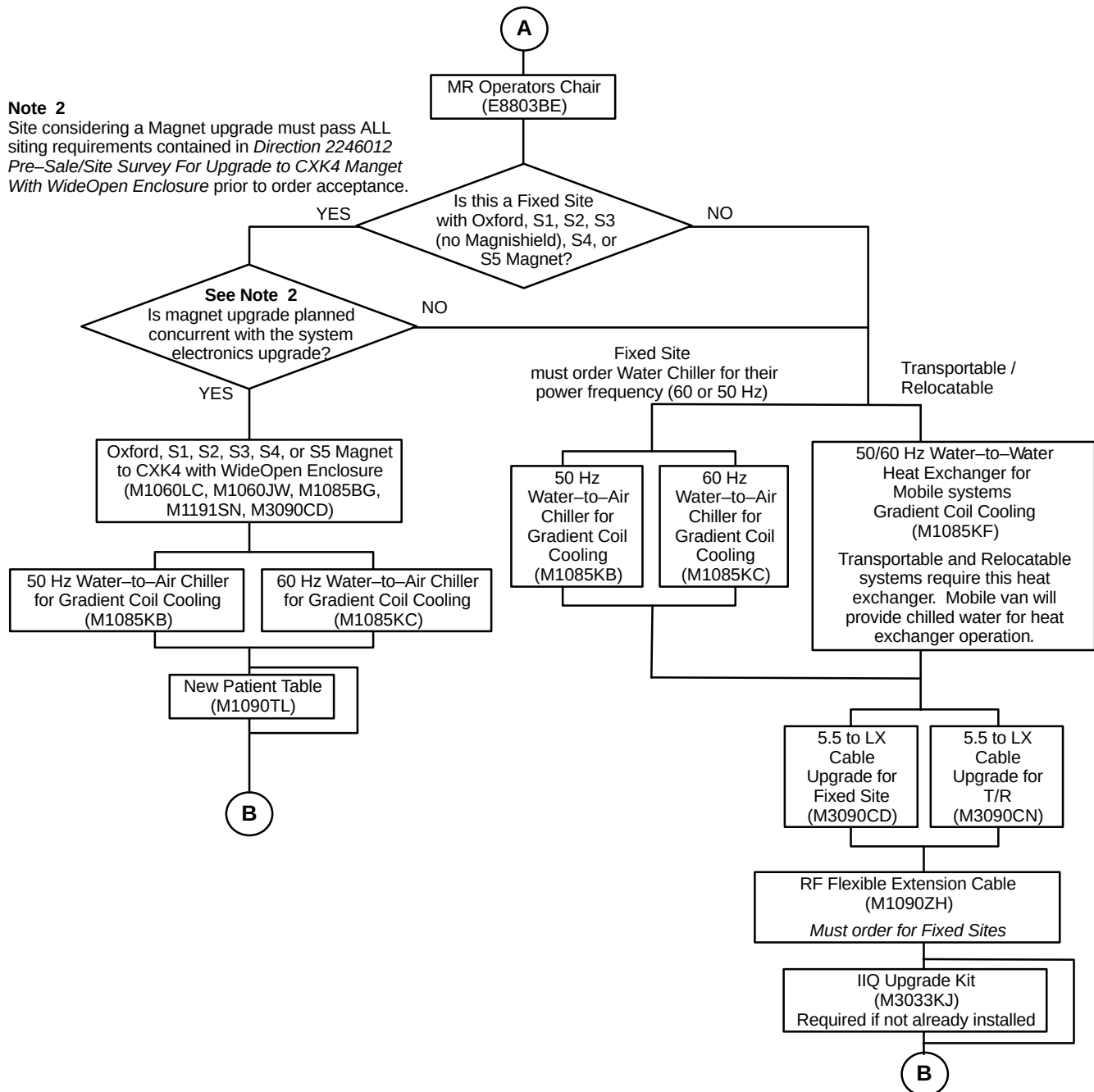


1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-5

1-2-2 Signa Horizon 5.X System (Continued)

Note 2

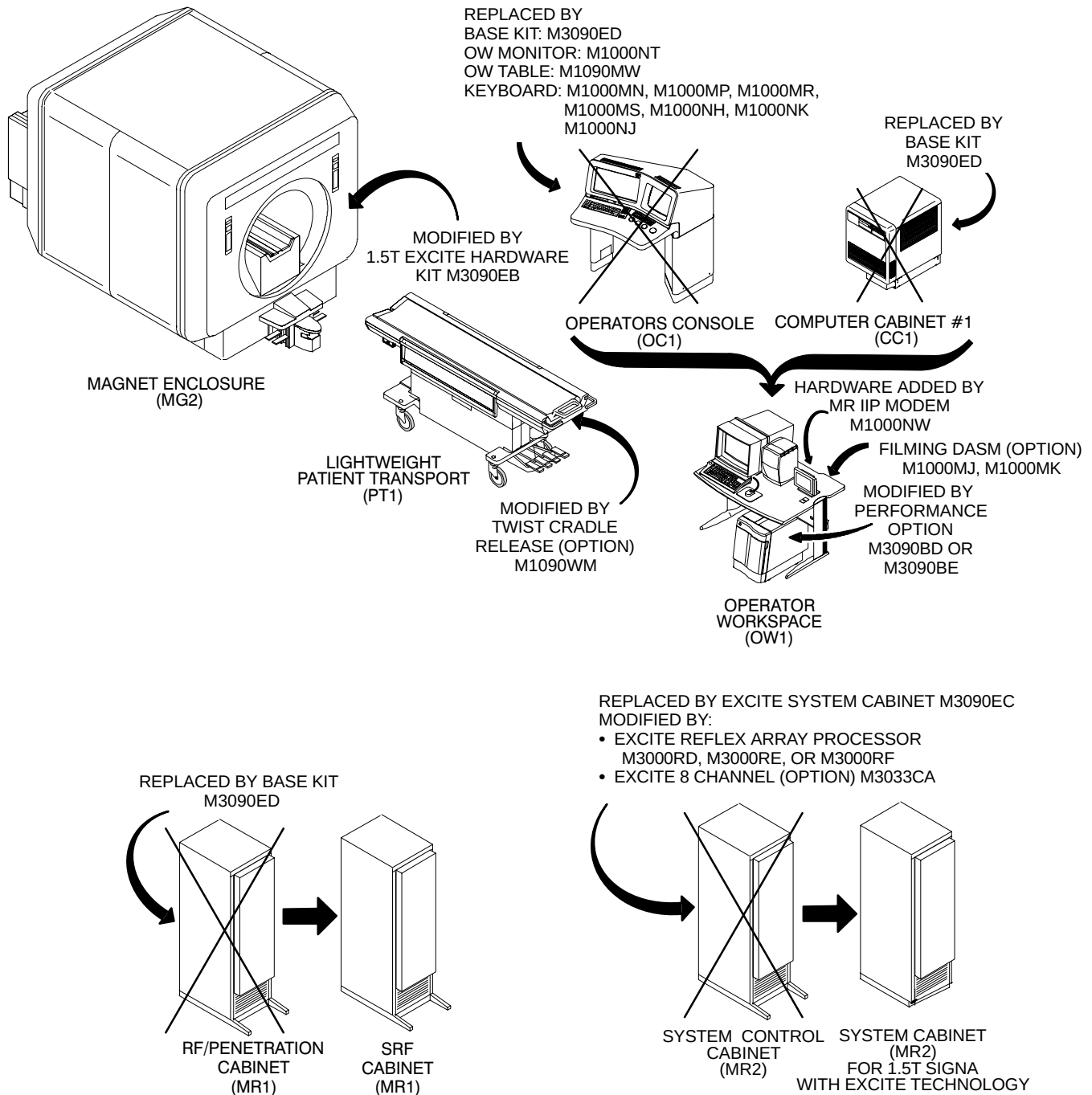
Site considering a Magnet upgrade must pass ALL siting requirements contained in *Direction 2246012 Pre-Sale/Site Survey For Upgrade to CXK4 Magnet With WideOpen Enclosure* prior to order acceptance.



SIGNA HORIZON 5.X TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS

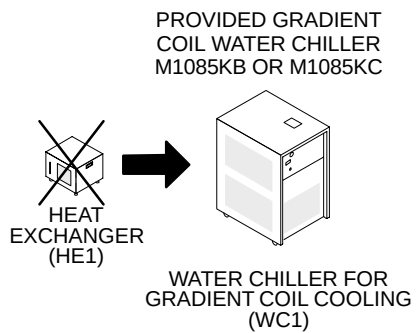
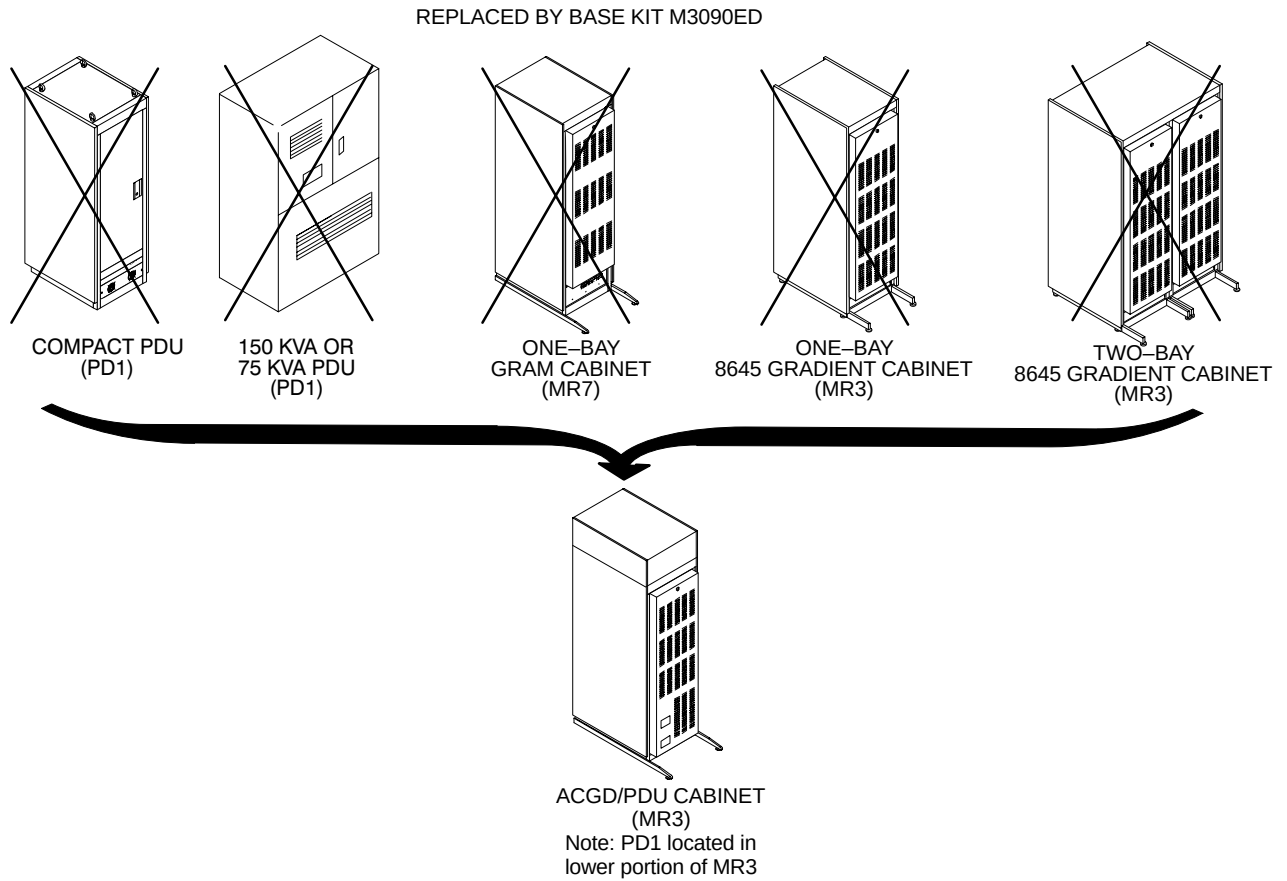
ILLUSTRATION 1-6

1-2-2 Signa Horizon 5.X System (Continued)



SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-8

1-2-2 Signa Horizon 5.X System (Continued)



SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-9

1-2-3 Signa Advantage 1.5T 5.X & 4.X

Illustrations 1-10 through 1-12 list the upgrade catalogs for 1.5T Signa Horizon 5.X software only system, 1.5T Signa Advantage 5.X hardware system, and 1.5T Signa 4.X system to 1.5T Signa LX with EXCITE Technology. Signa 5.1-5.4 and 4.X hardware systems are the following configurations:

- Signa Horizon 5.X Software only systems which have a 55 cm Body Coil
- Signa Advantage 5.X systems identified by the Genesis Computer, 1-bay System Cabinet, and 8607 Gradient Cabinet.
- Signa Advantage 4.X hardware systems are Signa Advantage 4.5 to 4.8 which can be identified by RF Cabinet model number 46-306900 and Signa Advantage 4.0 to pre-4.5 which can be identified by RF Cabinet model number 46-282900.

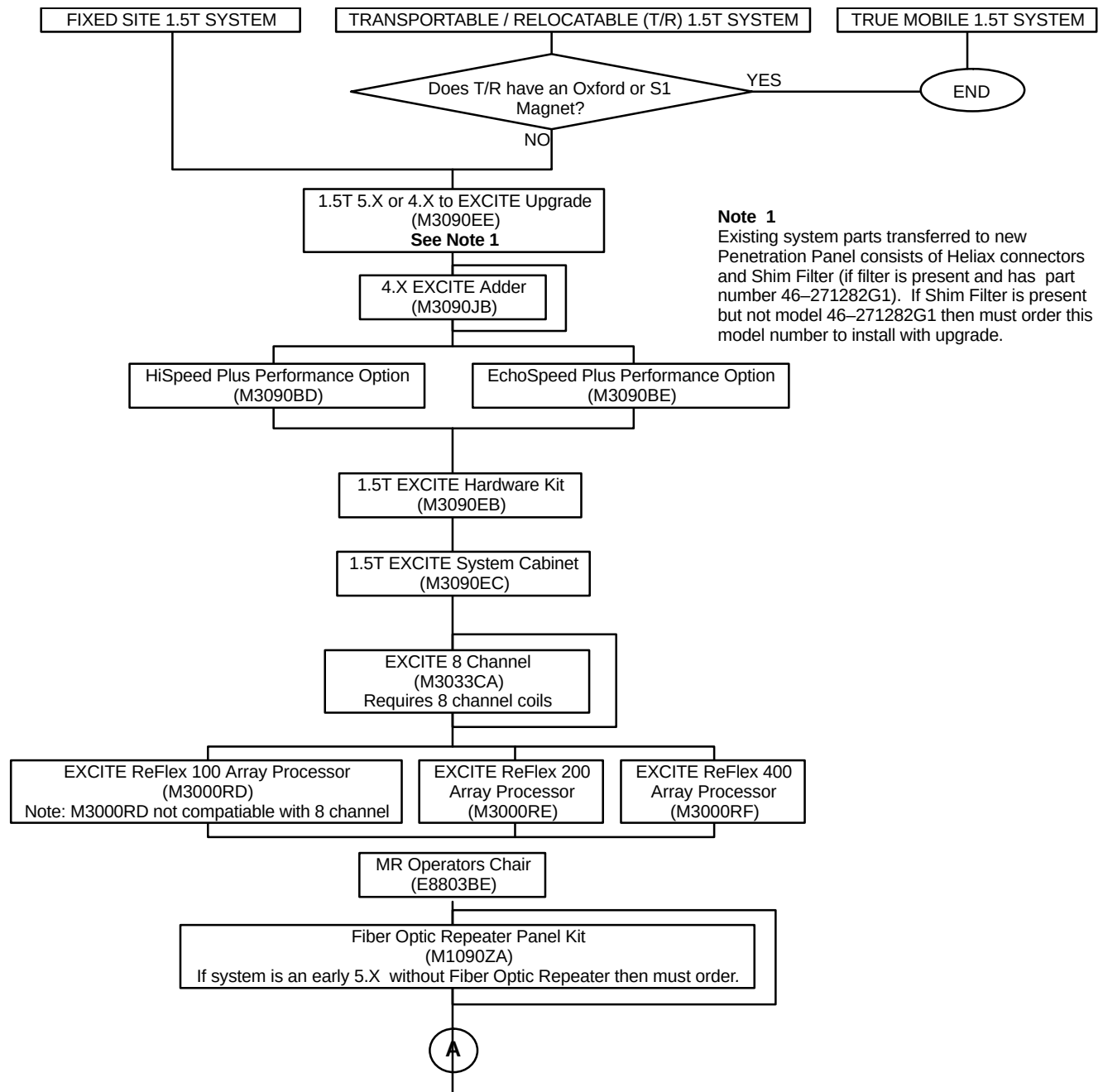
The flowchart shown in Illustrations 1-10 through 1-12 lists the Signa Horizon 5.X Software Only, Advantage 5.X and 4.X to 1.5T Signa LX with EXCITE Technolog upgrade catalogs. The equipment replaced and modified by the catalogs listed in the flowchart are shown in the illustrations as note below:

- Signa Horizon 1.5T 5.X Software Only, Advantage 5.X see Illustrations 1-13 and 1-14
- Signa Horizon 1.5T 4.X see Illustrations 1-15 and 1-16.

Note

Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

1-2-3 Signa Advantage 1.5T 5.X & 4.X (Continued)

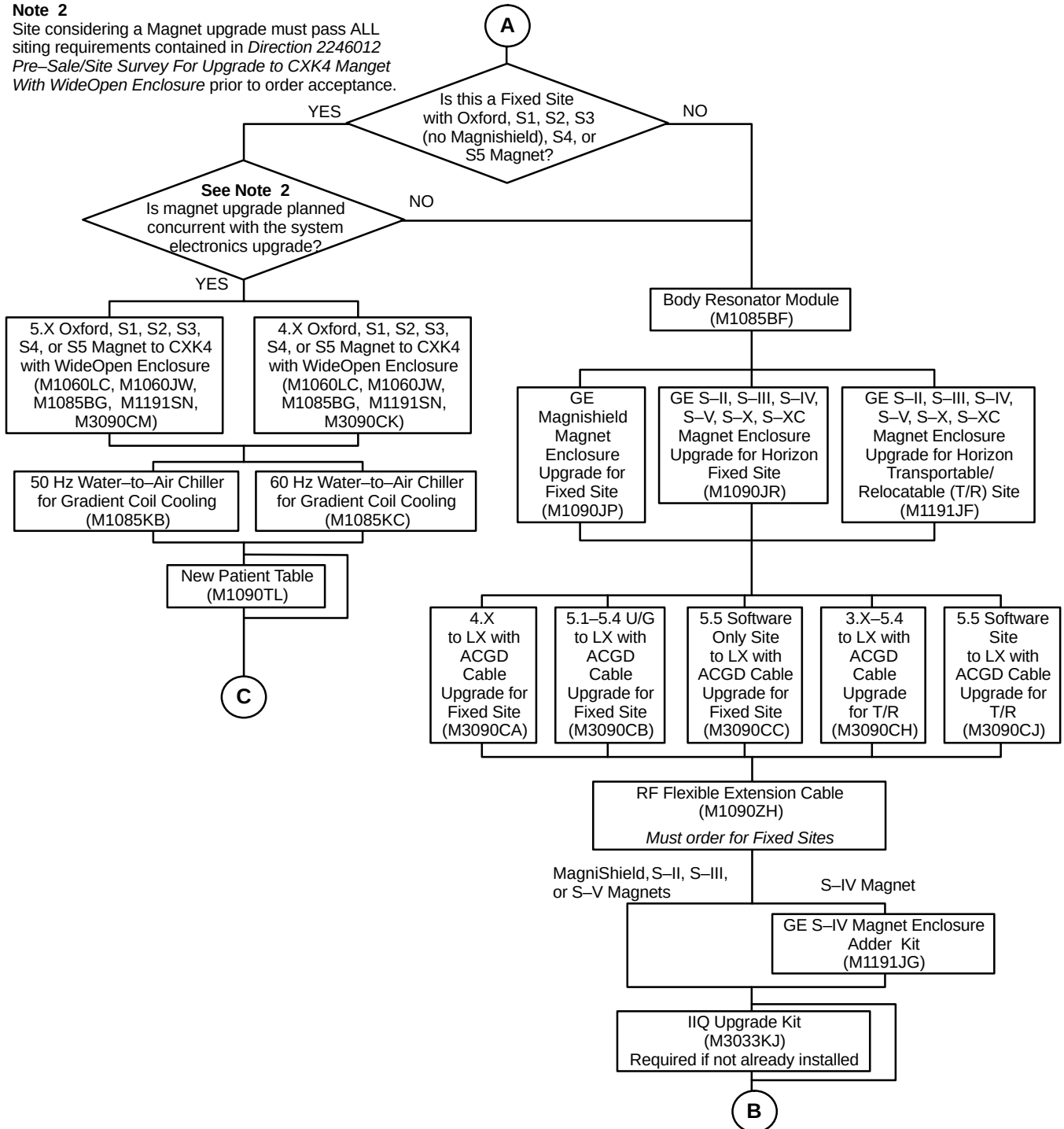


1.5T SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-10

1-2-3 Signa Advantage 1.5T 5.X & 4.X (Continued)

Note 2

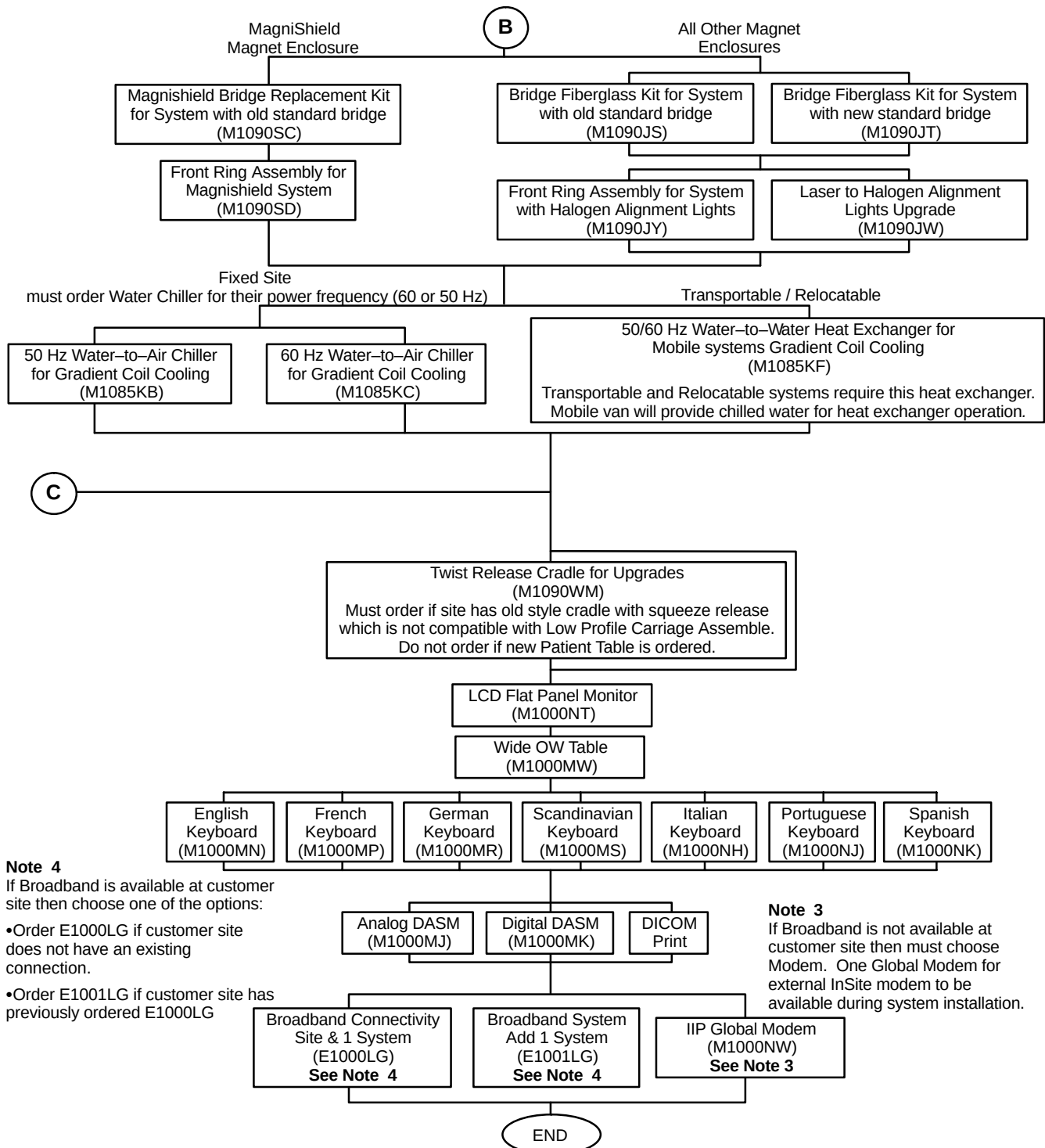
Site considering a Magnet upgrade must pass ALL siting requirements contained in *Direction 2246012 Pre-Sale/Site Survey For Upgrade to CXK4 Magnet With WideOpen Enclosure* prior to order acceptance.



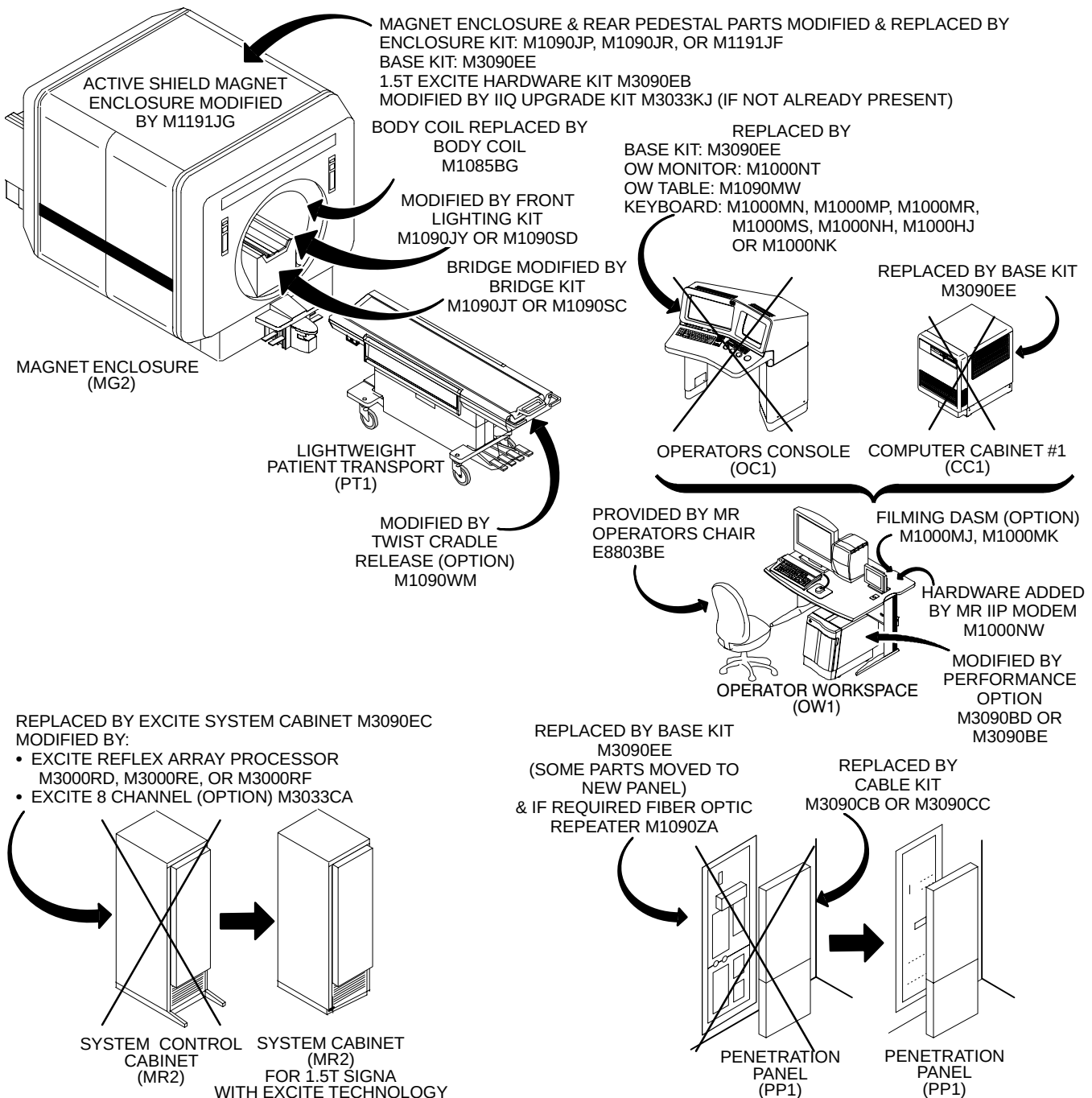
SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS

ILLUSTRATION 1-11

1-2-3 Signa Advantage 1.5T 5.X & 4.X (Continued)

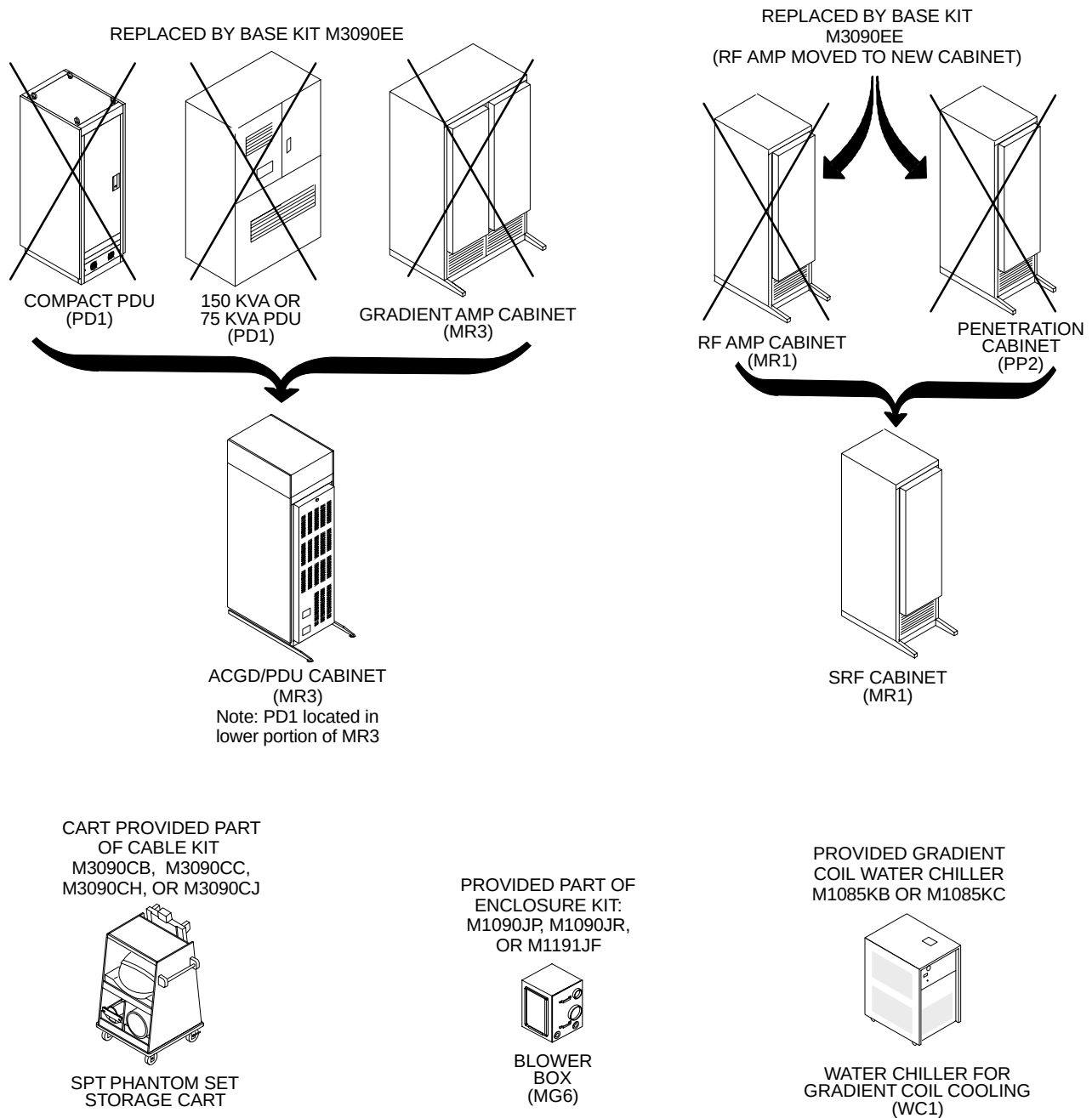


1-2-3 Signa Advantage 1.5T 5.X & 4.X (Continued)



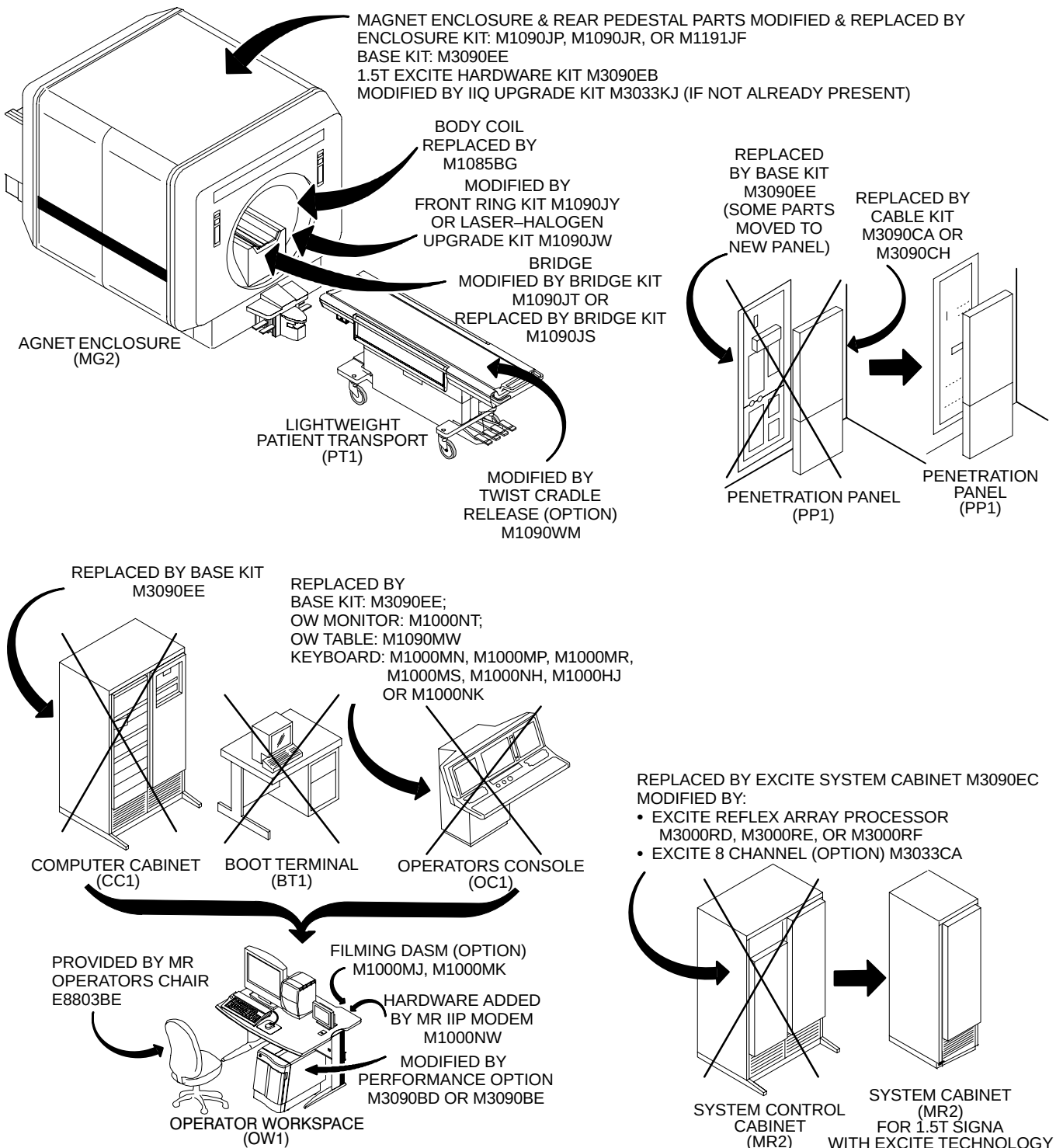
SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-13

1-2-3 Signa Advantage 1.5T 5.X & 4.X (Continued)



SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-14

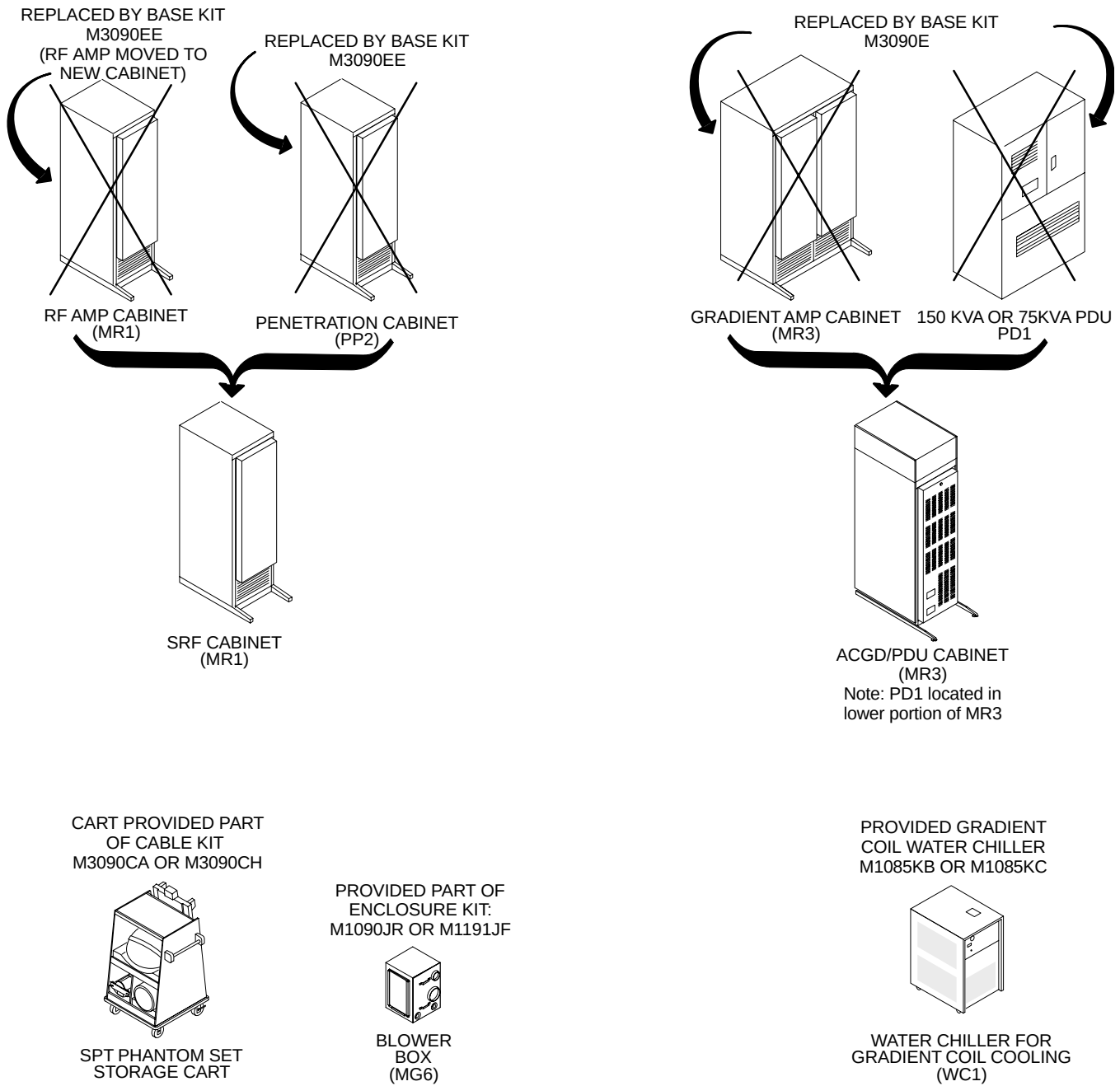
1-2-3 Signa Advantage 1.5T 5.X & 4.X (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-15

1-2-3 Signa Advantage 1.5T 5.X & 4.X (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-16

1-3 UPGRADES TO SIGNA LX WITH ACGD

Illustration 1-17 shows the system's major equipment after a Signa LX with ACGD upgrade is complete. The upgrade content is dependent on the configuration of the system being upgraded and the final Signa LX with ACGD product configuration. The following sub-sections provide an overview of the various Signa LX with ACGD upgrade hardware.



WARNING!

ALL SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE FACILITY TRANSFORMER AND FEEDER WIRES ARE CORRECTLY SIZED. ALSO THE MAIN DISCONNECT CONTROL NEEDS TO BE INVESTIGATED FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS. REFER SECTION 5 – POWER REQUIREMENTS.

SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS. REFER SECTION 5 – POWER REQUIREMENTS.

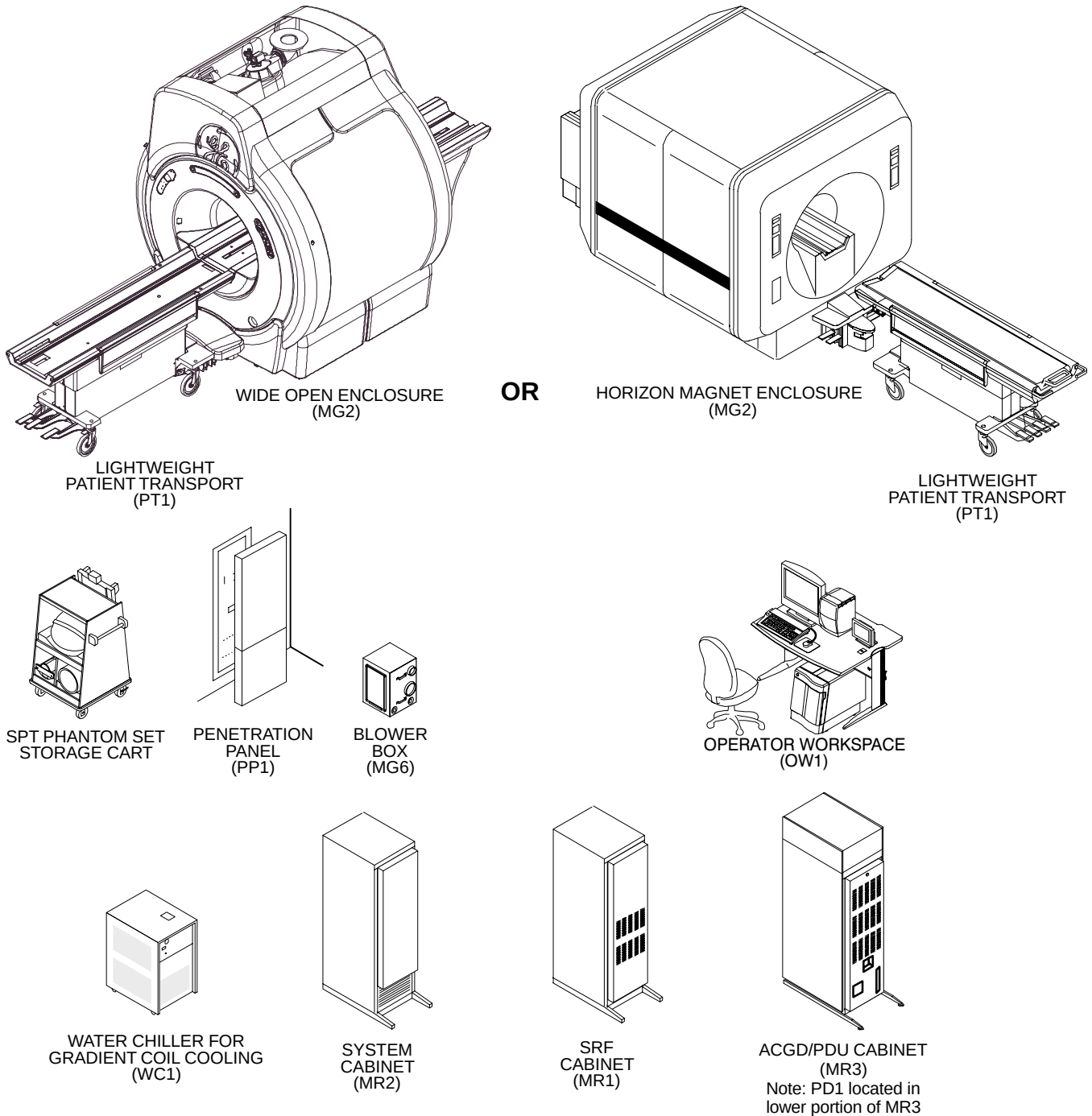
Note

Only catalogs which include hardware are documented in this publication. Software only catalogs are not included, existing system site parameters apply for software only upgrades.

Note

FOR USA, FIXED SITE ONLY CONSIDERING MAGNET UPGRADE: Refer to *Direction 22461012 Pre-Sales/Site Survey For Upgrade To CXK4 Magnet With WideOpen Enclosure* for the many environmental factors and logistics that need to be understood to qualify a site for a magnet upgrade. The Magnet Upgrade Option requires the MR system to be operating a Release 8.X prior or coincident with the Magnet Upgrade option.

1-3 UPGRADES TO SIGNA LX WITH ACGD (Continued)



SIGNA LX WITH ACGD SYSTEM MAJOR EQUIPMENT
ILLUSTRATION 1-17

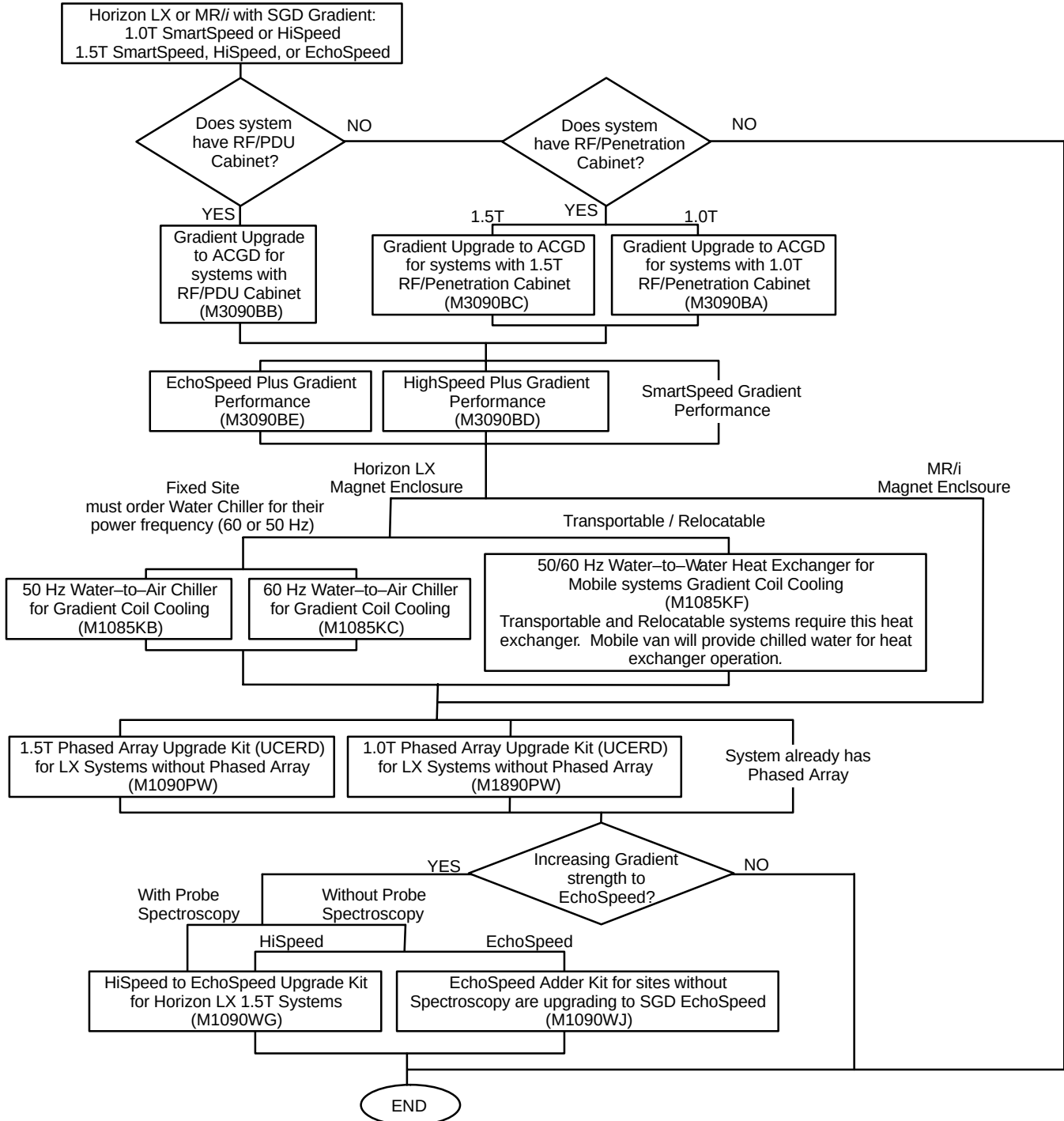
1-3-1 Signa Horizon LX With SGD & Signa MR/i System

■ Illustration 1-18 lists the upgrade catalogs for Signa Horizon LX with SGD and Signa MR/i with SGD system to Signa LX with ACGD. Illustration 1-19 shows the cabinets replaced and modified by the listed catalogs.

Note

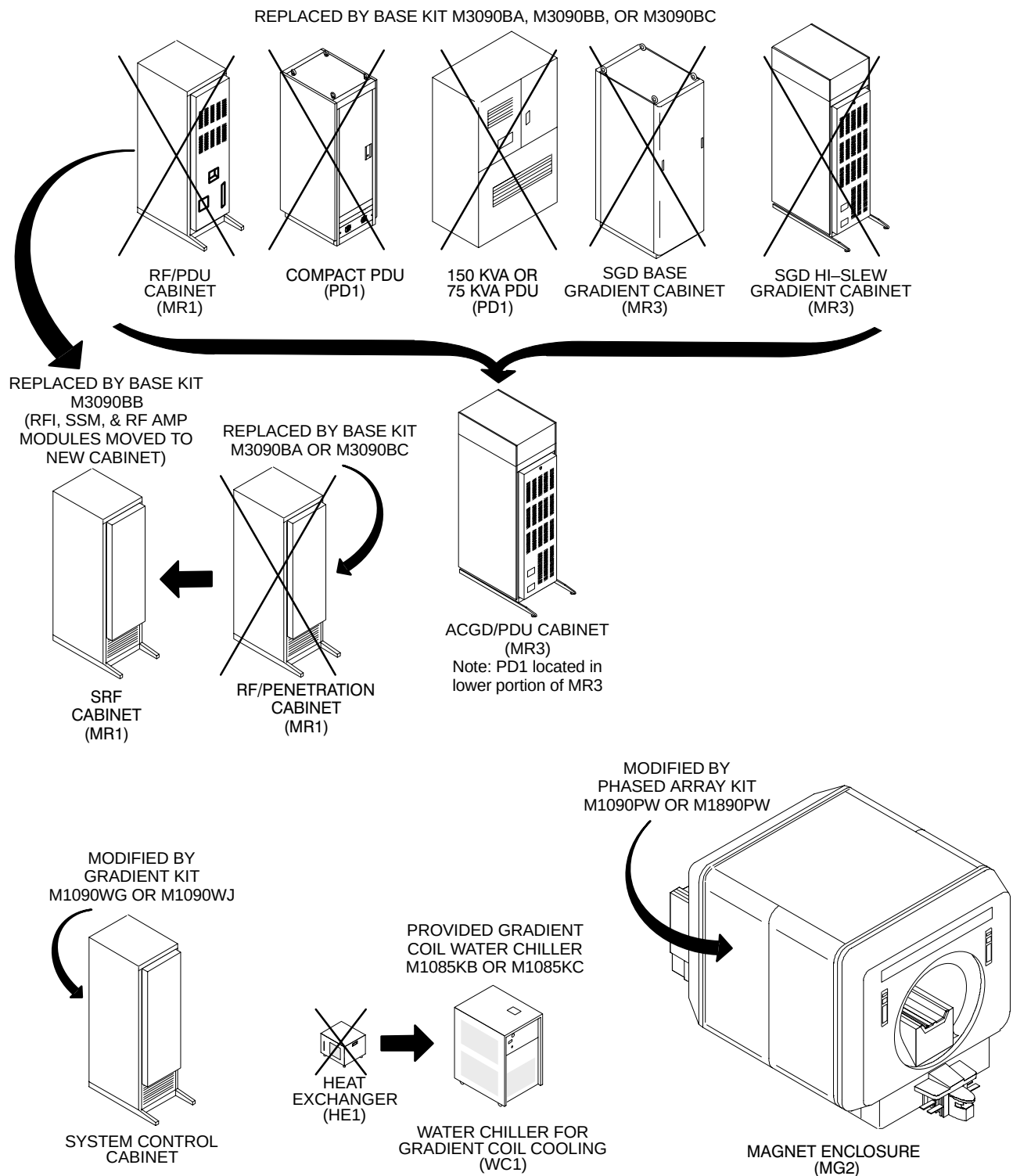
■ Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

1-3-1 Signa Horizon LX With SGD & Signa MR/i System (Continued)



1.5T SIGNA LX OR SIGNA MR/i WITH SGD GRADIENTS TO SIGNA LX WITH ACGD CATALOGS

1-3-1 Signa Horizon LX With SGD & Signa MR/i System (Continued)



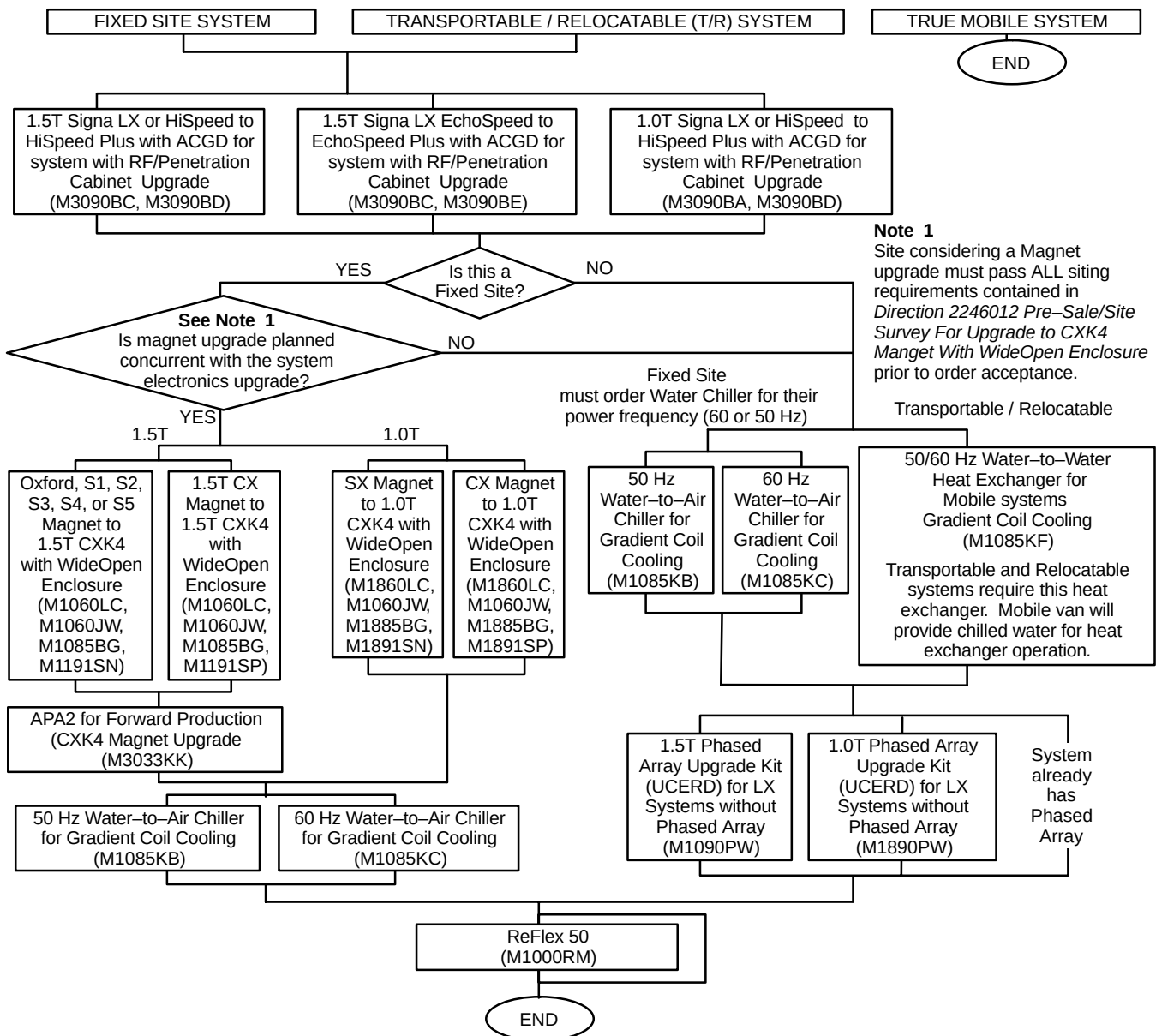
SIGNA HORIZON LX WITH SGD & SIGNA MR// SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-19

1-3-2 Signa Horizon LX With 8645 Gradients System

Illustration 1-20 lists the upgrade catalogs for Signa Horizon LX system with 8645 Gradients to Signa Horizon LX with ACGD. Illustration 1-21 shows the cabinets replaced and modified by the listed catalogs.

Note

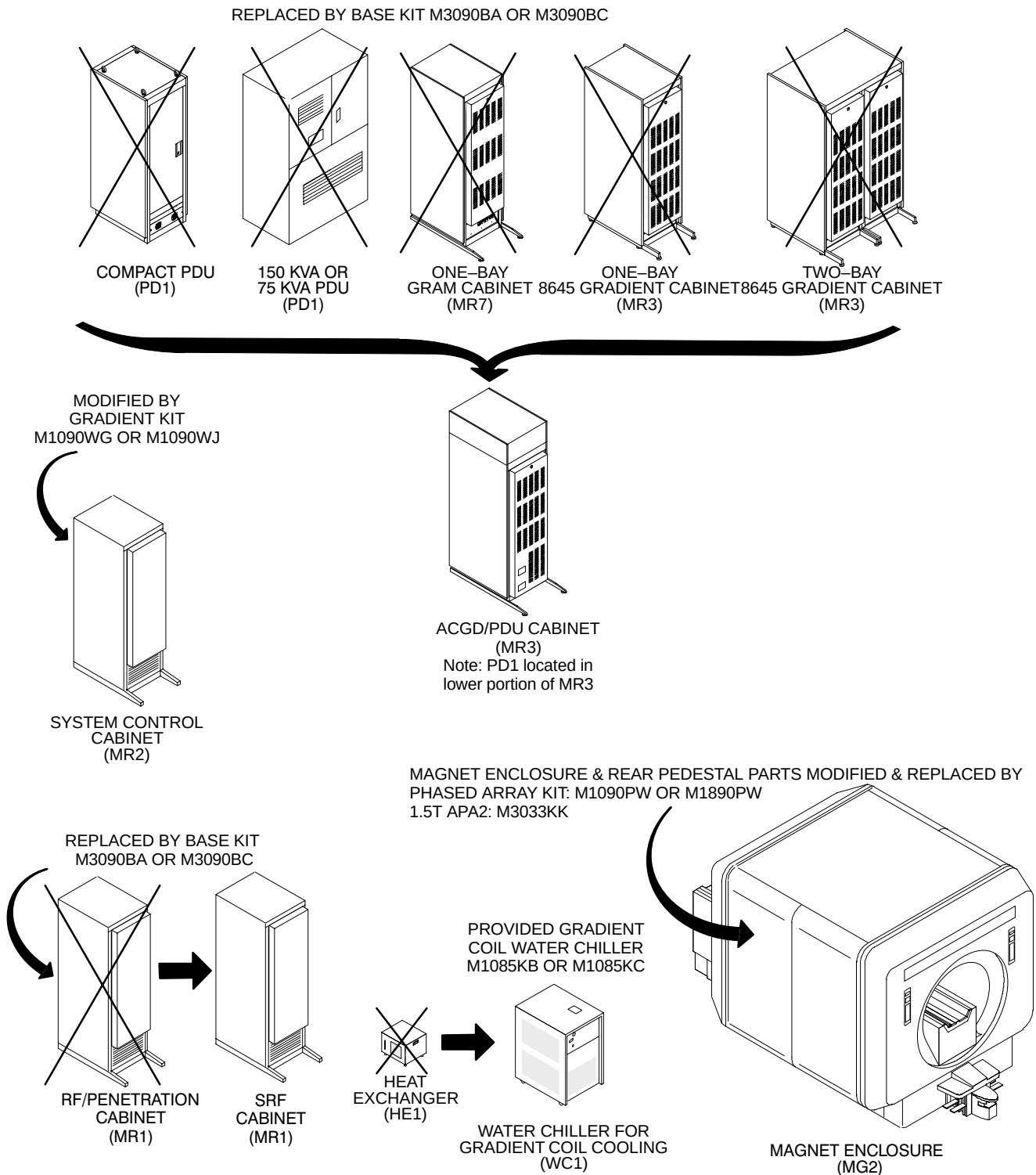
Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH ACGD CATALOGS

ILLUSTRATION 1-20

1-3-2 Signa Horizon LX With 8645 Gradients System (Continued)



SIGNA HORIZON LX SYSTEM UPGRADE MAJOR EQUIPMENT REPLACED & MODIFIED
ILLUSTRATION 1-21

1-3-3 Signa Horizon 5.X System

■ Illustrations 1-22 through 1-24 list the upgrade catalogs for Signa Horizon 5.X system to Signa Horizon LX with ACGD. Illustrations 1-25 and 1-26 show the cabinets replaced or modified and new equipment added by the listed catalogs.

Refer to Section 1-4 SIGNA HORIZON LX SYSTEM OPTIONS for additional option catalogs.

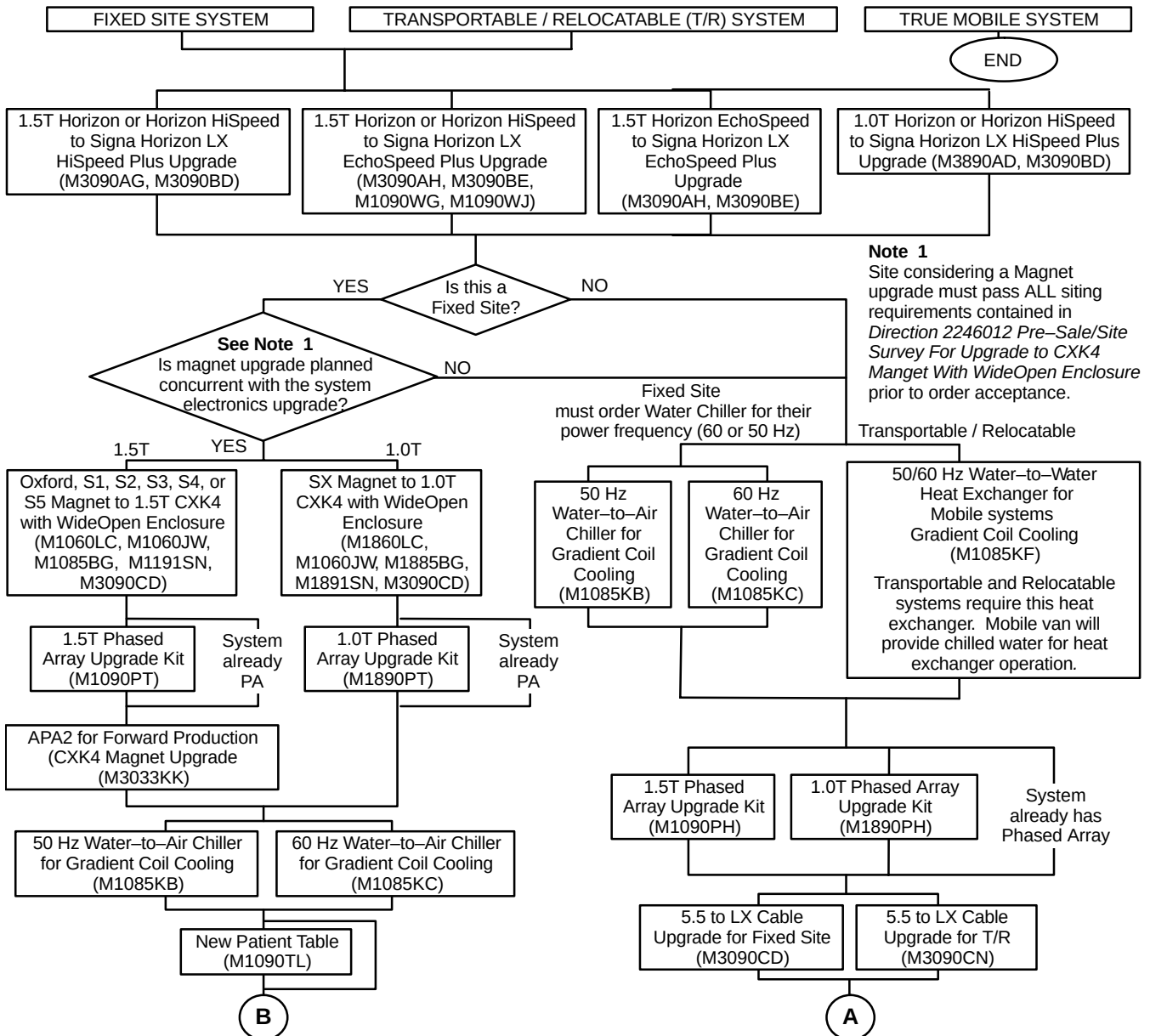
Note

Signa Horizon 5.5 Software only systems have a 55 cm Body Coil and require upgrade catalogs defined in Section 1-3-4, SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM UPGRADES.

Note

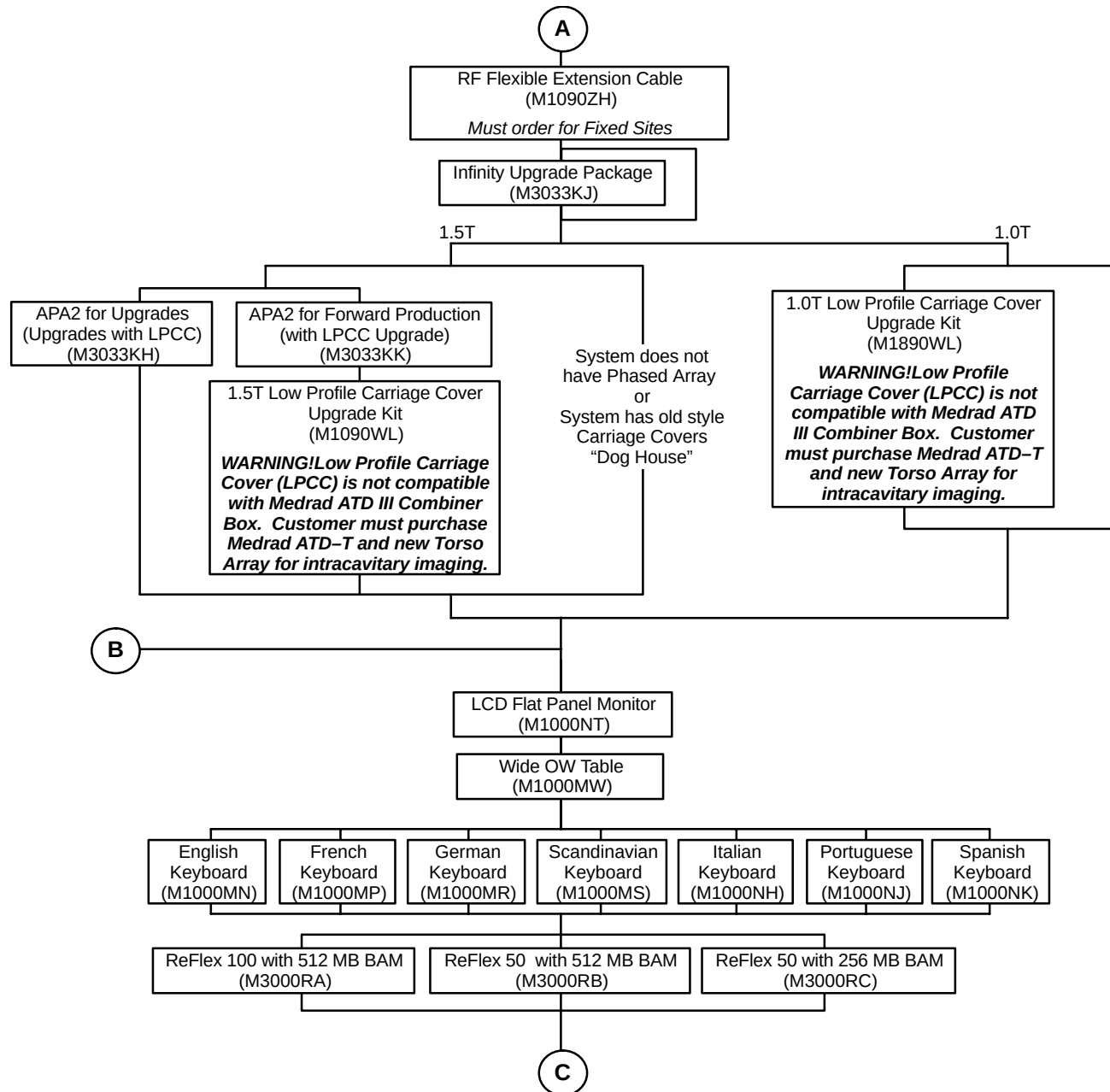
Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

1-3-3 Signa Horizon 5.X System (Continued)



1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1-22

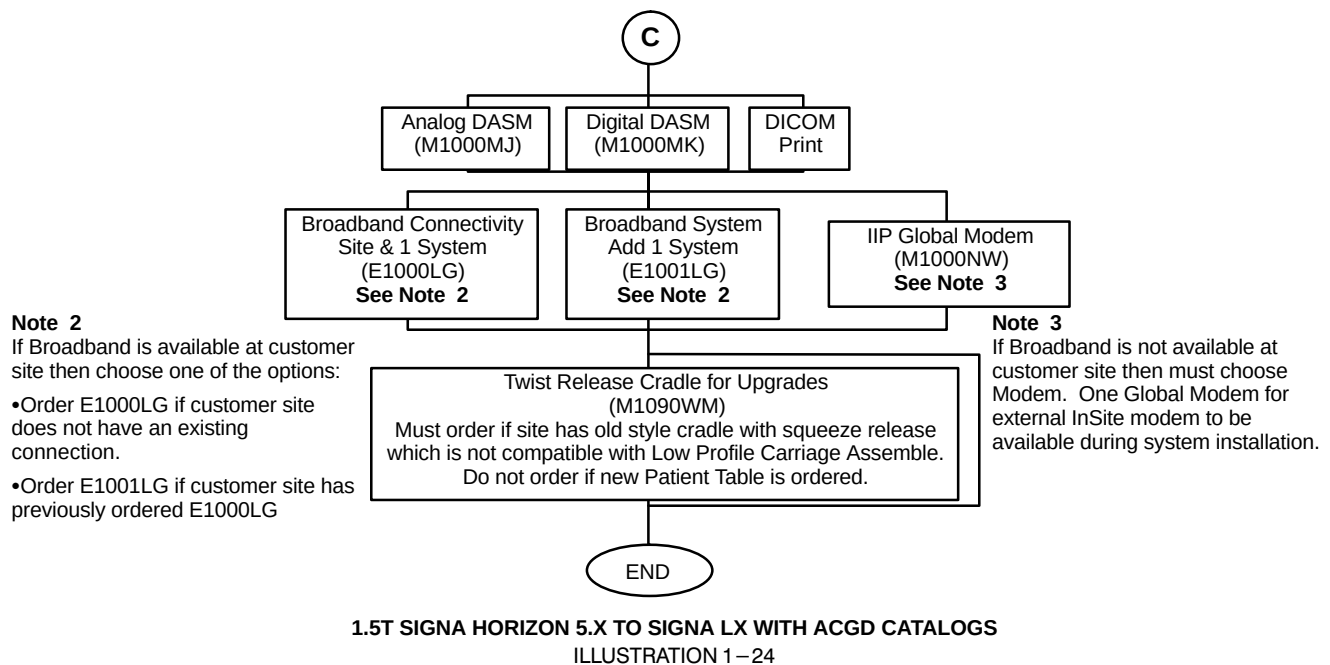
1-3-3 Signa Horizon 5.X System (Continued)



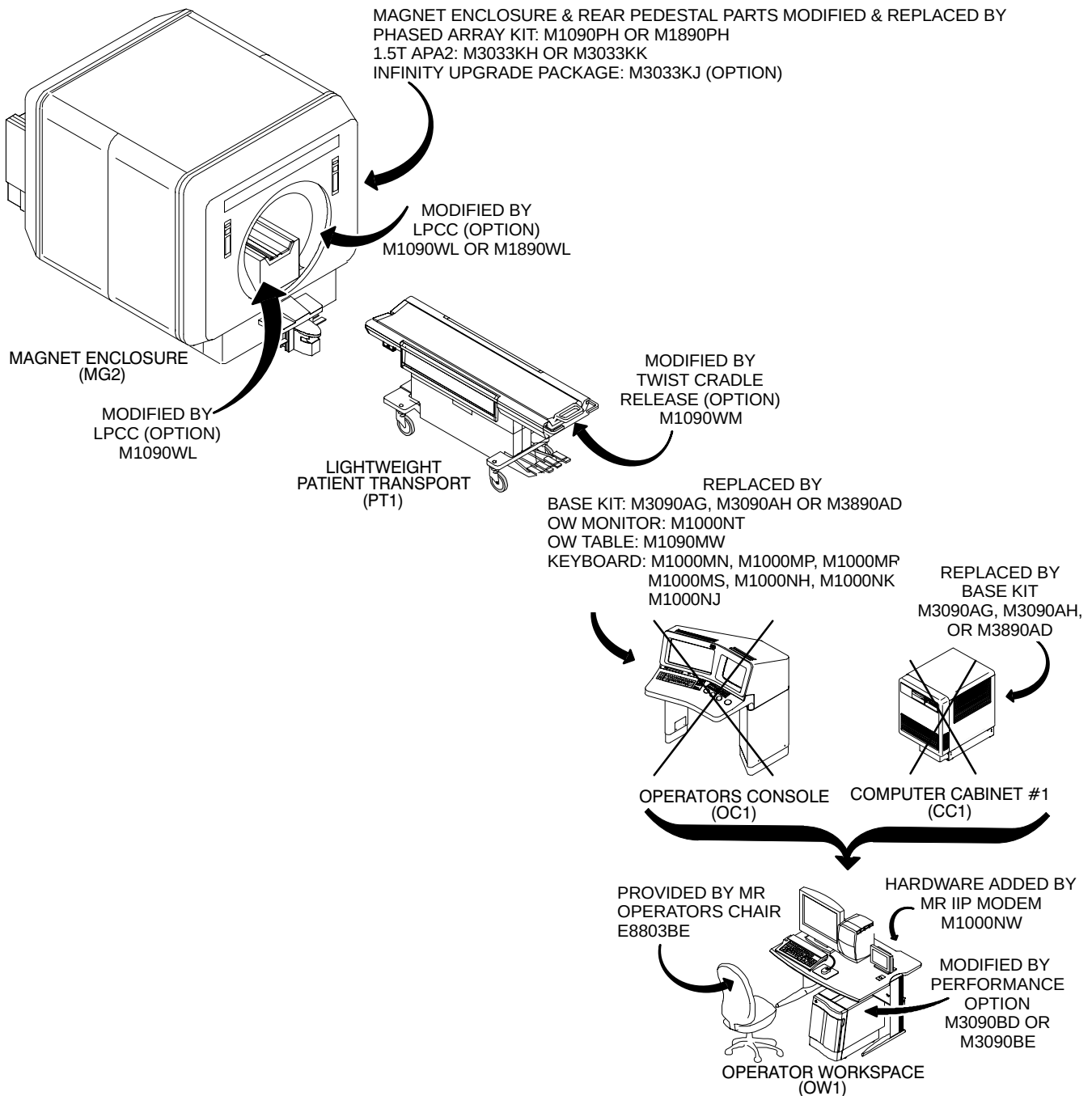
1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH ACGD CATALOGS

ILLUSTRATION 1-23

■ 1-3-3 Signa Horizon 5.X System (Continued)

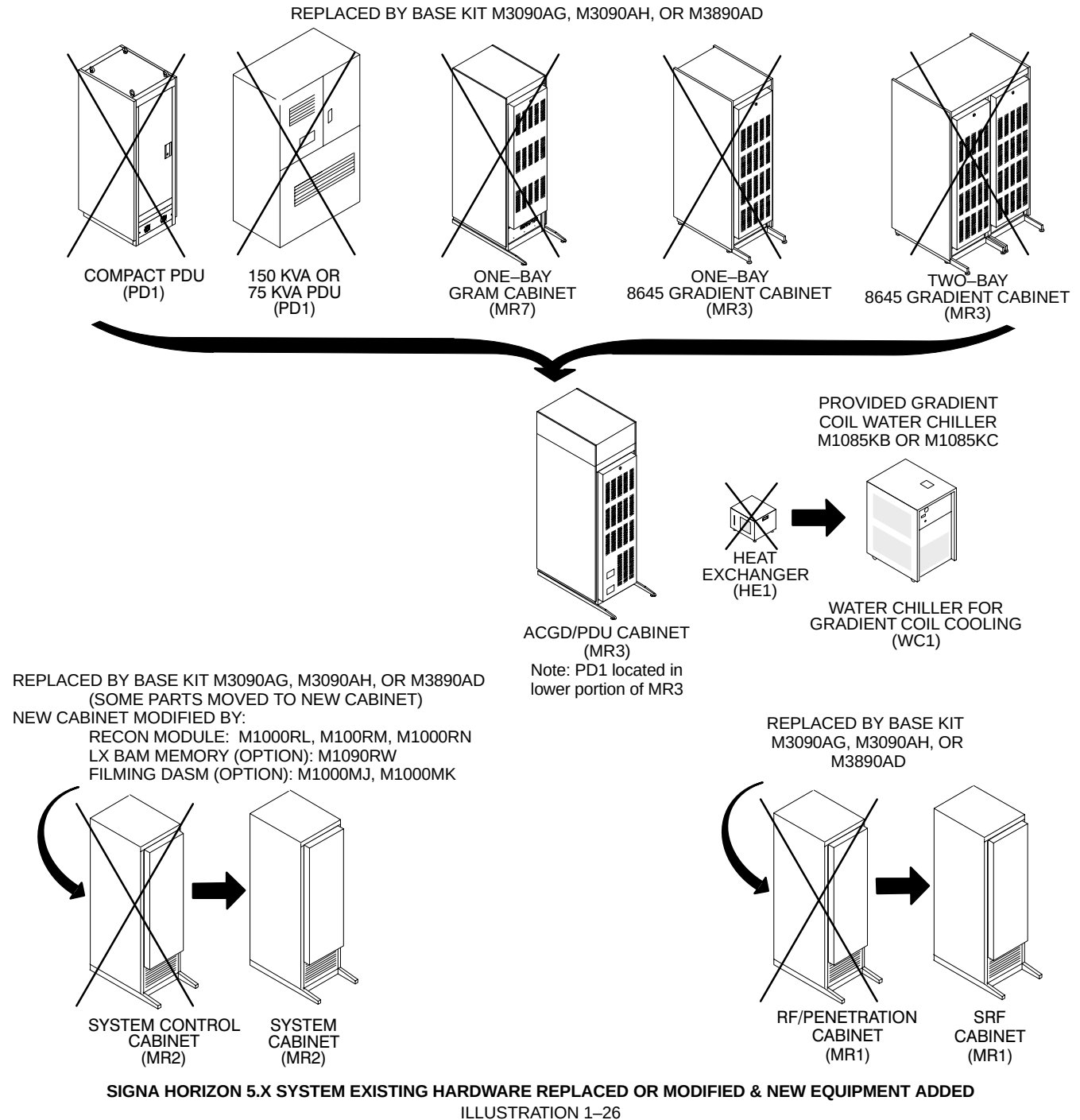


1-3-3 Signa Horizon 5.X System (Continued)



SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-25

1-3-3 Signa Horizon 5.X System (Continued)



1–3–4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System

Illustrations 1–27 through 1–29 lists the upgrade catalogs for Signa Advantage 5.X hardware system and Signa Horizon 5.X software only system to Signa LX with ACGD. Illustrations 1–30 and 1–31 show the cabinets replaced or modified and new equipment added by the listed catalogs. Signa 5.1–5.4 hardware systems are the following configurations:

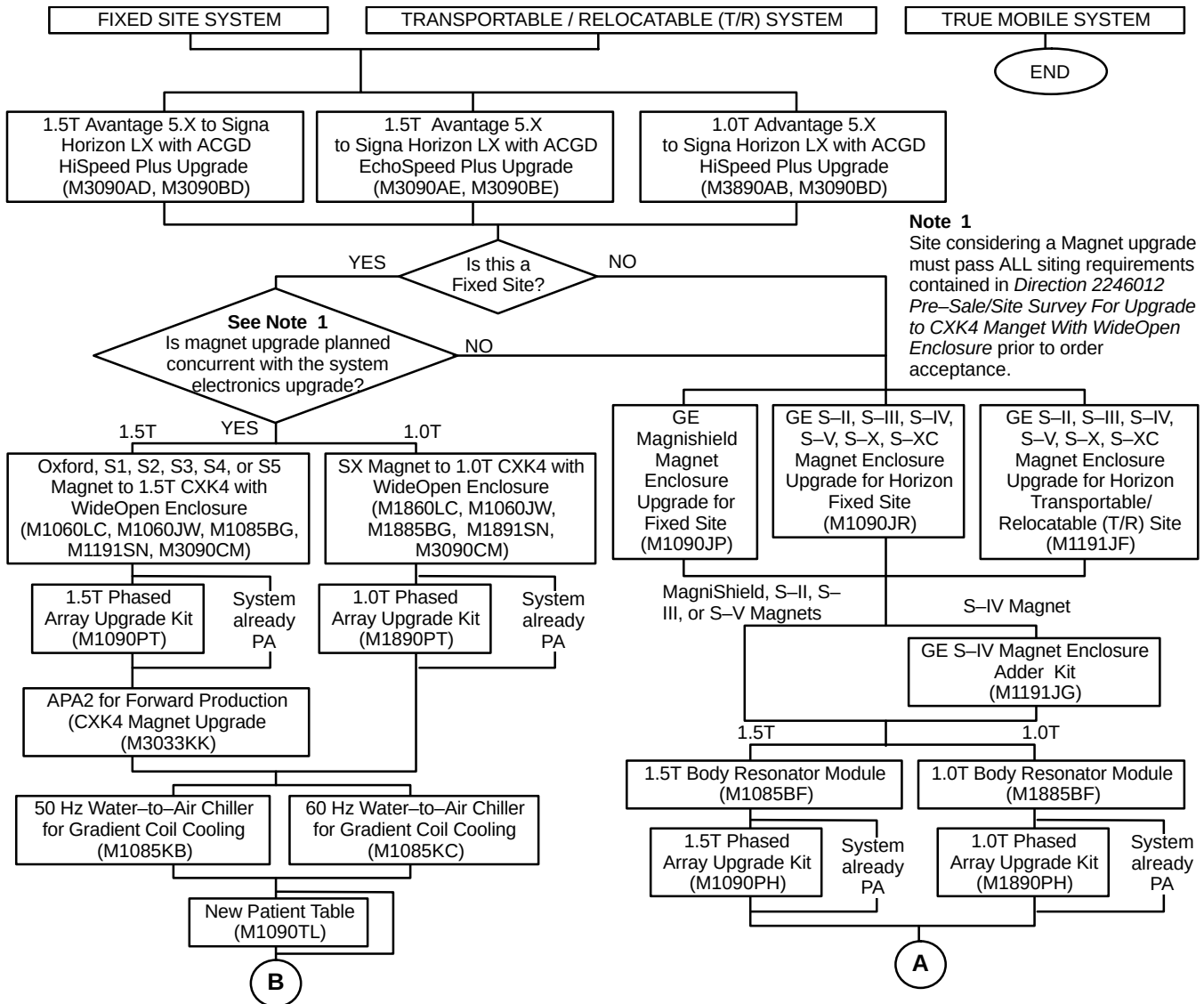
- Signa Horizon 5.X Software only systems which have a 55 cm Body Coil
- Signa Advantage 5.X systems identified by the Genesis Computer, 1-bay System Cabinet, and 8607 Gradient Cabinet.

Refer to Section 1–4 SIGNA HORIZON LX SYSTEM OPTIONS for additional option catalogs.

Note

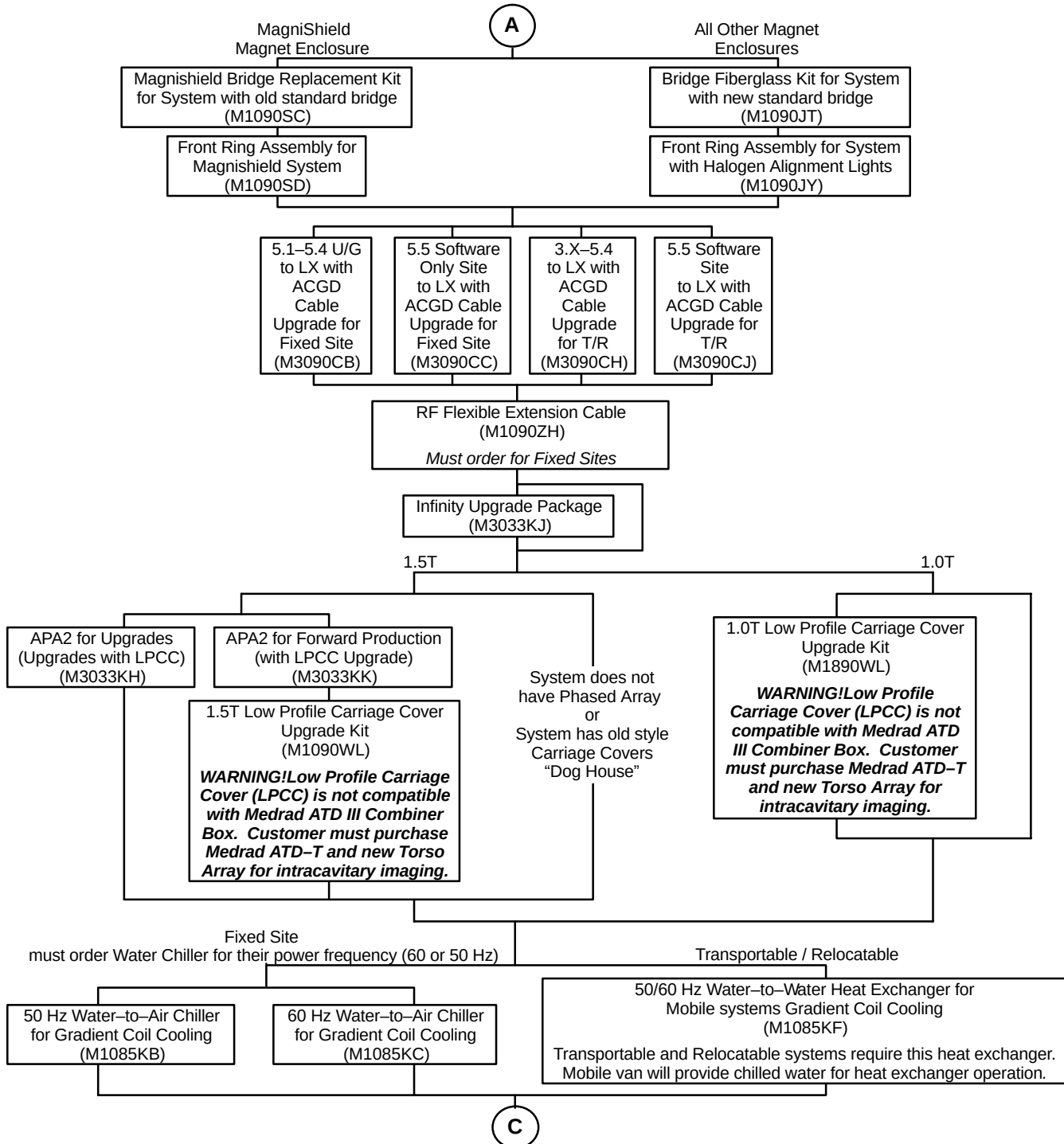
Systems with Oxford or S–1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

1-3-4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System – (Continued)



SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1-27

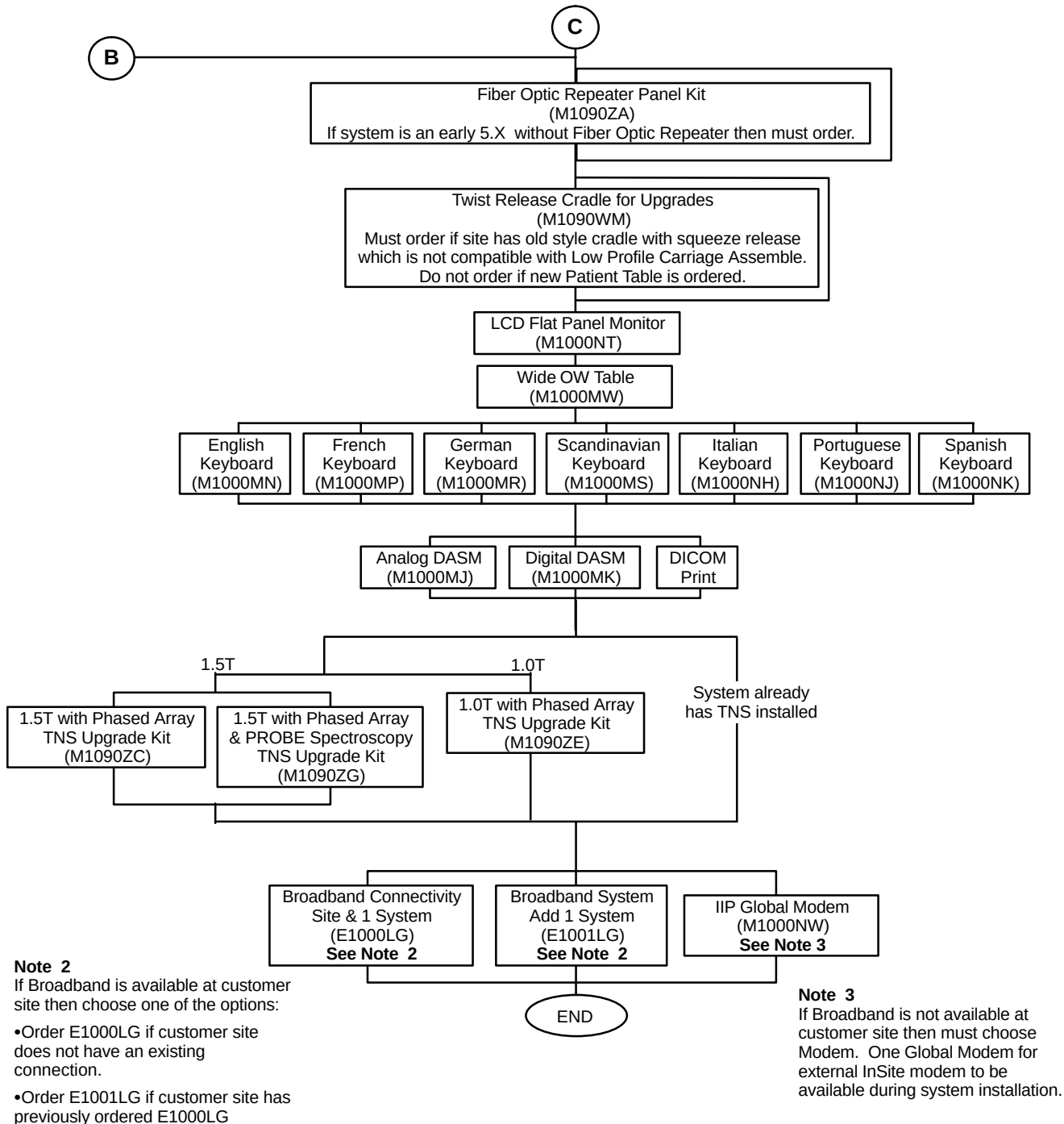
1-3-4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System – (Continued)



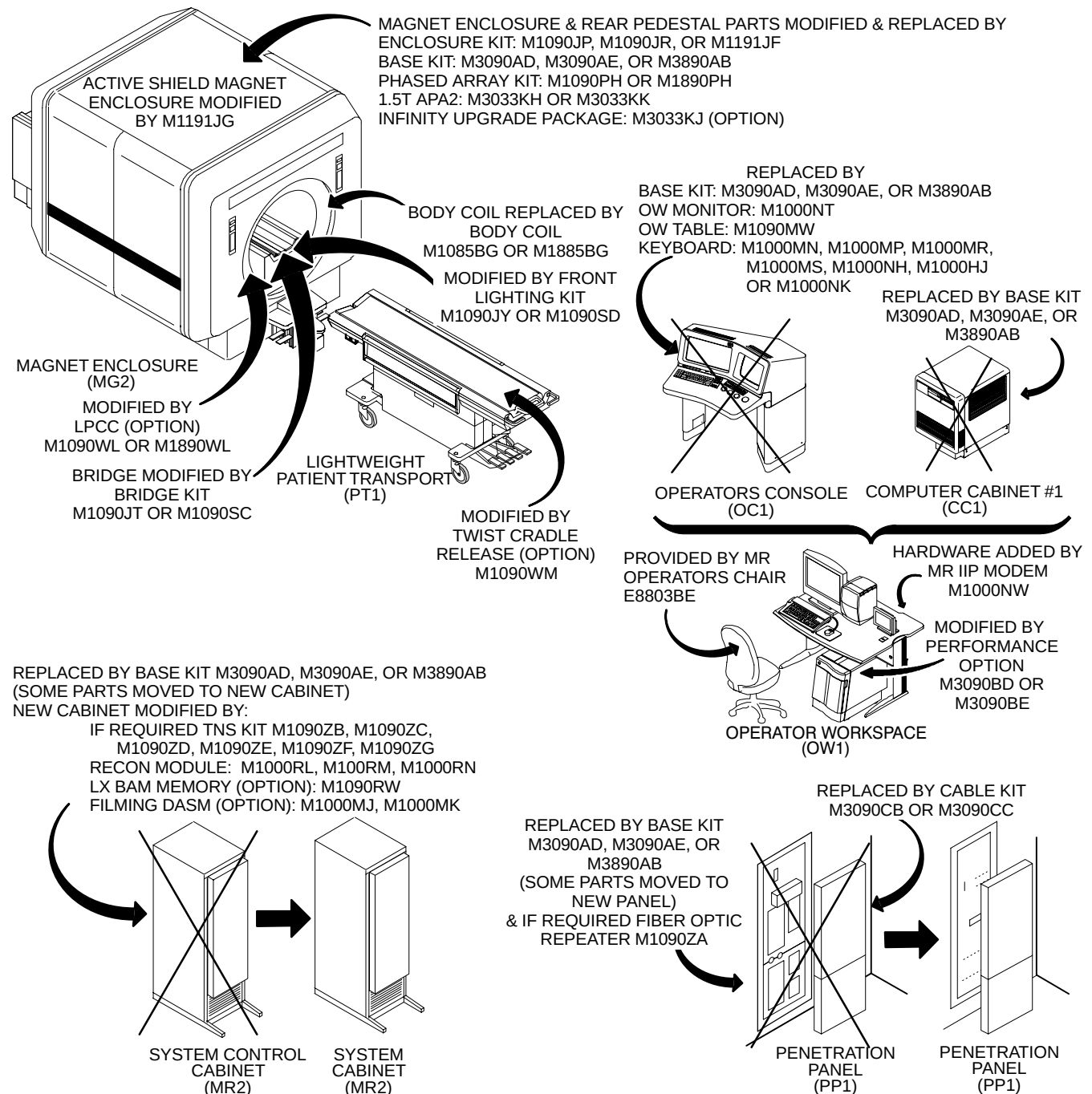
SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X TO SIGNA LX WITH ACGD CATALOGS

ILLUSTRATION 1-28

1-3-4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System – (Continued)



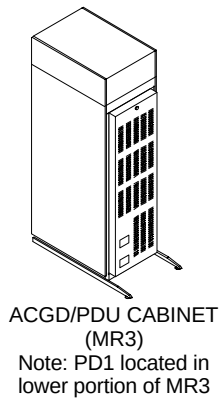
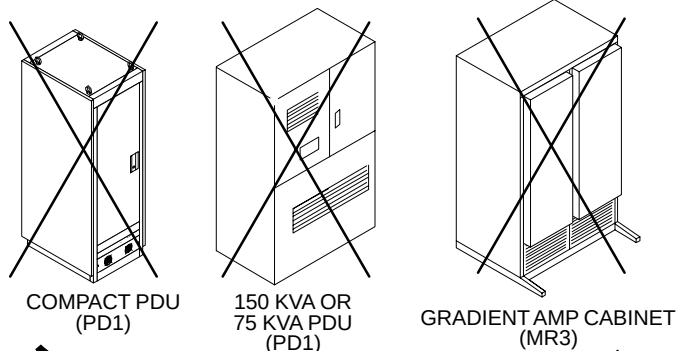
1-3-4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System – (Continued)



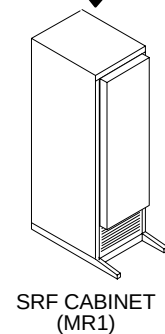
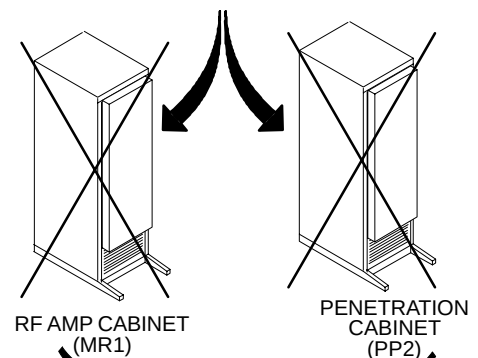
SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-30

1-3-4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System – (Continued)

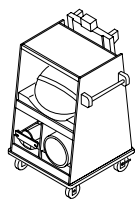
REPLACED BY BASE KIT M3090AD, M3090AE, OR M3809AB



REPLACED BY BASE KIT M3090AD, M3090AE, OR M3890AB (RF AMP MOVED TO NEW CABINET)



CART PROVIDED PART OF CABLE KIT M3090CB, M3090CC, M3090CH, OR M3090CJ



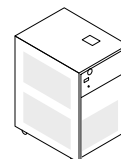
SPT PHANTOM SET STORAGE CART

PROVIDED PART OF ENCLOSURE KIT: M1090JP, M1090JR, OR M1191JF



BLOWER BOX (MG6)

PROVIDED GRADIENT COIL WATER CHILLER M1085KB OR M1085KC



WATER CHILLER FOR GRADIENT COIL COOLING (WC1)

SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-31

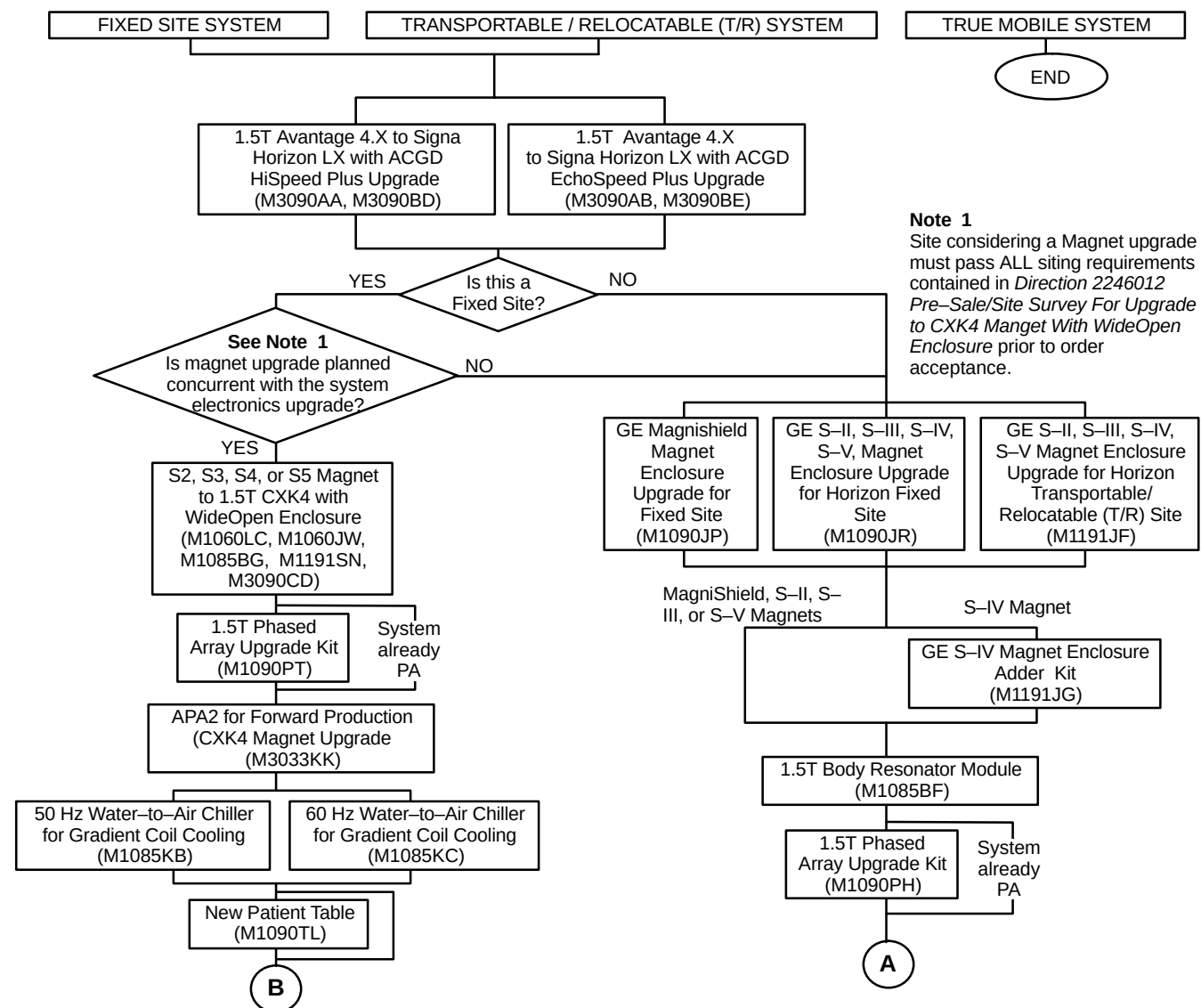
1-3-5 Signa Advantage 4.X System

Illustrations 1-32 through 1-34 lists the upgrade catalogs for Signa Advantage 4.X to Signa Horizon LX with ACGD. Signa Advantage 4.X hardware systems are Signa Advantage 4.5 to 4.8 which can be identified by RF Cabinet model number 46-306900 and Signa Advantage 4.0 to pre-4.5 which can be identified by RF Cabinet model number 46-282900. Illustrations 1-35 and 1-36 show the major equipment replaced or modified and new equipment added by the listed catalogs.

Refer to Section 1-4 SIGNA HORIZON LX SYSTEM OPTIONS for options catalogs.

Note

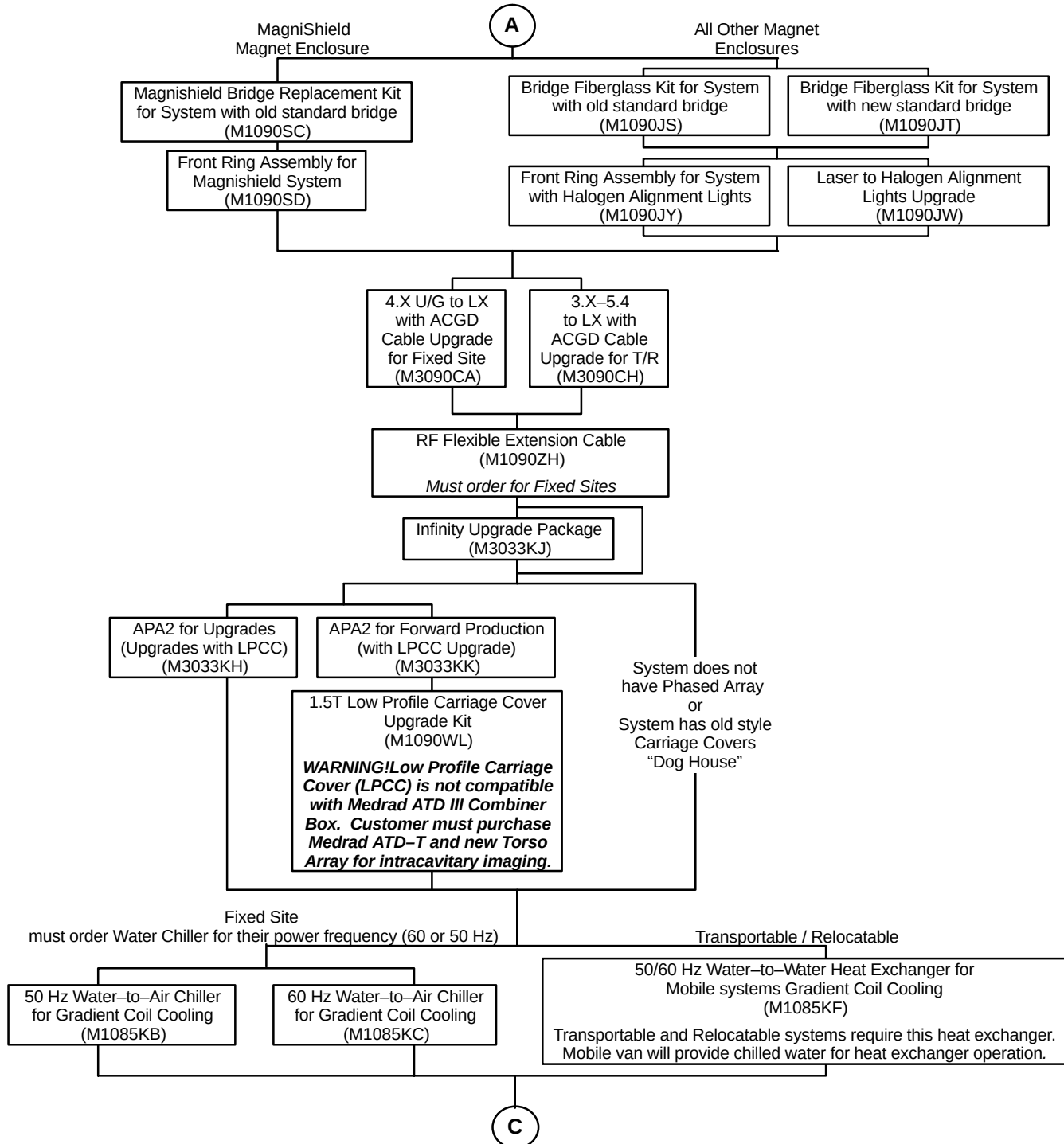
Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



SIGNA ADVANTAGE 4.X TO SIGNA LX WITH ACGD CATALOGS

ILLUSTRATION 1-32

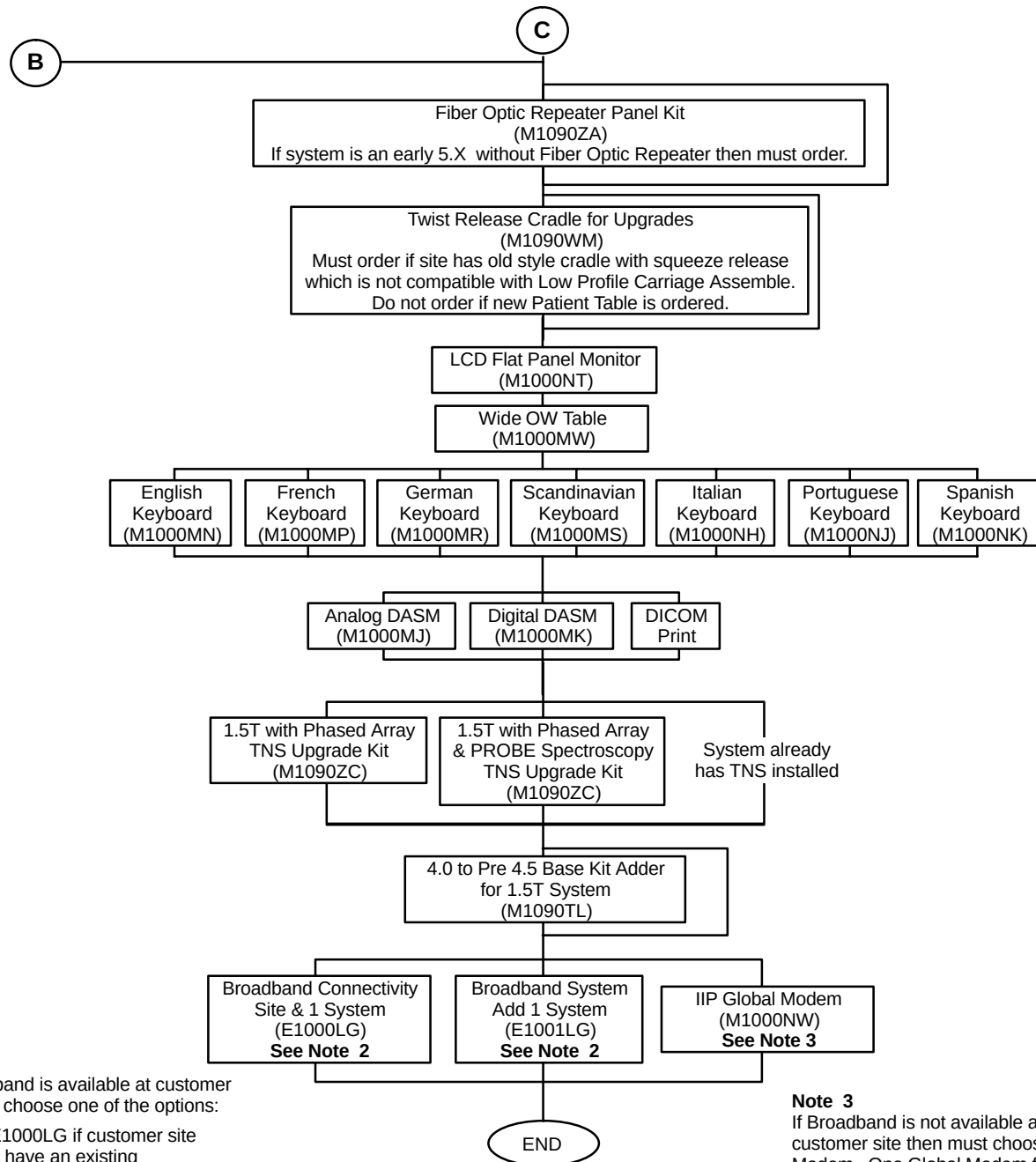
1-3-5 Signa Advantage 4.X System – (Continued)



SIGNA ADVANTAGE 4.X TO SIGNA LX WITH ACGD CATALOGS

ILLUSTRATION 1-33

1-3-5 Signa Advantage 4.X System – (Continued)



Note 2

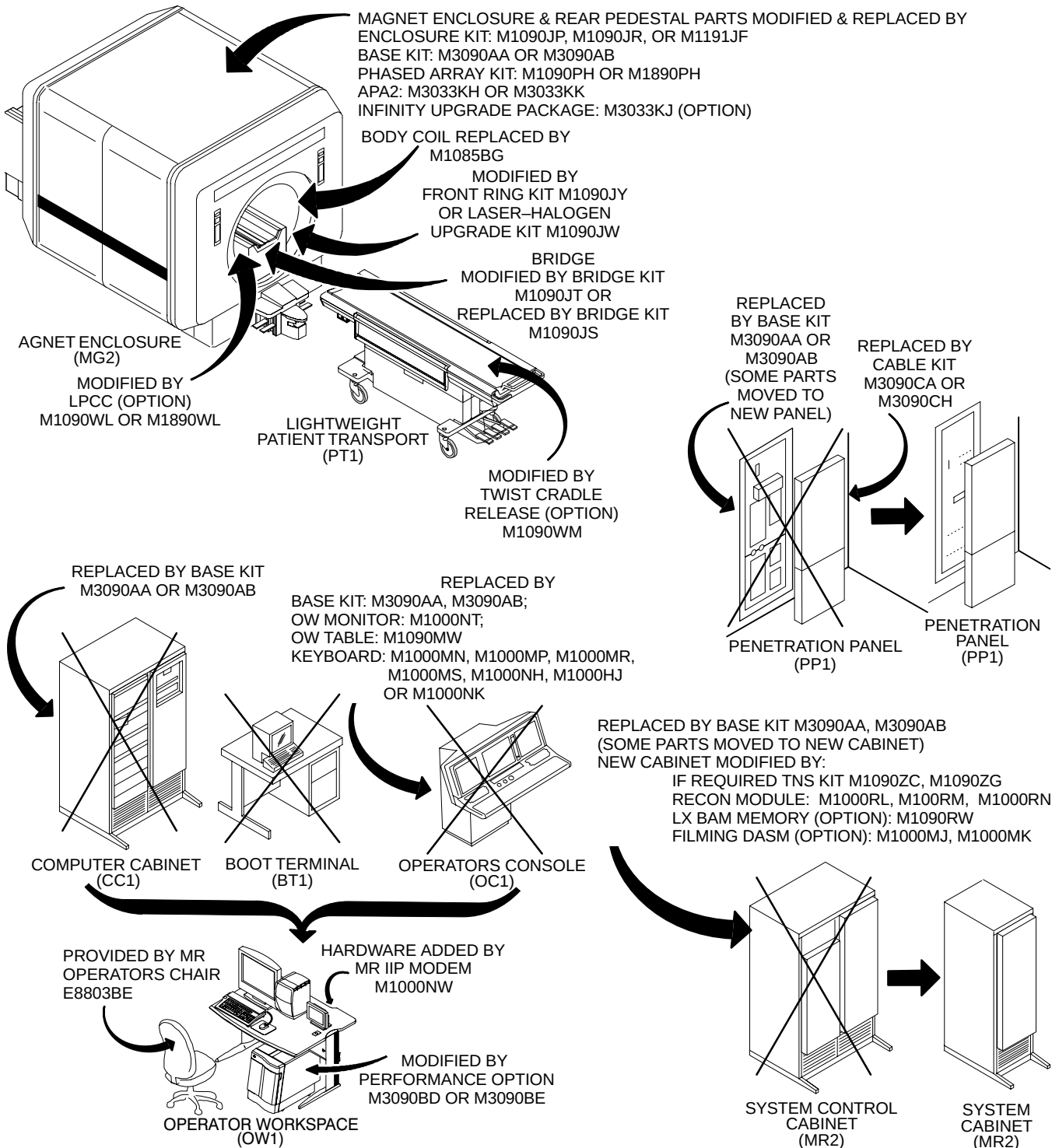
If Broadband is available at customer site then choose one of the options:

- Order E1000LG if customer site does not have an existing connection.
- Order E1001LG if customer site has previously ordered E1000LG

Note 3

If Broadband is not available at customer site then must choose Modem. One Global Modem for external InSite modem to be available during system installation.

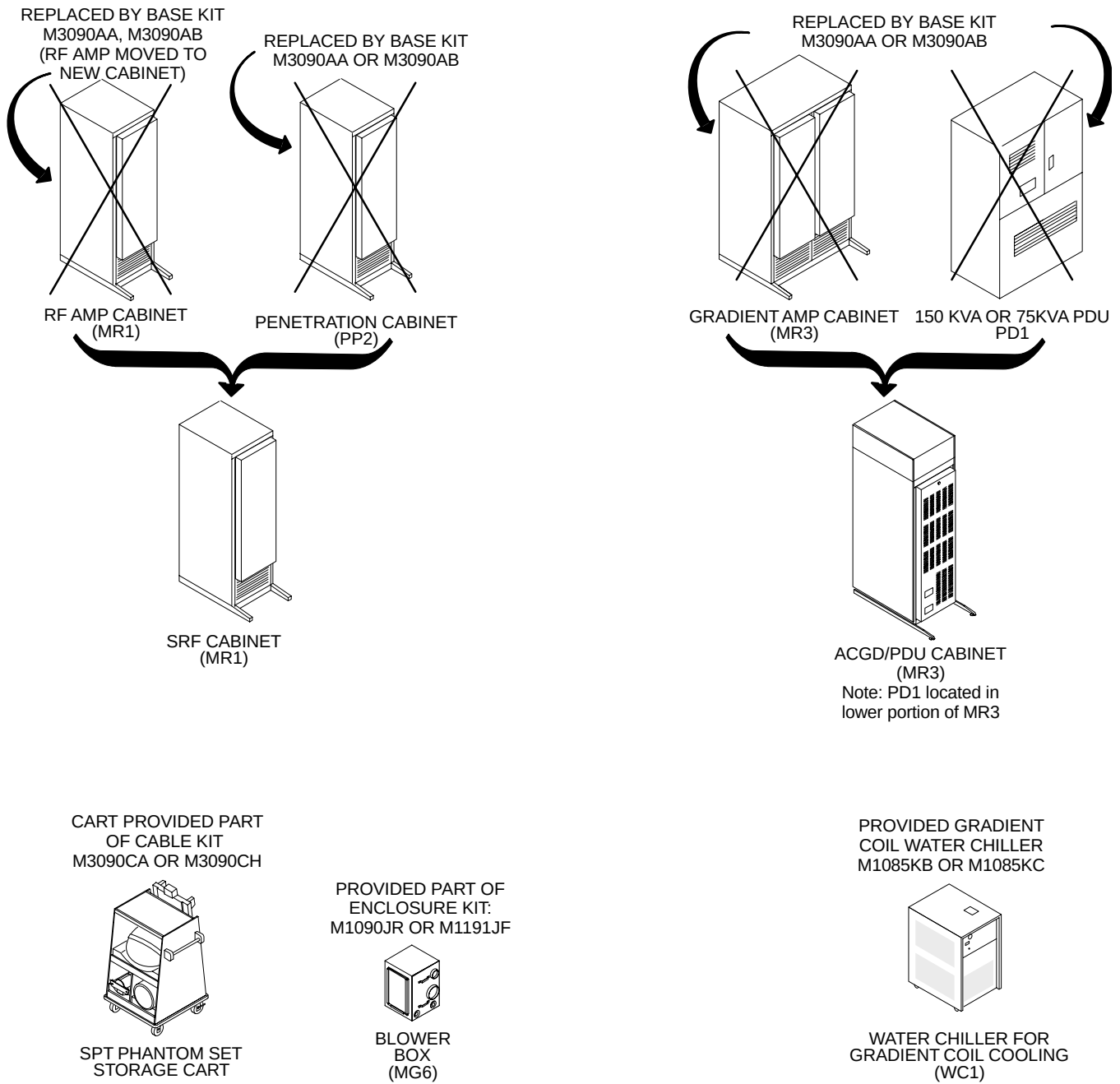
1-3-5 Signa Advantage 4.X System (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-3-5

1-3-5 Signa Advantage 4.X System (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-36

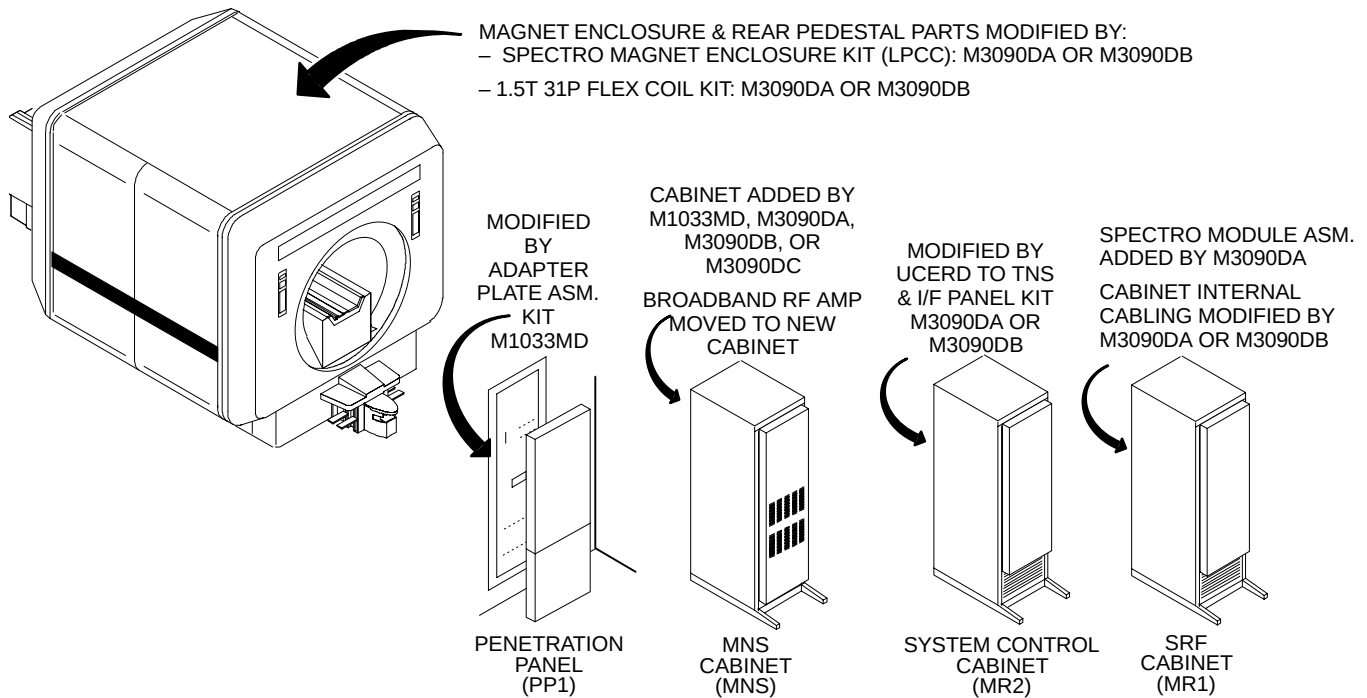
1-4 SIGNA LX SYSTEM OPTIONS

Signa LX system options which have site preparation impact include the following.

- VCR Interface (E8803BS, E8801YA, E8801YD)
- Remote Keyboard Option for use with Operator Workspace PC LCD:
 - English (E6901A)
 - German (E6901AA)
 - Spanish (E6901AB)
 - Italian (E6901AC)
 - French (E6901AD)
 - Scandinavian (E6901AE)
- Bar Code Reader (M1090PM) for use in conjunction with HIS/RIS software option.
- For 1.5T system only: Spectroscopy – Probe2000 (M1033JD) is compatible with all 1.5T Signa LX systems. Probe2000 includes both Single and Multi-Voxel Proton Spectroscopy techniques as well as Metabolic Map Spectral Display capabilities. The Resistive Shim Kit (M1040BH) is recommended for GE S-II or S-III magnets and should be included in initial site plan. The recommendation for the Resistive Shim Kit should be discussed with MR Spectroscopy Marketing which can be reached at (262) 521-6746.

1-4 SIGNA LX SYSTEM OPTIONS (Continued)

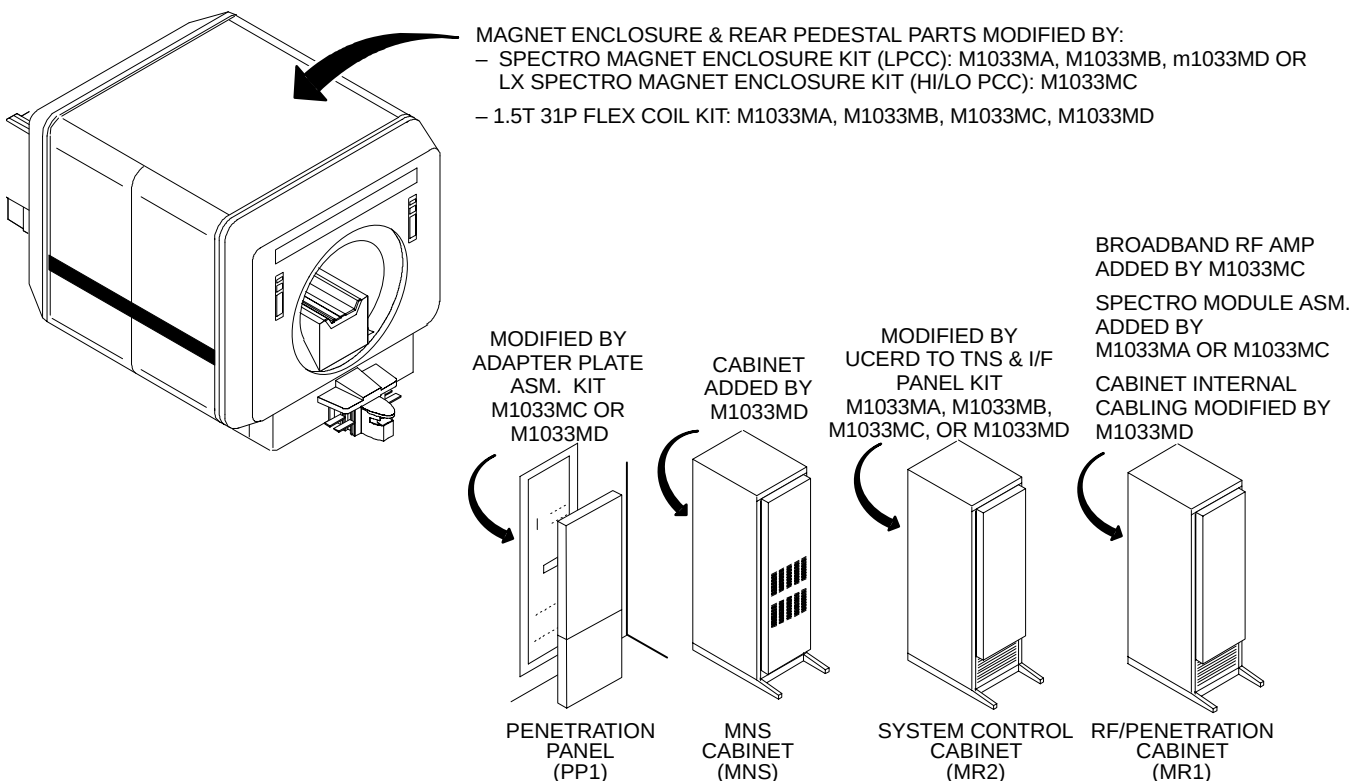
- For 1.5T system ACGD/PDU Gradient Cabinet: The following Multi-Nuclear Spectroscopy (MNS) options are available (refer to Illustration 1-37):
 - If system never had Multi-Nuclear Spectroscopy or did not retain all parts from the MNS Kit then need to install MNS Kit For Signa LX Systems That Have RF/PDU Cabinet or ACGD Cabinet (M1033MD). Kit requires UCERD (can be ordered M1033MF), RF/PDU Cabinet or ACGD Cabinet.
 - For Signa Advantage 4.X or Signa Advantage 5.4 system upgrading to Signa LX: If system already has Multi-Nuclear Spectroscopy and have all the parts on site then can install 8.X MNS Upgrade For Signa Advantage 4.X-5.4 (M3090DA).
 - For Signa Horizon 5.5 or later system upgrading to Signa LX: If system already has Multi-Nuclear Spectroscopy and have all the parts on site then can install 8.X MNS Upgrade For Signa Horizon 5.5 And Later (M3090DB).
 - Signa LX with Spectroscopy pre-RF/PDU Cabinet to ACGD Upgrade (M3090DC).



1.5T SYSTEM WITH ACGD/PDU CABINET: MNS OPTIONS EXISTING HARDWARE MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-37

1-4 SIGNA LX SYSTEM OPTIONS (Continued)

- For 1.5T system with SGD Cabinets: The following Multi-Nuclear Spectroscopy (MNS) options are available (refer to Illustration 1-37):
 - For Signa Advantage 4.X or Signa Advantage 5.4 system upgrading to Signa LX:
If system already has Multi-Nuclear Spectroscopy and have all the parts on site then can install 8.X MNS Upgrade For Signa Advantage 4.X–5.4 (M1033MA). Kit requires UCERD (can be ordered M1033MF), non-RF/PDU Cabinet, and non-ACGD Cabinet.
 - For Signa Horizon 5.5 or later system upgrading to Signa LX:
If system already has Multi-Nuclear Spectroscopy and have all the parts on site then can install 8.X MNS Upgrade For Signa Horizon 5.5 And Later (M1033MB). Kit requires UCERD (can be ordered M1033MF), non-RF/PDU Cabinet, and non-ACGD Cabinet.
 - If system never had Multi-Nuclear Spectroscopy or did not retain all parts from the MNS Kit then need to install MNS Kit for Signa LX Systems That Do Not Have RF/PDU Cabinet (M1033MC). Kit requires UCERD (can be ordered M1033MF), non-RF/PDU Cabinet, and non-ACGD Cabinet.
 - If system never had Multi-Nuclear Spectroscopy or did not retain all parts from the MNS Kit then need to install MNS Kit For Signa LX Systems That Have RF/PDU Cabinet or ACGD Cabinet (M1033MD). Kit requires UCERD (can be ordered M1033MF), RF/PDU Cabinet or ACGD Cabinet.



1.5T SYSTEM WITH SGD CABINET: MNS OPTIONS EXISTING HARDWARE MODIFIED & NEW EQUIPMENT ADDED

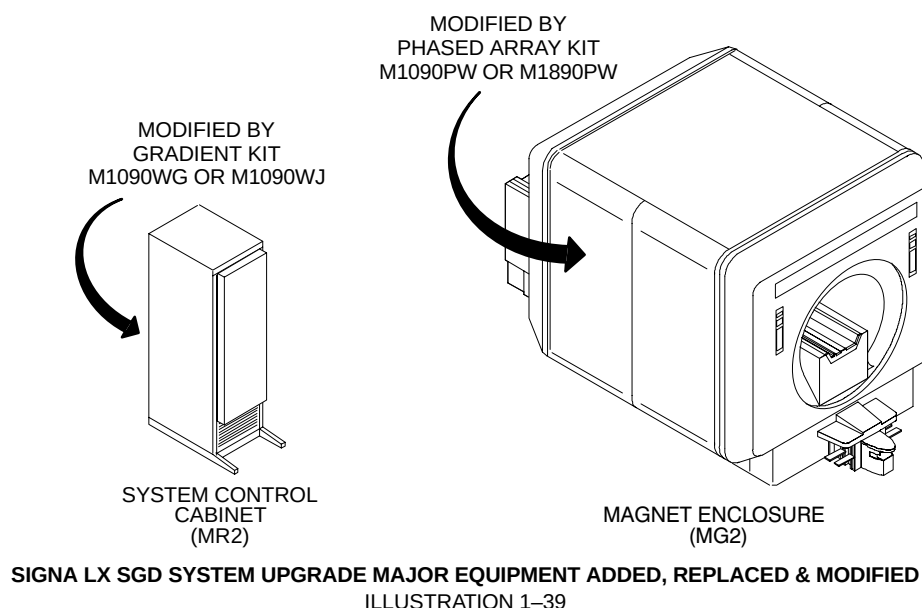
ILLUSTRATION 1-38

1-5 UPGRADES TO SIGNA LX WITH SGD SYSTEM

Table 1-1 lists the upgrade catalogs for Base Signa LX SGD or Signa LX SGD HiSpeed system within the Signa Horizon LX with SGD product line. Illustration 1-21 shows the cabinets added, replaced, and modified by the listed catalogs.

TABLE 1-1
SIGNA LX SGD SYSTEM WITH SGD UPGRADE CATALOGS

SYSTEM HARDWARE CONFIG	UPGRADE KIT	LX WITH SGD UPGRADES CATALOGS	
		HISPEED	ECHOSPEED
Base Horizon LX without Phased Array	Phased Array Kit	M1090PW – 1.5T Phased Array Upgrade Kit (UCERD) for LX Systems without Phased Array M1890PW – 1.0T Phased Array Upgrade Kit (UCERD) for LX Systems without Phased Array	M1090PW – 1.5T Phased Array Upgrade Kit (UCERD) for LX Systems without Phased Array
Horizon LX with SGD HiSpeed with Spectroscopy	Gradient Kit <i>Only required if increasing gradient strength</i>		M1090WG – HiSpeed SGD to EchoSpeed SGD Upgrade Kit for Horizon LX with SGD 1.5T Systems
Horizon LX with SGD without Spectroscopy	Gradient Kit <i>Only required if increasing gradient strength</i>		M1090WG – HiSpeed SGD to EchoSpeed SGD Upgrade Kit for Horizon LX with SGD 1.5T Systems M1090WJ – EchoSpeed Adder Kit for sites without Spectroscopy are upgrading to SGD EchoSpeed



1-6 SIGNA HORIZON 5.5 SYSTEM OPTIONS

- Phase Array Compatibility Kit for 1.5T system (M1090PC) or 1.0T system (M1890PA). The site impact of these kits is the system interconnects, for details refer to Section 6-2-7 M1090PC – 5.5 Phase Array Compatibility Kits For 1.5T System or Section 6-2-17 M1890PA – 5.5 Phased Array Compatibility Kits For 1.0T System.

1-7 TRANSPORTABLE/RELOCATABLE UPGRADE

Transportable and Relocatable MR system Horizon upgrade catalogs are listed in the previous upgrade sections. Note, the cable kits for Transportable/Relocatable include a complete set of system interconnect excluding magnet and option related cables.

In addition to the system, the van must also be upgraded to accommodate the Signa Horizon upgrades. The van/system upgrades can be preformed at the Van Manufacturer facility (preferred location) or at customer site. It is customer responsibility to arrange with the appropriate Van Manufacturer for any upgrade to be performed at site.

- Customer must contact Van Manufacturer to arrange for van/system upgrade at Van Manufacturer facility or customer site. Refer to Section 1-9, VAN MANUFACTURERS CONTACT INFORMATION.
- Customer is responsible for van upgrade costs with Van Manufacturer.
- Customer responsible for van transportation to and from Van Manufacturer facility for upgrades perform there.
- MR system check and calibration performance:
 - For van/system upgrades performed at Van Manufacturer facility, the installation will be checked by MR Manufacturing. Final calibrations will be performed by GE Field Service.
 - For van/system upgrades performed at customer site, the installation will be check and final calibrations will be performed by GE Field Service.

Note

Transportable or Relocatable with Oxford magnet can not be upgraded to Horizon configurations. The Oxford magnet can not be moved with the new body coil installed, serious magnet structural damage will result.

1-8 TRUE MOBILE UPGRADE**Note**

Signa Horizon configurations are not available for 1.5T True Mobile systems with Oxford Magnets due to structural limitations of the Oxford magnet in a mobile environment.

MR system Horizon upgrade catalogs for 1.0T True Mobiles are listed in the previous upgrade sections. Note, the cable kits for True Mobile include a complete set of system interconnect excluding magnet and option related cables.

In addition to the system, the van must also be upgraded to accommodate the Signa Horizon upgrades. The True Mobile van/system upgrade are preformed at the Van Manufacturer facility.

- Customer must contact Van Manufacturer to arrange for van/system upgrade at Van Manufacturer facility. Refer to Section 1-9, VAN MANUFACTURERS CONTACT INFORMATION.
- Customer is responsible for van upgrade costs with Van Manufacturer.
- Customer responsible for van transportation to and from Van Manufacturer facility.
- System check and complete calibrations are performed by MR Manufacturing.

1-9 VAN MANUFACTURERS CONTACT INFORMATION

- Ellis & Watts
4400 Glen Willow Lake Lane
Batavia, OH 45103
Telephone: 513-752-9000
FAX: 513-752-4983

Note

Customers with a MICA Relocatable system need to contact Ellis & Watts.

- PDC Facilities
700 Walnut Ridge Drive, PO Box 900
Hartland, WI 53029-0900
Telephone: 262-367-7700
FAX: 262-367-7744
- SMIT Mobile Equipment
Buys Ballotstraat 6
3261 La Oud-Beijerland
Holland
Telephone: (31) 186-6-14322
FAX: (31) 186-6-19367
- AK Specialty Vehicles
16745 South Lathrop Avenue
Harvey, IL 60426
USA
Telephone: 708-596-5066
FAX: 708-596-2480
AK Specialty Vehicles is also the contact for used Calumet Coach trailers.

SECTION 2 – ROOM LAYOUTS

TABLE OF CONTENTS

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
2-1	INTRODUCTION	2-3
2-2	TWO MAGNET SITE LAYOUT	2-4
2-3	MINIMUM DOOR/HALLWAY SIZES	2-4
2-3-1	Gradient Coil Assembly, Cradle, & Cart Requirements	2-6
2-4	CABLING CONSIDERATIONS	2-8
2-5	SPECIAL SITING CONSIDERATIONS	2-12
2-5-1	Power Distribution Unit (PDU)	2-12
2-5-2	Blower Box (MG6)	2-13
2-5-3	Water Chiller (WC1) For Gradient Coil Cooling	2-13
2-5-4	SPT Phantom Set Shipping/Storage Cabinet	2-13
2-5-5	Spectroscopy for 1.5T Magnet Only (Optional)	2-14
2-5-6	Magnet Monitor	2-14
2-5-7	System Monitoring & Support Connectivity Requirements	2-15
2-6	ARCHITECTURAL REMINDERS	2-15
2-7	FLOOR LOADING AND WEIGHTS	2-16
2-7-1	Operator Workspace Mounting Requirements For Signa LX	2-17
2-8	COMPONENT DIMENSIONS	2-18

2-1 INTRODUCTION

When laying out a floor plan there are special considerations that must be taken into account due to the magnetic field effect on equipment. The maximum magnetic field in which the upgrade supplied equipment can be located is listed in Table 2-1.

TABLE 2-1
PROXIMITY LIMITS

GAUSS (mT) LIMIT See Notes 1 & 3	EQUIPMENT	
0.5 GAUSS (0.05mT) OR LESS See Note 5	• Nuclear cameras	
1 GAUSS (0.1mT) OR LESS See Note 5	• Positron Emission Tomography scanner • Linear Accelerator • Cyclotrons • Accurate Measuring scale • Image intensifiers • Color TV	• Video display (color, B/W, monochrome) • CT scanner • Ultrasound • Lithotripter • Electron microscope • Advantage Workstation with CRT Monitor
3 GAUSS (0.3mT) OR LESS See Note 5	• Power transformers • Main electrical distribution transformers • Moving steel equipment such as: – Vehicular traffic – Fork lift trucks – Dumb waiters – Electric transport carts	– Loading dock (truck traffic) – Elevators – Escalators – Helicopters (See Note 4)
1 GAUSS (0.1mT) OR LESS	• Advantage Workstation with CRT Monitor	
10 GAUSS (1mT) OR LESS	• Operator Workspace Host Monitor (Color)	
30 GAUSS (3mT) OR LESS	• System Control Cabinet	• Heat Exchanger for Mobile System Gradient Coil Cooling
50 GAUSS (5mT) OR LESS	• Octane Computer • Operator Workspace Cabinet • LCD Color Monitor for OW (See note 6) • SRF Cabinet • ACGD/PDU Cabinet • Water Chiller for Gradient Coil Cooling • Main Disconnect Device	• Magnet Monitor • Telephones • Metal detector for screening • LCD Color Monitor for Advantage Workstation • MNS Cabinet
100 GAUSS (10mT) OR LESS	• Service Tool Magnet Power Supply Cabinet • Service Tool Shim Power Supply Cabinet	• Shield/Cryo Cooler Compressor Cabinet • Pneumatic Patient Alert Control Box
200 GAUSS (20mT) OR LESS	• Penetration Panel • Blower Box	• Magnet Rundown Unit
Note 1 Refer to SECTION 3, MAGNETIC FIELD CONSIDERATIONS, for magnet field plots. 2 Recommended limits given above are based on general MR site planning guidelines. Actual susceptibility of specific devices may vary significantly depending on electrical design orientation of the device relative to the magnetic field and the degree of interference considered unacceptable. 3 Recommended limits given above are based on general MR site planning guidelines. Actual susceptibility of specific devices may vary significantly depending on electrical design orientation of the device relative to the magnetic field and the degree of interference considered unacceptable. 4 Verify operating limits with provider of helicopter service 5 The 0.5 gauss (0.05 mT), 1 gauss (0.1 mT), and 3 gauss (0.3 mT) or less proximity limits apply to ALL magnets. For additional Active Shield magnet proximity limits refer to <i>Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 4-12 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC</i> 6 If gauss limit is more than indicated for OW with LCD Color Monitor then contact GE to determine impacts on OW equipment.		

2-2 TWO MAGNET SITE LAYOUT

Refer to ***Direction 2223170 Signa MR/i & CV/i Pre-Installation*** Section 2-3 TWO MAGNET SITE LAYOUT for two magnet site layout information.

2-3 MINIMUM DOOR/HALLWAY SIZES

Table 2-2 lists minimum actual clearance opening dimensions for doors and hallways required by new equipment installed as part of the various Signa LX Upgrades. Installation or replacement of components listed in Table 2-3 must be taken into consideration when determining delivery route hallway and door dimensions. Clearance for maneuvering around corners or turns must also be taken into consideration. Refer to SECTION 8, SHIPPING AND DELIVERY DATA, for component shipping dimensions.

TABLE 2-2
MINIMUM HALLWAY/DOOR DIMENSIONS

COMPONENT	MINIMUM HALLWAY/ DOOR WIDTH* in. (mm)	MINIMUM HALLWAY/ DOOR HEIGHT* in. (mm)	COMMENTS
Signa LX Operator Workspace Table	32 (813)	80 (2032)	
Signa LX Cabinets	36 (914)	80 (2032)	
Cryogen delivery route and Storage Room	43 (1092)	80 (2032)	Width requirements due to size of 500 liter dewars. Width and height requirements vary dependent on the dewars used. Check with cryogen supplier.
LCC Magnet	Refer to Note 1	Refer to Note 2	Refer to Table 2-3 for uncrated magnet dimensions.
RF Room Door	Refer to Section 7, RF SHIELDED ROOM	Refer to Section 7, RF SHIELDED ROOM	
Note * Minimum hallway and door dimensions are actual clearance openings. Width and height of rigging equipment is not included in above dimension. 1 Minimum width depends on access route to removable panels of RF shielded room wall. For straight path (i.e. no bends or turns) recommended to allow 6 in. (153 mm) on both sides of magnet. Appropriate calculations must be performed if turns exist along proposed magnet delivery route. Illustration NO TAG shows dimensions for 90° turn. 2 Final dimension is dependent on rigger equipment used, refer to <i>Direction 2223170 Signa MR/i & CV/i Pre-Installation</i> SECTION 8, SHIPPING AND DELIVERY DATA.			

2-3 MINIMUM DOOR/HALLWAY SIZES (Continued)

TABLE 2-3
UPGRADE COMPONENT DIMENSIONS FOR INSTALLATION/REPLACEMENT

COMPONENT	APPROXIMATE WEIGHT lbs (kg)	OVERALL DIMENSIONS W x D x H in. (mm)	COMMENTS
LCC Magnet (uncrated)	Refer to comments.	Refer to comments.	Refer to Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 8, SHIPPING AND DELIVERY DATA, for dimensions, illustrations and weights.
Split Bridge (LCC Magnet Enclosure)	40 (18)	21.5 x 77.3, x 7 (546 x 1969 x 18)	Room dimensions in front of LCC Magnet MUST allow for bridge installation/servicing and Gradient Coil Replacement, See Note 3.
Replacement Gradient (BRM) Coil Assembly on a Shipping Cradle/Cart	See Note 1	35 x 96.09 x 55.88 (889 x 2441 x 1420) See Note 2 & 3	Initial BRM Gradient Coil Assembly is shipped installed in the Magnet. Shipping/installation cart is used to install replacement coil assembly only. Refer to Section 2-3-1, Gradient Coil Assembly, Cradle, & Cart Requirements. The Short Room Gradient Coil Cart must be used to replace the BRM when the distance in front of the magnet is less than 117.6 inch (2987).
Enclosure Covers & End Bells	See Comments	See Comments	Refer to Section 2-8 COMPONENT DIMENSIONS, for details of individual assemblies and applicable catalogs.
Rear Pedestal	30 (13.6)	17 x 43 x 33 (432 x 1092 x 838)	
Note 1 The replacement BRM Gradient Coil Assembly weight is approximately 2300 lbs (1045 kg), the shipping cradle is 125 lbs (57 kg), and the Gradient Coil Assembly shipping/installation cart weighs 855 lbs (389 kg). Therefore total shipping weight is 3280 lbs (1490 kg). The coil assembly outside diameter x length dimensions are 35.1 x 73.5 in. (892 x 1867 mm). 2 RF/Gradient (BRM) Coil Assembly and shipping cart dimensions are with cart in lowest position. Cart can be adjusted to maximum height of 61.88 in. (1572 mm). 3 LCC and Cx Magnets MUST USE special Gradient Coil Replacement Tool Kit for replacing the Gradient Coil Assembly. The Gradient Coil Replacement Tool Kit is shipped in a wooden crate on casters. Utilization of Gradient Coil Replacement Tool requires 118 in. (2997 mm) clear space in front of magnet (same clearance as Bridge installation).			

2-3-1 Gradient Coil Assembly, Cradle, & Cart Requirements

Horizon upgrades will require the 55 cm Body Coil to be removed using the old 'blue' wooden coil cart.

■ Signa LX System without Magnet Upgrade

International sites upgrade shipments will have the RF/Gradient (BRM) Coil shipped on a wooden (disposable) cradle, see Illustration 2-1. A forklift or crane/hoist rated for 4000 lbs (1818 kg) is required to position the coil/cradle assembly onto the cart for installing the coil into the magnet, see Illustration 2-2. GE Service to provide the coil cart and accessories for International Sites, refer to Section 10-5 INSTALLATION EQUIPMENT.

USA sites ONLY upgrade shipments will receive the new RF/Gradient (BRM) coil on an aluminum (re-useable) cradle mated to coil cart with accessories. GE MR Manufacturing will provide cradle/cart which will be returned to MR Manufacturing for reuse.

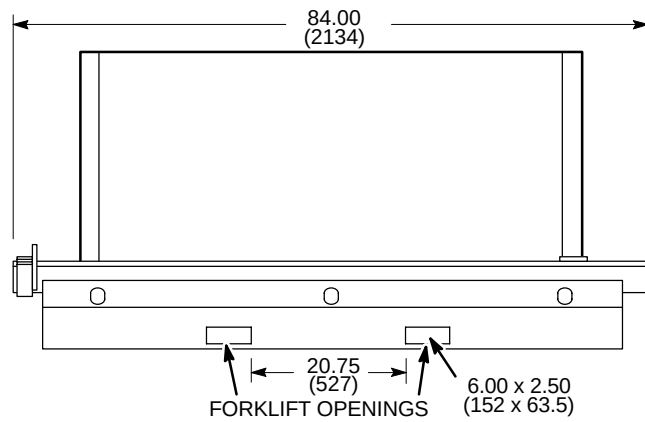
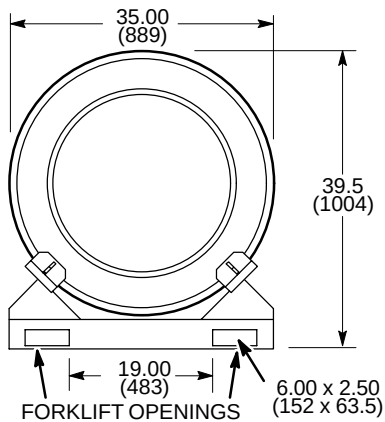
■ Systems Installing LCC Magnet Upgrade Concurrent With System Upgrade

Systems installing an LCC Magnet Upgrade at the same time as the System Upgrade will receive the LCC Magnet with the BRM Assembly installed inside the magnet bore and enclosure.

2-3-1 Gradient Coil Assembly, Cradle, & Cart Requirements (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: WITH BRM 2425 lbs (1102 kg)

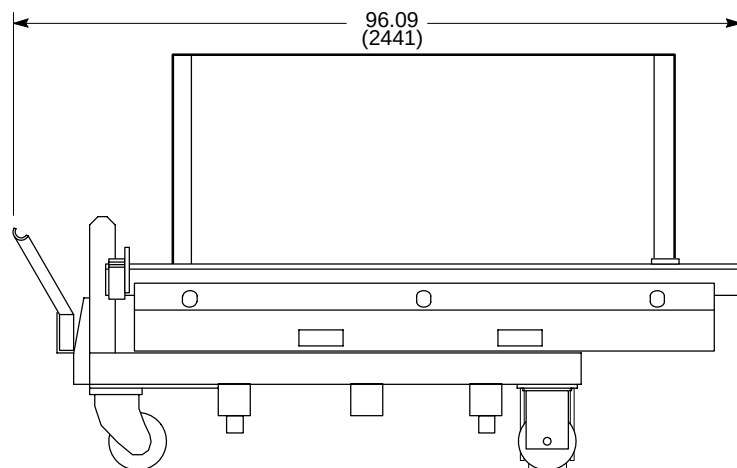
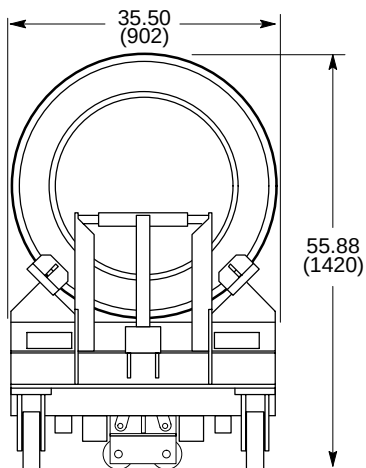


UPGRADE GRADIENT BRM COIL ASSEMBLY & CRADLE

ILLUSTRATION 2-1

NOTE:

- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT WITH BRM 3280 lbs (1491 kg)



SHORT ROOM REPLACEMENT GRADIENT BRM COIL ASSEMBLY & CRADLE ON CART

ILLUSTRATION 2-2

2-4 CABLING CONSIDERATIONS

Cable runs in the Magnet Room as well as throughout the system must be in accordance with local and national codes. Cabinets cable access areas are shown on illustrations in Section 2-8, COMPONENT DIMENSIONS. Some cable routing will be changed due to new equipment depending on system configuration being upgraded. Refer Section 6, INTERCONNECT DATA, for cables and their dimensions information.

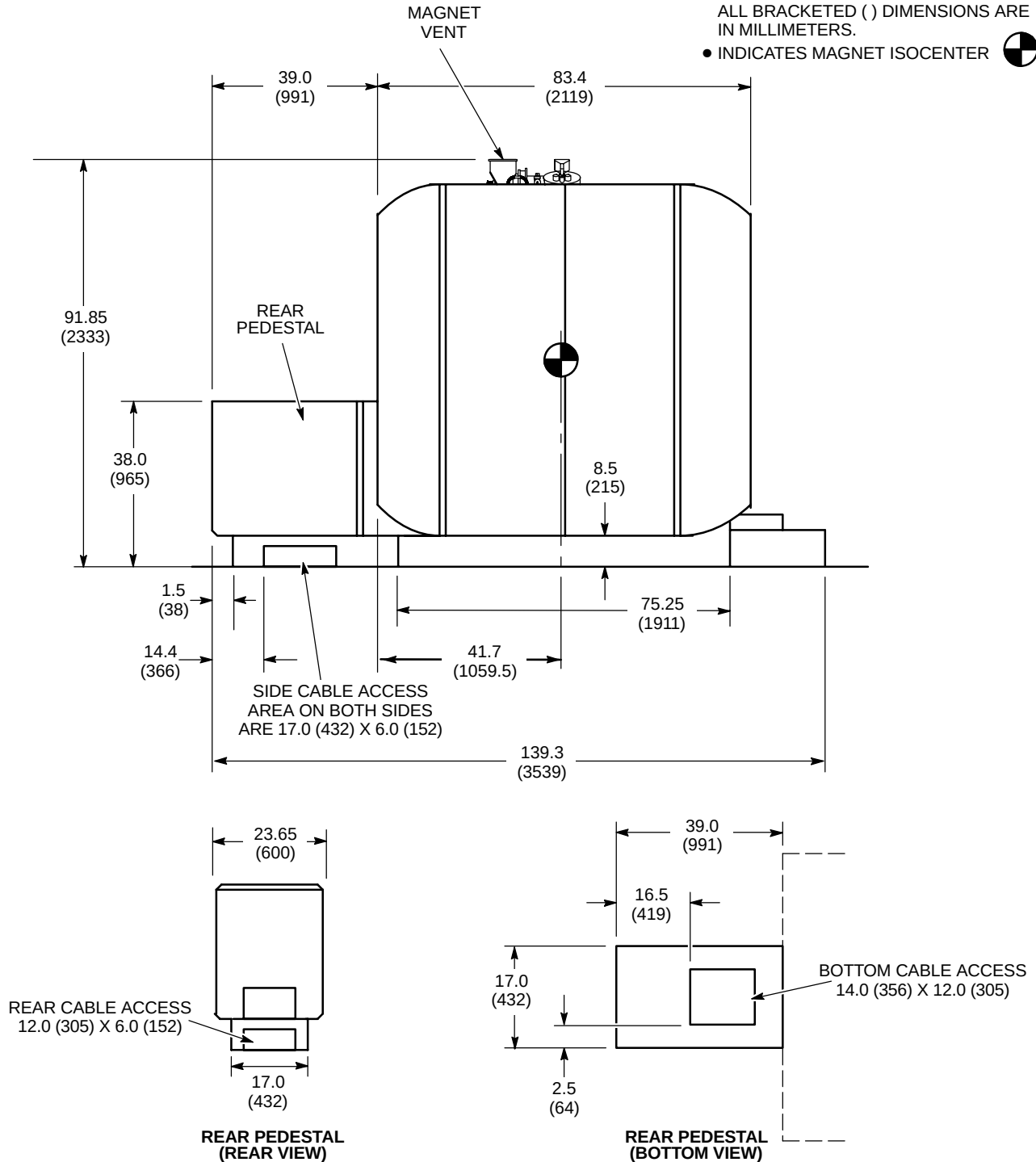
Note

Signa Horizon has an increase of cabling and hoses through the Rear Pedestal cable access areas of the Magnet Enclosure. These include the 2 Blower Box hoses, 2 new large Gradient cables, 2 water hoses for Body Coil, and the remainder of the Magnet Enclosure system interconnect cables. Sites with cable access from underneath the Rear Pedestal should have enough cable access area. Sites using the Rear Pedestal cable access area on rear or one side will need to plan to utilize more than one access opening. See Illustrations 2-3 through 2-5 for Magnet cable access areas.

2-4 CABLING CONSIDERATIONS (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE
IN MILLIMETERS.
- INDICATES MAGNET ISOCENTER



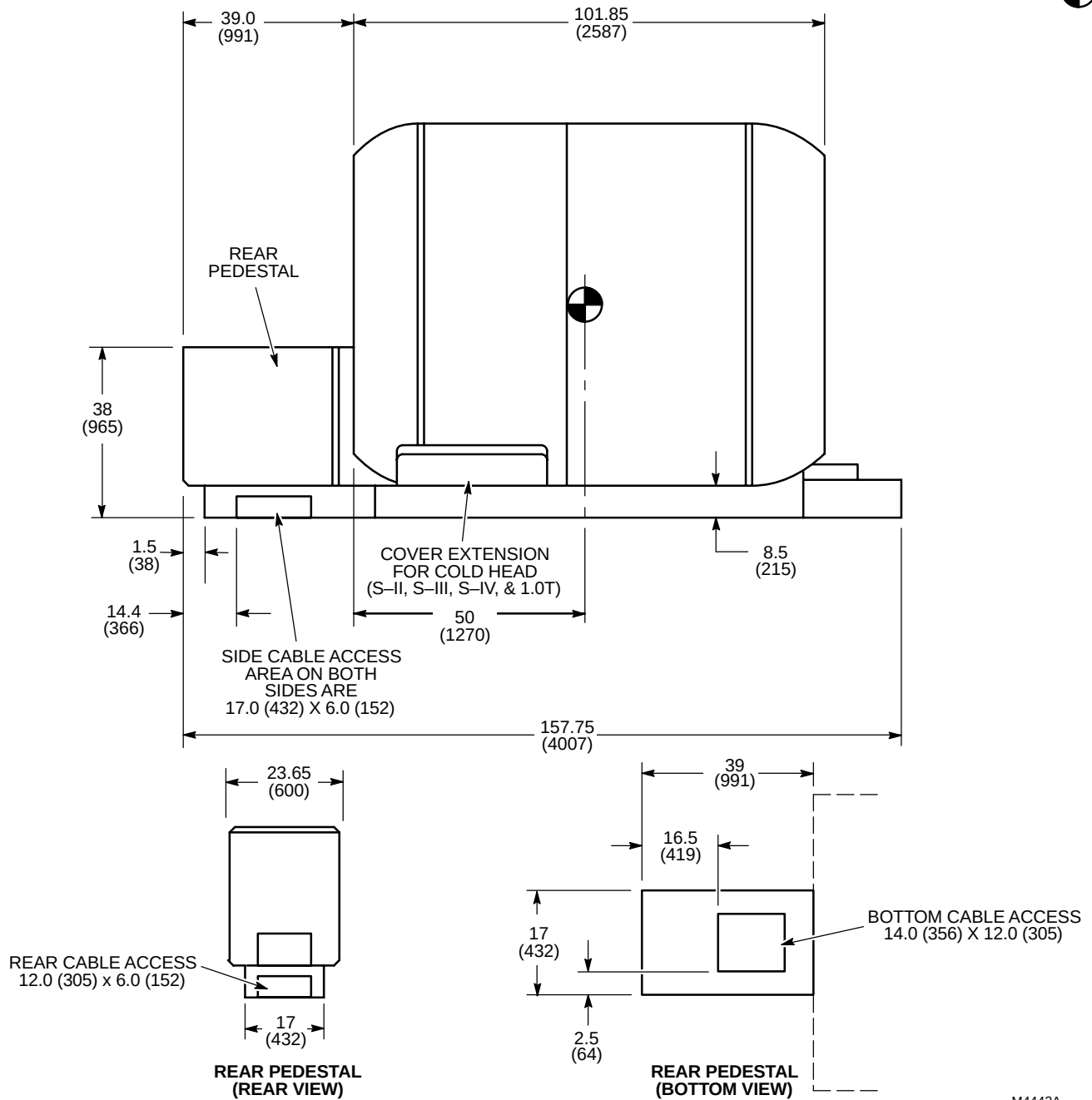
1.0T AND 1.5T Cx (ACTIVE SHIELD) MAGNET CABLE ACCESS

ILLUSTRATION 2-3

2-4 CABLING CONSIDERATIONS (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- INDICATES MAGNET ISOCENTER



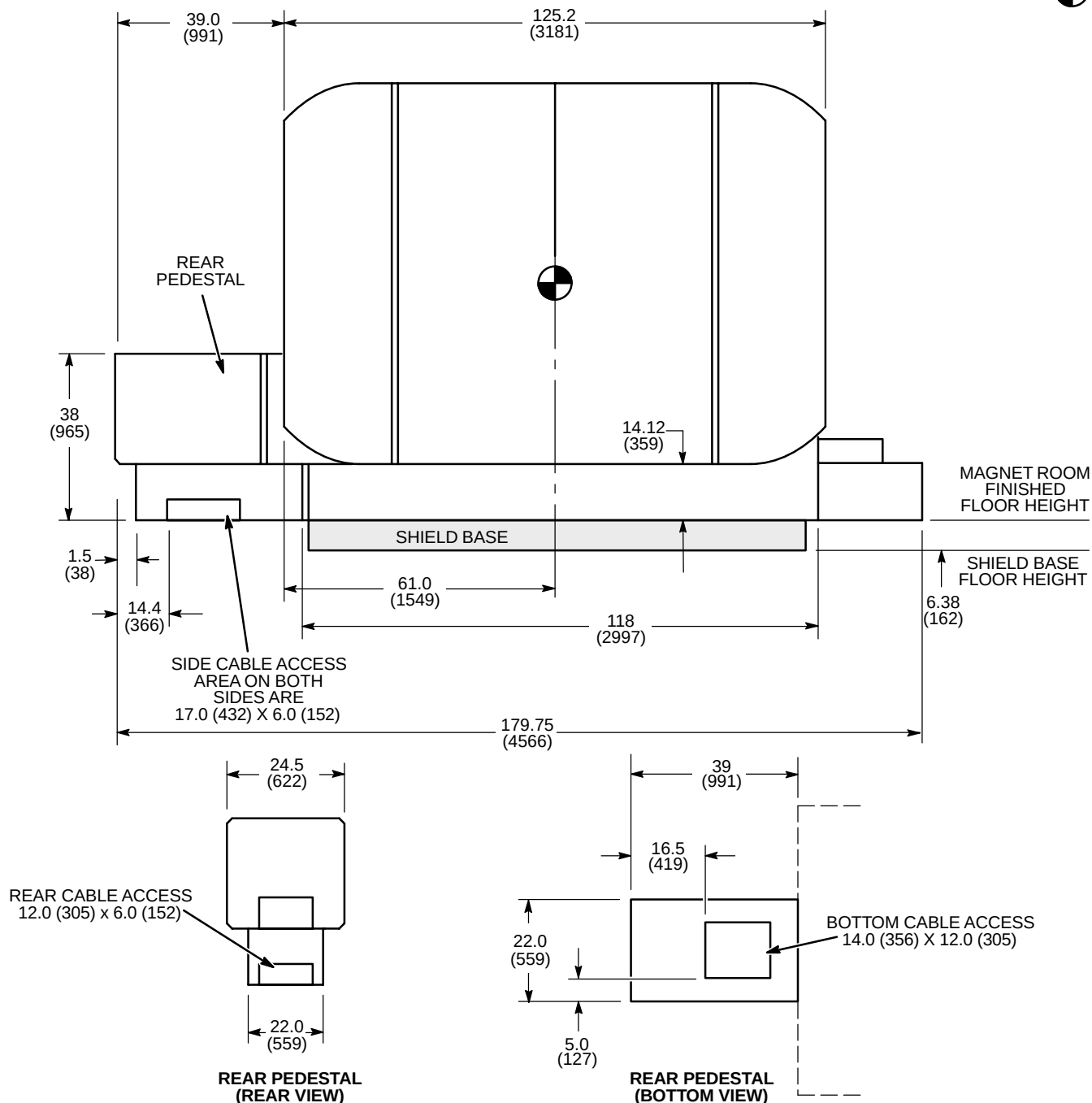
M4442A

1.0T & 1.5T ACTIVE SHIELD, 1.5T UNSHIELDED (S-II, S-III) MAGNET CABLE ACCESS
ILLUSTRATION 2-4

2-4 CABLING CONSIDERATIONS (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- INDICATES MAGNET ISOCENTER



1.5T MAGNET WITH MAGNISHIELD OPTION CABLE ACCESS

ILLUSTRATION 2-5

2-5 SPECIAL SITING CONSIDERATIONS

The following system components have special siting concerns which need to be considered.

2-5-1 Power Distribution Unit (PDU)

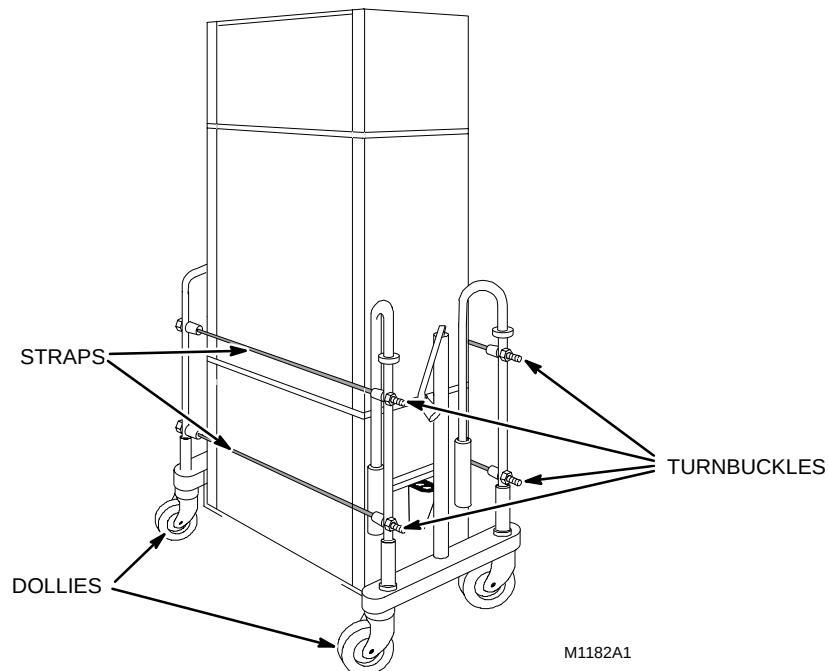
Sites that are upgrading to ACGD require the Full-size PDU, Compact PDU, or RF/PDU Cabinet be de-installed. The Compact PDU and RF/PDU Cabinet have built-in casters for ease of movement. The Full-size PDU does not have casters and weighs approximately 1000 lbs (455 kg). Sites with a Full-size (double bay) PDU must plan for removal of the PDU.

- **Sites located outside the USA:**

The Full-size PDU will have to be removed by riggers or other locally supplied equipment moving personnel/equipment.

- **Sites located in the USA:**

Can contact Techtrans at telephone 800-851-9929 to order a "MR PDU Double Bay Cabinet Dolly Set". The PDU can be moved with two shipping dollies placed one on each side of the PDU and clamped in place using straps with turnbuckles as shown in Illustration 2-6.



FULL-SIZE PDU SHIPPING DOLLIES

ILLUSTRATION 2-6

2-5-2 Blower Box (MG6)

The Blower Box (MG6) provides cooling air for the RF/Gradient Body Coil and the Patient Comfort Module in the Magnet Enclosure. The Blower Box will be mounted within the RF Shielded Room and connect to the Gradient Coil and the Patient Comfort Module by 6.5 inch (165.1 mm) OD and 4.5 inch (114.3 mm) OD flexible vinyl air ducting. **The Blower Box contains magnetic material and must be securely mounted to the floor of the Magnet Room or mounted to a raised support shelf on the Magnet Room wall or ceiling with support provided under the box. RF Shield integrity must be maintained for mounting the Blower Box within the Magnet Room.** The flexible vinyl air ducts route from the Blower Box through the Magnet Enclosure Rear Pedestal cable access and connect to the Gradient Coil. Refer to Illustrations 2-26 for Blower Box and mounting dimensions.

Note

Blower Box mounting requires customer supplied hardware (ie. lag bolts, screws, etc.) appropriate for the surface on which the box will be mounted.

2-5-3 Water Chiller (WC1) For Gradient Coil Cooling

Fixed Site System

A dedicated, closed loop, chilled water cooling system for the Gradient Coil is for fixed site systems utilizing ACGD Gradient driver. The chilled water cooling system consists of a water-to-air Water Chiller (WC1), two flexible vinyl hoses, and de-ionized water and anti-freeze (provided with the coil). See Illustration 2-25 for Water Chiller (WC1) dimension information. The two flexible hoses (supply and return) are to be routed from the Water Chiller (WC1) through waveguides in the Penetration Panel (PP1), through the Rear Pedestal (MG3), and connect to the rear of the Gradient Coil with supplied adjustable compression clamps. The Water Chiller will be powered from the PDU via a supplied power cord. Refer to Section 6, INTERCONNECT DATA, for additional cable information.

Note

The Fixed Site Water Chiller (WC1) for Gradient Coil Cooling must not be located below the Magnet Room floor level. The Water Chiller reservoir tank has a low pressure vent plug which relieves pressure from thermal expansion of water and activates when tank pressure reaches 3-5 psi (0.2-0.3 bar). The vent plug could allow fluid to leak if located below the MR system level. The reservoir tank vented plug can not be replaced with a non-vented type, damage to the tank may occur. The Water Chiller may be located one floor level above the MR system.

Mobile System (True Mobile, Transportable, Relocatables)

The Gradient Coil water cooling will be provided by the Mobile system Van chilled water system. Therefore, the Water Chiller (WC1) as shown in Illustration 2-25 is not used for Mobile configurations.

2-5-4 SPT Phantom Set Shipping/Storage Cabinet

System Performance Test provides the customer and GE Service with a means to quickly verify whether critical parameters affecting image quality are within specifications. The test uses a set of phantoms and a nesting plate for proper positioning of the phantoms on the Patient Table. The phantom set and nesting plate are provided in a cabinet which protects the pieces during shipment and storage at site. The cabinet is not magnetic therefore it can be stored inside the Magnet Room if so desired and moved to the Patient Table for ease of positioning the phantoms. See Illustration 2-38 for dimensions information.

2-5-5 Spectroscopy for 1.5T Magnet Only (Optional)

Upgrade hardware configuration is dependent on the current system cabinets as defined below. Refer to Section 6-7, SPECTROSCOPY INTERCONNECTIONS – 1.5T SYSTEM OPTION, cable length information and other cable data.

Multi-Nuclear Spectroscopy (MNS) For Systems With ACGD/PDU Cabinet

The MNS option provides the Spectroscopy Acquisition Option which consists of the MNS Cabinet which has mounting space for Broadband RF Amplifier and several module kits to modify the SRF Cabinet, Penetration Panel in the Magnet Room, and system cabling.

Multi-Nuclear Spectroscopy (MNS) For Systems With SGD Gradient Cabinet & RF/PDU Cabinet

The MNS option provides the Spectroscopy Acquisition Option which consists of the MNS Cabinet (contains Broadband RF Amplifier) and several module kits to modify the RF/PDU Cabinet, Penetration Panel in the Magnet Room, and system cabling.

Multi-Nuclear Spectroscopy (MNS) For Systems With SGD Gradient Cabinet & RF/Penetration Cabinet

The MNS option provides the Spectroscopy Acquisition Option which consists of the Broadband RF Amplifier (to be installed in the RF/Penetration Cabinet) and several module kits to modify the RF/PDU Cabinet, Penetration Panel in the Magnet Room, and system cabling.

2-5-6 Magnet Monitor

Refer to *Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 2-7-8 Magnet Monitor* for LCC Magnet Monitor and Magnet Monitor UPS option information.

2-5-7 System Monitoring & Support Connectivity Requirements

One of the system monitoring and support connectivity configurations listed in Table 2-4 must be provided for system installation and serviceability purposes. The network connection (if selected configuration) and telephone lines are to be provided and paid for by the customer.

TABLE 2-4
SYSTEM MONITORING & SUPPORT CONNECTIVITY REQUIREMENTS

CONFIGURATION	PHONE LINE	USE/LOCATION
Network connection & telephone lines	One <u>voice-grade</u> telephone line (voice line)	Available for Service Personnel use, located in the Control Room
	Network connection with a static IP address See Note 1	Access located near the Operator Workspace (OW) in the Control Room.
	One line must be a dedicated direct-distance-dialing <u>voice-grade</u> line (data line)	Access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring.
Multiple telephone lines	One <u>voice-grade</u> telephone line (voice line)	Available for Service Personnel use, located in the Control Room
	One line must be a dedicated direct-distance-dialing <u>voice-grade</u> line (data line)	Access located near the Operator Workspace (OW) in the Control Room. See Note 2 & 3
	One line must be a dedicated direct-distance-dialing <u>voice-grade</u> line (data line)	Access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring. See Note 2 & 3
Note 1 For North America & South America: One Internet accessible Virtual Private Network Connection with a static IP Address is required. For Europe: One ISDN Network Connection with a static IP Address is required. 2 A dedicated direct-distance-dialing <u>voice-grade</u> telephone line can be shared for Operator Workspace (OW) and Magnet Monitor (MSM) requirement through the use of a multiplexer box. The following multiplexer boxes are available for customer purchase. 46-328475P1 4 Line Phone Multiplexer box; 115 VAC input power 46-328475P3 4 Line Phone Multiplexer box; 220 VAC input power If the customer chooses not to purchase the multiplexer box then the customer must provide an additional line for each requirement as stated in this table. 3 If a Multiplexer Box is used then the Magnet Monitor MUST be Channel 1 to allow for call out after a power outage.		

2-6 ARCHITECTURAL REMINDERS

- The operator seated at the Operator Workspace must have an unobstructed view of the patient on the transport table when table is docked to the magnet.
- It is recommended that the Magnet Room viewing window be of fine mesh screening material (as opposed to a "honeycomb-type pattern") for better visibility of the patient from the Operator Workspace.
- Customer provided and paid for telephone lines must be supplied for system installation and serviceability purposes per **Section 2-5-7 System Monitoring & Support Connectivity Requirements**.
- Recommend protecting floors while moving heavy pieces of equipment (e.g. ACGD/PDU Cabinet, BRM Body Coil Assembly).

2-7 FLOOR LOADING AND WEIGHTS

This section contains loading considerations for Signa Horizon upgrade components. Listed in Table 2-5 are the weights, floor loading, and normal mounting methods for components.

TABLE 2-5
SIGNA HORIZON LX FLOOR LOADING

COMPONENT	NET WT lbs (kg)	OVERALL DIMENSIONS W x D x H in. (mm)	LOAD PATTERN in. (mm)	NORMAL MOUNTING METHOD
Cx Magnet, Enclosure, and Coils	See Note 1	87.5 x 83.4 x 91.9 (2223 x 2119 x 2333)	Not changed by Horizon upgrade.	Not changed by Horizon upgrades.
Oxford & S-Series Magnet, Enclosure, and Coils	See Note 1	90.62 x 157.75 x 96.36 (2302 x 4007 x 2448)	Not changed by Horizon upgrade.	Not changed by Horizon upgrades.
Magnet, Magnishield, Enclosure, and Coils	See Note 1	107 x 179.75 x 96.36 (2718 x 4566 x 2448)	Not changed by Horizon upgrade.	Not changed by Horizon upgrades.
Patient Table	No change to existing Lightweight Patient Transport.			
Blower Box	47 (21)	16 x 16 x 22 (406 x 406 x 559)	Square base 15 x 15 (381 x 381)	Anchor to floor, shelf, or wall.
Power Distribution Unit	Existing PDU is modified internally, floor loading not impacted.			
Water Chiller for Gradient Coil	374 (170)	21.65 x 27.95 x 37 (550 x 710 x 940)	Rectangular base 21.65 x 27.95 (550 x 710). Four casters 3 (76) dia. x .75 (19) wide.	Set on floor and rest on casters.
SRF Cabinet	525 (239)	24 x 40 x 70.7 (610 x 1016 x 1793)	Rectangular base 24 x 30 (610 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
Signa 1.5T MR/i with EXCITE Technology System Cabinet	468 (213)	23.25 x 42 x 76.5 (591 x 1067 x 1943)	Rectangular base 22 x 30 (584 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
1.5T & 1.0T Signa MR/i or 1.5T Signa CV/i System Cabinet	494 (225)	23.25 x 42 x 76.5 (591 x 1067 x 1943)	Rectangular base 22 x 30 (584 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
ACGD/PDU Cabinet	1844 (838)	23.5 x 37.25 x 85.6 (597 x 927 x 2174) See Note 3	Rectangular base 33.5 x 22 (851 x 559). Four casters each 3.0 (76.2) dia. x 1.82 (46.2) wide.	Set on floor on casters.
Operator Workspace Table with LCD Color Monitor (See Note 4)	175 (80)	54 x 43 x 52 (1372 x 1092 x 1321)	See Illustration 2-33..	Set on floor: Table on leveling pads & Cabinet rest on casters. Anchor Table to floor per Section 2-7-1.
Operator Workspace Cabinet (See Note 4)	192 (87)	18.5 x 29 x 26 (470 x 737 x 660)	See Illustration 2-34.	Set on floor and rest on casters.
(Continued)				

TABLE 2-5 (Continued)
SIGNA HORIZON LX FLOOR LOADING

COMPONENT	NET WT lbs (kg)	OVERALL DIMENSIONS W x D x H in. (mm)	LOAD PATTERN in. (mm)	NORMAL MOUNTING METHOD
MNS Cabinet Option	430 (195)	23.56 x 42 x 70.7 (598 x 1067 x 1797)	Rectangular base 33.5 x 22 (851 x 559).	Casters for location. Set on floor on four leveling pads.
Note 1 Weight of magnet with enclosure, shim coil, RF/Gradient Body Coil Assembly, and cryogenics is:				
Signa LX with BRM Coil				
–1.0T Cx Active Shield 11,856 lbs (5378 kg)				
–1.0T (S–X, S–XC) Active Shield 20,700 lbs (9409 kg)				
–1.5T Cx Active Shield 12,606 lbs (5719 kg)				
–1.5T (S–IV, S–V) Active Shield 20,600 lbs (9364 kg)				
–1.5T (S–III) Unshielded 18,420 lbs (8373 kg)				
–1.5T (S–II) Unshielded 22,020 lbs (10,010 kg)				
–1.5T (S–I) Unshielded 15,625 lbs (7102 kg)				
–1.5T (S–III)magnet and Magnishield 82,420 lbs (37,464 kg)				
2 Consult a structural engineer on method of calculating proper weight/unit area for floor loading.				
3 ACGD/PDU Cabinet height with Fan Module removed is 74.8 inches (1900 mm).				
4 The Operator Workspace Cabinet minimum area location is under the Workspace Table, see in Illustration 2–33. An alternate location is possible with the Operator Workspace Cabinet located against the right side of the Workspace Table.				

2-7-1 Operator Workspace Mounting Requirements For Signa LX

The Operator Workspace Table sets on the floor with the Workspace Cabinet positioned under the Table towards the right side, see Illustration 2-33. Note, the Workspace Table must be bolted to the floor for safe use of the table and equipment.

2-8 COMPONENT DIMENSIONS

To assist in completing your room layout, refer to Table 2-6 for list of component Illustrations and the upgrade catalog numbers which provide the new equipment. Included are items which may impact site layout or need to be taken into consideration when determining delivery route. Modifications to existing cabinets are not included, catalogs may list the same new equipment but differ due to other modification hardware.

TABLE 2-6
MR SYSTEM COMPONENT ILLUSTRATIONS LISTED BY UPGRADE CATALOG

UPGRADE CATALOG	ILLUSTRATION NAME	ILLUSTRATION NUMBER
M1000MN M1000MP M1000MR M1000MS M1000MW M1000NH M1000NJ M1000NK M1000NT	OPERATOR WORKSPACE (OW1)	2-33
M1033K M1033L	OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – OCTANE COMPUTER	2-36
M1033MD	MNS CABINET (MNS) – 1.5T OPTION	2-39
M1060JW	LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR CABINET (MS5)	2-24
M1060LC	1.0T AND 1.5T LCC (ACTIVE SHIELD) MAGNET LOAD PATTERN	2-22
	MAGNET MONITOR (MSM1)	2-23
M1085BG	UPGRADE RF/GRADIENT (BRM OR CRM) COIL & CRADLE ON CART	2-2
M1085KB M1085KC	WATER CHILLER (WC1) FOR GRADIENT COIL COOLING WATER	2-25
M1090TL	PATIENT TRANSPORT TABLE (PT1)	2-21
M1090JP	ENCLOSURE REAR END BELL ASSEMBLY – STANDARD MAGNET	2-12
	ENCLOSURE FRONT COVER – MAGNISHIELD MAGNET	2-18
	ENCLOSURE FRONT END BELL ASSEMBLY – MAGNISHIELD MAGNET	2-19
	ENCLOSURE REAR COVER – MAGNISHIELD MAGNET	2-20
	BLOWER BOX (MG6)	2-26
	REAR PEDESTAL	See Note 1
M1090JR	ENCLOSURE FRONT COVER – STANDARD	2-7
	ENCLOSURE FRONT END BELL ASSEMBLY – STANDARD MAGNET	2-8
	ENCLOSURE REAR COVER – STANDARD MAGNET	2-11
	ENCLOSURE REAR END BELL ASSEMBLY – STANDARD	2-12
	BLOWER BOX (MG6)	2-26
	REAR PEDESTAL	See Note 1
M1090JS M1090JT	BRIDGE – STANDARD	2-16
M1090SC	BRIDGE – MAGNISHIELD	2-17
M1090TL	PATIENT TABLE (PT1)	2-21
Note 1 Rear Pedestal is supplied with the indicated catalog numbers. Refer to Section 2-3, MINIMUM DOOR/HALLWAY SIZES, for dimension information.		
(Continued)		

2-8 COMPONENT DIMENSIONS (Continued)

TABLE 2-6 (Continued)
MR SYSTEM COMPONENT ILLUSTRATIONS LISTED BY UPGRADE CATALOG

UPGRADE CATALOG	ILLUSTRATION NAME	ILLUSTRATION NUMBER
M1191JF	ENCLOSURE FRONT COVER – STANDARD MAGNET	2-7
	ENCLOSURE FRONT END BELL ASSEMBLY – STANDARD MAGNET	2-8
	ENCLOSURE REAR END BELL ASSEMBLY – STANDARD MAGNET	2-12
	BLOWER BOX (MG6)	2-26
	REAR PEDESTAL	See Note 1
	ENCLOSURE REAR COVER – MOBILE RF TUBE EXTENSION	2-15
M1860LC	1.0T AND 1.5T LCC (ACTIVE SHIELD) MAGNET LOAD PATTERN	2-22
	MAGNET MONITOR (MSM1)	2-23
M1885BG	UPGRADE RF/GRADIENT BRM COIL & CRADLE ON CART	2-2
M3090AA	ACGD/PDU CABINET (MR3)	2-27
	SYSTEM CONTROL CABINET (MR2)	2-29
	SRF CABINET (MR1)	2-30
	PENETRATION PANEL (PP1)	2-31
	OPERATOR WORKSPACE (OW1)	2-33
	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
	SPT PHANTOM SET SHIPPING/STORAGE CABINET	2-38
M3090AD	ACGD/PDU CABINET (MR3)	2-27
	SYSTEM CABINET (MR2)	2-29
	SRF CABINET (MR1)	2-30
	PENETRATION PANEL (PP1)	2-31
	OPERATOR WORKSPACE (OW1)	2-33
	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
M3090AH	ACGD/PDU CABINET (MR3)	2-27
M3090AH	SYSTEM CABINET (MR2)	2-29
M3090AH	SRF CABINET (MR1)	2-30
M3090AH	OPERATOR WORKSPACE (OW1)	2-33
M3090AH	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
M3090BA	ACGD/PDU CABINET (MR3)	2-27
M3090BA	SRF CABINET (MR1)	2-30
M3090BB	ACGD/PDU CABINET (MR3)	2-27
M3090BB	SRF CABINET (MR1)	2-30
M3090BC	ACGD/PDU CABINET (MR3)	2-27
M3090BC	SRF CABINET (MR1)	2-30
M3090AE	ACGD/PDU CABINET (MR3)	2-27
M3090AE	SYSTEM CABINET (MR2)	2-29
M3090AE	SRF CABINET (MR1)	2-30
M3090AE	PENETRATION PANEL (PP1)	2-31
M3090AE	OPERATOR WORKSPACE (OW1)	2-33
M3090AE	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
Note 1 Rear Pedestal is supplied with the indicated catalog numbers. Refer to Section 2-3, MINIMUM DOOR/HALLWAY SIZES, for dimension information.		
(Continued)		

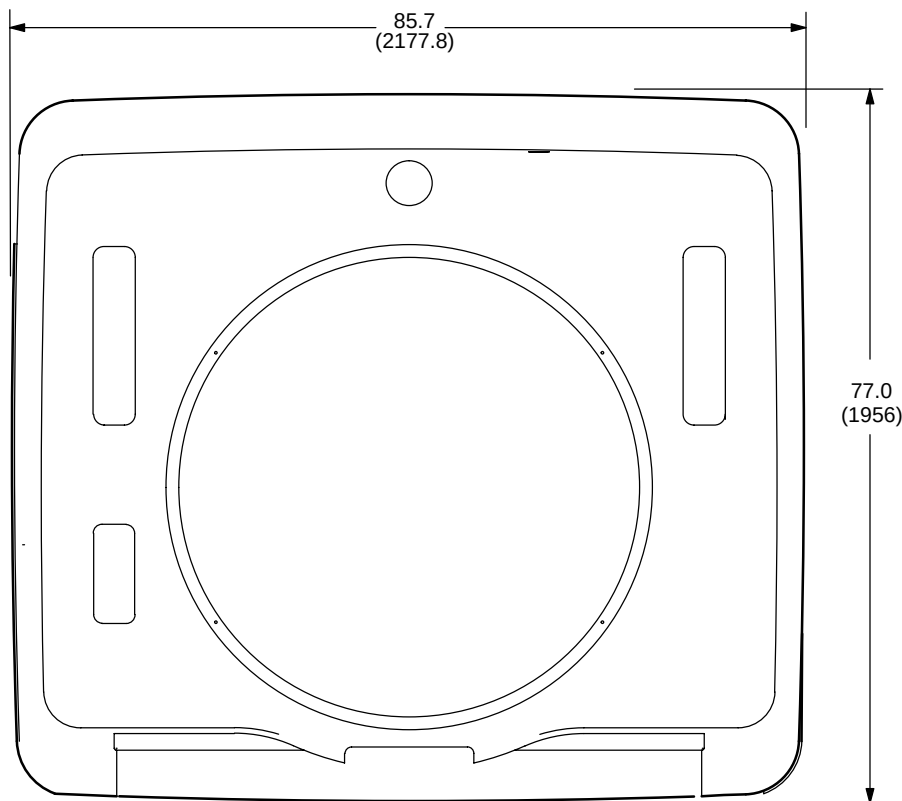
2-8 COMPONENT DIMENSIONS (Continued)

TABLE 2-6 (Continued)
MR SYSTEM COMPONENT ILLUSTRATIONS LISTED BY UPGRADE CATALOG

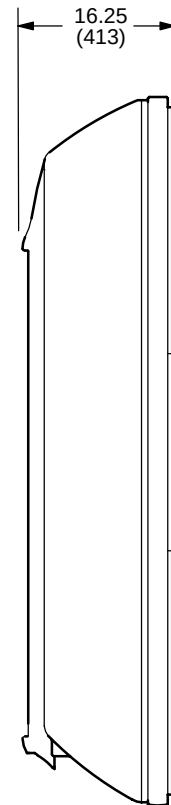
UPGRADE CATALOG	ILLUSTRATION NAME	ILLUSTRATION NUMBER
M3090AG	ACGD/PDU CABINET (MR3)	2-27
	SYSTEM CABINET (MR2)	2-29
	SRF CABINET (MR1)	2-30
	OPERATOR WORKSPACE (OW1)	2-33
	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
M3090CA	PENETRATION PANEL COVER	2-32
	SPT PHANTOM SET SHIPPING/STORAGE CABINET	2-38
M3090CB	PENETRATION PANEL COVER	2-32
	SPT PHANTOM SET SHIPPING/STORAGE CABINET	2-38
M3090CC	PENETRATION PANEL COVER	2-32
	SPT PHANTOM SET SHIPPING/STORAGE CABINET	2-38
M3090CH	SPT PHANTOM SET SHIPPING/STORAGE CABINET	2-38
M3090CJ	SPT PHANTOM SET SHIPPING/STORAGE CABINET	2-38
M3090CK M3090CM	PENETRATION PANEL COVER	2-32
	SPT PHANTOM SET SHIPPING/STORAGE CABINET	2-38
M3090DA M3090DB M3090DC	MNS CABINET (MNS) – 1.5T OPTION	2-39
M3090EC	SYSTEM CABINET (MR2) FOR 1.5T SIGNA MR/i WITH EXCITE TECHNOLOGY	2-28
M3090ED M3090EE	ACGD/PDU CABINET (MR3)	2-27
	SRF CABINET (MR1)	2-30
	OPERATOR WORKSPACE (OW1)	2-33
	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
M3890AB	ACGD/PDU CABINET (MR3)	2-27
	SYSTEM CABINET (MR2)	2-29
	SRF CABINET (MR1)	2-30
	PENETRATION PANEL (PP1)	2-31
	OPERATOR WORKSPACE (OW1)	2-33
	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
M3890AD	ACGD/PDU CABINET (MR3)	2-27
	SYSTEM CABINET (MR2)	2-29
	SRF CABINET (MR1)	2-30
	OPERATOR WORKSPACE (OW1)	2-33
	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
Note 1 Catalog provides a kit to modify existing system equipment to the configuration shown in the illustration. 2 Rear Pedestal is supplied with the indicated catalog numbers. Refer to Section 2-3, MINIMUM DOOR/HALLWAY SIZES, for dimension information.		

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 125 lbs (57 kg)



FRONT VIEW

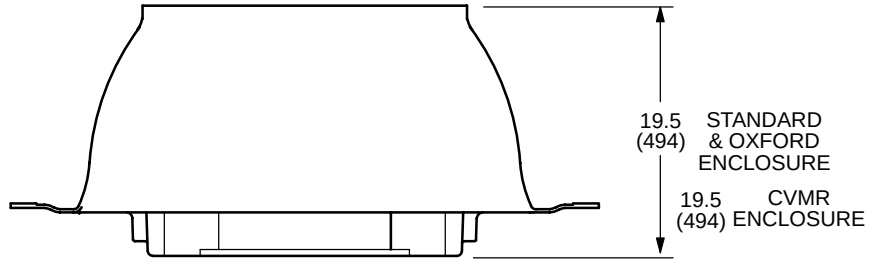


SIDE VIEW

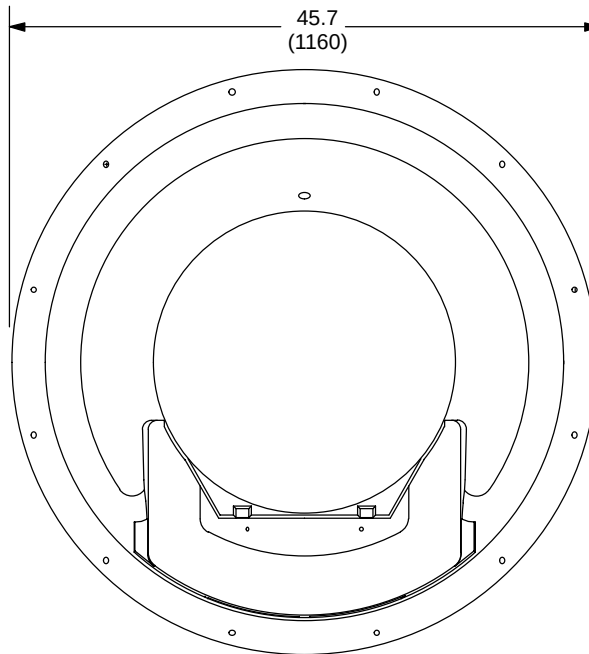
ENCLOSURE FRONT COVER – STANDARD
ILLUSTRATION 2-7

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 42 lbs (19 kg)
- STANDARD ENCLOSURE END BELL SHOWN.
OXFORD ENCLOSURE END BELL HAS SAME
DIMENSIONS WITH MINOR FIT AND
APPEARANCE DIFFERENCES.



TOP VIEW

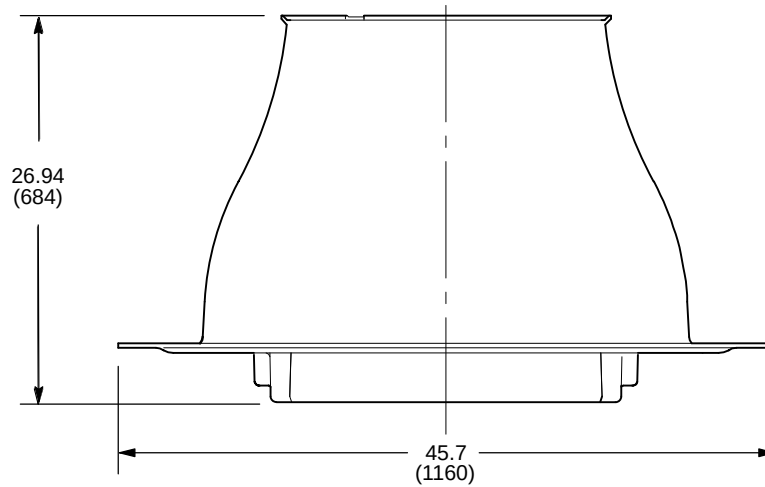


FRONT VIEW

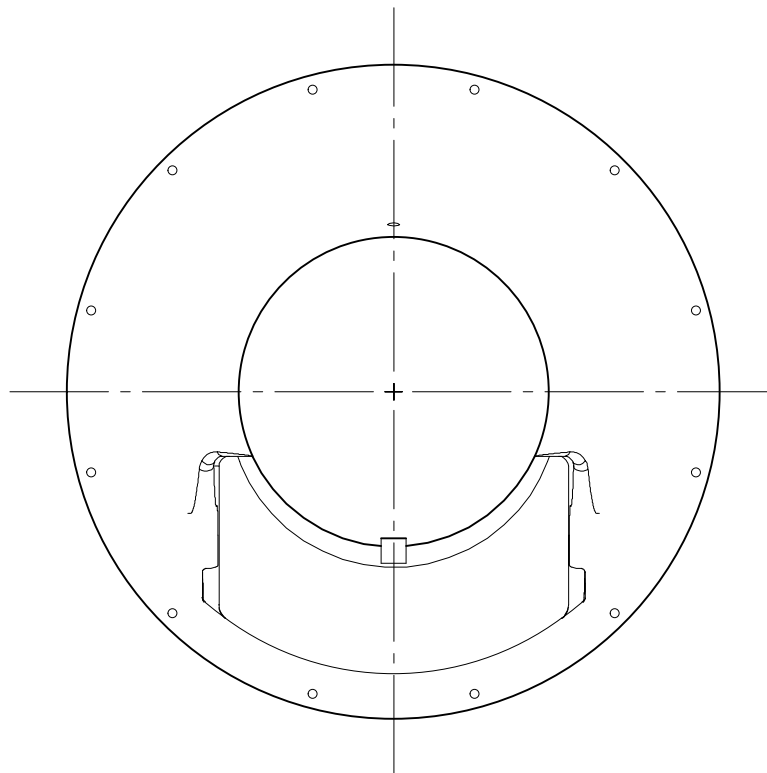
ENCLOSURE FRONT END BELL ASSEMBLY – STANDARD MAGNET
ILLUSTRATION 2–8

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 45 lbs (21 kg)



TOP VIEW

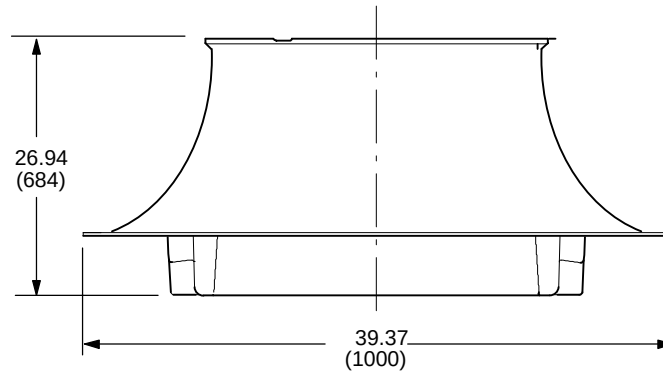


FRONT VIEW

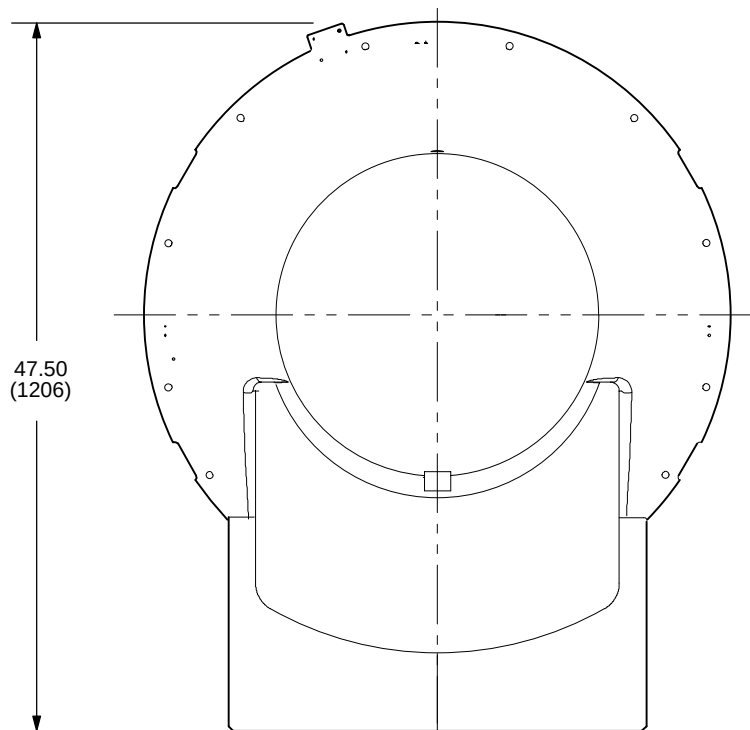
ENCLOSURE FRONT END BELL ASSEMBLY FOR S-II THROUGH S-V MAGNETS – SIGNA CVMR
ILLUSTRATION 2-9

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 45 lbs (21 kg)



TOP VIEW



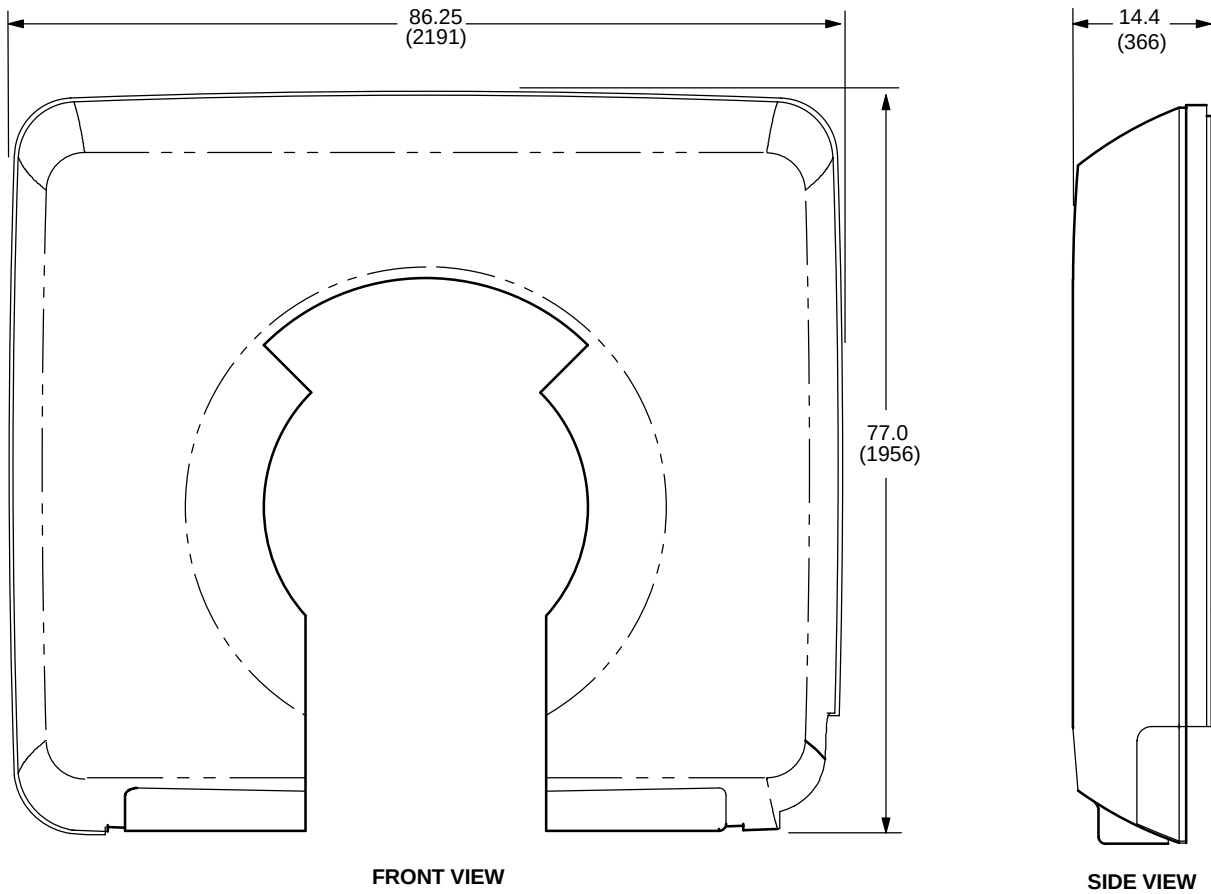
FRONT VIEW

M

ENCLOSURE FRONT END BELL ASSEMBLY FOR Cx & LCC MAGNETS
ILLUSTRATION 2-10

NOTE:

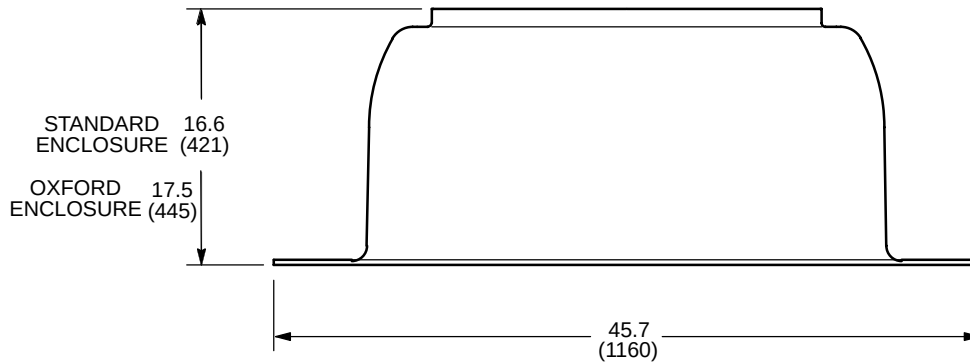
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 140 lbs (64 kg)



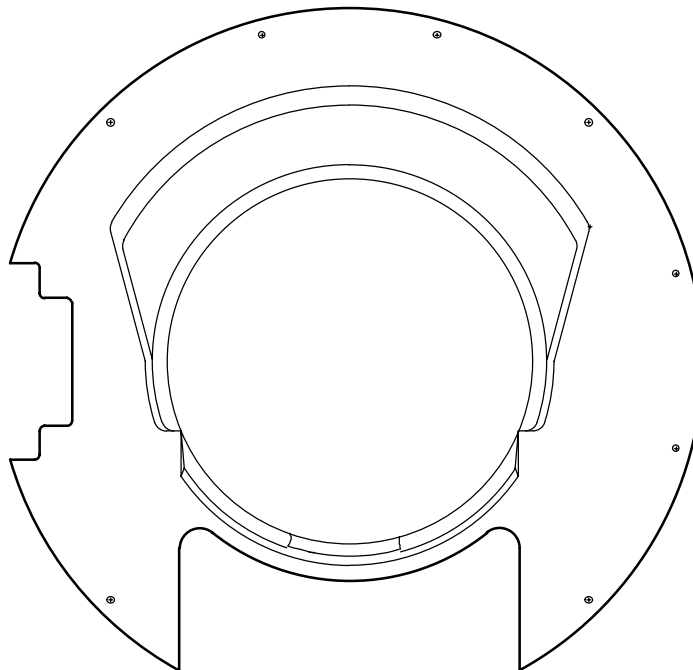
ENCLOSURE REAR COVER – STANDARD
ILLUSTRATION 2-11

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 45 lbs (21 kg)
- STANDARD ENCLOSURE END BELL SHOWN.
OXFORD ENCLOSURE END BELL HAS MINOR
FIT AND APPEARANCE DIFFERENCES IN
ADDITION TO DIMENSION DIFFERENCE
NOTED.



TOP VIEW

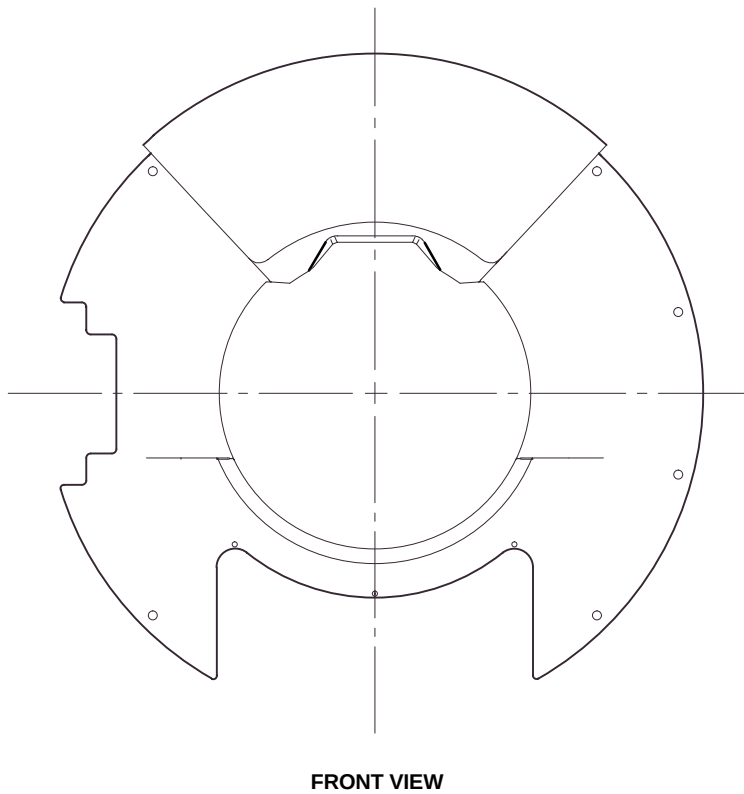
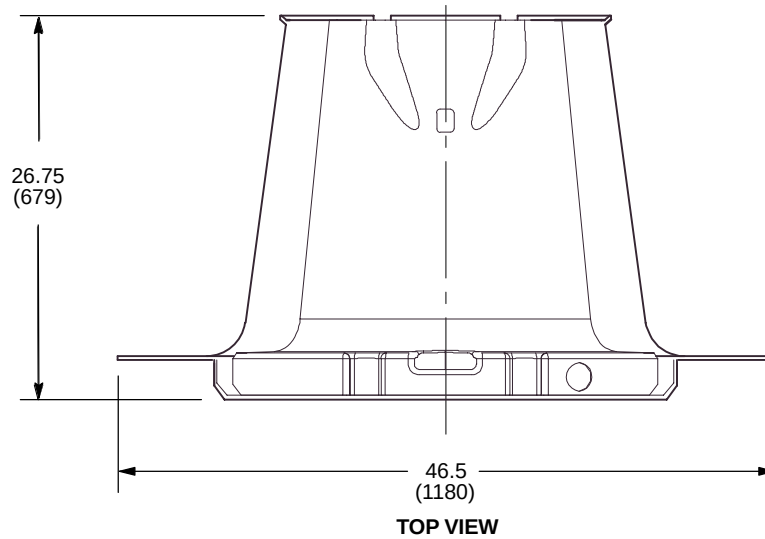


FRONT VIEW

ENCLOSURE REAR END BELL ASSEMBLY – STANDARD MAGNET
ILLUSTRATION 2-12

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 45 lbs (21 kg)

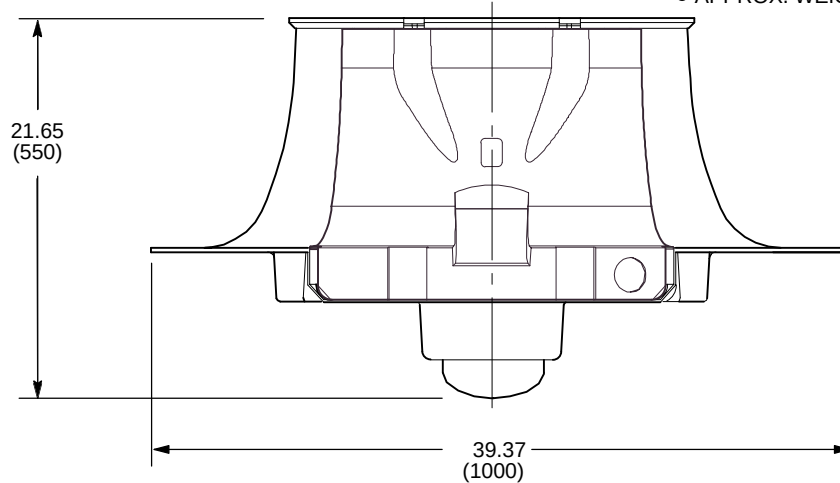


ENCLOSURE REAR END BELL ASSEMBLY FOR S-II THROUGH S-V MAGNETS
ILLUSTRATION 2-13

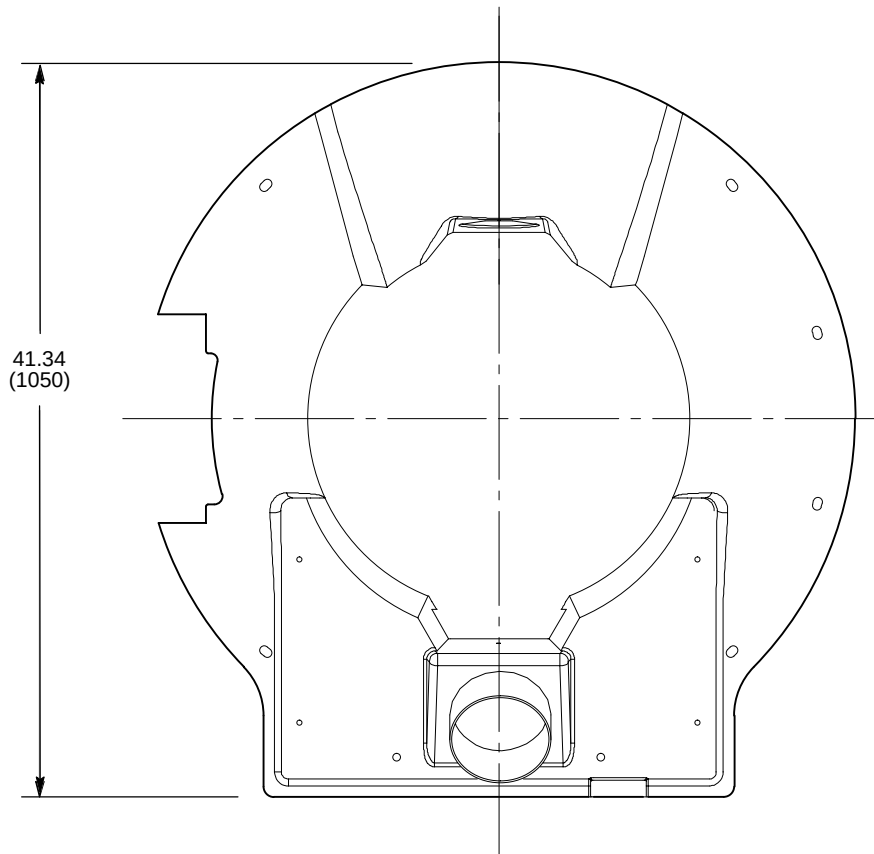
M

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN
MILLIMETERS.
- APPROX. WEIGHT: 45 lbs (21 kg)



TOP VIEW



FRONT VIEW

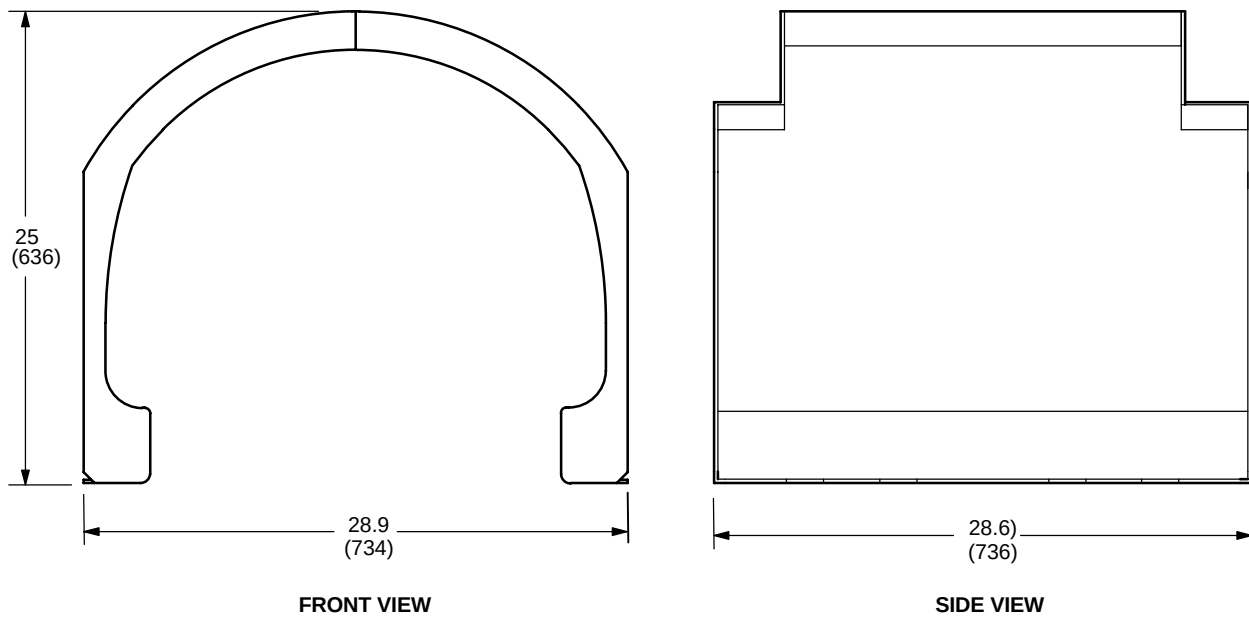
ENCLOSURE REAR END BELL ASSEMBLY FOR Cx & LCC MAGNETS

ILLUSTRATION 2-14

M

NOTE:

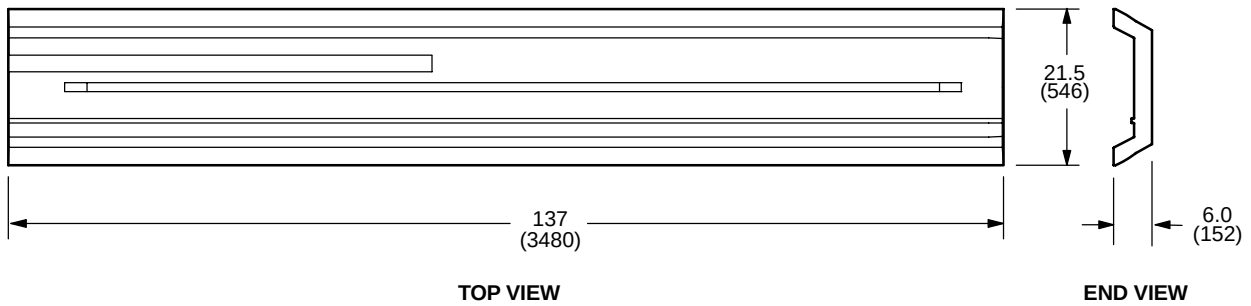
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 25 lbs (11 kg)



ENCLOSURE REAR COVER - MOBILE RF TUBE EXTENSION
ILLUSTRATION 2-15

NOTE:

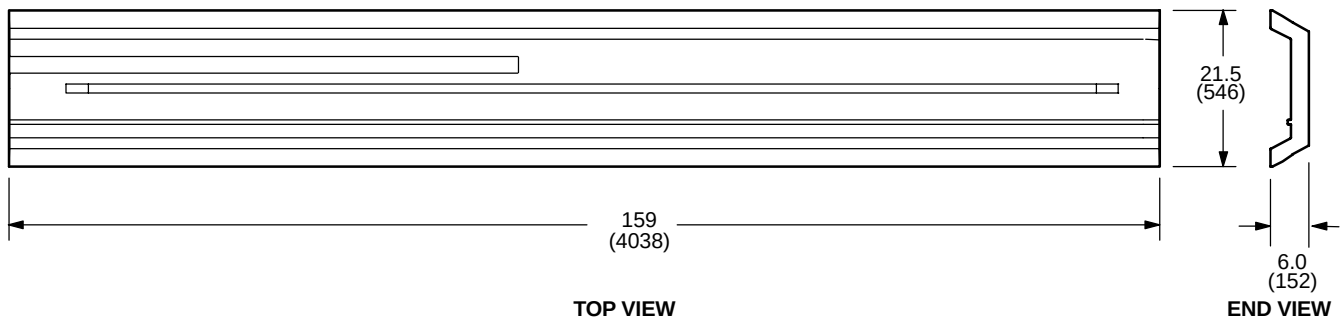
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 75 lbs (34 kg)



BRIDGE – STANDARD ENCLOSURE
ILLUSTRATION 2-16

NOTE:

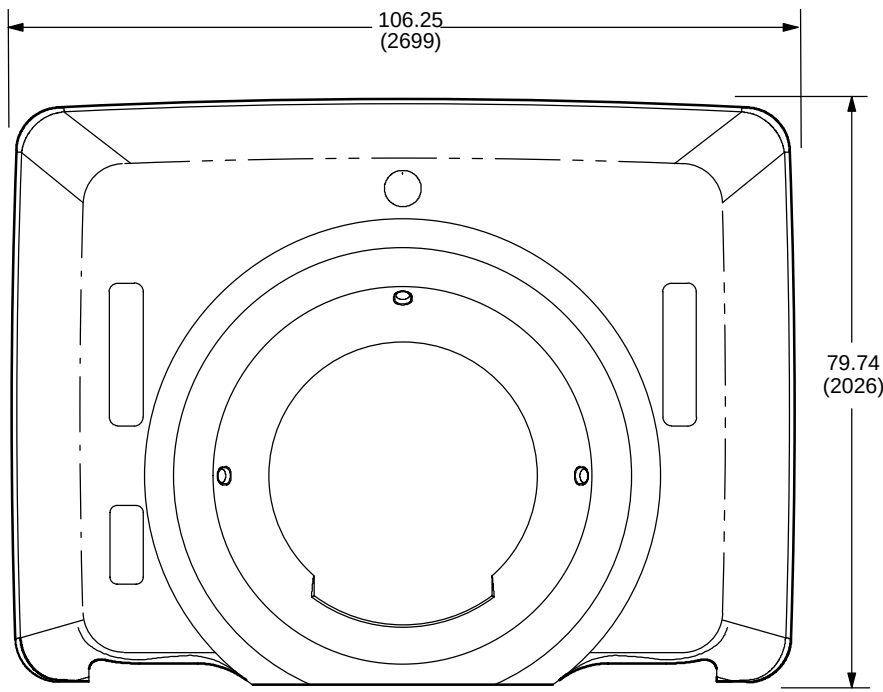
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 90 lbs (39 kg)



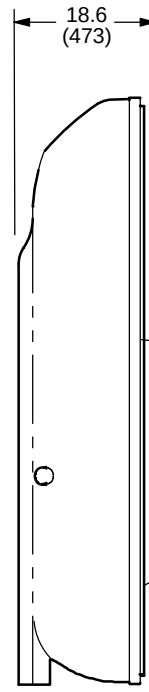
BRIDGE – MAGNISHIELD ENCLOSURE
ILLUSTRATION 2-17

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 170 lbs (77 kg)



FRONT VIEW

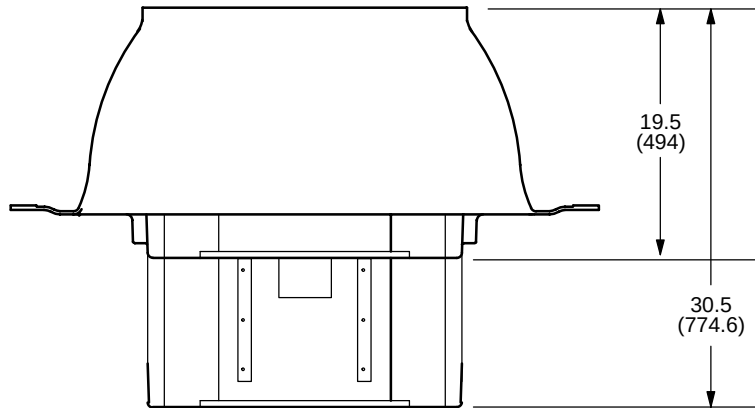


SIDE VIEW

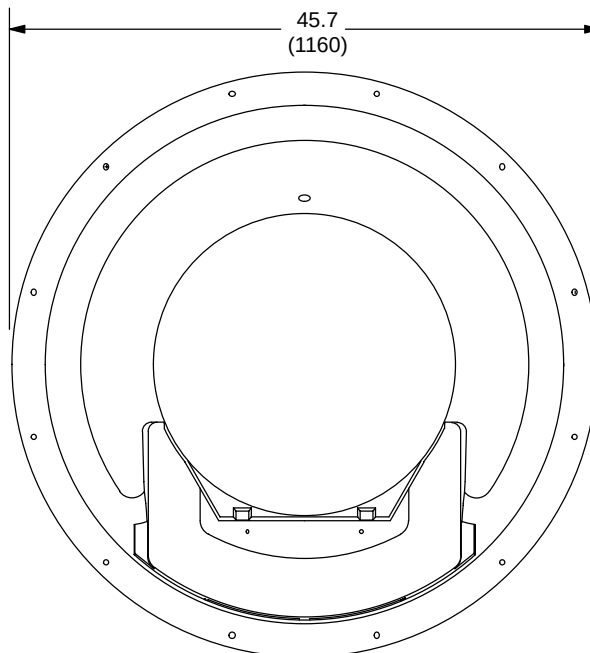
ENCLOSURE FRONT COVER – MAGNISHIELD MAGNET
ILLUSTRATION 2-18

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 55 lbs (22 kg)
- MAGNISHIELD ENCLOSURE FRONT END BELL CONSISTS OF STANDARD ENCLOSURE FRONT END BELL AND MAGNISHIELD END BELL EXTENSION INSTALLED DURING UPGRADE.



TOP VIEW



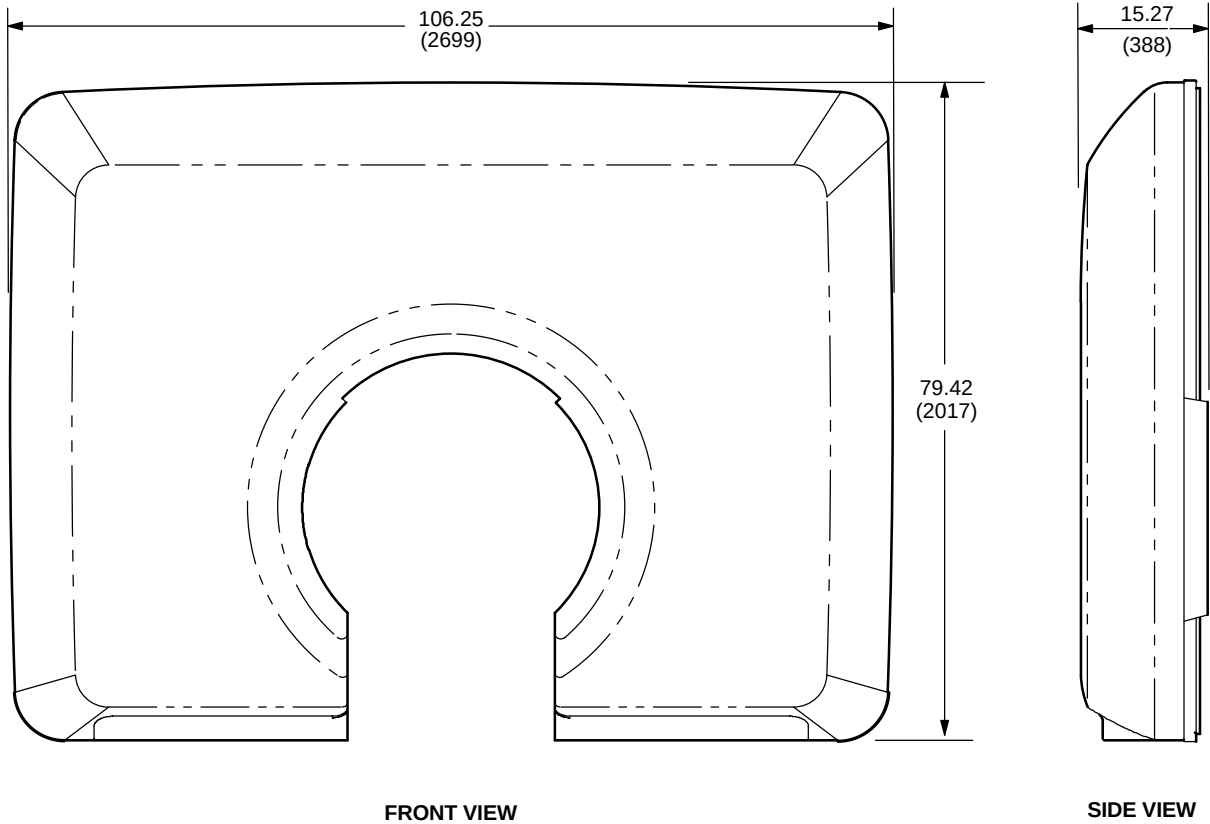
FRONT VIEW

ENCLOSURE FRONT END BELL ASSEMBLY – MAGNISHIELD

ILLUSTRATION 2-19

NOTE:

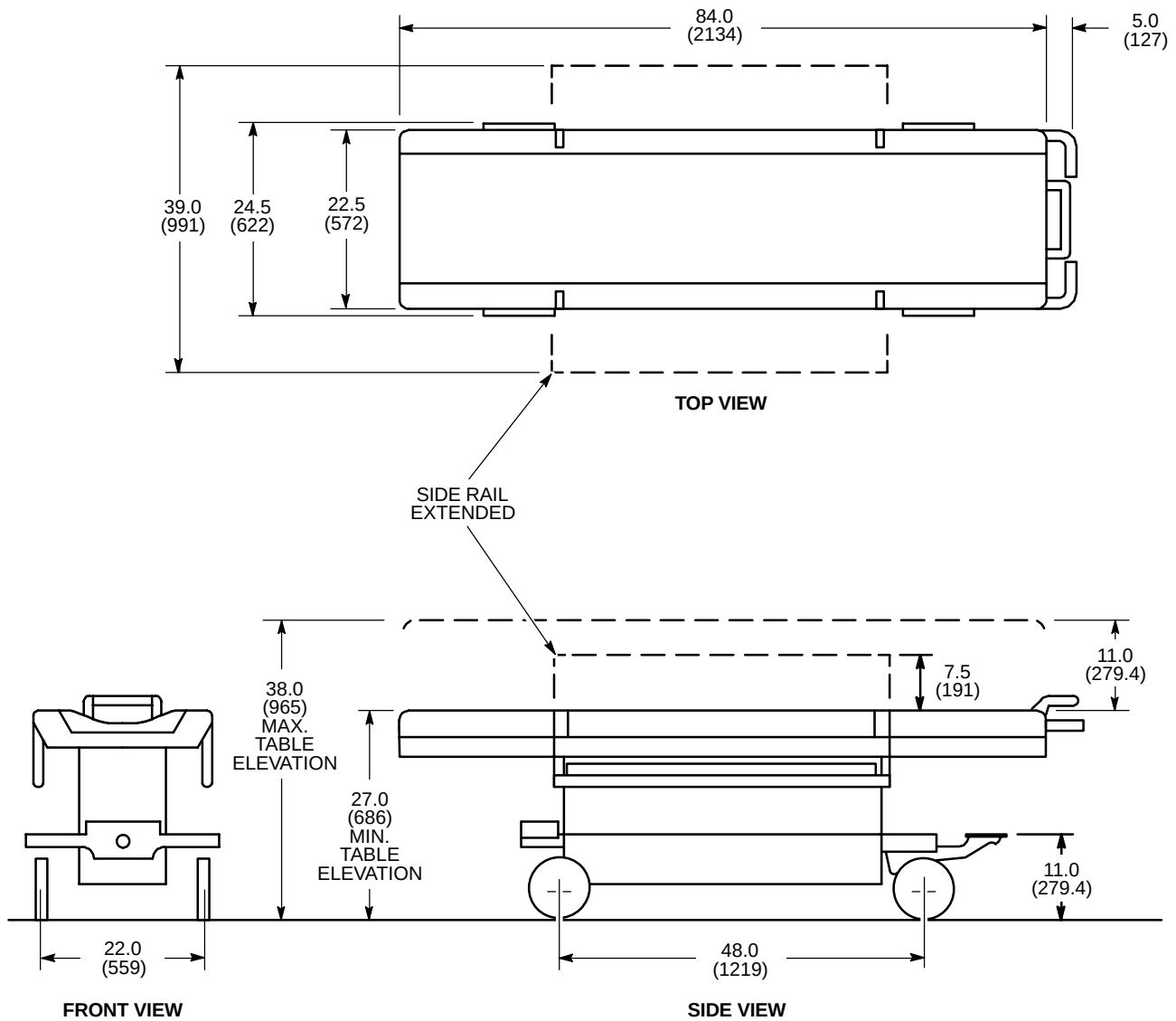
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 170 lbs (77 kg)



ENCLOSURE REAR COVER – MAGNISHIELD MAGNET
ILLUSTRATION 2-20

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- WEIGHT: 280 lbs (127 kg)



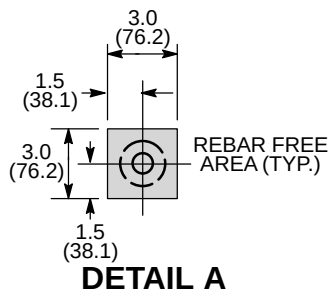
M1081A2

PATIENT TABLE (PT1)
ILLUSTRATION 2-21

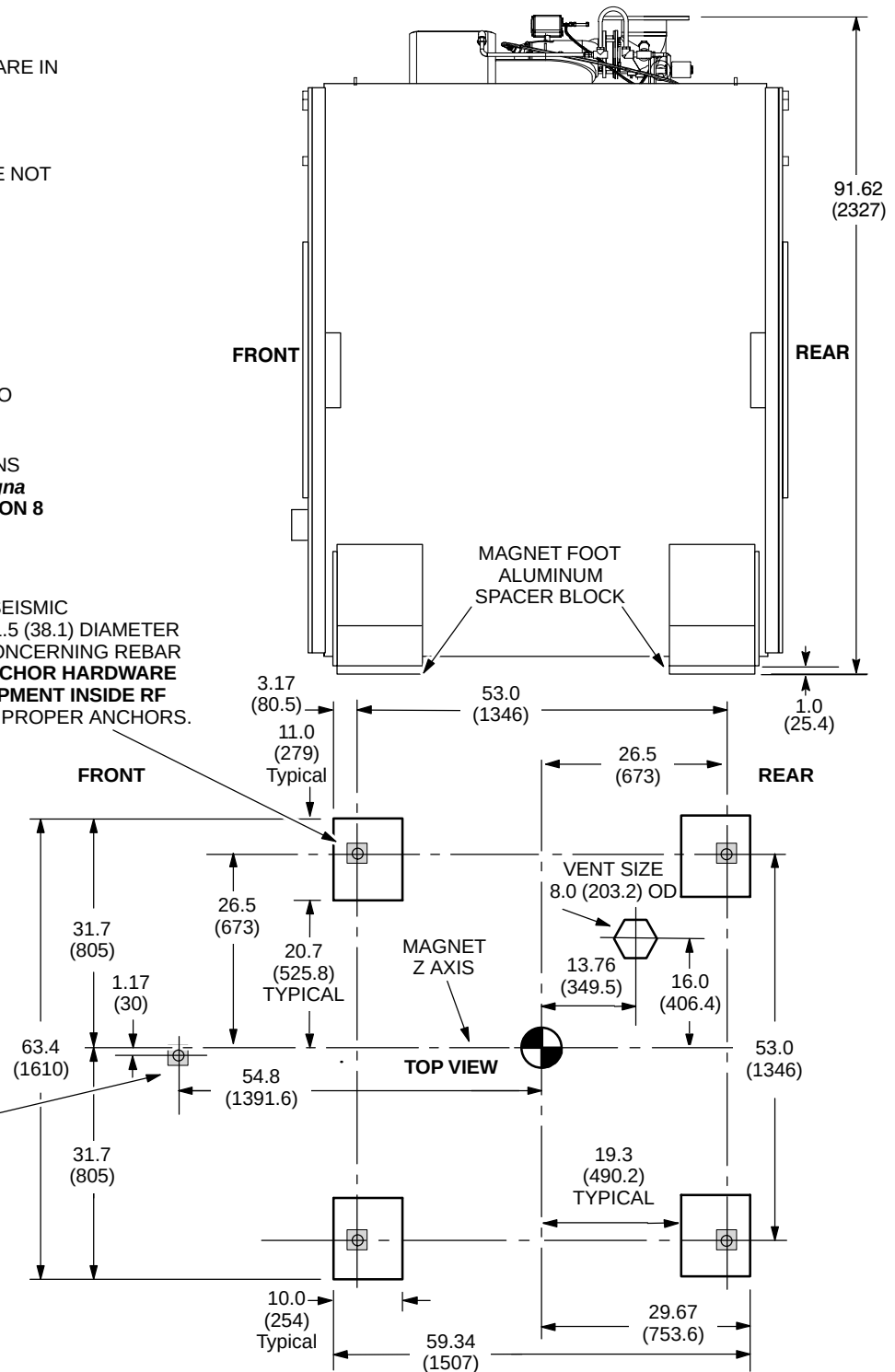
NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- LIFTING/JACKING BEAMS ARE ATTACHED TO SIDES OF MAGNET. REFER TO ILLUSTRATION 8-1.
- ALUMINUM LEVELING PLATES ARE NOT SHOWN.
- APPROX. WEIGHT OF MAGNET, CRYOGENS, COILS, AND ENCLOSURE WITH BRM
1.0T: 11,856 lbs (5378 kg)
1.5T: 12,606 lbs (5719 kg)
FOR SIGNA CV/i WITH CRM
1.5T: 12,806 lbs (5821 kg)
- STEEL REBAR MUST NOT BE POSITIONED IN SHADED AREAS TO PREVENT INTERFERENCE WITH MOUNTING BOLTS.
- FOR MAGNET MOVING DIMENSIONS REFER TO **DIRECTION 2223170 Signa MR/i & CV/i Pre-Installation SECTION 8 SHIPPING AND DELIVERY DATA**

SEISMIC AND NON-SEISMIC
MAGNET MOUNTING HOLES (4) 1.5 (38.1) DIAMETER
SEE NOTE ABOVE & DETAIL A CONCERNING REBAR
REFER TO **SECTION NO TAG ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM** TO DETERMINE PROPER ANCHORS.

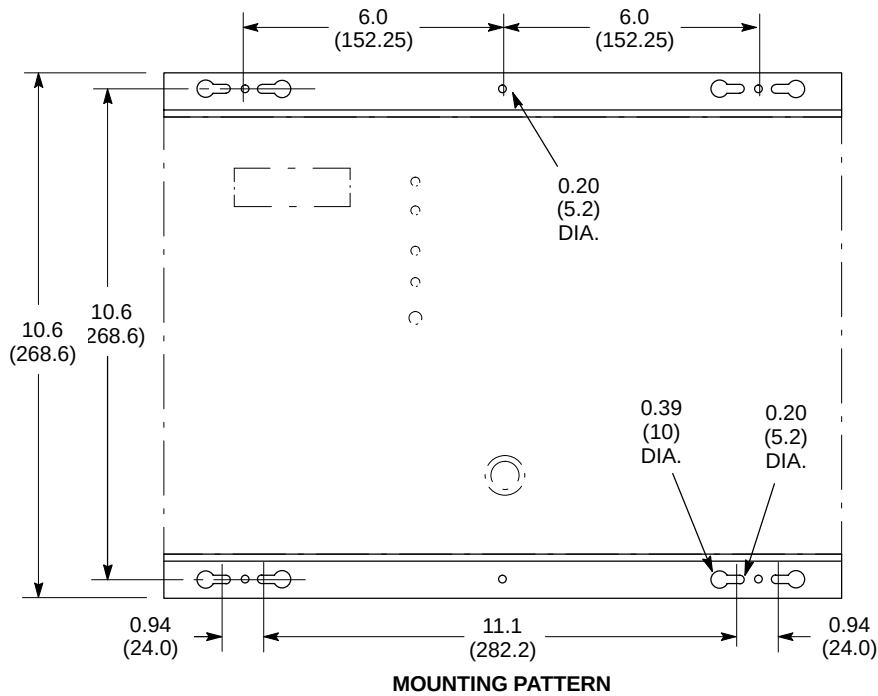


M10 ANCHOR FOR
TABLE DOCK MOUNTING,
LOCATED OFFSET FROM
MAGNET CENTER LINE.
FOR ADDITIONAL
MOUNTING INFORMATION
REFER TO SECTION
NO TAG
**ANCHOR HARDWARE
REQUIREMENTS FOR MR
EQUIPMENT INSIDE RF
SHIELDED ROOM.**



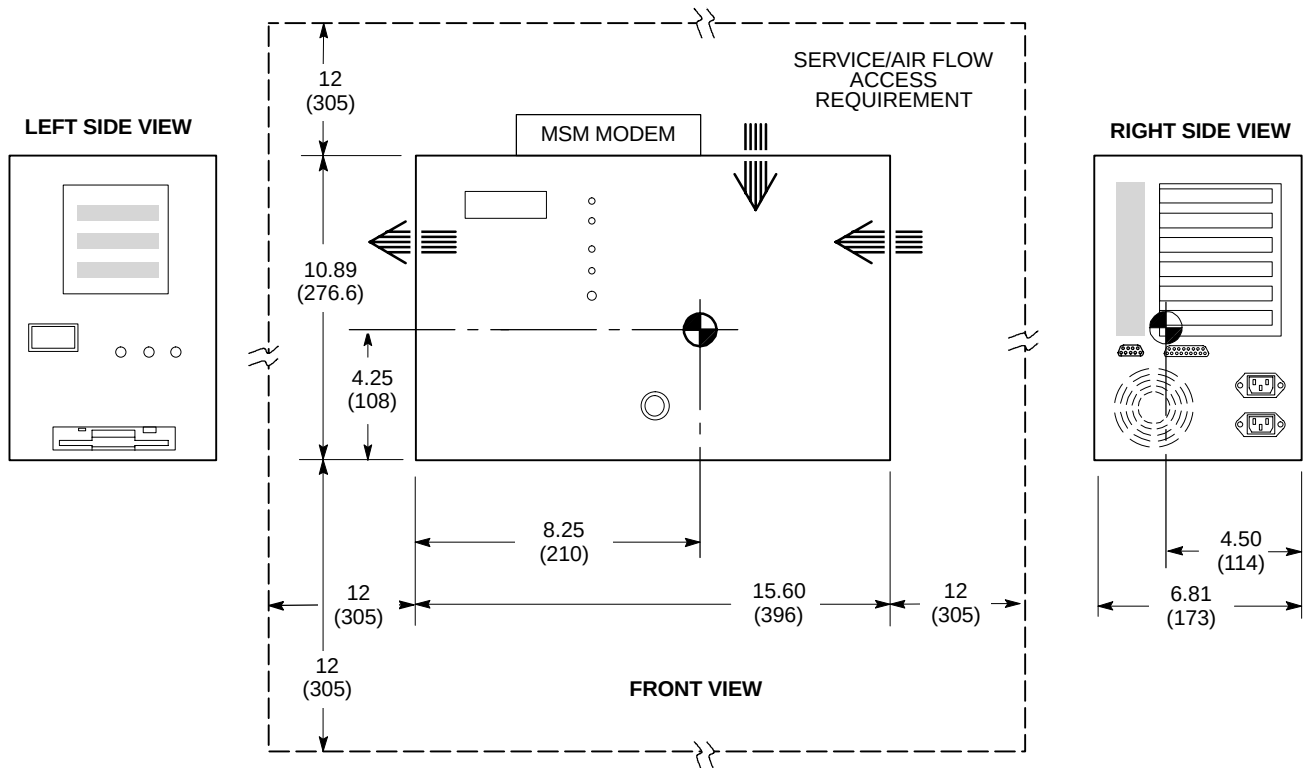
1.0T AND 1.5T LCC (ACTIVE SHIELD) MAGNET LOAD PATTERN

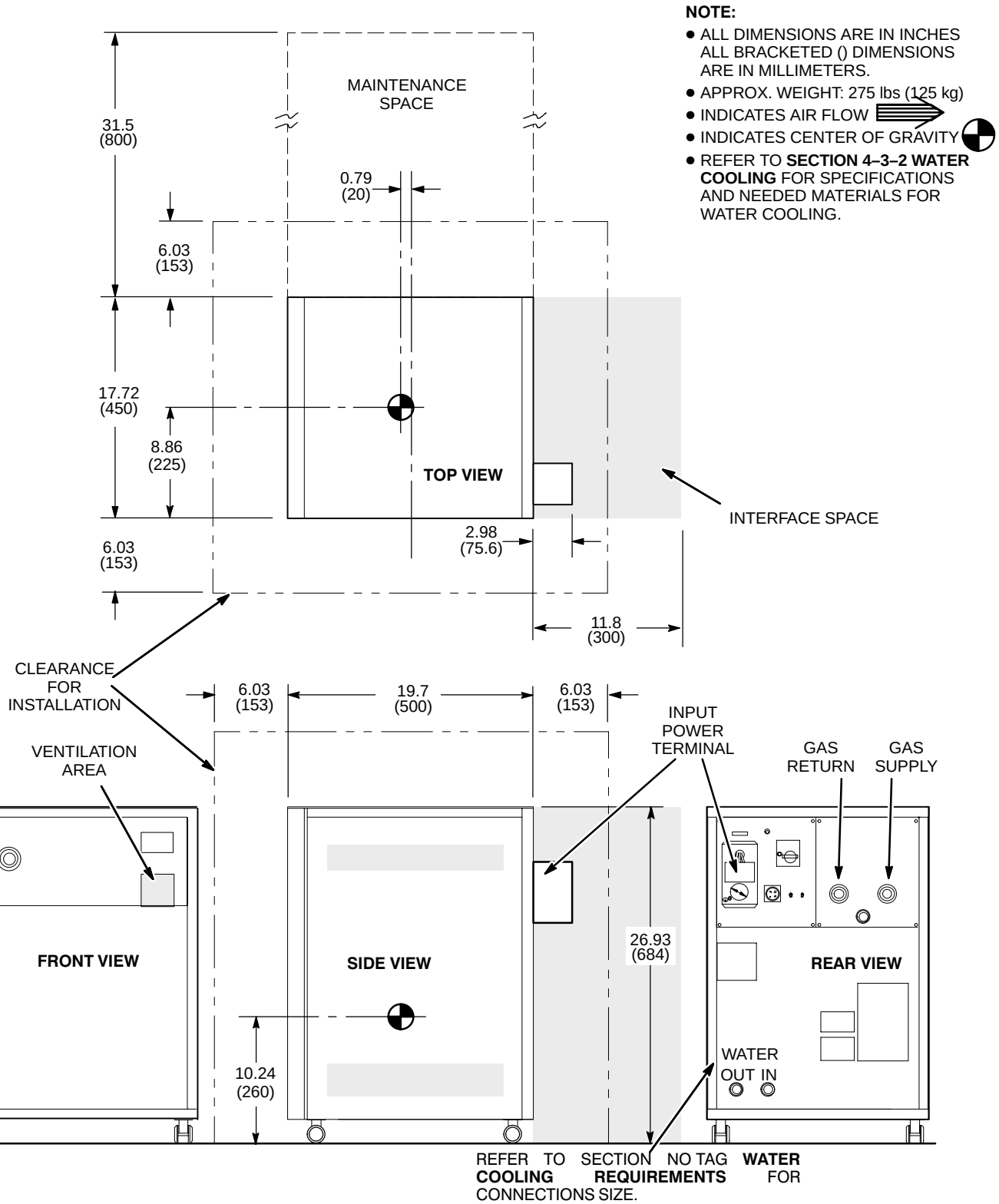
ILLUSTRATION 2-22



NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 22 lbs (10 kg)
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY



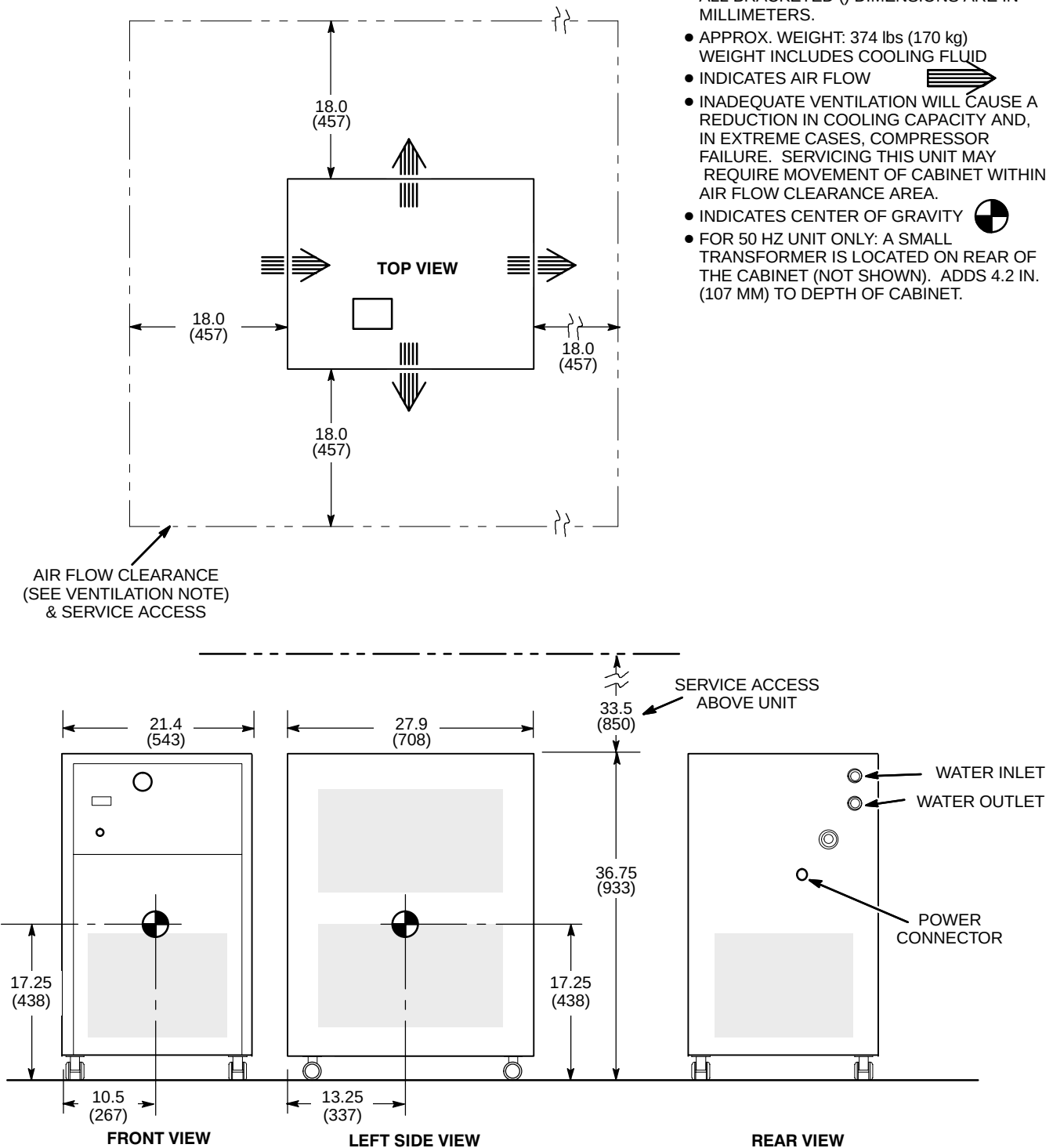


LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR CABINET (MS5)

ILLUSTRATION 2-24

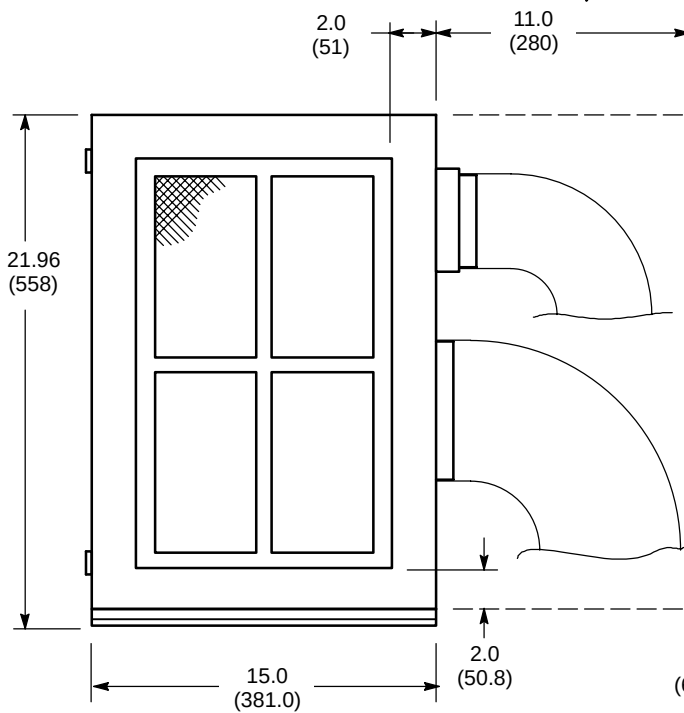
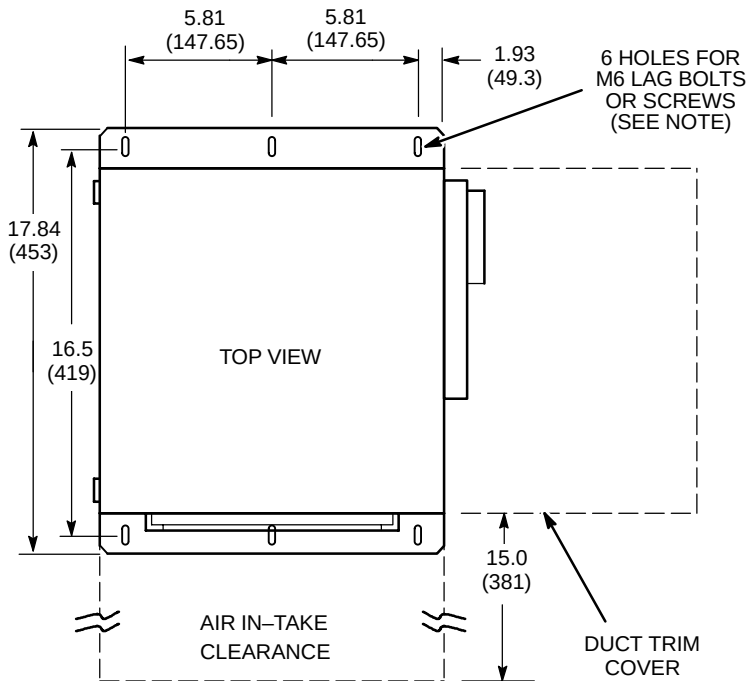
NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 374 lbs (170 kg)
WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW 
- INADEQUATE VENTILATION WILL CAUSE A REDUCTION IN COOLING CAPACITY AND, IN EXTREME CASES, COMPRESSOR FAILURE. SERVICING THIS UNIT MAY REQUIRE MOVEMENT OF CABINET WITHIN AIR FLOW CLEARANCE AREA.
- INDICATES CENTER OF GRAVITY 
- FOR 50 HZ UNIT ONLY: A SMALL TRANSFORMER IS LOCATED ON REAR OF THE CABINET (NOT SHOWN). ADDS 4.2 IN. (107 MM) TO DEPTH OF CABINET.

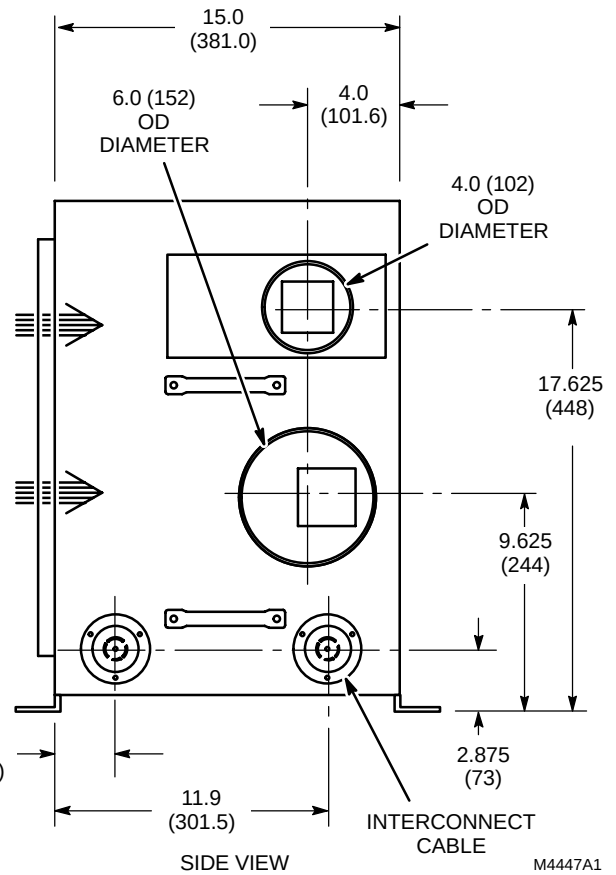


WATER CHILLER (WC1) FOR GRADIENT COIL COOLING WATER

ILLUSTRATION 2-25



FRONT VIEW



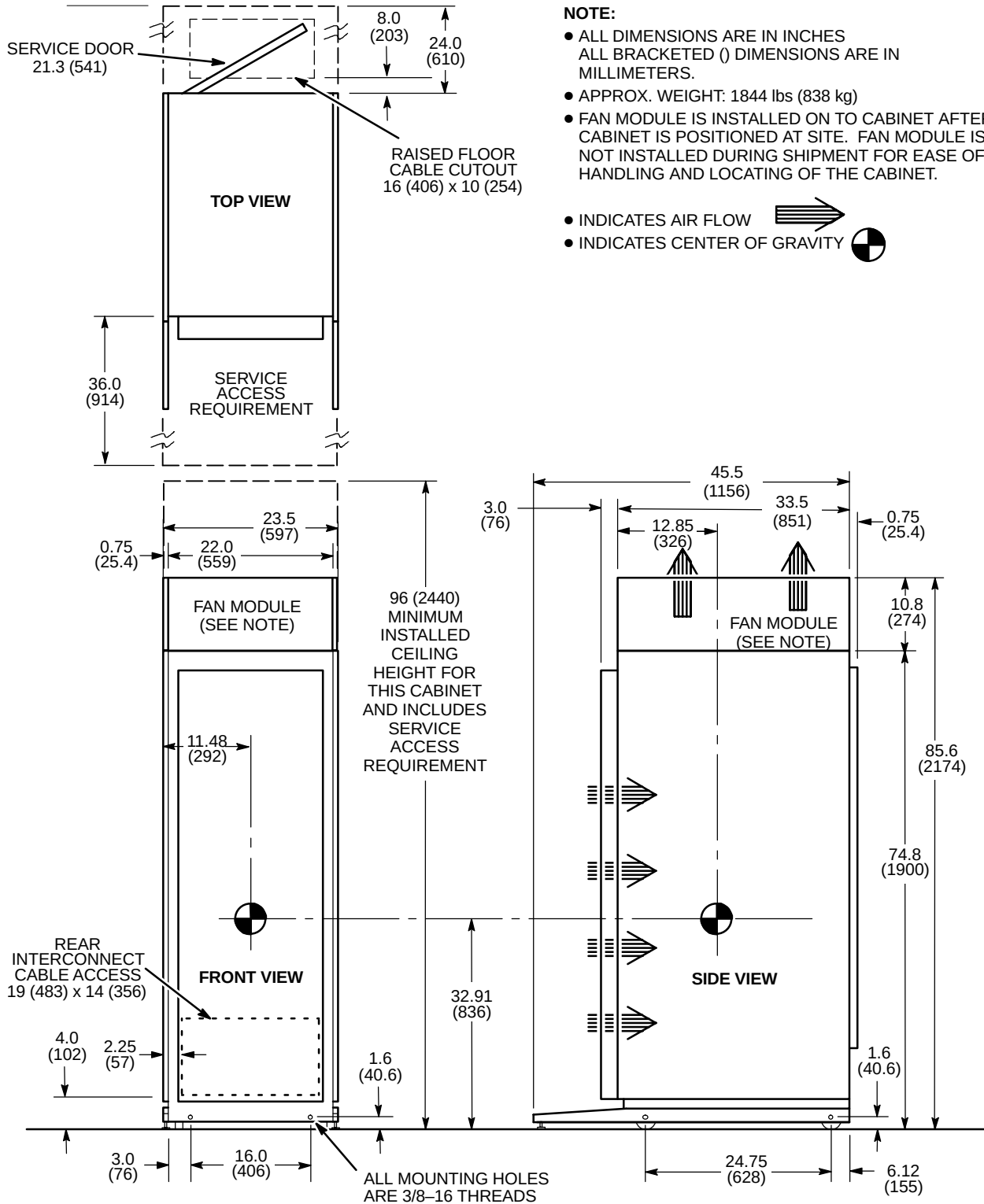
SIDE VIEW

BLOWER BOX (MG6)
ILLUSTRATION 2-26

NOTE:

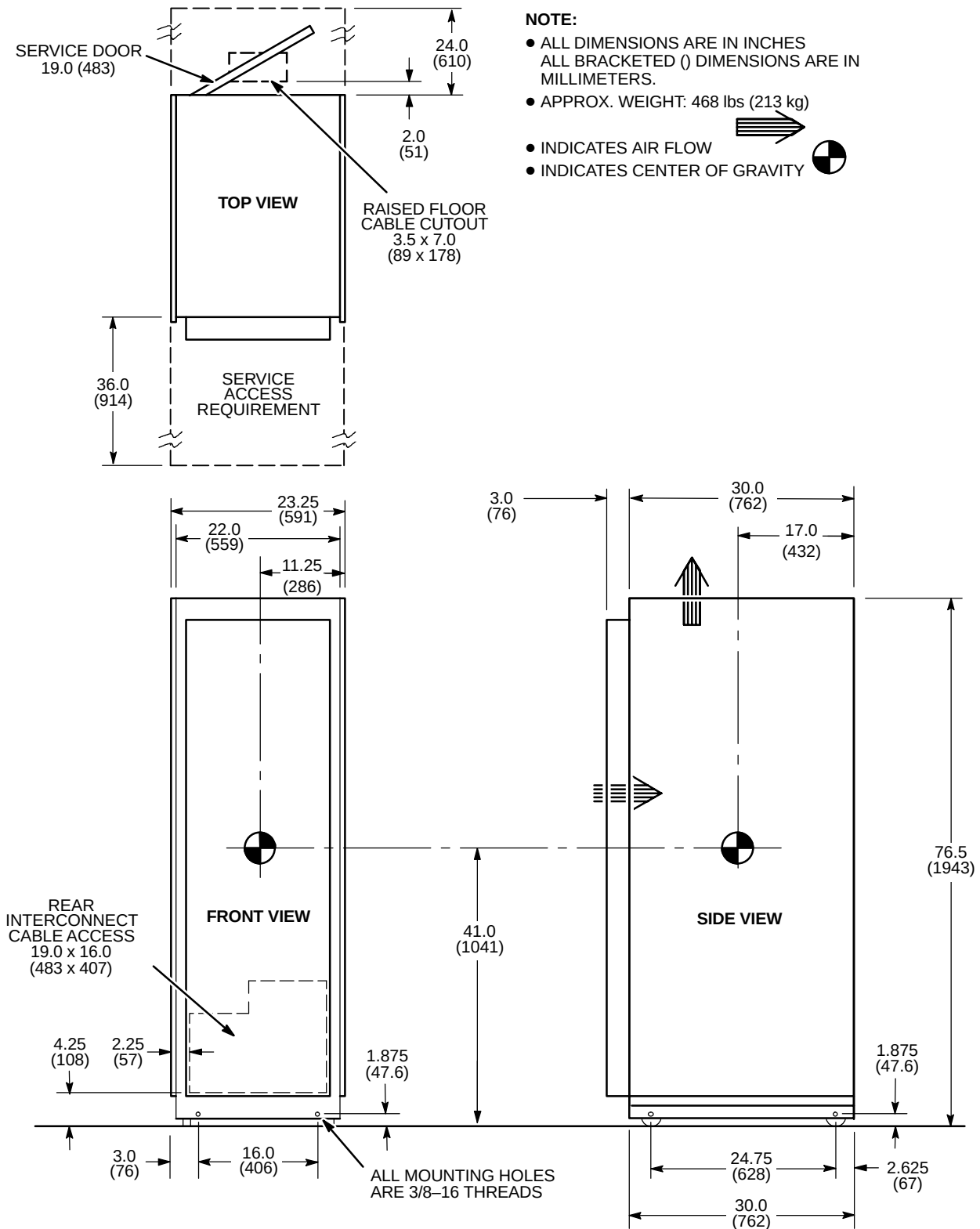
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 47 lbs (21 kg)
- BLOWER BOX MUST BE ANCHORED DUE FERROUS COMPONENTS.
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 

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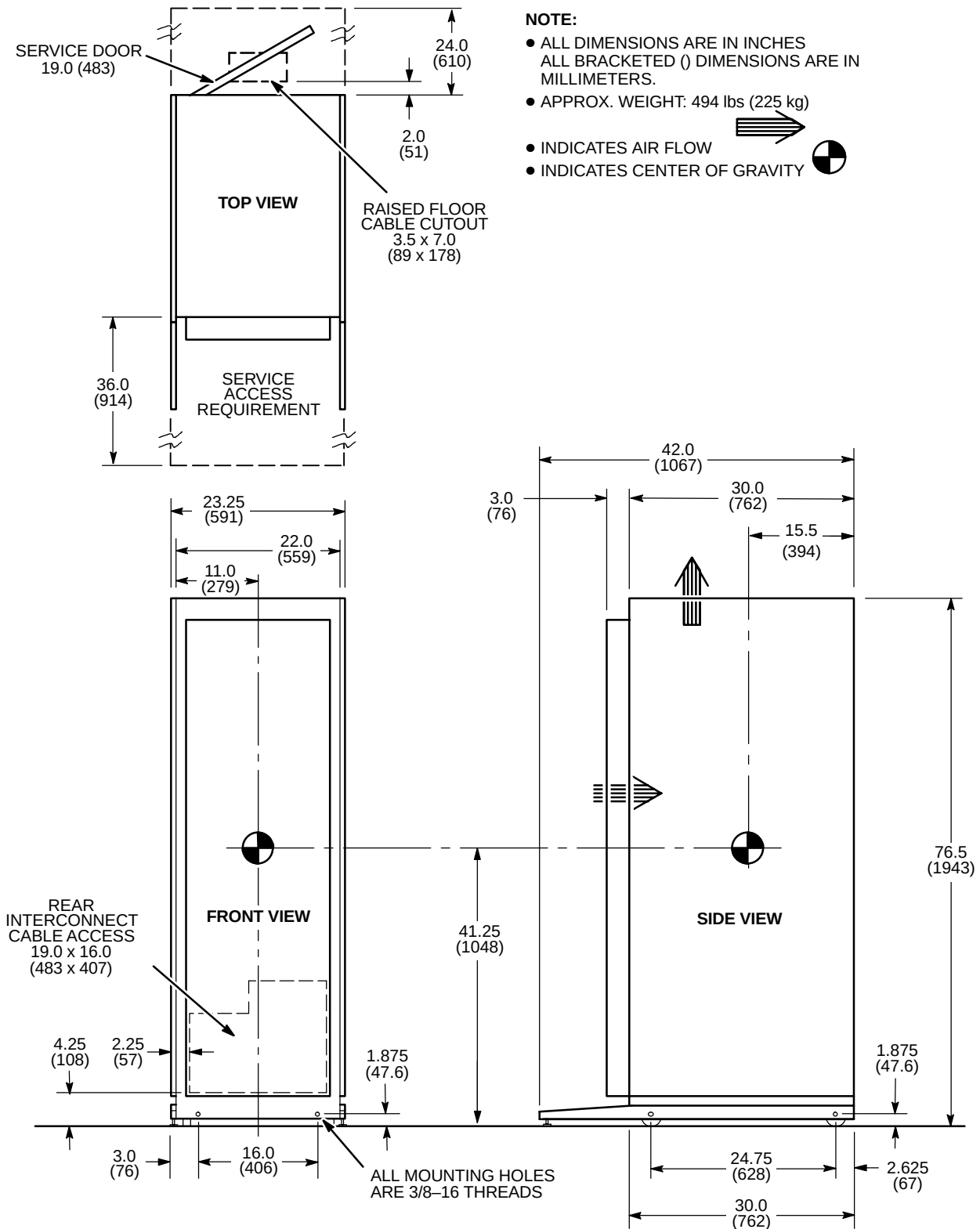
ACGD/PDU CABINET (MR3)

ILLUSTRATION 2-27



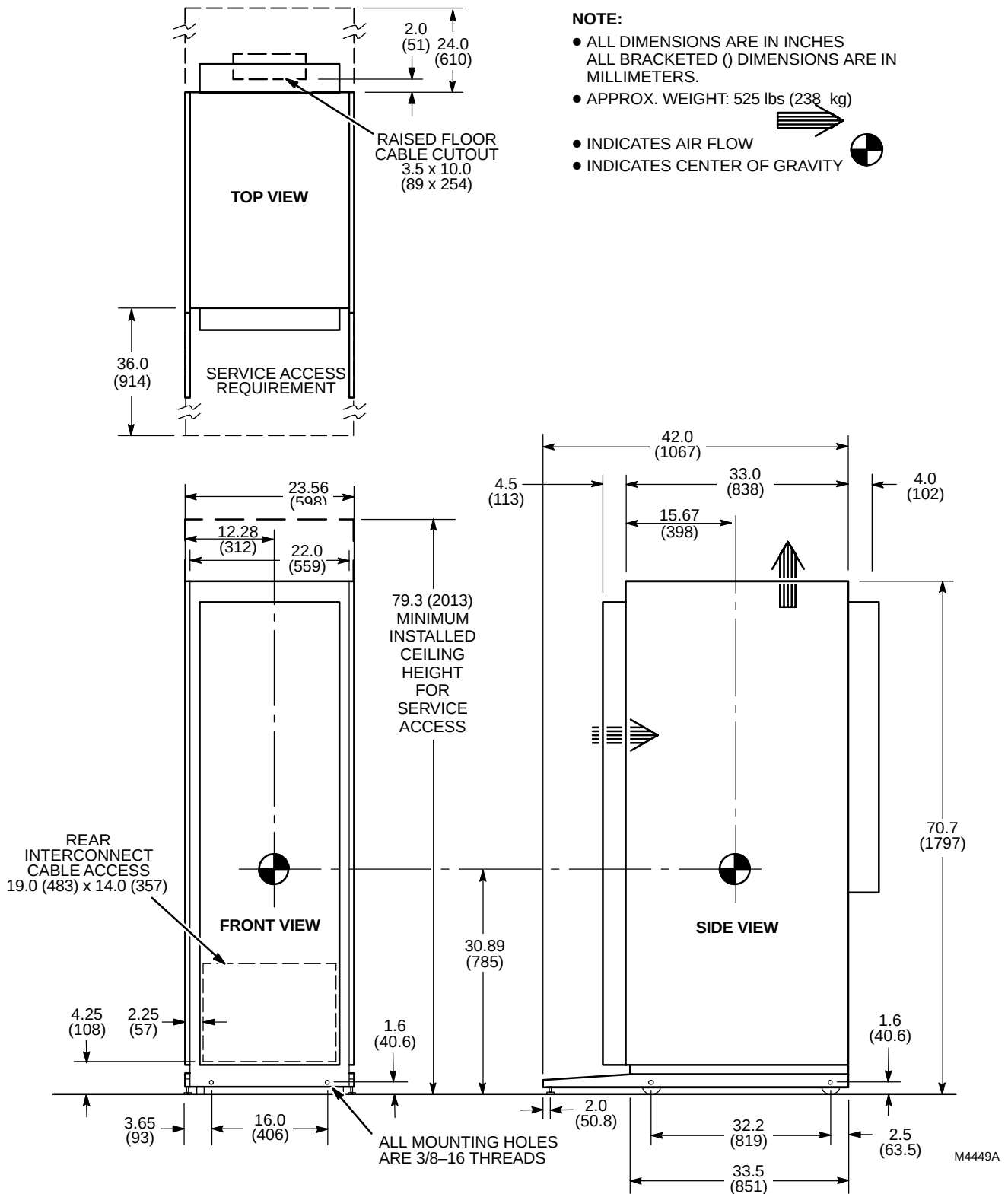
SYSTEM CABINET (MR2) FOR 1.5T SIGNA MR/i WITH EXCITE TECHNOLOGY

ILLUSTRATION 2-28



SYSTEM CABINET (MR2) FOR 1.5T SIGNA MR/i, 1.0T SIGNA MR/i, & 1.5T SIGNA CV/i

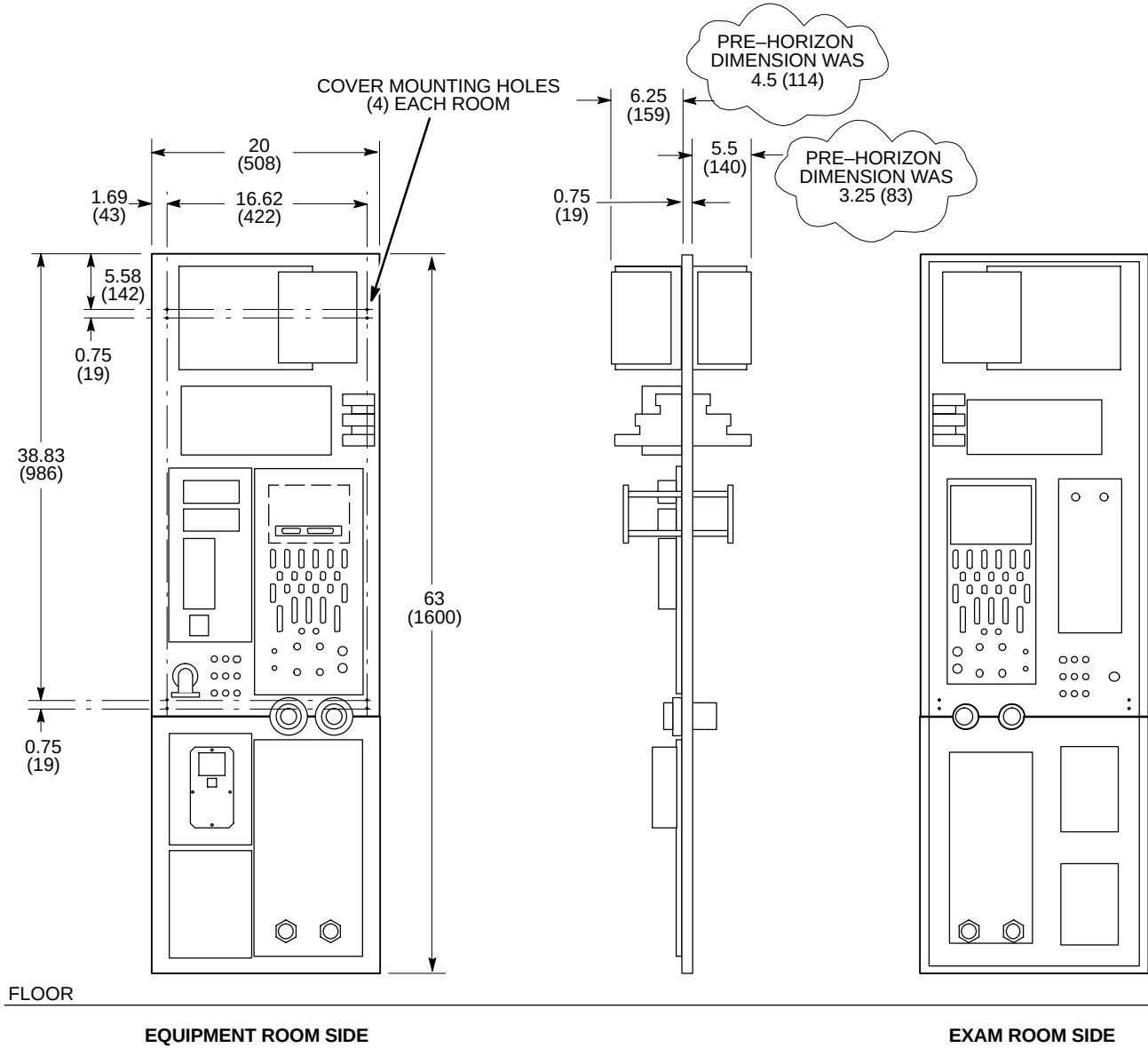
ILLUSTRATION 2-29



SRF CABINET (MR1)
ILLUSTRATION 2-30

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- APPROX. WEIGHT: 52 lbs (23.6 kg)

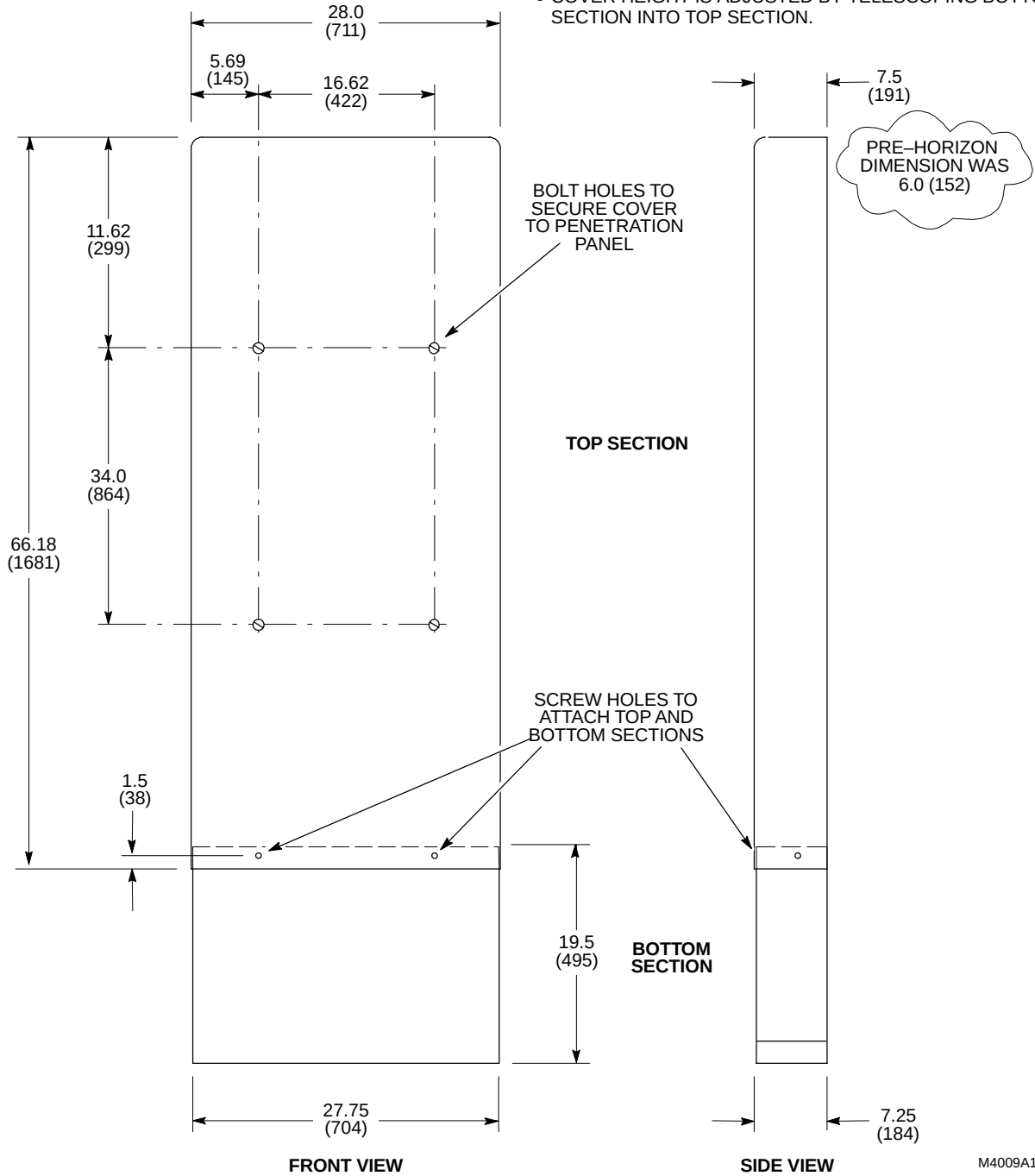


PENETRATION PANEL (PP1)
ILLUSTRATION 2-31

M4494A

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- COVER USED ON BOTH EXAM ROOM AND EQUIPMENT ROOM SIDES OF PENETRATION PANEL. TWO COVERS ARE PROVIDED.
- COVER HEIGHT IS ADJUSTED BY TELESOPING BOTTOM SECTION INTO TOP SECTION.

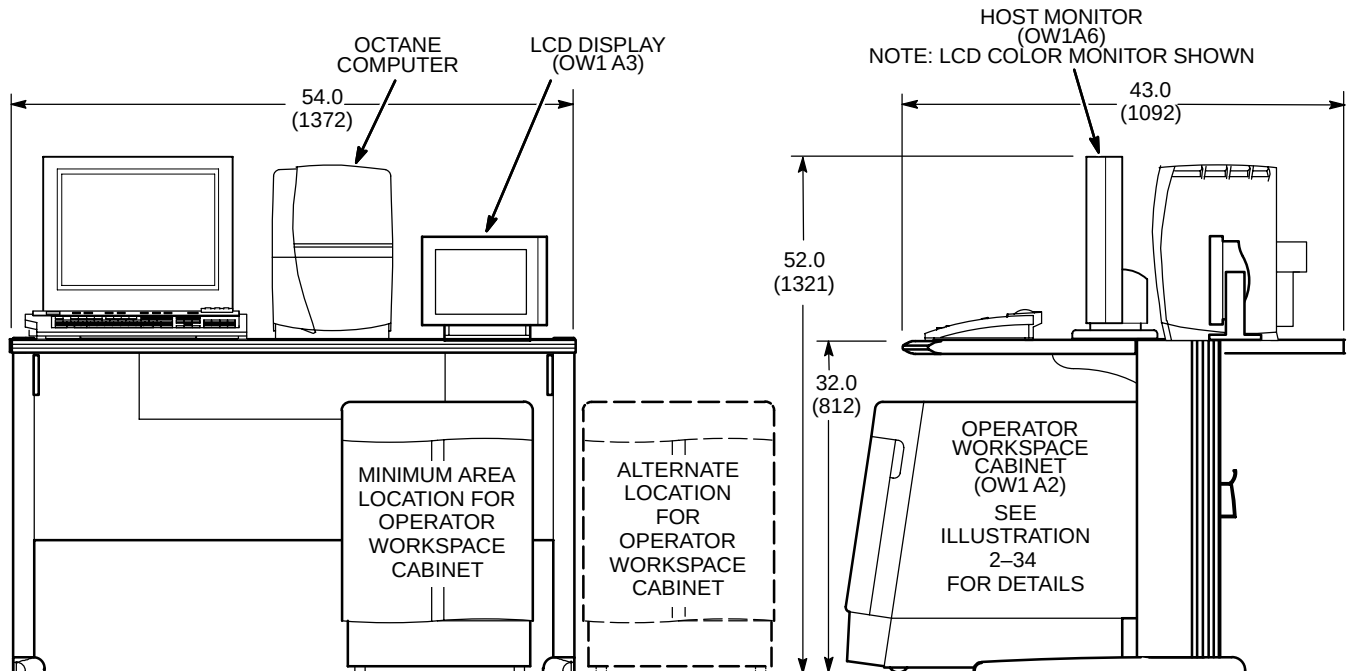
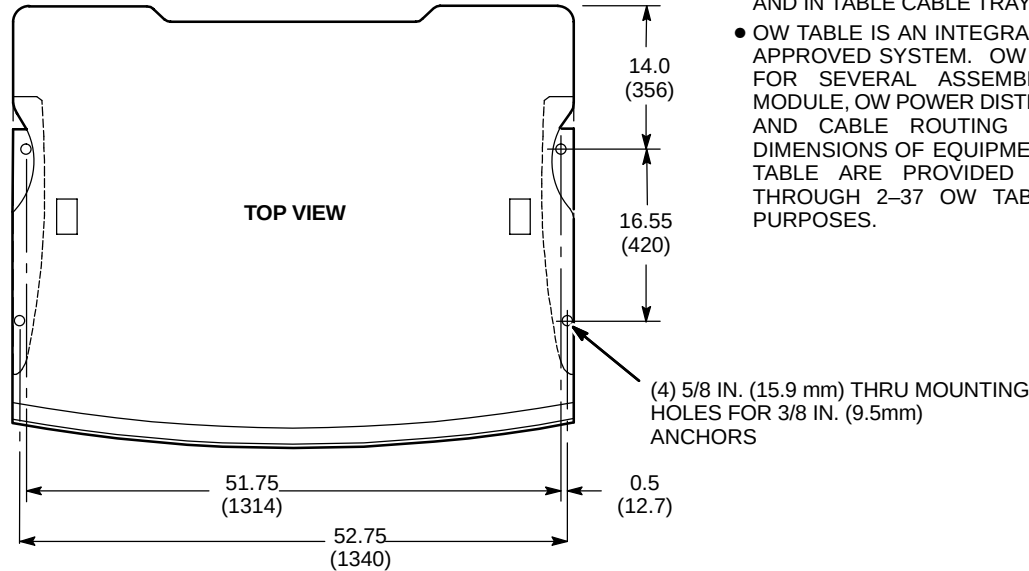


M4009A1M

PENETRATION PANEL COVER
ILLUSTRATION 2-32

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- ASSEMBLIES WHICH MOUNT TO UNDERSIDE OF TABLE AND IN TABLE CABLE TRAY ARE NOT SHOWN.
- OW TABLE IS AN INTEGRAL PART OF THE REGULATORY APPROVED SYSTEM. OW TABLE PROVIDES MOUNTING FOR SEVERAL ASSEMBLIES (E.G. OW INTERFACE MODULE, OW POWER DISTRIBUTION BOX, MODEM, DASM) AND CABLE ROUTING FOR OW INTERCONNECTS. DIMENSIONS OF EQUIPMENT LOCATED ON TOP OF OW TABLE ARE PROVIDED IN ILLUSTRATIONS NO TAG THROUGH 2-37 OW TABLE TOP LAYOUT PLANNING PURPOSES.



FRONT VIEW

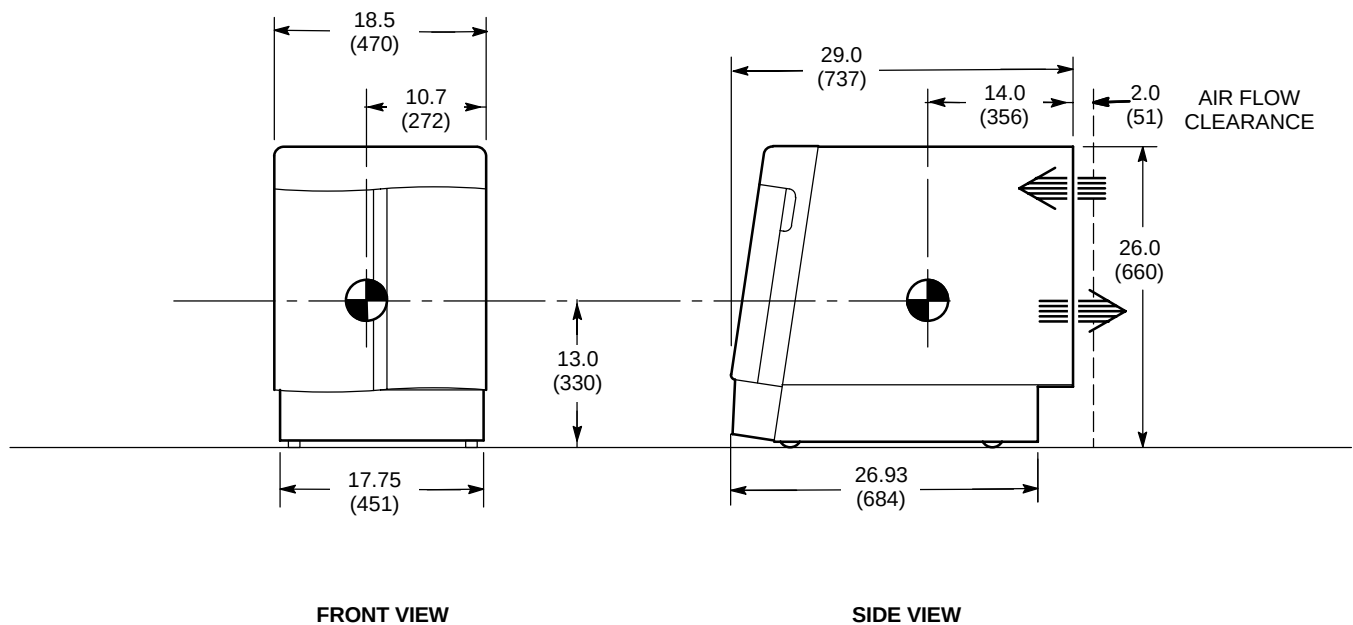
SIDE VIEW

OPERATOR WORKSPACE (OW1)

ILLUSTRATION 2-33

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 192 lbs (87 kg)
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 

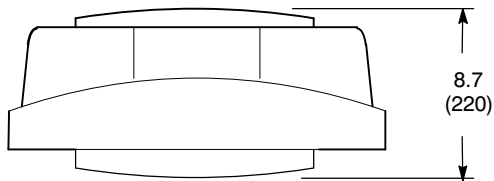


OPERATOR WORKSPACE CABINET (OW1 A2)

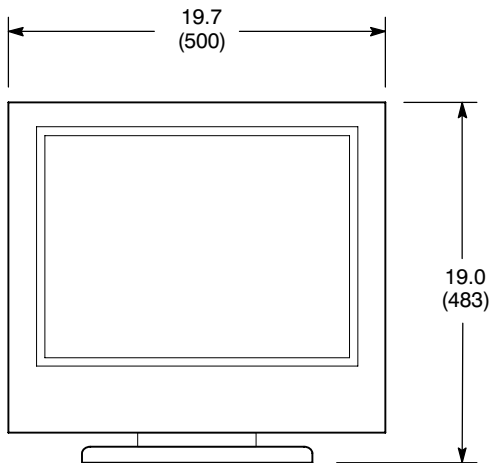
ILLUSTRATION 2-34

NOTE:

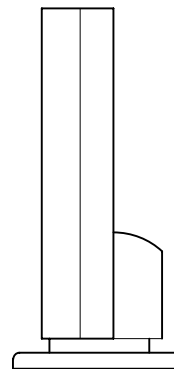
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



TOP VIEW



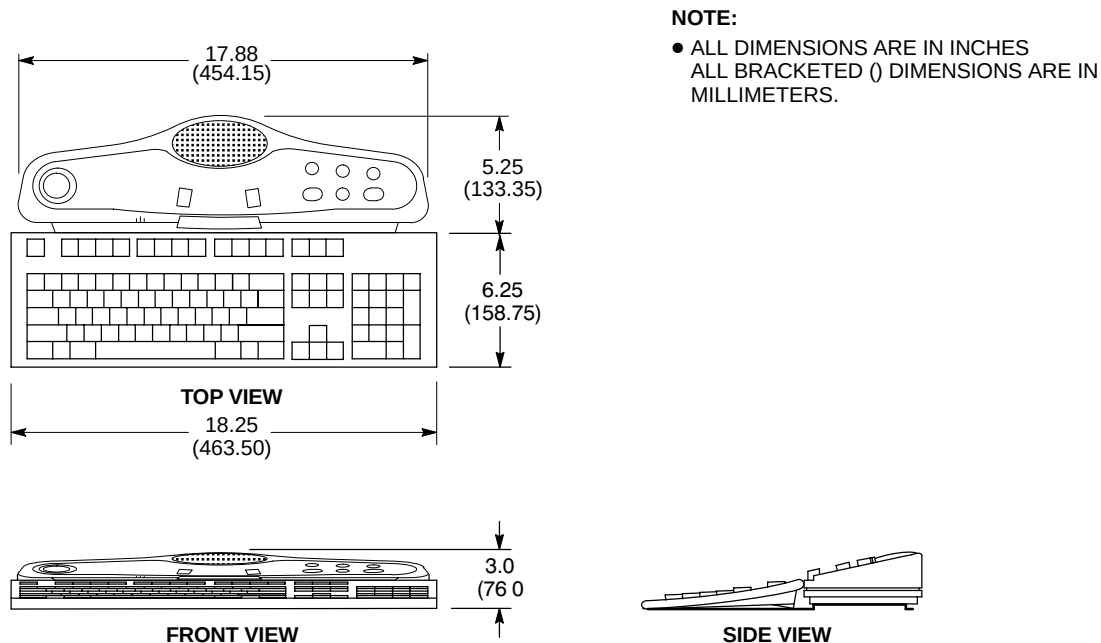
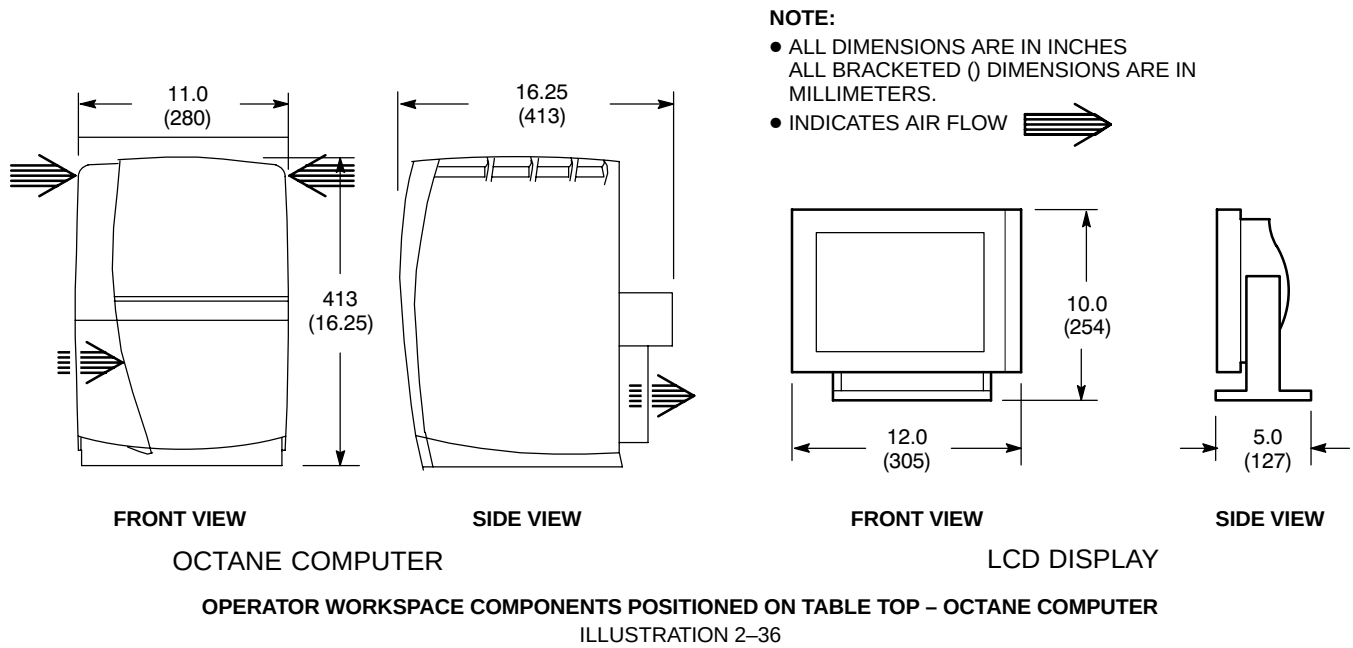
FRONT VIEW



SIDE VIEW

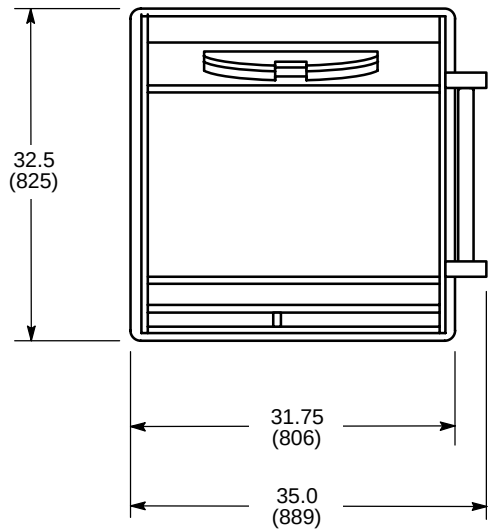
OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – LCD COLOR MONITOR OPTION

ILLUSTRATION 2-35

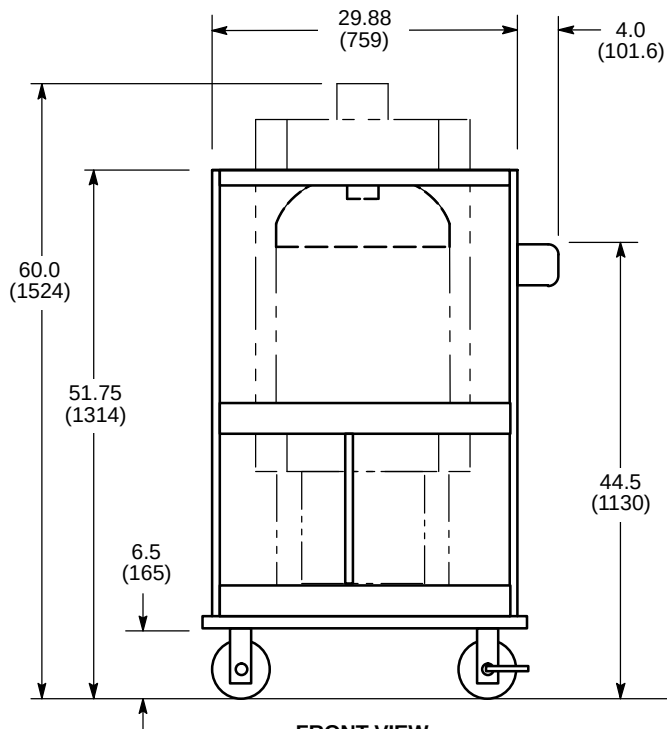


NOTE:

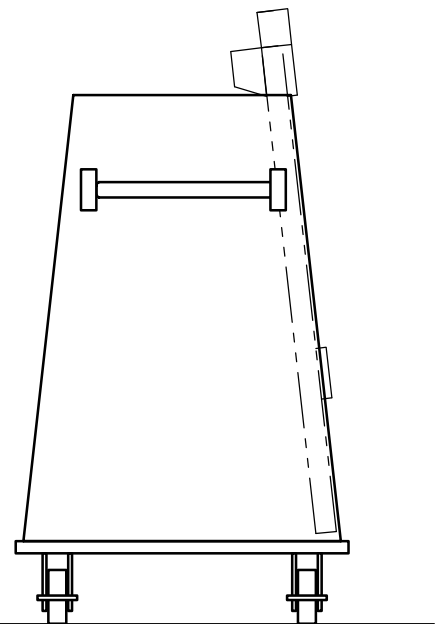
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE
IN MILLIMETERS.
- APPROX. WEIGHT INCLUDING
PHANTOMS: 350 lbs (159 kg)



TOP VIEW



FRONT VIEW

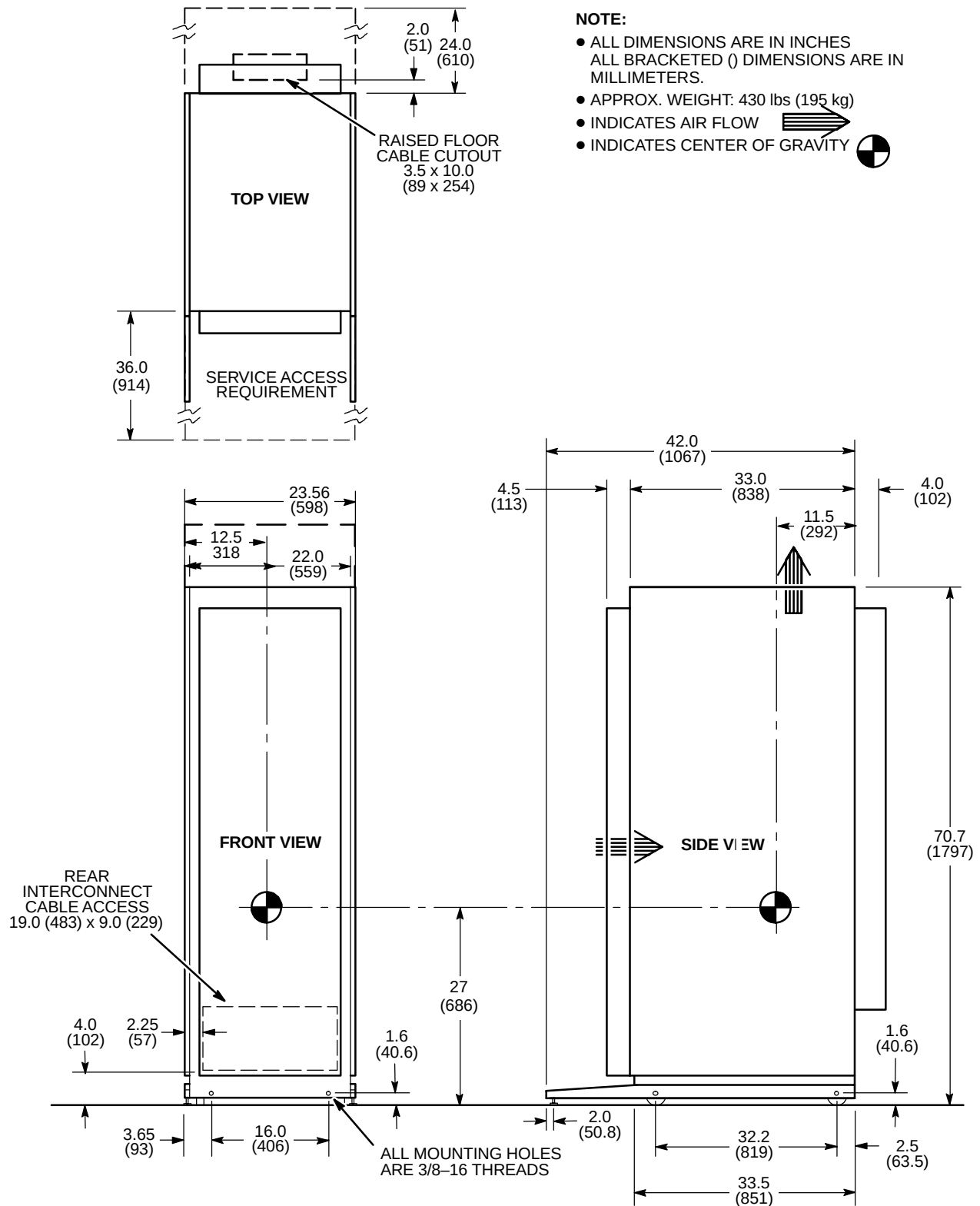


SIDE VIEW

M4454A1

SPT PHANTOM SET SHIPPING/STORAGE CABINET

ILLUSTRATION 2-38



MNS CABINET (MNS) - 1.5T OPTION

ILLUSTRATION 2-39

SECTION 3 – MAGNETIC FIELD CONSIDERATIONS

Note

The magnet and magnetic field considerations of the system is not modified by a Signa LX Upgrade. There are no changes from previous system Magnetic Field Considerations.

SECTION 4 – SITE ENVIRONMENT

TABLE OF CONTENTS

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
4-1	INTRODUCTION	4-3
4-2	TEMPERATURE AND HUMIDITY SPECIFICATIONS	4-3
4-3	COOLING REQUIREMENTS	4-4
4-3-1	Air Cooling	4-4
4-3-2	Water Cooling	4-6
4-4	ALTITUDE	4-6
4-5	LIGHTING	4-6
4-6	ACOUSTICS	4-7
4-7	ROOM VENTILATION	4-11
4-8	CRYOGENIC VENTING	4-11
4-9	LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS	4-12
4-10	CONSTRUCTION MATERIALS	4-12
4-11	VIBRATION	4-12
4-11-1	Definition	4-12
4-11-2	Types Of Vibration Image Quality Issues	4-12
4-11-3	Specifications For 1.5T S-II, & S-III (Unshielded) Magnets and 1.5T S-III Magnet With Magnishield	4-13
4-11-4	Specifications For 1.5T S-IV, S-V, & CX And 1.0T S-X & CX Magnets	4-14
4-11-5	Specifications For 1.5T And 1.0T LCC Magnets Without Gradient Isolation	4-14
4-11-6	Specifications For 1.5T And 1.0T LCC Magnets With Gradient Isolation	4-15

4-1 INTRODUCTION

The rating and duty cycles of all subsystems are applicable only if the room environment is maintained as specified in the following sections. The environment must be constantly maintained (i.e. holidays, weekends, etc.) to prevent exceeding these restrictions. Subjecting the equipment to consistent excessive temperatures and humidity above specifications may shorten the life of the internal electrical components.

4-2 TEMPERATURE AND HUMIDITY SPECIFICATIONS**Note**

If specifications in this section are not strictly adhered to, failures of the ACGD modules in the ACGD/PDU Cabinet may occur.

Use the specifications listed in Table 4-1 for designing your HVAC (heating, ventilation, and air conditioning) system. Proper insulation and moisture barrier should be installed within the environmental controlled space (e.g. area above drop ceiling) for humidity, condensation, and temperature control.

To help prevent a patient from feeling uncomfortably warm during a scan, ensure that the magnet room temperature does not exceed 70°F (21.1°C) maximum.

TABLE 4-1
TEMPERATURE AND HUMIDITY SPECIFICATIONS FOR SYSTEM WITH ACGD/PDU CABINET

AREA	TEMPERATURE RANGE °F (°C)	TEMPERATURE CHANGE °F/Hr (°C/Hr)	HUMIDITY (%)	HUMIDITY CHANGE (%/Hr)	MAX. ROOM GRADIENT °F (°C)
Equipment Room at Inlet to Equipment	59–89.6* (15–32)*	5 (3)	30–75*	5	5 (3)**
Magnet Room	59–69.8 (15–21)	5 (3)	30–60*	5	5 (3)
Operator's Control Room	59–89.6* (15–32)*	5 (3)	30–75*	5	5 (3)
Note * Non-condensing humidity with 50% nominal at 65°F (18.3°C). ** Room temperature gradient specification applies from floor to height of top discharge of equipment cabinets.					

4-3 COOLING REQUIREMENTS

4-3-1 Air Cooling

Table 4-2 contains the heat output of the upgrade equipment listed in the typical site location. These values do not include people, lights, magnet equipment cabinets, and non-MR equipment. Actual site average values will vary depending on system use (ie. protocols used, patient load, etc.). Note any variations of equipment location for your site when calculating your cooling requirements for each room.

Requirements

Due to the large variation in Equipment Room heat loads, the air conditioner compressor should be equipped with unloaders or hot gas bypass to prevent moisture stripping of the evaporator coils.

The Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room temperature specification as listed in Section 4-2, TEMPERATURE AND HUMIDITY SPECIFICATIONS. It is recommended that cool inlet air be directed towards the Enclosure Rear Pedestal, location of coil/bore cooling fans intake.

Recommendations

A dedicated air conditioner with a dual compressor is preferred to avert shutdowns during repair of the primary air conditioner.

It is recommended that a temperature and humidity recorder be used during preinstallation and during actual installation and placed near the Gradient Cabinet air inlets to establish the true criteria. Refer to cooling table calculator in this section for each room's cooling requirements.

GE recommends the use of a 12 inch high raised flooring system for the equipment room (10 inch minimum clearance from floor slab to underside of access flooring). Care must be taken in locating the air conditioning supply vents in the floor. The air conditioning supply vents should be located directly in front of the cabinet inlets.

TABLE 4-2
MAXIMUM MR AIR COOLING TABLE

MR COMPONENT	SIGNA HORIZON 1.5T MAX HEAT OUTPUT†						SIGNA HORIZON 1.0T MAX HEAT OUTPUT†					
	MAGNET ROOM (SEE NOTE 1)		EQUIPMENT ROOM		OPERATOR/ CONTROL AREA		MAGNET ROOM (SEE NOTE 1)		EQUIPMENT ROOM		OPERATOR/ CONTROL AREA	
	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT
RF/Gradient Body Coil Assembly (BRM or CRM), Shim Coils	14,000	4100					10,925	3200				
Blower Box	3415	1000					3415	1000				
SRF Cabinet			8530	2500					8530	2500		
ACGD/PDU Cabinet			34,120	10,000					34,120	10,000		
System Cabinet			6140	1800					6140	1800		
Water Chiller for Gradient Coil			14,000	4100					14,000	4100		
MNS Cabinet*			2730	800			Not Applicable					
Shield Cooler Compressor			See Note 2						See Note 2			
Operator Workspace (See Note 3) with LCD Color Monitor					4095	1200					4095	1200
Main Disconnect (See Note 4)			900	264					900	264		
*Advantage Workstation for Signa CVMR Option					See Note 5						Not Applicable	
Existing Magnet Equipment Cabinets (ie. Magnet Power Supply, Shim Power Supply) as applicable to system												
Note * Optional equipment. † Maximum heat output is defined for temperature and humidity as defined in Table 4-1. 1 Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room temperature specification as listed in Section 4-2, TEMPERATURE AND HUMIDITY SPECIFICATIONS. It is recommended that cool inlet air be directed towards the Blower Box intake which contains coil/patient cooling fans. 2 Shield Cooler Compressor Cabinet requires water cooling to dissipate heat output. Heat dissipation to air is negligible. 3 New Operator Workspace equipment includes the following: LCD Color Monitor, Workspace Cabinet, Mouse and Mouse Pad, LCD Panel, Keyboard, and interface modules mounted to Workspace Table 4 Values are for Main Disconnect Control option available from GEMS. Heat output may be different if customer supplied control is used. 5 Refer to Direction 2261309-100 Advantage Workstation 3.1 Ultra60 Pre-Installation.												

4-3-2 Water Cooling

Magnet equipment water cooling requirements are unchanged.

Gradient Coil

The Gradient Coil is air and water cooled. A water-to-air Water Chiller (WC1) is provided with the fixed site system. De-ionized water and anti-freeze mixture is supplied with the Gradient Coil for use in the closed loop system.

Note

The Water Chiller is a dedicated closed loop unit for Gradient Coil water cooling **only** and not to be shared with other equipment, refer to Section 4-3-3 Water Chiller (WC1) For Gradient Coil Cooling.

Note

The Gradient Coil has GE provided flexible hose (0.5 in. (12.7 mm) inside diameter) and adjustable compression clamps for connections. Refer to Section 6, INTERCONNECT DATA, for flexible hose routing information.

For Mobile systems (True Mobile, Transportable, Relocatable), the Gradient Coil water cooling will be provided by the Water-to-Water Heat Exchanger, the Van chilled water system will provide chilled water for the Heat Exchanger operation.

Shield/Cryo Cooler Compressor

If an upgrade to an LCC magnet is being performed with the Signa system upgrade then refer to **Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 4-3-2 Water Cooling** for Shield/Cryo Cooler Compressor water cooling requirements.

4-4 ALTITUDE

If an upgrade to an LCC magnet is being performed with the Signa system upgrade then refer to **Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 4-4 ALTITUDE** for additional information and specifications.

4-5 LIGHTING

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to **Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 4-5 LIGHTING** for additional information and specifications.

4-6 ACOUSTICS

Note

Design improvements in the MR system (RF/Gradient Body Coil, Magnet, RF and Gradient subsystems) balances the impact of the higher performance scan protocols such that there may be slight increases in the acoustics output. The following information is supplied for design of new installations but can also be used to assess the expected performance of the upgrade MR system.

The following acoustic information is provided for site planning and architectural design activities to address acoustics to meet local regulations and customer requirements. For more information about recommended safety procedures regarding patient exposure to MR-generated acoustic levels, see the system operator manual.

4-6-1 Background

A typical MR suite has two types of acoustic noise issues. The first is the acoustics of the room where patients and technicians are impacted by the noise of the MR system as the gradients are pulsed. The second is noise transmitted to other spaces via airborne and structure borne paths.

Airborne

The airborne transmission path entails the excitation of air within the magnet room; the resonator module consisting of the magnet, RF coil, and gradient coil generates acoustic noise similar to an intense loud speaker. The airborne noise passes through walls via any openings, i.e. small holes, cracks, HVAC ducts, and waveguides, into surrounding spaces within and possibly beyond the confinements of the building. Acoustic energy can transmit across distances of significant length.

Examples of airborne acoustic issues may include the following (not limited to only these):

- MR Operator exposure at Operator Workstation
- Image reading rooms adjacent to Magnet Room, may be separated by hallways
- Secretarial, offices, meeting rooms, patient rooms (ICU, exam, primary care, etc.)
- Adjacent residential areas/spaces
- In-house library facilities

Structureborne

The structureborne transmission path is the result of mechanical excitation of the floor/building structure causing the building to vibrate. The vibration of the surfaces at surrounding spaces then radiates as acoustic noise. Acoustic energy can transmit across distances of significant length.

Note

Less than 2 to 3% of installed base sites have experienced structureborne acoustic issues.

Examples of structureborne acoustic issues may include the following (not limited to only these):

- Areas directly above or below the Magnet Room, may not always be an issue
- Image reading rooms adjacent to Magnet Room, may be separated by hallways
- Secretarial, offices, meeting rooms, patient rooms (ICU, exam, primary care, etc.)
- Adjacent residential areas/spaces
- In-house library facilities

4-6-2 System Acoustic Noise Levels

Any GE factory-installed protocol can be modified by operators, which can increase or decrease acoustic SPL (Sound Pressure Level); or operators may create their own protocol which could produce a higher or lower acoustic SPL as stated under **Operating Conditions** Condition 1 below. Typical scans generate acoustic levels as stated under **Operating Conditions** Condition 2 below. In addition, the exposure times are completely under operator control. Consequently, hearing protection is required for all people in the Magnet Room during scans to prevent hearing impairment, acoustic levels may exceed 99dBA. Again, for more information about recommended safety procedures regarding patient exposure to MR-generated acoustic noise, see the MR system operator manual.

Ambient Conditions

To reduce any background noise due to cabinet blowers, etc., acoustical ceilings, walls, and floors are recommended. The following are typical noise level readings:

- Operator Area 55 dBA
- Equipment Room 75 dBA

Operating Conditions

Condition 1

MR scanners under “worst-case” operating conditions, could generate acoustic levels (as measured at the magnet iso-center) as follows:

Average SPL 118 dBA SPL= Sound Pressure Level
 Peak 128 dB
 Frequency Range 20 to 20k Hz

Condition 2

MR scanners for many typical clinical scanning scenarios though, generate acoustic levels (as measured at the magnet iso-center) somewhat lower as follows:

Average SPL 95 to 110 dBA
 Peak 110 to 120 dB
 Frequency Range 20 to 20k Hz

As recent history has shown an evolution towards more powerful (and hence louder) gradient subsystems, architects should consider the acoustic levels stated in the “worst case” Condition 1, mentioned above. Note that high-field Signa systems have the ability to run scanning protocols which can generate acoustic levels over the entire human perceptible frequency range (20 to 20k Hz), therefore attenuation over this entire range must be considered for site design.

4-6-3 Design Guidelines

Magnet Room

Noise generated by the MR system is inherent to the operation of the system, refer to Section 4-6-2 System Acoustic Noise Levels. The sound quality (human perception) within the Magnet Room can be modified by including sound absorbing materials to make the room sound more subdued and less harsh. The measured sound levels via a sound level meter will not change. However, the measured sound levels can be reduced only when the sound level generated by the MR System is reduced.

Sound quality improvements can be achieved by the following:

- Use ceiling tiles with fiberglass panels having a 2 inch (51mm) thickness set into the standard T-bar grid system.
- Adding fiberglass panels to the side walls covering approximately 20% of the side wall surface area. The panels should focus on covering the top half of the side walls. Panels could take many different and decorative shapes to improve the sterile look of the rooms. Typically panels might be on the order of 4ft x 6ft (1.2m x 1.8m) with a thickness of 4 inches (102mm) or equivalent. Panels shape could vary to produce mosaic effects to meet the customer preference. Any decorative materials used to cover the wall panels must be porous so that sound waves can pass through with ease. In principle, a person should be able to breath through the material with ease. Fire retardant cloth should be used. The NRC (Noise Reduction Coefficient) of the panels should be 0.95 or better when mounted against a hard surface such as drywall or concrete.

Inter-Spacial Areas

Acoustic noise control to mitigate noise from being transmitted to other spaces often amounts to paying attention to small details while working with ordinary construction materials. The key objectives are to eliminate all cracks and gaps in the wall construction while making sure that the doors, walls, floor, and ceiling have adequate transmission loss via mass or special double wall construction along with good fitting massive doors.

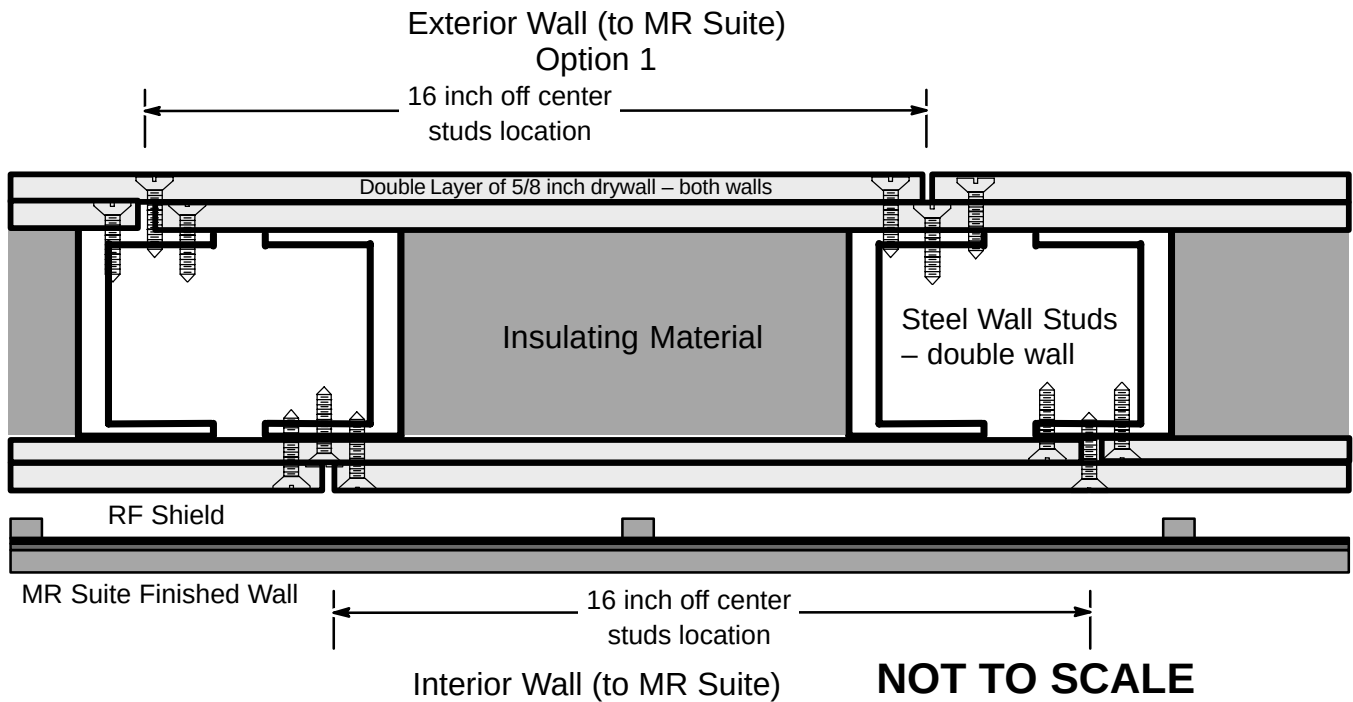
The entire Magnet must be surrounded by walls with substantial mass and/or double wall construction so that noise is contained in the room and not allowed to pass through into nearby spaces. Wall junctions must be sealed with acoustical sealant so that noise waves do not escape from the room. In principle, if the room were filled with smoke and under a positive pressure, no smoke would leak from the room.

Wall Construction

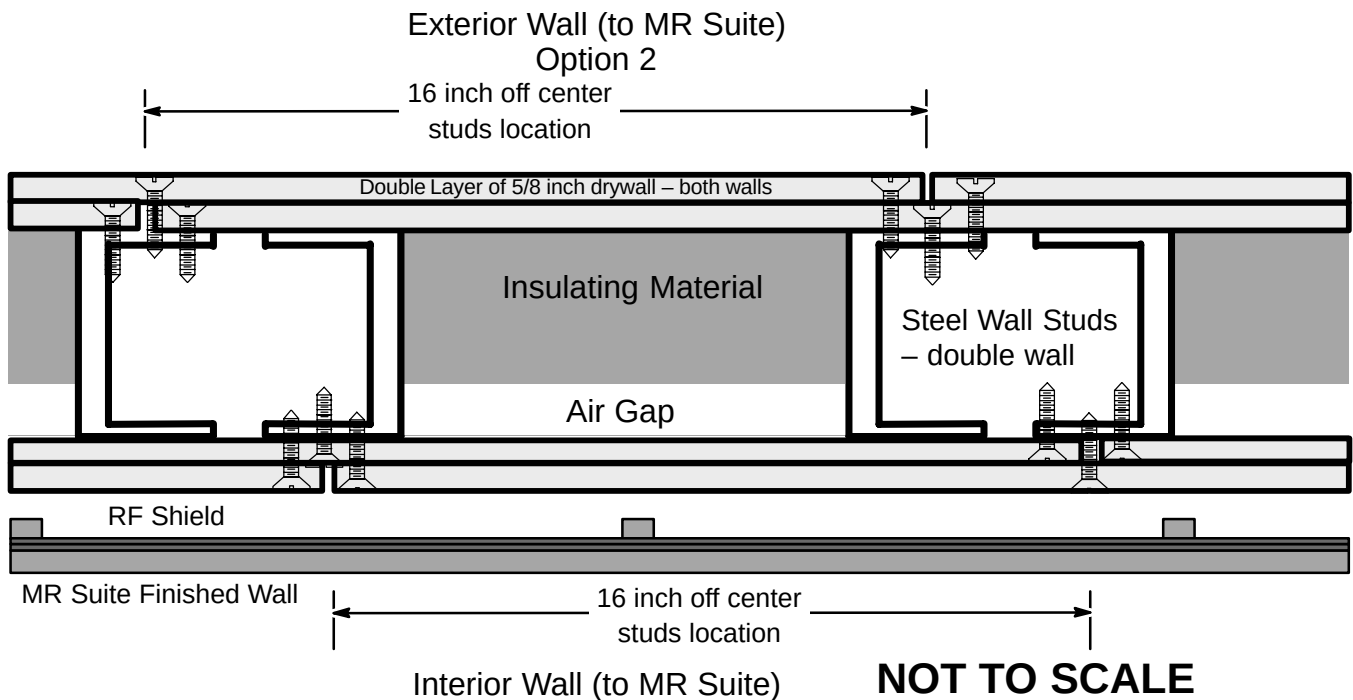
Wall Construction will entail ordinary building materials in a careful configuration.

- The preferred wall would have an ASTM STC 50 construction which entails the use of standard wall construction of steel studs (typically 3-5/8 inch (92 mm)) with 2 layers of Type X drywall (typically 5/8 (16 mm)) on each side totaling 4 layers and fiberglass batt in the stud cavity. All drywall must be overlapped by 6 inches (152 mm) or more. Beads of (USG) Acoustical Caulking (non-hardening) would be used around the entire perimeter of the drywall. Any form of wall penetration should be avoided. Any necessary wall penetrations must be sealed using combination of Acoustical Caulking (non-hardening) and fiberglass batt material. See examples of wall construction shown in Illustrations 4-1 and 4-2.
- The top of the wall must join the ceiling/floor above so that no cracks or gaps occur. If metal pan is used on the ceiling/floor (above), then flute seals would be used to seal the gaps between the drywall and the pan. Alternately drywall can be cut out to fit into the flutes. Acoustical caulking (non-hardening) will be used to seal the remaining cracks and gaps.

4-6-3 Design Guidelines (Continued)



EXAMPLE OF WALL CONSTRUCTION FOR AIRBORNE NOISE CONTROL - OPTION 1
ILLUSTRATION 4-1



EXAMPLE OF WALL CONSTRUCTION FOR AIRBORNE NOISE CONTROL - OPTION 2
ILLUSTRATION 4-2

4-6-3 Design Guidelines (Continued)High Bay RF Room

A high bay RF Room is a self contained RF Room which has open air space between the RF Room ceiling and the building floor above. The air space is an acoustic transmission path. Acoustic energy must be reduced to minimize this transmission of energy through this path.

In cases where the Magnet is to be installed in a high bay, it may be most effective to enclose the RF Room with its own drywall and steel stud room. The key difference being a ceiling assembly that mimics the sidewall construction to contain noise.

- Normal high STC stud walls from above would be used to support a ceiling assembly constructed of structural C channel with two layers of drywall on each side (total of 4 layers) with fiberglass batt in the cavity.
- Penetrations should be avoided via the use of surface mounted lights. HVAC and ducts passing through the ceiling, party wall or side walls would require acoustic noise attenuation in the form of inline silencers. Gaps and cracks would be sealed between the ceiling, party wall or vertical side walls and the cryogen vent plumbing. In essence the Magnet would be enclosed in a drywall "doghouse".

Miscellaneous Plumbing, RF Windows and RF Doors

Other construction details are equally important to mitigate noise transmission to meet the intended goal.

- Pipes (gas or water) and electrical conduit or Magnet Room signal cables must be sealed where they penetrate the walls or ceiling. A heavy mastic material such as Duxseal™ is appropriate.
- RF windows should be purchased as window/frame units with an STC rating obtained from laboratory testing per ASTM standards. STC 50 to 60 windows are needed. The installation must include proper sealing to avoid sound leaks.
- RF doors should be selected to provide and STC 50 to 50 to quell the noise. Contact RF Shield Room supplier for selection of RF doors that meet the local acoustic codes and site acoustic requirements. RF door seals must be selected to prevent small gaps around the door perimeter and at the door threshold. RF door seals would either require periodic replacement or a door seal that would last the life of the Magnet Room.

4-7 ROOM VENTILATION

If an upgrade to an LCC magnet is being performed with the Signa system upgrade then refer to ***Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 4-7 ROOM VENTILATION*** for requirements and recommendations.

4-8 CRYOGENIC VENTING


WARNING!

IF AN UPGRADE TO AN LCC MAGNET IS BEING PERFORMED WITH THE SIGNA SYSTEM UPGRADE THEN REFER TO *DIRECTION 2223170 SIGNA MR/i & CV/i PRE-INSTALLATION SECTION 4-8 CRYOGENIC VENTING* FOR REQUIREMENTS AND RECOMMENDATIONS FOR LCC MAGNET CRYOGENIC VENTING.

4-9 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS

If an upgrade to an LCC magnet is being performed with the Signa system upgrade then refer to ***Direction 2223170 Signa MR/i & CV/i Pre-Installation SECTION 4-12 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS*** for requirements and recommendations..

4-10 CONSTRUCTION MATERIALS

Systems installing an upgrade to LCC Magnet must meet the requirements in ***Direction 2223170 Signa MR/i & CV/i Pre-Installation SECTION 4-13 CONSTRUCTION MATERIALS*** for requirements and recommendations..

4-11 VIBRATION

4-11-1 Definition

Improvements in MR imaging technology has increased imaging capabilities. Certain MR procedures require a stable environment to achieve high resolution image quality. **Environmental site vibration has the ability to affect the magnet phase stability and ultimately image quality.** The vibration effects on image quality can be minimized early in the site planning for Signa Horizon upgrades.

The magnet may be sensitive to vibration in the frequency range of 0.5 to 45 Hz depending on the amplitude of the vibration. In the physical area where the MR system is located, every precaution must be taken to ensure that the vibration is minimized. In the magnet siting area, the structural stability and behavioral characteristics can be assessed. **The stability tests (HSS) currently used by local GE Field Service to evaluate the existing system performance and image quality can be used to assess the vibration environment. Sites which currently pass their system stability criteria (HSS) may proceed with the upgrade. Sites which have marginal stability performance require vibration measurements to be performed to identify vibration source(s).** Then it is the customer's responsibility to contract a vibration consultant to perform vibration measurements as defined in Appendix A, MR SITE VIBRATION TEST GUIDELINES. With the vibration consultant, local GE Field Service and/or Installation Specialist must verify the elimination/reduction of all identified sources do improve the vibration environment. GE can assist in interpreting marginal site test results and predicting the impact on system performance. However it is ultimately the customer/architect/engineer responsibility to design a site solution.

4-11-2 Types Of Vibration Image Quality Issues

Vibration capable of affecting MR image quality can be classified as either steady state or transient vibration. Steady state vibration typically refers to disturbances caused by rotating machinery. Examples of machinery known to have previously generated vibration image quality problems are exhaust fans, air conditioning blower units, compressors, pumps (air and water), etc. Steady state vibration would be measurable during long time intervals. Transient vibrations are typically a function of the building structure or the building foundation. Transient vibration would typically be associated with vehicular traffic, pedestrian motion, patient transport, door slamming, etc. Transient vibration would typically occur only during short periods of time. The transient event would typically decay from a high vibration amplitude to lower levels in short periods of time.

Steady State Vibration Issues

1. Power down all sources exceeding vibration guidelines.
2. Run MR system stability test.
 - a. If all sources have been properly identified the MR system will meet GE specifications.
 - b. If GE specifications are exceeded, continue searching for additional vibration sources.

4-11-2 Types Of Vibration Image Quality Issues (Continued)

Transient Vibration Issues

Vibration measurements and system stability tests must be performed during the same period of time. Only when the vibration event(s) and phase stability deviation(s) occur at the same time will transient vibration be confirmed as source of system stability.

1. Measure the MR system stability and vibration, simultaneously, during normal system scanning hours. Transient vibration and transient phase stability deviation exceedance(s) must occur at the exact same time for transient vibration to be an imaging concern.
2. If transient vibration produces imaging concerns, the vibration consultant must identify the source(s) of the transient in which case additional testing is needed. Schedule and perform additional vibration testing during periods of time when the suspected transient vibration issues can be controlled, limited, or eliminated. Evaluate the MR system performance during periods of time per the following.
 - a. If all sources have been properly identified the MR system will meet GE specifications.
 - b. If GE specifications are exceeded, continue searching for additional vibration sources.

Magnet Siting Requirement

Improper mechanical coupling of the magnet to the floor can in some circumstances create unique vibration characteristics. Magnets improperly mechanically coupled to the floor may experience steady state and/or transient vibration system stability issues. When the system stability tests do not meet acceptance criteria the vibration testing defined in Appendix A will help to identify if the magnet is properly mounted to the floor. The customer is not limited to the GE options for mounting the magnet: site conditions may necessitate alternative methods.

4-11-3 Specifications For 1.5T S-II, & S-III (Unshielded) Magnets and 1.5T S-III Magnet With Magnishield

Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed **10⁻⁵ g rms** maximum single frequency **above ambient baseline** from **0.5 to 45 Hz**. In order to ensure that any discrete signal represents a real mechanical vibration source, the signal must have a bandwidth that typifies dynamic system response.

Transient Vibration

Time history transient levels exceeding **500 micro-g, zero to peak** must be fully analyzed to assess impact to the building structure. The building (spectral) response immediately following the **500 micro-g** trigger level must not cause the site environment to exceed the **Steady State Vibration** level defined above.

4-11-4 Specifications For 1.5T S-IV, S-V, & CX And 1.0T S-X & CX Magnets

Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed the following maximum single frequency components **above ambient baseline**:

- 1×10^{-5} g rms 0–15 Hz
- 5×10^{-5} g rms 15–20 Hz
- 1×10^{-5} g rms 20–35 Hz
- 2×10^{-4} g rms 35–45 Hz

In order to ensure that any discrete signal represents a real mechanical vibration source, the signal must have a bandwidth that typifies dynamic system response.

Transient Vibration

Time history transient levels exceeding **500 micro-g, zero to peak** must be fully analyzed to assess impact to the building structure. The building (spectral) response immediately following the **500 micro-g** trigger level must not cause the site environment to exceed the **Steady State Vibration** levels defined above.

4-11-5 Specifications For 1.5T And 1.0T LCC Magnets Without Gradient Isolation

Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed the following maximum single frequency components **above ambient baseline**:

- 1×10^{-5} g rms 0–10 Hz
- 5×10^{-5} g rms 10–30 Hz
- 1×10^{-5} g rms 30–38 Hz
- 2×10^{-4} g rms 38–45 Hz

In order to ensure that any discrete signal represents a real mechanical vibration source, the signal must have a bandwidth that typifies dynamic system response.

Transient Vibration

Time history transient levels exceeding **500 micro-g, zero to peak** must be fully analyzed to assess impact to the building structure. The building (spectral) response immediately following the **500 micro-g** trigger level must not cause the site environment to exceed the **Steady State Vibration** levels defined above.

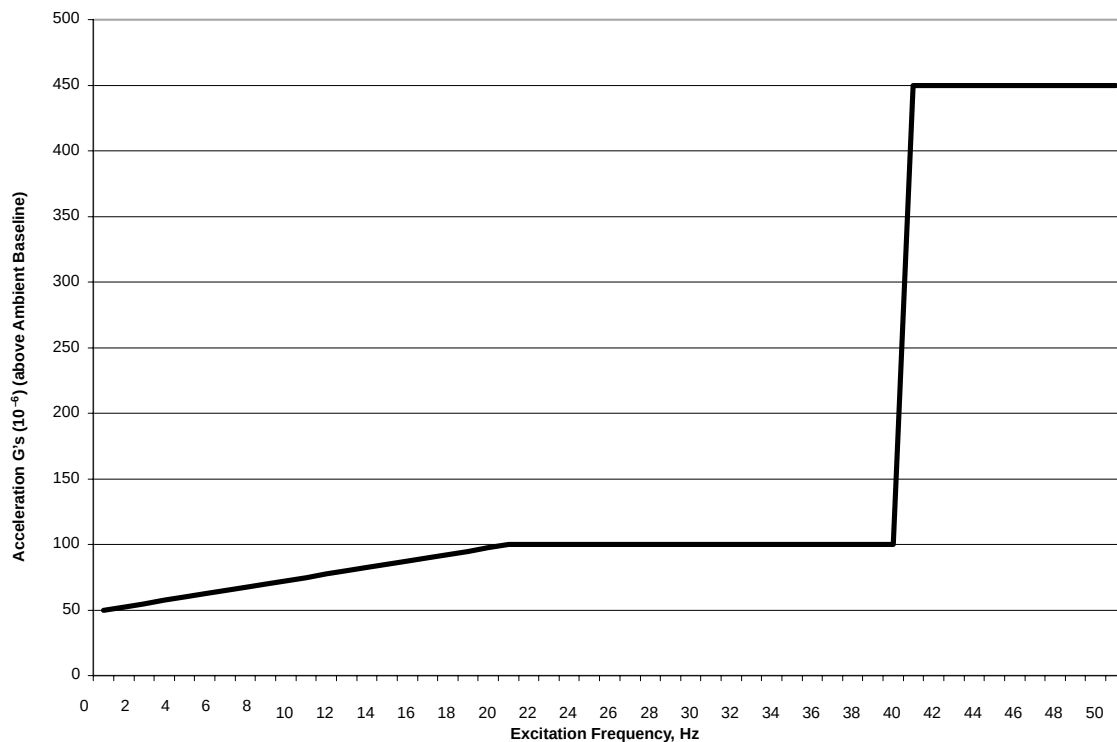
4-11-6 Specifications For 1.5T And 1.0T LCC Magnets With Gradient Isolation

Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed the following maximum single frequency components **above ambient baseline** (refer to Illustration 4-3):

- 5×10^{-5} g rms at 0 Hz ramping to 10×10^{-5} g rms at 20 Hz
- 10×10^{-5} g rms 20-40 Hz
- 45×10^{-5} g rms 40-50 Hz

In order to ensure that any discrete signal represents a real mechanical vibration source, the signal must have a bandwidth that typifies dynamic system response.



LCC MAGNET WITH GRADIENT ISOLATION STEADY STATE VIBRATION SPECIFICATIONS

ILLUSTRATION 4-3

Transient Vibration

Time history transient levels exceeding **500 micro-g, zero to peak** must be fully analyzed to assess impact to the building structure. The building (spectral) response immediately following the **500 micro-g** trigger level must not cause the site environment to exceed the **Steady State Vibration** levels defined above.

SECTION 5 – POWER REQUIREMENTS

TABLE OF CONTENTS

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
5-1	INTRODUCTION	5-3
5-2	CRITICAL POWER REQUIREMENTS	5-4
5-3	POWER DISTRIBUTION SYSTEM	5-6
5-3-1	Main Disconnect Control (MDC)	5-6
5-3-2	System Power Distribution Unit	5-9
5-4	GROUNDING	5-10
5-4-1	Facility Ground	5-10
5-4-2	System Ground	5-12
5-5	GROUND FAULT PROTECTION	5-14

5-1 INTRODUCTION

The existing system PDU is replaced during the upgrade to ACGD. The MR upgrade includes the new Power Distribution module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3) which distributes power to most MR system components and provide the proper voltage configuration for the ACGD modules. Refer to Table 5-1 for the required customer power specifics. Refer to Section 5-2 CRITICAL POWER REQUIREMENTS for specifications of required facility input for the system.

WARNING!

ALL SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE FACILITY TRANSFORMER AND FEEDER WIRES ARE CORRECTLY SIZED. ALSO THE MAIN DISCONNECT CONTROL NEEDS TO BE INVESTIGATED FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS.

SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS.

Note

Location of the new PDU in the ACGD/PDU Cabinet relative to the existing system PDU may require replacement of the customer provided feeder wires from the MDP to the PDU. Refer to requirements and information throughout this section (Section 5 – POWER REQUIREMENTS). Local and national codes may require an electrician to provide and/or install new feeder wires to the PDU.

TABLE 5-1
REQUIRED CUSTOMER POWER

MR COMPONENT	VOLTAGE (VAC)	FREQUENCY	PHASE	MAX. AMPS	COMMENTS
Power Distribution Unit (PD1) Module located in the lower portion of ACGD/PDU Cabinet (MR3) SEE NOTE 1	480/415/400/ 380/208/200	50/60 $\pm$ 3 Hz	(3+GND) See Comments	See Note 2	Recommend input configuration: 3 phase Grounded WYE with Neutral and Ground (5 wire system). Note, Neutral must be terminated prior to PDU or inside the Main Disconnect Panel and not brought to the ACGD/PDU Cabinet. Optional input configuration: 3 phase DELTA with Ground (4 wire) input, recommend corner Grounded Delta configuration.
Note 1 Power phase conductors, neutral (if present), and ground conductor must be routed inside the same raceway, cable tray, trench cable, or cord per National Electric Code (NEC) 2002 Articles 250.134, 300.3, 517.13 or 1999 Articles 250-134, 300-3, 517-13. 2 Maximum amps dependent on voltage selected. Refer to Section 5-2, CRITICAL POWER REQUIREMENTS, for configuration.					

5-2 CRITICAL POWER REQUIREMENTS

The facility shall provide a Main Disconnect Control (MDC) with Low Voltage Low Energy local and multi-point remote control capability in the feeder lines that supply input power to the Shield Cooler Cabinet (if applicable for system magnet configuration) and the Power Distribution Unit (PD1). All work is to be done in accordance with national and local electrical codes. Refer to Section 5-3, POWER DISTRIBUTION SYSTEM, for MDC capability and set up.

WARNING!

ALL SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE FACILITY TRANSFORMER AND FEEDER WIRES ARE CORRECTLY SIZED. ALSO THE MAIN DISCONNECT CONTROL NEEDS TO BE INVESTIGATED FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS.

SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS.

Configuration:

- Recommend input configuration 3 phase Grounded WYE with Neutral and Ground (5 wire system). Note, Neutral must be terminated prior to or inside the Main Disconnect Panel and not brought to the ACGD/PDU Cabinet.
- Optional input configuration 3 phase DELTA with Ground (4 wire) input, recommend corner Grounded Delta configuration.

Frequency: 50 ± 3 Hz or 60 ± 3 Hz

System PDU Voltage: 200/208/380/400/415/480 VAC $\pm 10\%$

Note

A second circuit breaker in the Main Disconnect is required for Shield/Cryo Cooler Cabinet. Refer to Section 5-3 Main Disconnect Control.

Shield Cooler Compressor Voltage (Water Cooled type): 380/400/415 VAC 50 Hz or 460/480 VAC 60Hz

Daily Voltage Variation: $\pm 10\%$ from nominal under worst case line and load regulation.

Voltage Transients: Maximum transient voltage above nominal waveshape not to exceed 200 V for the nominal voltages 380, 400, 415, or 480 Vrms.

Voltage Transients: Phase-to-phase voltages must be within 2% of the lowest phase-to-phase voltage. Maximum allowable transient voltage above or below nominal waveshape not to exceed 200 V at a maximum duration of 1 cycle and frequency of 10 times per hour.

Phase Balance: Difference between the highest phase line-to-line voltage and the lowest phase line-to-line voltage must not exceed 2%.

5-2 CRITICAL POWER REQUIREMENTS (Continued)**Facility Ground:**

- Main facility ground wire to be minimum 1/0 AWG copper or same size as power feeder wire, which ever is larger.
- Main facility ground wire to be insulated.
- Ground impedance to earth at power source to be 2 ohms or less.
- Main facility ground wire to be bonded at every distribution box in an approved grounding block.

Maximum Momentary Demand: The power demands for the system with ACGD are specified as a function of the duration of the power demand. Table 5-2 lists points on the curve.

The power system feeding the Signa system with ACGD must be designed to meet the specifications of less than 4% regulation when loaded at the 5.0 second allowable consumption. For short intervals the Signa system with ACGD power demands can exceed the 5.0 second value and the line voltage delivered to the system will sag below the 4% regulation. The Signa system with ACGD is designed to tolerate these **short** voltage sags.

TABLE 5-2
SYSTEM WITH ACGD POWER DEMANDS DETAILS

TIME INTERVAL	PDU POWER DEMAND	SHIELD COOLER COMPRESSOR POWER DEMAND	TOTAL SYSTEM POWER DEMAND
Continuous	45 KVA	9 KVA	54 KVA
5.0 seconds	~56.2 KVA	9 KVA	~65.2 KVA
1.0 seconds	~65 KVA	9 KVA	~74 KVA
0.10 seconds	~65 KVA	9 KVA	~74 KVA
0.04 seconds	~65 KVA See Note 1	9 KVA	~74 KVA
Note 1 The PDU draw on the line will not exceed list values. The ACGD Power Supply may provide up to 170 KVA for 0.003 seconds from supply internal capacitance but the supply will recharge capacitors at a power level less than 65 KVA.			

Average (while scanning) Power Demand: 54 KVA consisting of 45 KVA for PDU and 9KVA (continuous operation) for Shield/Cryo Cooler Cabinet.

Standby (no scan) Power Demand: 13.4 KVA at 0.9 lagging Power Factor including 4.4 KVA for PDU and 9KVA (continuous operation) for Shield/Cryo Cooler Cabinet.

5-3 POWER DISTRIBUTION SYSTEM

■ 5-3-1 Main Disconnect Control (MDC)

WARNING!

ALL SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE MAIN DISCONNECT CONTROL FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS.

SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS.

Required Main Disconnect Control shall consist of a three-pole, 600 VAC circuit breaker trip rated for the appropriate current. The short-circuit current interrupting rating of the breaker must be sized to accommodate fault current available.

Note

The MDC circuit for the PDU shall have an undervoltage device that automatically trips the breaker when the applied coil voltage drops to a specific percent of the rated value.

Note

The MDC circuit for the Shield Cooler Compressor Cabinet shall provide auto restart upon return of normal power to maximize proper uninterrupted magnet operation and minimize cryogen consumption of the system.

WARNING!

THE MDC CIRCUIT FOR THE SHIELD COOLER COMPRESSOR CABINET AUTO RESTART FUNCTION MUST BE CONTROLLED BY THE EMERGENCY OFF FUNCTION.

■ Main Disconnect Control is to be located so the panel is visible to Power Distribution Unit (PD1) service personnel. Panel position above the floor must be per National Electric Code (NEC) 2002 Article 404.8 or NEC 1999 Article 380-8 or other national or local code.

A recommended method for providing a Main Disconnect Control with multi-point remote control capability is shown in Illustration 5-1.

5-3-1 Main Disconnect Control (MDC) (Continued)

Time delay fuses are recommended on the lines providing power to the protective disconnect device and must be rated according to the cable size. The circuit breaker ahead of the fuses must be capable of handling the magnetizing inrush current of the Shield Cooler Compressor and transformer of the PDU module (PD1) in the ACGD/PDU Cabinet (MR3).

Check local and national codes to determine if an interlock to the air-conditioning unit in the Computer/Equipment Room is required in the protective disconnect set-up.

Customer supplied emergency off buttons are to be mounted in the Magnet Room near each exit 60 in. (1524 mm) from the floor and connected to the protective disconnect device in order to disable the power to the PDU in emergency situations. This button is to be clearly labeled "Emergency Off" and visible to personnel in the Magnet Room. It is important that the button be labeled "off" and not "stop" since there exists an "Emergency Stop" button in the Signa system which powers down different equipment than the emergency off button installed by the facility. An additional emergency off button should be mounted in the computer/equipment room near the exit door.

Customer supplied main disconnect controller (MDC) must be lockable to meet OSHA requirements for power Lockout/Tagout requirements. An optional MDC is available from GE providing for the disconnect of the facility power to the PDU, including the emergency off buttons, contacts for an interlock to the air-conditioning units in the computer/equipment room, and the Shield Cooler Compressor Cabinet power circuit with auto restart and emergency off functions.

The Main Disconnect Control Panel must be listed and labeled by a Nationally Recognized Testing Lab (NRTL) such as Underwriters Laboratory (UL) in accordance with 2002 National Electric Code (NEC) Article 110.2 or 1999 NEC Article 110-2.

Note

The maximum conductor the GE available option MDC can accept is #3/0 AWG (95 mm²). For feeders larger than 3/0 AWG (83 mm²) the wires must be reduced (ie. splice, junction box, etc.) to 3/0 AWG (83 mm²) within 10 feet (3 meters) of MDC. It is important to note the maximum cable wire from the MDC to the PDU must not be larger than the maximum wire the PDU will accept, refer to Section 5-3-2 System Power Distribution Unit.

■ 5-3-2 System Power Distribution Unit

The PDU Module in the lower portion of ACGD/PDU Cabinet has an integrated filter for a level of power conditioning.

■ The largest allowable phase conductor the PDU will accept is 2/0 AWG (70 mm²). Larger feeder wires can be connected to the MDC with 2/0 AWG (70 mm²) between the MDC and PDU.

Note

The ground conductor shall be minimum size of 1/0 AWG copper or the same size as the feeder wire, which ever is larger. Lug connector for the ground wire is to be provided by the contractor, recommended Amp Inc. number 36919 lug.

Note

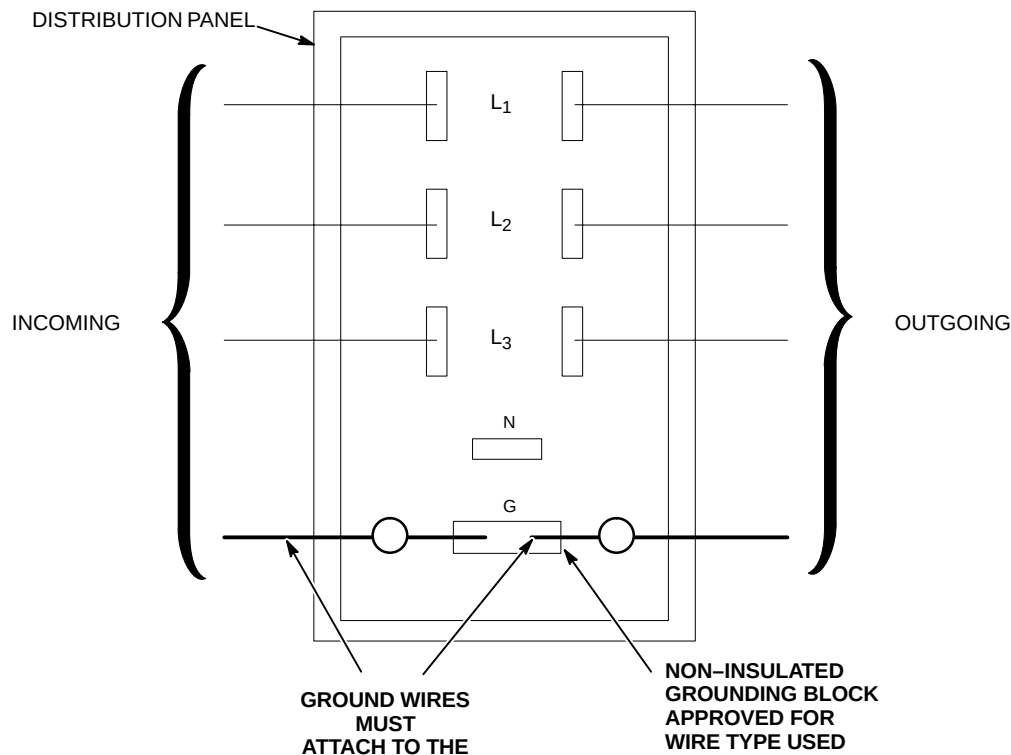
Neutral, if present, must be terminated prior to or inside the Main Disconnect Panel and not brought to the PDU Module in the lower portion of ACGD/PDU Cabinet (MR3).

5-4 GROUNDING

5-4-1 Facility Ground

The ground for the MR system shall originate at the system power source, ie. transformer or first access point of power into a facility, and be continuous to the MR system power disconnect in the room. This ground can be spliced with "High Compression Fittings" and should be terminated at each distribution panel it passes through. When the ground is broken for a connection to a panel, the ground shall be connected into an approved non-insulated grounding block with the incoming and outgoing ground in this same grounding block, which is then connected to the steel panel, never using the steel or other material of the panel as the grounding block. See Illustration 5-2.

The connection at the power source shall be at the grounding point of the "Neutral – Ground" if a "Wye" transformer is used, or the grounded phase of a grounded "Delta", or typical grounding points of separately derived system. In the case of an external facility, it shall be bonded to the facility ground point at the service entrance.



GROUND CONNECTION AT DISTRIBUTION PANEL

ILLUSTRATION 5-2

M4301A

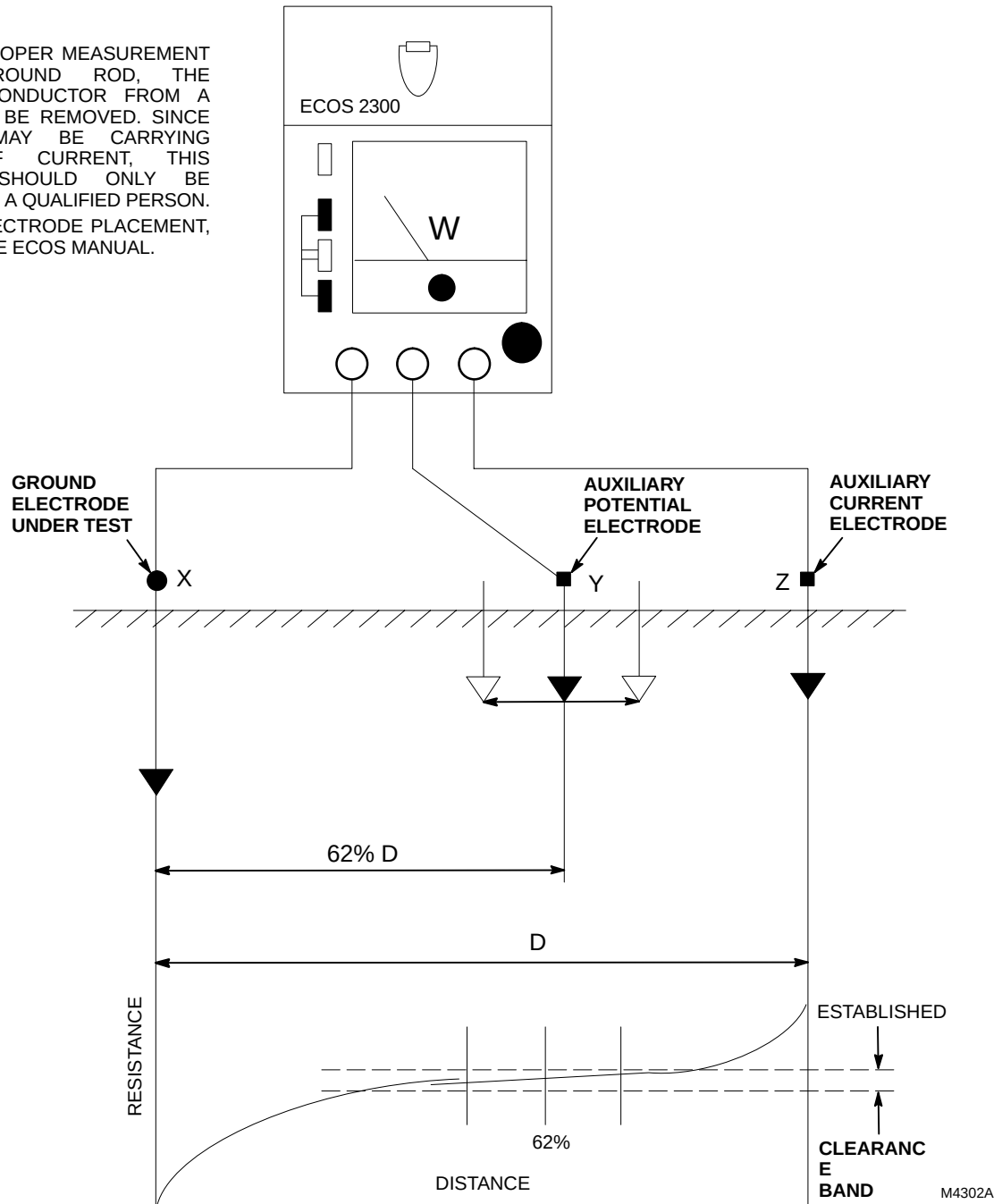
Ground Wire

The ground wire shall be copper wire with a minimum of AWG 1/0 or the same size as the power feeders whichever is larger. This means that if there is a primary feeder to a distribution panel of 500 MCM with a secondary feeder to the MR system of AWG 2 wire, the ground to the distribution panel shall be 500 MCM with an AWG 1/0 to the MR system. The ground wire impedance from the MR system disconnect, including the ground rod, shall not have an impedance greater than 2 ohms to earth as measured by one of the applicable techniques described in Section 4 of ANSI/IEEE Standard 142 – 1982. Refer to Illustration 5-3 for typical equipment and methods to measure the different portions of the 2 ohm impedance.

5-4-1 Facility Ground (Continued)

NOTE:

- TO ENSURE PROPER MEASUREMENT OF THE GROUND ROD, THE GROUNDING CONDUCTOR FROM A FACILITY MUST BE REMOVED. SINCE THIS WIRE MAY BE CARRYING AMOUNTS OF CURRENT, THIS PROCEDURE SHOULD ONLY BE PERFORMED BY A QUALIFIED PERSON.
- DETAILS OF ELECTRODE PLACEMENT, ETC., ARE IN THE ECOS MANUAL.



STANDARD HOOK-UP TO SYSTEM UNDER TEST FOR GROUND ROD IMPEDANCE
ILLUSTRATION 5-3

■ 5-4-2 System Ground

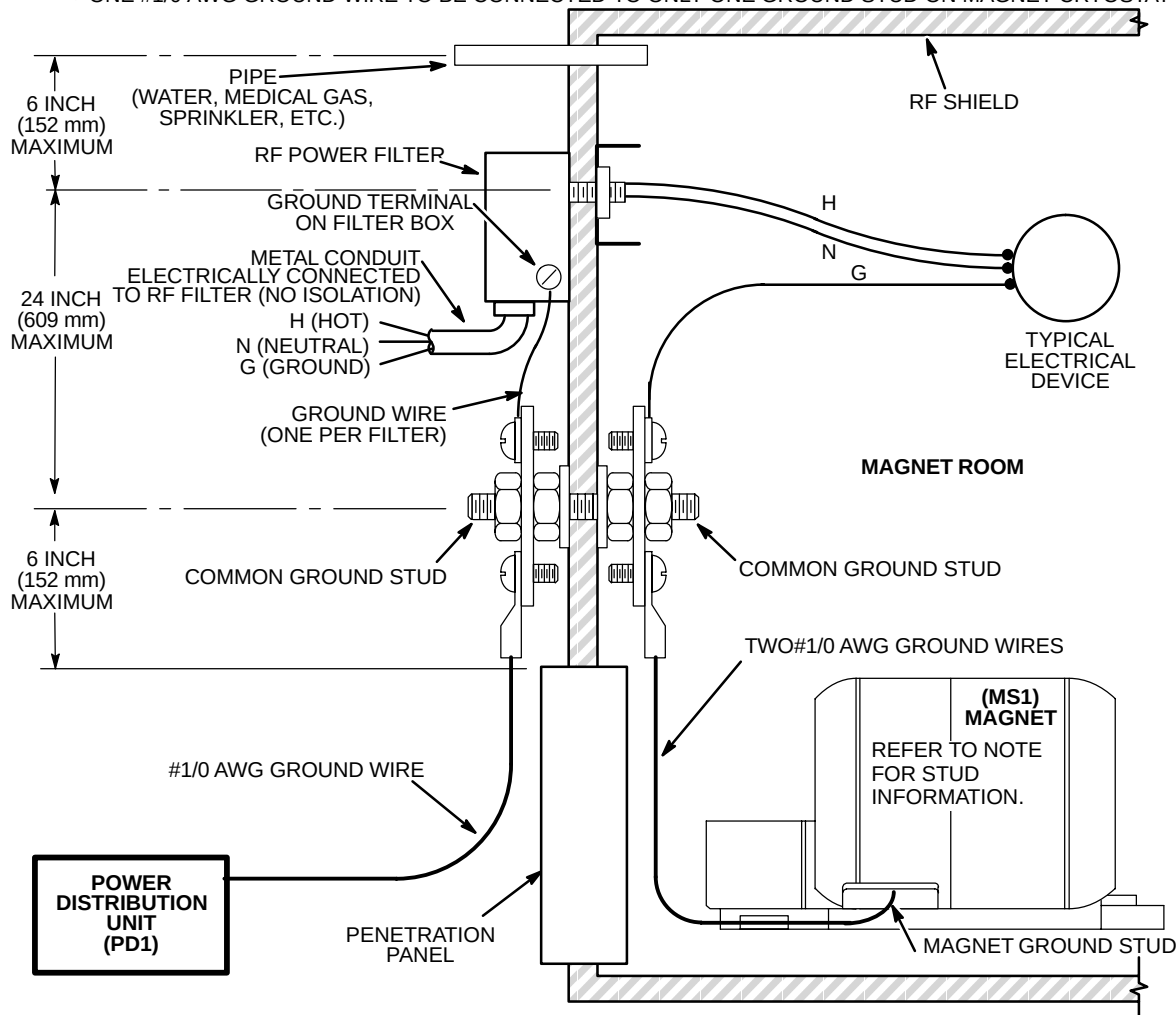
The three major grounding points in the MR system are: the system ground point (bus) in the System PDU (PD1), the enclosure ground points (ground studs located in each cabinet or enclosure), and the RF Shielded Room common ground point. **This RF Shielded Room common ground point is to be located within 6 in. (152 mm) of the GE supplied penetration panel.** Refer to Section 7, RF SHIELDED ROOM, for a further description of the RF shielded room common ground point.

To ensure patient safety and system performance, the conditions defined in Illustration 5-4 must be met when running power lines into the Magnet Room.

Any modifications or non-MR equipment grounds added to the MR ground system must be approved by your GE Service Representative in order to ensure safety and performance.

5-4-2 System Ground (Continued)

- NOTE:**
- ALL ITEMS SHOWN ARE CUSTOMER SUPPLIED EXCEPT POWER DISTRIBUTION UNIT, MAGNET, AND TWO #1/0 AWG GROUND WIRES BETWEEN MAGNET GROUND STUD AND RF COMMON GROUND POINT.
 - **RESISTANCE BETWEEN ANY TWO GROUNDED DEVICES MUST NOT EXCEED 0.1 OHM TO ENSURE EQUAL POTENTIAL GROUND SYSTEM WITHIN MAGNET ROOM.**
 - **LOCATE FILTERS WITHIN 2 FEET (600 mm) OF RF COMMON GROUND STUD WHICH MUST BE LOCATED WITHIN 6 INCHES (152 mm) OF PENETRATION PANEL.**
 - ALL EXTERNAL CONDUIT MUST BE METAL AND ELECTRICALLY CONNECTED TO THE RF POWER FILTERS (IE. NO ISOLATION) UNLESS THE FILTERS ARE LOW VOLTAGE (<30 VOLTS).
 - RF POWER FILTERS OF 30 VOLTS OR LESS MAY BE LOCATED ANYWHERE ON THE RF SHIELD **PROVIDED THE INCOMING CONDUIT IS NON-METALLIC OR IS DIELECTRICALLY ISOLATED AND WITH NO GROUND WIRE. IF THE INCOMING CONDUIT IS METALLIC AND/OR HAVE A GROUND WIRE, THESE FILTERS MUST ALSO BE LOCATED WITHIN 24 INCHES (609 mm) OF THE RF COMMON GROUND STUD.**
 - ALL CONDUITS IN THE RF ROOM MUST BE METAL. STEEL IS ACCEPTABLE PROVIDED IT IS ADEQUATELY ANCHORED.
 - ALL ELECTRICAL DEVICES (IE. OUTLETS, LIGHT FIXTURES, ETC.) MUST HAVE A GROUND WIRE FROM ITS POWER SOURCE AND BE GROUNDED TO RF ROOM SHIELD AT THE RF COMMON GROUND STUD AS SHOWN BELOW.
 - ALL METALLIC PIPES ENTERING THE RF ROOM, EXCLUDING CRYOGENIC VENT AND FLOOR DRAINS, MUST BE LOCATED WITHIN 30 INCHES (762 mm) OF THE RF COMMON GROUND.
 - ONE #1/0 AWG GROUND WIRE TO BE CONNECTED TO ONLY ONE GROUND STUD ON MAGNET CRYOSTAT OR FOOT.



MR MAGNET ROOM GROUNDING REQUIREMENTS AND TYPICAL DIAGRAM

ILLUSTRATION 5-4

5-5 GROUND FAULT PROTECTION

MR suites and radiology departments are considered health care facilities pursuant to National Electric Code (NEC) 2002 Article 517.2 definitions or NEC1999 Article 517–3 definitions and as such must be powered from sources that comply with the ground fault requirements of NEC 2002 Article 517.17 or NEC 1999 Article 517–17. NEC 2002 Article 517.17(A) or NEC 1999 Article 517–17(a) states “Where ground fault is required for the operation of the service disconnecting means or feeder disconnecting means as specified in NEC 230.95 or 215.10, an additional step of ground fault protection shall be provided in the next level of feeder disconnecting means downstream towards the load.”

NEC 2002 Article 230.95 or 215.10 or NEC 1999 Article 230–95 or 215–10 requires ground fault protection on service disconnecting means rated 1000 Amps or more on solidly grounded WYE services over 150 volts to ground but not over 600 volts phase to phase.

The two or more levels of ground fault shall be coordinated to provide selectivity between each level of ground fault such that a ground fault on the load side of the feeder would cause the feeder and not the service disconnect to open on a ground fault. Six cycles of separation between the different levels of ground fault tripping is required for the system to be considered selective in accordance with NEC 2002 Article 517.17(B) or NEC 1999 Article 517–17(b).

Check national and local electrical codes.

SECTION 6 – INTERCONNECT DATA

TABLE OF CONTENTS

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
6-1	INTRODUCTION	6-3
6-1-1	Component Designators	6-4
6-1-2	Definition of Terms	6-5
6-2	SYSTEM INTERCONNECTS ADDED BY UPGRADES	6-6
6-2-1	Introduction	6-6
6-2-2	M1033L – Indigo to Octane II with EXCITE Upgrade	6-7
6-2-3	M1060JW – Shield/Cryo Cooler Compressor	6-7
6-2-4	M1060LC – 1.5T CXK4 Magnet Upgrade with Wide Open Enclosure	6-8
6-2-5	M1090JP & M1090JR – Enclosure Upgrade For Fixed Site	6-8
6-2-6	M1090JW – Laser to Halogen Alignment Lights Upgrade	6-9
6-2-7	M1090PC – 5.5 Phased Array Compatibility Kits for 1.5T System	6-9
6-2-8	M1090PH – Pre LX to LX Without Phase Array To 4 channel Phase Array Adder Kit For 1.5T System	6-9
6-2-9	M1090PW – 1.5T Phased Array Upgrade Kit (UCERD) For LX Systems Without Phase Array	6-10
6-2-10	M1090SJ – 5.5 To LX Upgrade For Fixed Site	6-10
6-2-11	M1191JF – Enclosure Upgrade For Transportable/Relocatable	6-10
6-2-12	M1191JM – 5.5 To LX Cable Upgrade for True Mobile with Rear Mounted Penetration Panel	6-10
6-2-13	M1191JN – 5.5 To LX Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel	6-10
6-2-14	M1191SN – Collector Kit for Upgrade to 1.5T LCC Magnet	6-11
6-2-15	M1191SP – Collector Kit for Upgrade to 1.5T LCC Magnet	6-13
6-2-16	M1860LC – 1.0T CXK4 Magnet Upgrade with Wide Open Enclosure	6-13
6-2-17	M1890PA – 5.5 Phased Array Compatibility Kits For 1.0T System	6-13
6-2-18	M1890PH – Pre LX to LX Without Phase Array to 4 channel Phase Array Adder Kit For 1.0T System	6-13
6-2-19	M1890PW – 1.0T Phased Array Upgrade Kit (UCERD) for LX Systems without Phased Array	6-13
6-2-20	M1891SN – Collector Kit for Upgrade to 1.0T LCC Magnet	6-14
6-2-21	M1891SP – Collector Kit for Upgrade to 1.0T LCC Magnet	6-15
6-2-22	M3090BA – 1.0T ACGD Upgrade for system with SGD Gradient Cabinet & RF/Penetration Cabinet	6-16
6-2-23	M3090BB – ACGD Upgrade for system with SGD Gradient Cabinet & RF/PDU Cabinet	6-16
6-2-24	M3090BC – 1.5T ACGD Upgrade for system with SGD Gradient Cabinet & RF/Penetration Cabinet	6-16

TABLE OF CONTENTS (Continued)

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
6-2-25	M3090CA – 4.X To LX With ACGD Cable Upgrade For Fixed Site	6-17
6-2-26	M3090CB – 5.1-5.4 To LX With ACGD Cable Upgrade For Fixed Site	6-21
6-2-27	M3090CC – 5.5 Software Only Site To LX With ACGD Cable Upgrade For Fixed Site	6-23
6-2-28	M3090CD – 5.5 To LX With ACGD Cable Upgrade For Fixed Site	6-24
6-2-29	M3090CH – 3.X-5.4 to LX with ACGD Cable Upgrade for Transportable/ Relocatable (T/R) with Rear Mounted Penetration Panel	6-25
6-2-30	M3090CJ – 5.5 Software Site to LX with ACGD Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel	6-25
6-2-32	M3090CM – 5.X To LX with ACGD Cable Upgrade for Fixed Site	6-28
6-2-33	M3090CN – 5.5 To LX Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel	6-30
6-2-34	M3090EB – 1.5T EXCITE Hardware Kit	6-30
6-3	ADVANTAGE WORKSTATION INTERCONNECTS	6-31
6-4	INTERCONNECTS RE-ROUTED BY HORIZON LX WITH ACGD UPGRADES	6-31
6-5	INTERCONNECTS REMOVED BY LX WITH EXCITE TECHNOLOGY UPGRADES	6-32
6-6	INTERCONNECTS REMOVED BY HORIZON LX WITH ACGD UPGRADES	6-34
6-6-1	Signa Horizon LX System With SGD & RF/PDU Cabinet	6-34
6-6-2	Signa Horizon LX System With SGD & RF/Penetration Cabinet	6-35
6-6-3	Signa Horizon LX System With 8645 Gradients & RF/Penetration Cabinet	6-37
6-6-4	Signa Horizon 5.X System	6-39
6-6-5	Signa Horizon 5.X Software & Signa Advantage 5.X System	6-42
6-7	INTERCONNECTS REMOVED BY SIGNA HORIZON 5.X TO SIGNA LX COMPUTER ONLY UPGRADE	6-52
6-8	REAR PEDESTAL AND ENCLOSURE CABLE ACCESS	6-54
6-9	SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH ACGD	6-55
6-9-1	M3090DA – 4.X-5.4 MNS Upgrade Kit For ACGD Site	6-55
6-9-2	M3090DB – 5.5 MNS Upgrade Kit For ACGD Site	6-56
6-9-3	M3090DB – LX MNS Upgrade Kit For ACGD Site	6-56
6-10	SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH SGD	6-57
6-10-1	M1033MC – MNS Kit For Signa LX (8.X) Systems That Do Not Have RF/PDU Cabinet	6-57
6-10-2	M1033MD – MNS Kit For Signa LX (8.X) Systems That Have RF/PDU Cabinet ..	6-58

6-1 INTRODUCTION

Section 6, INTERCONNECT DATA, addresses system interconnections which are impacted by the various Horizon LX upgrade catalogs. It's subsections are broken down as follows:

- 6-1 INTRODUCTION
 - general upgrade information, component designations
- 6-2 SYSTEM INTERCONNECTS ADDED BY UPGRADES
 - cable interconnects for the system provided by Signa LX With EXCITE Technology Upgrade catalogs
 - cable interconnects for the system provided by Signa LX with ACGD Upgrade catalogs
- 6-3 SIGNA CVMR ADVANTAGE WORKSTATION INTERCONNECTS
 - cable interconnects for Signa CVMR Option Advantage Workstation Package
- 6-4 INTERCONNECTS RE-ROUTED BY HORIZON LX WITH ACGD UPGRADES
 - existing system cable interconnects re-labeled and re-routed during Horizon LX with ACGD Upgrades
- 6-6 SYSTEM INTERCONNECTS REMOVED BY HORIZON LX WITH ACGD UPGRADES
 - existing system cable interconnects removed during Horizon LX Upgrades
- 6-7 INTERCONNECTS REMOVED BY SIGNA HORIZON 5.X TO SIGNA LX COMPUTER ONLY UPGRADE
 - cables interconnects removed and discarded during upgrade to Signa Horizon LX Computer only configuration with no gradient subsystem change
- 6-8 REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
 - need to plan Rear Pedestal cable access and routing due to increase of cabling and hoses
- 6-9 SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH ACGD
 - additional cable interconnects for the system provided by Spectroscopy Option Upgrade catalogs for systems with ACGD Cabinet
- 6-10 SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH SGD
 - additional cable interconnects for the system provided by Spectroscopy Option Upgrade catalogs for systems with SGD Cabinet

Note

Magnet subsystem interconnects are not impacted by Horizon upgrades and therefore are not listed in this section.

6-1-1 Component Designators

The Component Designator System is used as the means of identifying system components in a consistent manner. All subsystem cabinets and other components are referred to by their component designators in the diagrams and tables of this section. (For example, the System Cabinet is referred to as MR2.) Refer to Table 6-1 for component designators impacted by Signa Horizon LX upgrades.

TABLE 6-1
COMPONENT DESIGNATION

APPLICABLE SYSTEM	COMPONENT DESIGNATOR	DESCRIPTION
Signa LX with ACGD	MG2	Magnet Enclosure
	MG3	Magnet Rear Pedestal
	MG6	Blower Box
	MR1	SRF Cabinet
	MR2	System Cabinet
	MR2	System Cabinet for Signa LX With EXCITE Technology or Signa LX
	MR3	ACGD/PDU Cabinet has PD1 in lower portion cabinet
	MS4	Magnet Rundown Unit
	PD1	Power Distribution Unit (PDU) is a module in lower portion of ACGD/PDU Cabinet
	PP1	Penetration Panel
Signa LX with ACGD	WC1	Gradient Coil Water Chiller
1.5T System Option	MNS	MNS Cabinet
System CVMR Option	AW	Advantage Workstation
Signa Horizon LX with SGD, Signa Horizon, Signa Advantage or Signa Obsolete Equipment	CC1	Computer Cabinet
	CC2	Auxiliary Computer Cabinet
	CM1	Cardiac Monitor
	HE1	Heat Exchanger
	MR1	RF/PDU Cabinet
		RF/Penetration Cabinet
		RF Cabinet
	MR3	SGD Base Gradient Cabinet for Base System or
		SGD Hi-Slew Gradient Cabinet for HiSpeed or EchoSpeed
		8645 Gradient Cabinet
	MR4	Gradient Amp Cabinet
		Gradient Resistor Box (may exist on Pre-Performance Plus systems)
	MR7	GRAM Cabinet
	OC1	Operator Console
	PD1	Power Distribution Unit (PDU) is a module in lower portion of RF/PDU Cabinet
		Full Size (150 KVA or 75 KVA) or Compact Power Distribution Unit
	PP2	Penetration Cabinet

6-1-2 Definition of Terms

The definition of terms used in following sections tables are:

- Group Number: identifying number referenced to bundles (i.e. groups) of cables as shown in interconnect diagrams
- Between Units (From/To): component designators as found in tables
- Usable Length: total length of a cable MINUS any required take up within cabinets
- Run Number: unique number assigned to each GE-supplied cable

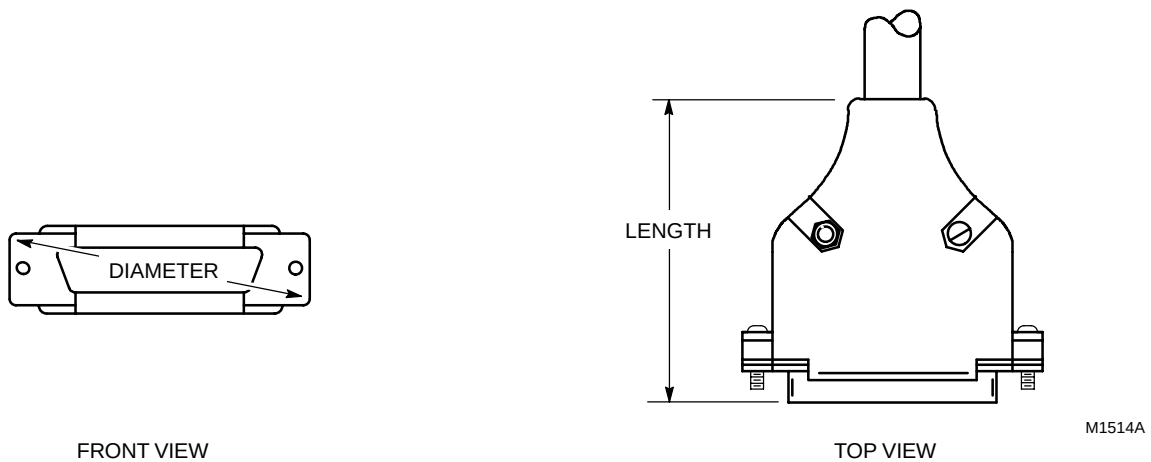
Note

The run number must be used when making special cable orders.

- Cable Diameter: diameter of an individual cable
- Plug Pulling Diameter x Length: cable plug dimensions

Note

In some cases, a cable has more than one connector on an end. These cables will have the number of connectors following the english dimensions of the plug pulling diameter times length (e.g. '2.0x3.25 -x2' means there are 2 connectors with dimensions of 2.0 in. diameter and 3.25 in. length). Of course, the same number of connectors apply to the metric dimensions as well.



SUBMINIATURE-D CONNECTOR PLUG PULLING DIMENSIONS
ILLUSTRATION 6-1

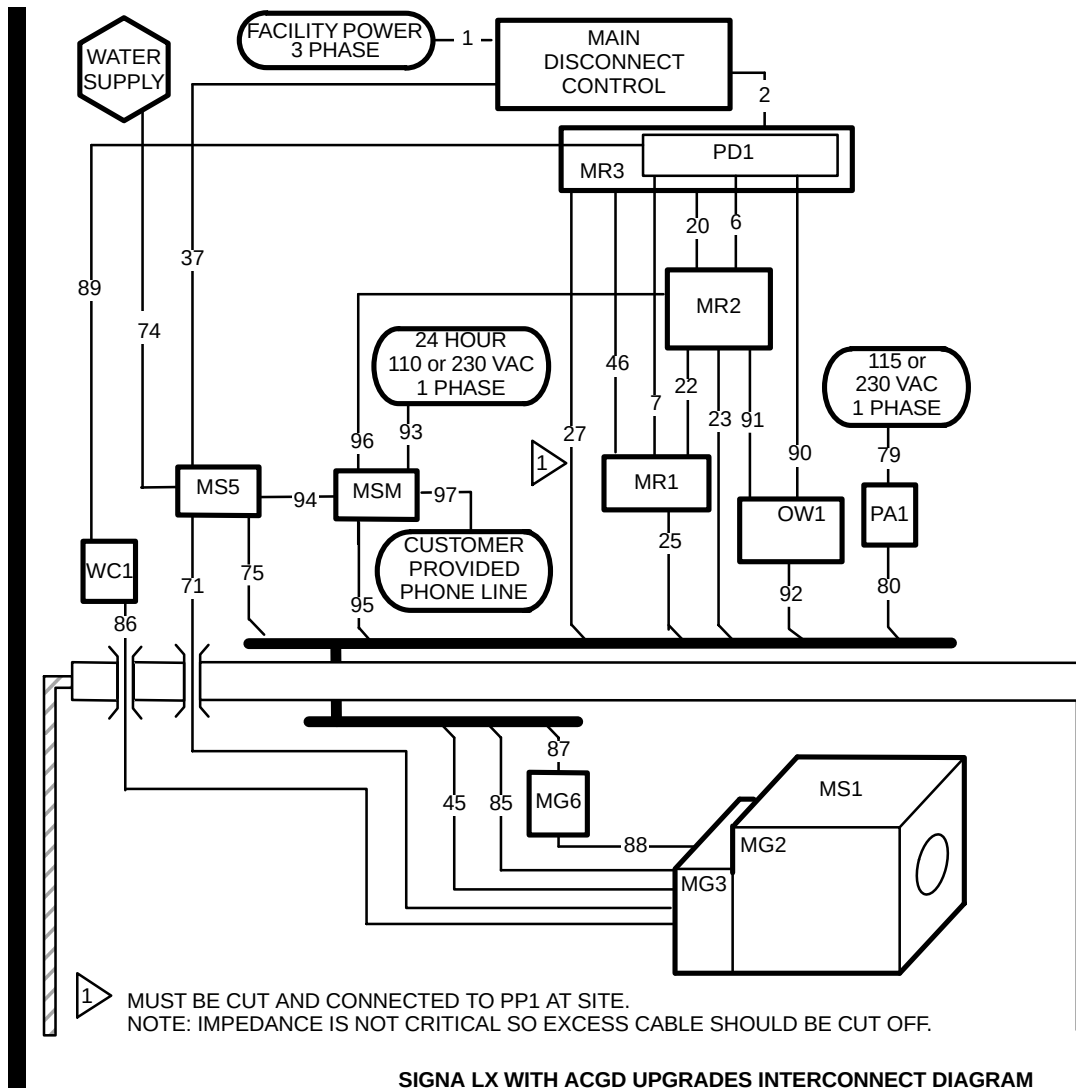
6-2 SYSTEM INTERCONNECTS ADDED BY ACGD UPGRADES

6-2-1 Introduction

Illustration 6-2 interconnect diagrams shows the groups to which cables are added by the various Signa LX upgrade catalogs. The following sub-sections list the cables provided by the various Signa LX upgrade catalogs. Conduit or pipe is not recommended for cable runs since the system uses many prefabricated cables with large connectors. Unless otherwise specified, cables and components listed in tables in this section are supplied by GE.

Note

Need to verify customer supplied Group1 (Facility Power to Main Disconnect Control) and Group 2 (Main Disconnect Control) are properly sized for specifications in Section 5, Power Requirements.



6-2-2 M1033L – Indigo to Octane II with EXCITE Upgrade

M1033L provides interconnects for an Indigo to Octane II with EXCITE Upgrade listed in Table 6-2.

TABLE 6-2
CABLES PROVIDED BY M1033L

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
91	MR2	OW1	51 (15.5)	1032	0.525 (13.3)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-2-3 M1060JW – Shield/Cryo Cooler Compressor

M1060JW provides interconnects for Shield/Cryo Cooler Compressor for the LCC Magnet listed in Table 6-3.

TABLE 6-3
CABLES PROVIDED BY M1060JW

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
37	MDP	MS5	29.8 (9.1) minus takeup at MDP	—	1.29 (32.75)	Hard Wired	Hard Wired	Both ends have liquid tight connector which wires pass through for hard wire connections.
45	PP1	MG3	30 (9.15)	624	0.26 (6.6)	Ring Terminals	1.00x1.38 (25.4x35.1)	
71	MS5	MS1	38.5 (11.7)	621 622	1.65 (41.9) 1.65 (41.9)	2.00x3.75 (50.8x95.3) 2.00x3.75 (50.8x95.3)	2.00x3.75 (50.8x95.3) 2.00x3.75 (50.8x95.3)	Runs 621 and 622 are continuous helium lines routed through PP1. Bend radius of Runs 621 and 622 is 8 in. (203.2 mm). Cable diameter includes foam insulation installed on lines.
75	MS5	PP1	42 (12.8)	623	0.21 (5.28)	1.00x1.38 (25.4x35.1)	Ring Terminals	

6-2-4 M1060LC – 1.5T CXK4 Magnet Upgrade with Wide Open Enclosure

M1060LC provides interconnects for the LCC Magnet listed in Table 6-4.

TABLE 6-4
CABLES PROVIDED BY CATALOGS M1060LC & M1860LC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG2/3	30 (9.15)	828	0.31 (7.75)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				829	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
48	MS4	MS1	86 (26.2)	606	0.30 (7.6)	0.65x1.85 (16.5x47.0)	0.65x1.85 (16.5x47.0)	
93	24 Hour Facility Power	MSM	6 (1.8) minus takeups at each end	—	0.375 (9.5)	1.2x3.00 (30.5x76.2)	1.25x2.5 (31.8x63.5)	
94	MSM	MS5	54 (16.4) minus takeup at MSM	826	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
95	MSM	PP1	54 (16.4) minus takeup at MSM	824	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				825	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
96	MSM	MR2	54 (16.4) minus takeup at MSM	823	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	

6-2-5 M1090JP & M1090JR – Enclosure Upgrade For Fixed Site

The enclosure catalogs for fixed sites provide the Blower Box to Magnet Enclosure air hoses. Refer to Table 6-5 for details.

TABLE 6-5
CABLES PROVIDED BY CATALOGS M1090JP, M1090JR, & M1191JF

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
88	MG6	MG3	50 (15.2) minus takeup, See Notes	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The 6.5 in. (165.1 mm) hose has bend radius of 4 in. (101.6 mm). The 4.5 in. (104.3 mm) hose has bend radius of 2 in. (50.8 mm). The flexible vinyl hoses are cut to length during installation. Total length provided is 50 ft (15.2 m). Refer to Section 6-8, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				—	4.5 (104.3)	Flexible vinyl hose	Flexible vinyl hose	

6-2-6 M1090JW – Laser to Halogen Alignment Lights Upgrade

M1090JW provides interconnect for the Halogen Alignment Lights listed in Table 6-4.

TABLE 6-6
CABLES PROVIDED BY CATALOG M1090JW

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG2/3	30 (9.15)	385	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	

6-2-7 M1090PC – 5.5 Phased Array Compatibility Kits for 1.5T System

M1090PC provides interconnects for Signa Horizon LX 1.5T System without Phased Array. Refer to Table 6-7 interconnect details.

TABLE 6-7
CABLES PROVIDED BY M1090PC, M1090PH, M1890PA, M1890PH

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
23	MR2	PP1	31 (19.45)	488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
25	MR1	PP1	21 (6.40)	486	0.36 (9.1)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				487	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				726	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
45	PP1	MG3	30 (9.15)	491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	

6-2-8 M1090PH – Pre LX to LX Without Phase Array to 4 channel Phase Array Adder Kit For 1.5T System

M1090PH provides interconnects for Signa Horizon LX 1.5T System without Phased Array upgrade to 4 channel Phased Array. Cables provided by catalog M1090PH are the same as catalog M1090PC. Refer to Table 6-7 for M1090PH cables details.

6-2-9 M1090PW – 1.5T Phased Array Upgrade Kit (UCERD) For LX Systems Without Phase Array

M1090PW provides interconnects for Signa Horizon LX 1.5T System without Phased Array upgrade to 4 channel Phased Array. Cables provided by catalog M1090PW are the same as catalog M1090PC. Refer to Table 6-7 for M1090PH cables details.

6-2-10 M1090SJ – 5.5 To LX Cable Upgrade For Fixed Site

Catalog M1090SJ provides fixed site system interconnect cables for 5.5 Signa Horizon configuration to Signa Horizon LX. Also included in this catalog are Horizon LX upgrade power and ground cables. Refer to Table 6-14 for M1090SJ cables details.

TABLE 6-8
CABLES PROVIDED BY CATALOG M1090SJ

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals	Ring Terminals	Power Cable, #10 AWG 4 wires
				048	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire, #10 AWG 1 wire
91	MR2	OW1	51 (15.5)	789	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				790	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
				791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
92	PP1	OW1	53 (16.1)	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-2-11 M1191JF – Enclosure Upgrade For Transportable/Relocatable

Cables provided by catalog M1191JF are the same as catalogs M1090JP and M1090JR. Refer to Table 6-5 for M1191JF cables details.

6-2-12 M1191JM – 5.5 To LX Cable Upgrade for True Mobile with Rear Mounted Penetration Panel

All signal cables are replaced during a True Horizon LX Mobile upgrade, PDU to cabinet power cables are retained and re-routed as necessary. Upgrade plans are available from the van manufacturer.

6-2-13 M1191JN – 5.5 To LX Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel

All signal cables are replaced during a Transportable or Relocatable Mobile Horizon LX upgrade, PDU to cabinet power cables are retained and re-routed as necessary. Upgrade plans are available from the van manufacturer.

6-2-14 M1191SN – Collector Kit for Upgrade to 1.5T LCC Magnet

M1191SN provides interconnects for the 1.5T LCC Magnet listed in Table 6-9.

TABLE 6-9
CABLES PROVIDED BY CATALOGS M1191SN

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
25	MR1	PP1	21 (6.40)	235	0.82 (20.8)	1.44x2.00 (36.6x50.8)	1.44x2.00 (36.6x50.8)	Used for 1.5T system only. Has a minimum bend radius of 10 in. (254 mm).
				236	0.59 (15.0)	0.92x2.10 (23.4x53.3)	2.00x2.10 (50.8x53.3)	Used for 1.5T system only. Has a minimum bend radius of 5 in. (127 mm).
27	MR3	PP1	33 (10.0)	762	1.21 (30.7)	Hard wired	Ring Terminals	These runs are cut and connected to PP1 at site; total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 85 for Magnet Room routing of these interconnects.
				763	1.21 (30.7)	Hard Wired	Ring Terminals	
				764	1.21 (30.7)	Hard Wired	Ring Terminals	
42	PP1	E01	84 (25.6) minus takeup at EO1	297	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	
45	PP1	MG2/3	55 (16.76)	257	1.10 (27.9)	1.40x3.40 (35.6x86.4)	1.40x3.40 (35.6x86.4)	Used for 1.5T system only. Has a minimum bend radius of 10 in. (254 mm).
			55 (16.76)	258	0.64 (16.3)	0.90x2.50 (22.9x63.5)	0.90x2.50 (22.9x63.5)	Used for 1.5T system only. Has a minimum bend radius of 5 in. (127 mm).
				262	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				282	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				318	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				387	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
(Continued)								

6-2-14 M1191SN – Collector Kit for Upgrade to 1.5T LCC Magnet

TABLE 6-9 (Continued)
CABLES PROVIDED BY CATALOGS M1191SN

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG2/3	55 (16.76)	700	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				715	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				716	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				778	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				779	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				780	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				781	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				782	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
78	RF COM- MON GND STUD	MS1 GND STUD	60 (18.29) minus takeup at RF Common GND Stud	040	0.464 (11.79)	Hard Wired	Ring Terminal	1 ground wire.
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs are cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. For Equipment Room routing of these interconnects refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm).
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See Notes	045	0.195 (4.95)	1/32X0.23 (33.5X5.8)	1/32X0.23 (33.5X5.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
				784	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
				786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	
89	PD1	WC1	67 (20.3)	065	0.635 (16.1)	Hard Wired	Hard Wired	#10 AWG / 3 wire Power cable

6-2-15 M1191SP – Collector Kit for Upgrade to 1.5T LCC Magnet

M1191SP provides interconnects for the LCC Magnet listed in Table 6-10.

TABLE 6-10
CABLES PROVIDED BY CATALOGS M1191SP & M1891SP

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
86	WC1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
88	MG6	MG3	13 (3.95) minus takeup	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The flexible vinyl hoses are cut to length during installation.

6-2-16 M1860LC – 1.0T CXK4 Magnet Upgrade with Wide Open Enclosure

Cables provided by catalog M1860LC for 1.0T system are the same as catalog M1090PC for 1.5T system. Refer to Table 6-4 for cables details.

6-2-17 M1890PA – 5.5 Phased Array Compatibility Kits For 1.0T System

Cables provided by catalog M1890PA for 1.0T system are the same as catalog M1090PC for 1.5T system. Refer to Table 6-7 for M1890PA cables details.

6-2-18 M1890PH – Pre LX to LX Without Phase Array to 4 channel Phase Array Adder Kit For 1.0T System

M1890PH provides interconnects for Signa Horizon LX 1.5T System without Phased Array upgrade to 4 channel Phased Array. Cables provided by catalogs M1090PH are the same as catalog M1090PC. Refer to Table 6-7 for M1090PH cables details.

6-2-19 M1890PW – 1.0T Phased Array Upgrade Kit (UCERD) for LX Systems without Phased Array

M1890PW provides interconnects for Signa Horizon LX 1.5T System without Phased Array upgrade to 4 channel Phased Array. Cables provided by catalogs M1090PW are the same as catalog M1090PC. Refer to Table 6-7 for M1090PH cables details.

6-2-20 M1891SN – Collector Kit for Upgrade to 1.0T LCC Magnet

M1191SN provides interconnects for the 1.0T LCC Magnet listed in Table 6-11.

TABLE 6-11
CABLES PROVIDED BY CATALOGS M1891SN

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
25	MR1	PP1	21 (6.40)	745	0.50 (12.7)	0.91x2.50 (23.0x63.5)	0.91x2.50 (23.0x63.5)	Used for 1.0T system only.
				748	0.44 (11.0)	0.63x2.01 (15.9x51.1)	0.63x2.01 (15.9x51.1)	Used for 1.0T system only.
27	MR3	PP1	33 (10.0)	762	1.21 (30.7)	Hard wired	Ring Terminals	These runs are cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 85 for Magnet Room routing of these interconnects.
				763	1.21 (30.7)	Hard Wired	Ring Terminals	
				764	1.21 (30.7)	Hard Wired	Ring Terminals	
42	PP1	E01	84 (25.6) minus takeup at EO1	297	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	
45	PP1	MG2/3	55 (16.76)	262	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				282	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				318	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				387	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				700	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				715	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				716	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
(Continued)								

6-2-14 M1891SN – Collector Kit for Upgrade to 1.0T LCC Magnet

TABLE 6-11 (Continued)
CABLES PROVIDED BY CATALOGS M1891SN

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45 (continue)	PP1	MG2/3	55 (16.76)	727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	Used for 1.0T system only.
				746	0.50 (12.7)	0.91x2.50 (23x63.5)	0.91x2.45 (23x62.2)	
				747	0.440 (11.2)	0.63x2.01 (15.9x51.1)	0.63x2.01 (15.9x51.1)	
				778	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				779	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				780	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				781	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				782	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
78	RF COM- MON GND STUD	MS1 GND STUD	60 (18.29) minus takeup at RF Common GND Stud	040	0.464 (11.79)	Hard Wired	Ring Terminal	1 ground wire.
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs are cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. For Equipment Room routing of these interconnects refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm).
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See Notes	045	0.195 (4.95)	1/32X0.23 (33.5X5.8)	1/32X0.23 (33.5X5.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
				784	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
				786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	
89	PD1	WC1	67 (20.3)	065	0.635 (16.1)	Hard Wired	Hard Wired	#10 AWG / 3 wire Power cable

6-2-21 M1891SP – Collector Kit for Upgrade to 1.5T LCC Magnet

Cables provided by catalog M1891SP for 1.0T system are the same as catalog M1091SP for 1.5T LCC Magnet. Refer to Table 6-10 for cables details.

6-2-22 M3090BA – 1.0T ACGD Upgrade for system with SGD Gradient Cabinet & RF/Penetration Cabinet

Catalog M3090BA provides ACGD upgrade power and ground cables. Refer to Table 6-13 for cables details.

TABLE 6-12
CABLES PROVIDED BY CATALOG M3090BA, M3090BB, & M3090BC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	32 (9.75)	968	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
7	MR3/ PD1	MR1	32 (9.76)	966	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
				967	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
90	MR3/ PD1	OW1	51 (15.5)	969	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires

6-2-23 M3090BB – ACGD Upgrade for system with SGD Gradient Cabinet & RF/PDU Cabinet

Cables provided by catalogs M3090BB are the same as catalogs M3090BA. Refer to Table 6-13 for cables details.

6-2-24 M3090BC – 1.5T ACGD Upgrade for system with SGD Gradient Cabinet & RF/Penetration Cabinet

Cables provided by catalogs M3090BB are the same as catalogs M3090BA. Refer to Table 6-13 for cables details.

6-2-25 M3090CA – 4.X To LX With ACGD Cable Upgrade For Fixed Site

Catalog M3090CA provides fixed site system interconnect cables for Signa Advantage 4.X configuration to Signa Horizon LX with ACGD. Also included in this catalog are Horizon LX with ACGD upgrade power and ground cables, fiber optic cables, and the Gradient Coil Water Cooling flexible water lines. Refer to Table 6-13 for cables details.

TABLE 6-13
CABLES PROVIDED BY CATALOG M3090CA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	32 (9.75)	968	0.92 (23.4)	2.22x3.84 (56.4x97.6)	Ring Terminals	Power cable, #8 AWG 5 wires
7	MR3/ PD1	MR1	32 (9.76)	966	0.92 (23.4)	2.22x3.84 (56.4x97.6)	Ring Terminal	Power cable, #8 AWG 5 wire
				967	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wire
20	MR2	MR3	24 (7.31)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
22	MR2	MR1	24 (7.31)	702	0.35 (8.90)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Run 708 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
				708	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	
				774	0.44 (11.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
23	MR2	PP1	41 (12.50)	711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (51 mm).
(Continued)								

6-2-25 M3090CA – 4.X To LX With ACGD Cable Upgrade For Fixed Site (Continued)

TABLE 6-13 (Continued)
CABLES PROVIDED BY CATALOG M3090CA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
25	MR1	PP1	21 (6.40)	044	0.20 (5.1)	1.32x0.23 (33.5x5.8)	1.32x0.23 (33.5x5.8)	
				768	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				769	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				770	0.31 (7.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				771	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				772	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				773	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				775	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				776	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				777	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				783	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
				785	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
27	MR3	PP1	31 (9.45)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs are cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 85. These cables have a minimum bend radius of 8 in. (203 mm).
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
(Continued)								

6-2-25 M3090CA – 4.X To LX With ACGD Cable Upgrade For Fixed Site (Continued)

TABLE 6-13 (Continued)
CABLES PROVIDED BY CATALOG M3090CA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG3	55 (16.76)	711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend radius of 2 in. (51 mm). Refer to Section 6–8, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				715	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				716	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				778	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				779	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				780	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				781	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				782	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
			pneumatic tubing	0.25 (6.4)	pneumatic tubing	pneumatic tubing	Pneumatic Patient Alert connection, refer to Group 80 for route.	
46	MR1	MR3	59 (17.98)	709	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 709 is a flexible conduit containing fiber optic cables.
80	PA1	MG2	97 (29.6) minus takeup at PA1	—	0.25 (6.4)	pneumatic tubing	pneumatic tubing	This pneumatic tubing is continu- ously routed from PA1 through PP1 and MG3 to MG3.
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs are cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm). Refer to Section 6–8, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
(Continued)								

6-2-25 M3090CA – 4.X To LX With ACGD Cable Upgrade For Fixed Site (Continued)

TABLE 6-13 (Continued)
CABLES PROVIDED BY CATALOG M3090CA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
86	WC1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See Notes	045	0.195 (4.95)	1/32X0.23 (33.5X5.8)	1/32X0.23 (33.5X5.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
				784	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	
				786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	
89	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
90	MR3/ PD1	OW1	51 (15.5)	048	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire, #10 AWG 1 wire
				969	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires
91	MR2	OW1	51 (15.5)	789	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8).
				818	0.44 (11.2)	2.30x2.00 (57.2x50.8)	See Notes	
				791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				836	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
92	PP1	OW1	53 (16.1)	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-2-26 M3090CB – 5.1-5.4 To LX With ACGD Cable Upgrade For Fixed Site

Catalog M3090CB provides fixed site system interconnect cables for Signa Advantage 5.1-5.4 configurations to Signa LX with ACGD. Also included in this catalog are Horizon LX with ACGD upgrade power and ground cables, fiber optic cables, and the Gradient Coil Water Chiller flexible water lines. Refer to Table 6-14 for cables details.

TABLE 6-14
CABLES PROVIDED BY CATALOGS M3090CB & M3090CC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	32 (9.75)	968	0.92 (23.4)	2.22x3.84 (56.4x97.6)	Ring Terminals	Power cable, #8 AWG 5 wires
7	MR3/ PD1	MR1	32 (9.76)	966	0.92 (23.4)	2.22x3.84 (56.4x97.6)	Ring Terminal	Power cable, #8 AWG 5 wires
				967	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
20	MR2	MR3	24 (7.31)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
22	MR2	MR1	24 (7.31)	708	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 708 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
25	MR1	PP1	21 (6.40)	044	0.20 (5.1)	1.32x0.23 (33.5x5.8)	1.32x0.23 (33.5x5.8)	
				768	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				769	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				770	0.31 (7.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				771	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				775	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				776	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				777	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				783	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
				785	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
(Continued)								

6-2-26 M3090CB – 5.1-5.4 To LX With ACGD Cable Upgrade For Fixed Site (Continued)

TABLE 6-14 (Continued)
CABLES PROVIDED BY CATALOGS M3090CB & M3090CC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
27	MR3	PP1	31 (9.45)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs are cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 85. These cables have a minimum bend radius of 8 in. (203 mm).
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
45	PP1	MG3	55 (16.76)	711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend radius of 2 in. (51 mm). Refer to Section 6-8, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				778	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				779	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				780	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				781	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				782	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				pneumatic tubing	0.25 (6.4)	pneumatic tubing	pneumatic tubing	
46	MR1	MR3	59 (17.98)	709	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 709 is a flexible conduit containing fiber optic cables.
80	PA1	MG2	97 (29.6) minus takeup at PA1	—	0.25 (6.4)	pneumatic tubing	pneumatic tubing	This pneumatic tubing is continuously routed from PA1 through PP1 and MG3 to MG3.
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs are cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm). Refer to Section 6-8, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	

(Continued)

6-2-26 M3090CB – 5.1-5.4 To LX With ACGD Cable Upgrade For Fixed Site (Continued)

TABLE 6-14 (Continued)
CABLES PROVIDED BY CATALOGS M3090CB & M3090CC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
86	WC1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See Notes	045	0.195 (4.95)	1/32X0.23 (33.5X5.8)	1/32X0.23 (33.5X5.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
				784	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	
				786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	
89	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
90	MR3/ PD1	OW1	51 (15.5)	969	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires
				048	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire, #10 AWG 1 wire
91	MR2	OW1	51 (15.5)	789	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8).
				818	0.44 (11.2)	2.30x2.00 (57.2x50.8)	See Notes	
				791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				836	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
92	PP1	OW1	53 (16.1)	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-2-27 M3090CC – 5.5 Software Only Site To LX With ACGD Cable Upgrade For Fixed Site

Cables provided by catalog M3090CC is the same as catalog M3090CB. Refer to Table 6-14 for cables details.

6-2-28 M3090CD – 5.5 To LX With ACGD Cable Upgrade For Fixed Site

Catalog M3090CD provides fixed site system interconnect cables for 5.5 Signa Horizon configuration to Signa Horizon LX with ACGD Gradient. Also included in this catalog are Horizon LX upgrade power and ground cables. Refer to Table 6-15 for cables details.

TABLE 6-15
CABLES PROVIDED BY CATALOG M3090CD

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	32 (9.75)	968	0.92 (23.4)	2.22x3.84 (56.4x97.6)	Ring Terminals	Power cable, #8 AWG 5 wires
7	MR3/ PD1	MR1	32 (9.76)	966	0.92 (23.4)	2.22x3.84 (56.4x97.6)	Ring Terminal	Power cable, #8 AWG 5 wire
				967	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wire
90	MR3/ PD1	OW1	51 (15.5)	969	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires
				048	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire, #10 AWG 1 wire
91	MR2	OW1	51 (15.5)	789	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8).
				818	0.44 (11.2)	2.30x2.00 (57.2x50.8)	See Notes	
				791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				836	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
92	PP1	OW1	53 (16.1)	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-2-29 M3090CH – 3.X-5.4 to LX with ACGD Cable Upgrade for Transportable/ Relocatable (T/R) with Rear Mounted Penetration Panel

All signal cables are replaced during a Transportable or Relocatable Mobile Horizon LX upgrade. PDU to cabinet power cables are replaced but signal cables are re-labelled and re-routed as necessary. Upgrade plans are available from the van manufacturer.

6-2-30 M3090CJ – 5.5 Software Site to LX with ACGD Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel

All signal cables are replaced during a Transportable or Relocatable Mobile Horizon LX upgrade. PDU to cabinet power cables are replaced but signal cables are re-labelled and re-routed as necessary. Upgrade plans are available from the van manufacturer.

6-2-31 M3090CK – 4.X To LX With ACGD Cable Upgrade For Fixed Site

Catalog M3090CK provides fixed site system interconnect cables for Signa Advantage 4.X configuration to Signa Horizon LX with ACGD. Refer to Table 6-16 for cables details.

TABLE 6-16
CABLES PROVIDED BY CATALOG M3090CK

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	32 (9.75)	968	0.92 (23.4)	2.22x3.84 (56.4x97.6)	Ring Terminals	Power cable, #8 AWG 5 wires
7	PD1	MR1	32 (9.76)	038	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				966	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
				967	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
20	MR2	MR3	24 (7.31)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
22	MR2	MR1	24 (7.31)	702	0.35 (8.90)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Run 708 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
				708	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	
				774	0.44 (11.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
23	MR2	PP1	41 (12.50)	711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (51 mm).
(Continued)								

6-2-31 M3090CK – 4.X To LX With ACGD Cable Upgrade For Fixed Site (Continued)

TABLE 6-16 (Continued)
CABLES PROVIDED BY CATALOG M3090CK

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
25	MR1	PP1	21 (6.40)	044	0.20 (5.1)	1.32x0.23 (33.5x5.8)	1.32x0.23 (33.5x5.8)	
				768	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				769	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				770	0.31 (7.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				771	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				772	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				773	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				775	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				776	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				777	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				783	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
				785	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
45	PP1	MG3	55 (16.76)	711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend radius of 2 in. (51 mm).
				pneumatic tubing	0.25 (6.4)	pneumatic tubing	pneumatic tubing	Refer to Section 6-8, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS Pneumatic Patient Alert connection, refer to Group 80 for route.
46	MR1	MR3	59 (17.98)	709	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 709 is a flexible conduit containing fiber optic cables.
80	PA1	MG2	97 (29.6) minus takeup at PA1	—	0.25 (6.4)	pneumatic tubing	pneumatic tubing	This pneumatic tubing is continuously routed from PA1 through PP1 and MG3 to MG3.
(Continued)								

6-2-31 M3090CK – 4.X To LX With ACGD Cable Upgrade For Fixed Site (Continued)

TABLE 6-16 (Continued)
CABLES PROVIDED BY CATALOG M3090CK

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See Notes	045	0.195 (4.95)	1/32X0.23 (33.5X5.8)	1/32X0.23 (33.5X5.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
89	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
90	MR3/ PD1	OW1	51 (15.5)	048	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire, #10 AWG 1 wire
				969	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires
91	MR2	OW1	51 (15.5)	789	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8).
				818	0.44 (11.2)	2.30x2.00 (57.2x50.8)	See Notes	
				791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				836	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
92	PP1	OW1	53 (16.1)	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-2-32 M3090CM – 5.X To LX with ACGD Cable Upgrade for Fixed Site

Catalog M3090CM provides fixed site system interconnect cables for 5.X Signa Horizon configuration to Signa LX with ACGD Gradient. Also included in this catalog are LX upgrade power and ground cables. Refer to Table 6-17 for cables details.

TABLE 6-17
CABLES PROVIDED BY CATALOG M3090CM

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
6	PD1	MR2	32 (9.75)	968	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
7	PD1	MR1	32 (9.76)	038	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				966	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
				967	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
20	MR2	MR3	24 (7.31)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
22	MR2	MR1	24 (7.31)	708	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 708 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
25	MR1	PP1	21 (6.40)	044	0.20 (5.1)	1.32x0.23 (33.5x5.8)	1.32x0.23 (33.5x5.8)	
				768	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				769	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				770	0.31 (7.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				771	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				775	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				776	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				777	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				783	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
				785	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
(Continued)								

6-2-32 M3090CM – 5.X To LX with ACGD Cable Upgrade for Fixed Site (Continued)

TABLE 6-17 (Continued)
CABLES PROVIDED BY CATALOG M3090CM

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG3	55 (16.76) 55 (16.76)	711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend radius of 2 in. (51 mm). Refer to Section 6-8, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
46	MR1	MR3	59 (17.98)	709	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 709 is a flexible conduit containing fiber optic cables.
80	PA1	MG2	97 (29.6) minus takeup at PA1	—	0.25 (6.4)	pneumatic tubing	pneumatic tubing	This pneumatic tubing is continuously routed from PA1 through PP1 and MG3 to MG3.
87	PP1	MG6	32 (9.76) minus takeup at MG6, See Notes	045	0.195 (4.95)	1/32X0.23 (33.5X5.8)	1/32X0.23 (33.5X5.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
89	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
90	PD1	OW1	51 (15.5)	048	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire, #10 AWG 1 wire
				969	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires
91	MR2	OW1	51 (15.5)	789	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8).
				791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				818	0.44 (11.2)	2.30x2.00 (57.2x50.8)	See Notes	
				836	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
92	PP1	OW1	53 (16.1)	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-2-33 M3090CN – 5.5 To LX Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel

All signal cables are replaced during a Transportable or Relocatable Mobile Horizon LX upgrade. PDU to cabinet power cables are replaced but signal cables are re-labelled and re-routed as necessary. Upgrade plans are available from the van manufacturer.

6-2-34 M3090EB – 1.5T EXCITE Hardware Kit

Catalog M3090EB provides additional cables for EXCITE Technology hardware upgrade. Refer to Table 6-18 for cables details.

TABLE 6-18
CABLES PROVIDED BY CATALOG M3090EB

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
20	MR2	MR3	14 (4.27)	1034	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1034 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (50 mm).
23	MR2	PP1	31 (9.45)	488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1025	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1026	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1027	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1028	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1029	0.34 (8.6)	1.6x2.00 (40.6x50.8)	1.6x2.00 (40.6x50.8)	
				1030	0.525 (13.3)	2.8x2.00 (71.1x50.8)	2.8x2.00 (71.1x50.8)	
				1045	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1045 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).
(Continued)								

6-2-34 M3090EB – 1.5T EXCITE Hardware Kit (Continued)

TABLE 6-18 (Continued)
CABLES PROVIDED BY CATALOG M3090EB

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG2/3	30 (9.15)	491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				1036	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				1037	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1038	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1039	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1040	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1041	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				1042	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
91	MR2	OW1	51 (15.5)	791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				1031	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	
				1044	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	

6-3 ADVANTAGE WORKSTATION INTERCONNECTS

Refer to Direction 2261309-100 Advantage Workstation Ultra60 Pre-Installation for AW interconnect information.

6-4 INTERCONNECTS RE-ROUTED BY HORIZON LX WITH ACGD UPGRADES

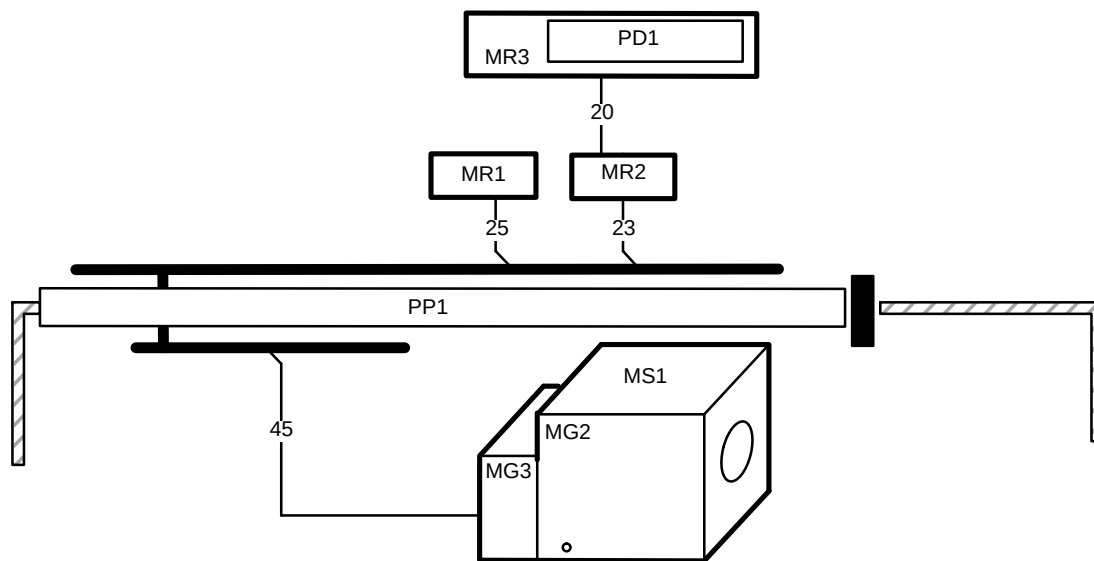
Upgrades to Signa Horizon LX with ACGD replaces the existing system PDU (Full-size, Compact, or PDU in RF/PDU Cabinet) with the PDU located in the lower portion of the ACGD/PDU Cabinet. Therefore, all connections to the PDU may need to be re-routed depending on site layout of the upgraded system.

Note

GE Installation Specialist & site Field Engineer need to consider impact of upgrade site layout on re-routed cables especially if any cables (e.g. power cables) were cut to length for current installed system.

6-5 INTERCONNECTS REMOVED BY LX WITH EXCITE TECHNOLOGY UPGRADES

Illustration 6-3 and Table 6-19 lists the existing Signa LX system system cables which are removed and discarded during upgrade to Signa LX With EXCITE Technology configuration. If upgrade to ACGD/PDU Cabinet is also being performed then refer to Section 6-6 **INTERCONNECTS REMOVED BY HORIZON LX WITH ACGD UPGRADES** for additional removed and discarded interconnects.



1.5T SYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6-3

6-5 INTERCONNECTS REMOVED BY LX WITH EXCITE TECHNOLOGY UPGRADES (Continued)

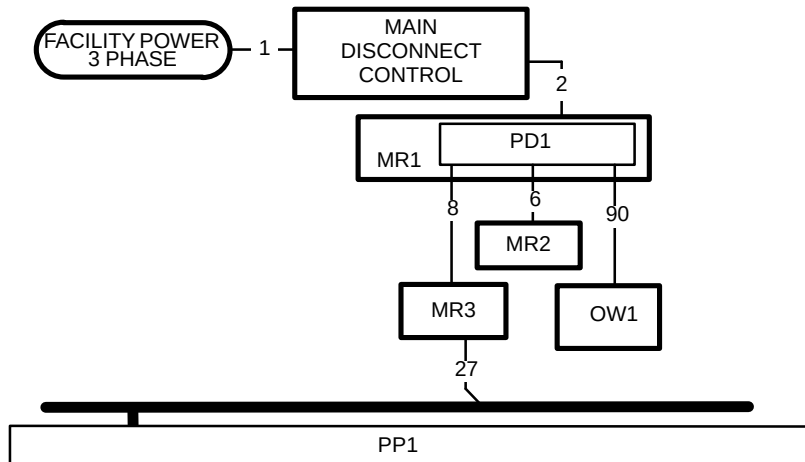
TABLE 6-19
SIGNA LX SYSTEM CABLES REMOVED BY EXCITE TECHNOLOGY UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
20	MR2	MR3	14 (4.27)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (50 mm).
23	MR2	PP1	31 (9.45)	488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	Used for system with Phased Array only.
				711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).
25	MR1	PP1	21 (6.40)	486	0.36 (9.1)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Used for system with Phased Array only.
				487	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	Used for system with Phased Array only.
				726	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	Used for system with Phased Array only.
45	PP1	MG2/3	30 (9.15)	260/ 700	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Used for system with Phased Array only.
				491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	Used for system with Phased Array only.
				727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	Used for system with Phased Array only.

6-6 INTERCONNECTS REMOVED BY HORIZON LX WITH ACGD UPGRADES

6-6-1 Signa Horizon LX System With SGD & RF/PDU Cabinet

Illustration 6-4 and Table 6-20 lists the existing Signa Horizon LX system with SGD & RF/PDU Cabinet system cables which are removed and discarded during upgrade to Signa Horizon LX with ACGD configuration.



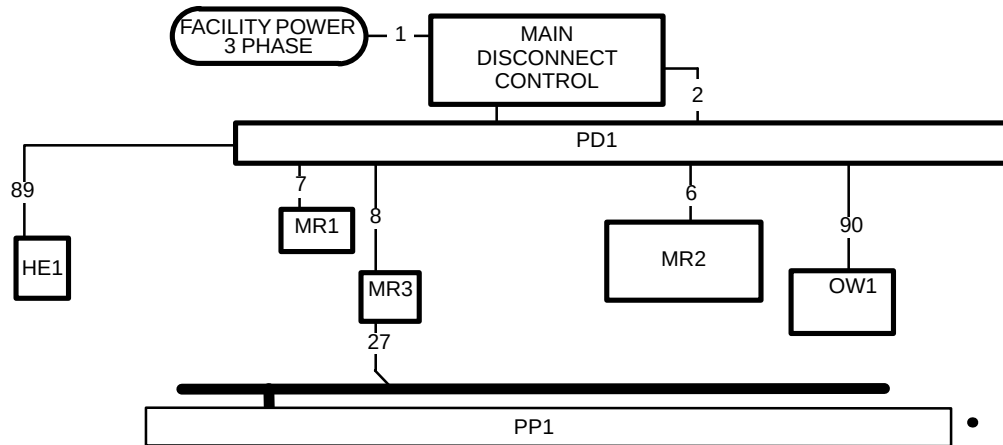
SIGNA HORIZON LX WITH SGD & RF/PDU CABINET TO ACGD UPGRADES REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-4

TABLE 6-20
SIGNA HORIZON LX WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	22 (6.70)	030	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.
27	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires

6-6-2 Signa Horizon LX System With SGD & RF/Penetration Cabinet

Illustration 6-5 and Table 6-21 lists the existing Signa Horizon LX system with SGD & RF/Penetration Cabinet system cables which are removed and discarded during upgrade to Signa Horizon LX with ACGD configuration.



SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET TO ACGD UPGRADES REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-5

TABLE 6-21
SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	22 (6.70)	030	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
7	PD1	MR1	22 (6.70)	009	0.92 (23.4)	Ring Terminal	Ring Terminal	Power Cable, #8 AWG 5 wires
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.

6-6-2 Signa Horizon LX System With SGD & RF/Penetration Cabinet (Continued)

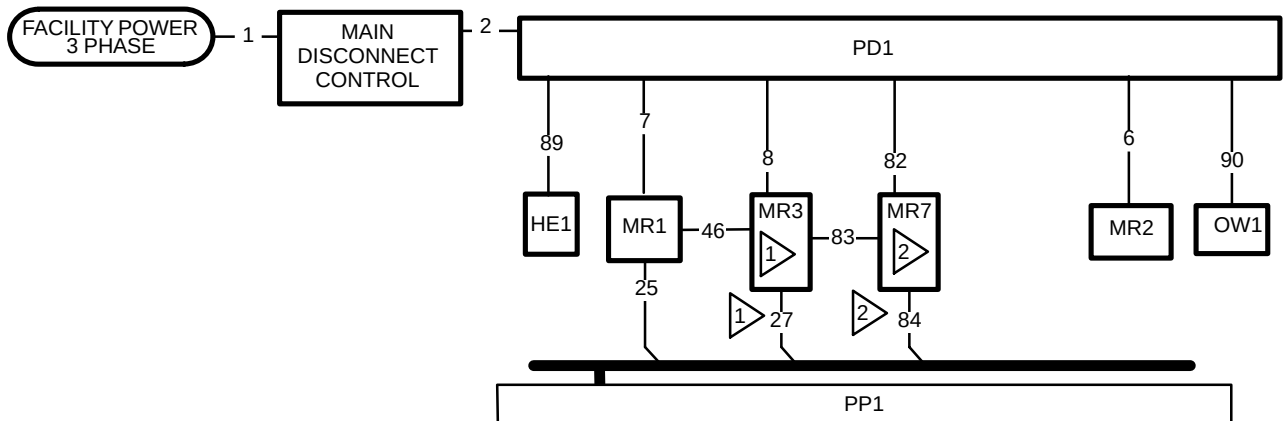
TABLE 6-21 (Continued)

SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
27	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
89	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires

6-6-3 Signa Horizon LX System With 8645 Gradients & RF/Penetration Cabinet

Illustration 6-6 and Table 6-22 lists the existing Signa Horizon LX system with 8645 Gradients & RF/Penetration Cabinet system cables which are removed and discarded during upgrade to Signa Horizon LX with ACGD configuration. Base system configuration used 8645 Gradient Cabinet. HiSpeed and EchoSpeed system configurations used 8645 Gradient Cabinet and GRAM Cabinet.



- 1 BASE SYSTEM CONFIGURATION USED 8645 GRADIENT CABINET (MR3) & CABLE GROUP 27 RUNS.
- 2 HISPEED & ECHOSPEED SYSTEM CONFIGURAITONS USED 8645 GRADIENT CABINET (MR3), GRAM CABINET (MR7), AND CABLE GROUPS 82, 83, & 84 RUNS. THESE RUNS ARE REMOVED ONLY IF GRAM CABINET (MR7) IS PART OF THE EXISTING SYSTEM BEING UPGRADED.

**SIGNA HORIZON LX WITH 8645 GRADIENTS & RF/PENENTRATION CABINET TO ACGD UPGRADES
REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-6**

TABLE 6-22
SIGNA HORIZON LX WITH 8645 GRADIENTS & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	22 (6.70)	030	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
7	PD1	MR1	22 (6.70)	009	0.92 (23.4)	Ring Terminal	Ring Terminal	Power Cable, #8 AWG 5 wires
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.

6-6-3 Signa Horizon LX System With 8645 Gradients & RF/Penetration Cabinet (Continued)

TABLE 6-22 (Continued)

SIGNA HORIZON LX WITH 8645 GRADIENTS & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

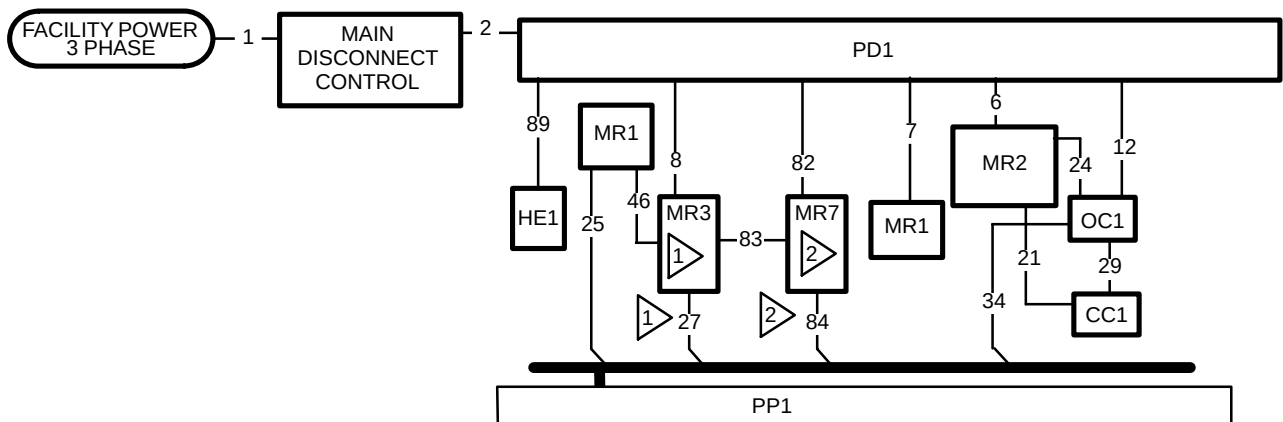
GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
27 See Note 1	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
82 See Note 2	PD1	MR7	22 (6.70)	043	0.92 (23.4)	Ring Terminals	2.50x4.00 (63.5x102)	Power cable, #8 AWG 5 wires
83 See Note 2	MR1	MR7	9 (4.6)	753	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				754	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				755	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				756	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				757	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				758	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				759	1.64 (41.7)	Ring Terminals	Ring Terminals	
				760	1.64 (41.7)	Ring Terminals	Ring Terminals	
				761	1.64 (41.7)	Ring Terminals	Ring Terminals	
84 See Note 2	MR7	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
89	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires

Note 1 Base system configuration used 8645 Gradient Cabinet (MR3) & cable group 27 Runs.

2 HiSpeed & EchoSpeed system configuraitons used 8645 Gradient Cabinet (MR3), GRAM Cabinet (MR7), and cable groups 82, 83, & 84 Runs. These Runs are removed onlu if GRAM Cabinet (MR7) is part of the existing system being upgraded.

6-6-4 Signa Horizon 5.X System

Illustration 6-7 and Table 6-23 lists the existing Signa Horizon 5.X system system cables which are removed and discarded during upgrade to Signa Horizon LX with ACGD configuration. Base system configuration used 8645 Gradient Cabinet. HiSpeed and EchoSpeed system configurations used 8645 Gradient Cabinet and GRAM Cabinet.



- 1 BASE SYSTEM CONFIGURATION USED 8645 GRADIENT CABINET (MR3) & CABLE GROUP 27 RUNS.
- 2 HISPEED & ECHOSPEED SYSTEM CONFIGURAITONS USED 8645 GRADIENT CABINET (MR3), GRAM CABINET (MR7), AND CABLE GROUPS 82, 83, & 84 RUNS. THESE RUNS ARE REMOVED ONLY IF GRAM CABINET (MR7) IS PART OF THE EXISTING SYSTEM BEING UPGRADED.

SIGNA HORIZON 5.X TO ACGD UPGRADES REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-7

TABLE 6-23
SIGNA HORIZON 5.X SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
7	PD1	MR1	22 (6.70)	009	0.92 (23.4)	Ring Terminal	Ring Terminal	Power Cable, #8 AWG 5 wires
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.
12	PD1	OC1	52 (15.8)	023	0.188 (4.78)	Ring Terminal	Ring Terminal	Ground wire, #10 AWG 1 wire
				050	1.02 (25.9)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires

6-6-4 Signa Horizon 5.X System (Continued)

TABLE 6-22 (Continued)

SIGNA HORIZON LX WITH 8645 GRADIENTS & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
21	MR2	CC1	11 (3.35)	064	0.78 (19.8)	2.00x3.75 (50.8x95.3)	2.00x3.75 (50.8x95.3)	Run 064 is power for CC1 from MR2.
				719	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
				787	0.48 (12.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
27 See Note 1	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
24	MR2	OC1	64 (19.51)	707	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				718	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
29	CC1	OC1	74 (22.56)	800	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				801	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				802	0.43 (10.9)	1.60x1.90 (40.6x48.3)	1.60x1.90 (40.6x48.3)	
				803	0.242 (6.15)	0.57x1.07 (14.5x27.2)	0.57x1.07 (14.5x27.2)	
				804	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
34	OC1	PP1	56 (17.1)	704	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
82 See Note 2	PD1	MR7	22 (6.70)	043	0.92 (23.4)	Ring Terminals	2.50x4.00 (63.5x102)	Power cable, #8 AWG 5 wires
Note 1 Base system configuration used 8645 Gradient Cabinet (MR3) & cable group 27 Runs. 2 HiSpeed & EchoSpeed system configurations used 8645 Gradient Cabinet (MR3), GRAM Cabinet (MR7), and cable groups 82, 83, & 84 Runs. These Runs are removed only if GRAM Cabinet (MR7) is part of the existing system being upgraded.								

6-6-4 Signa Horizon 5.X System (Continued)

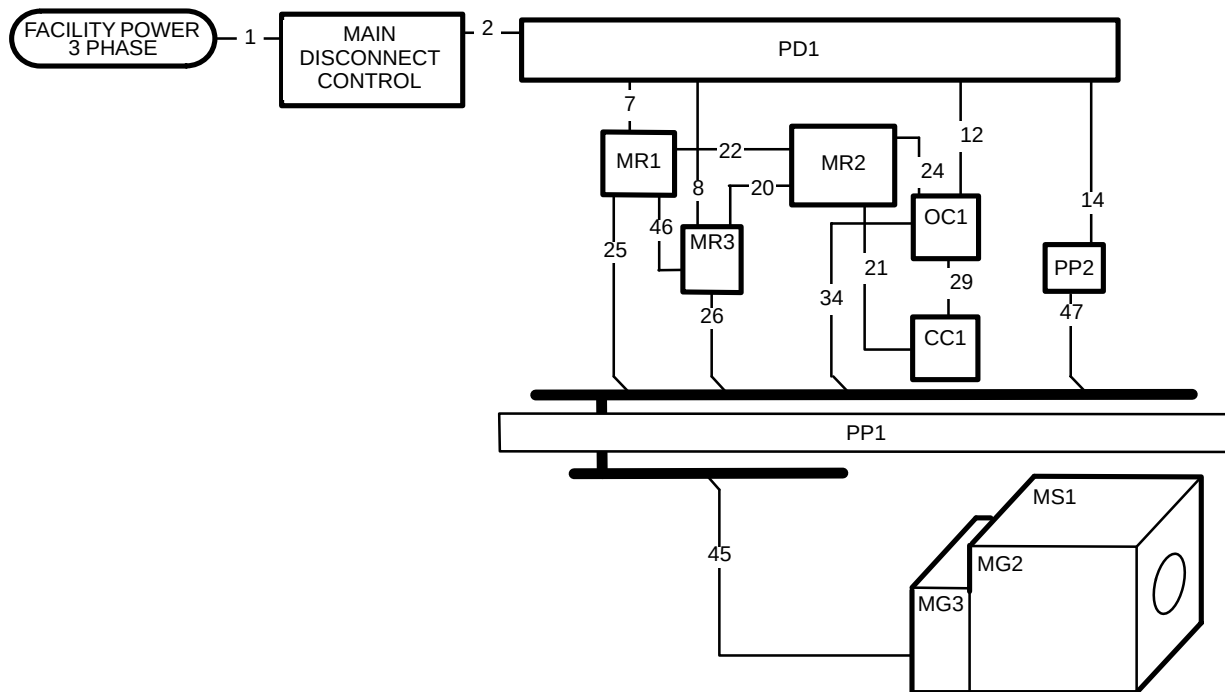
TABLE 6-22 (Continued)

SIGNA HORIZON LX WITH 8645 GRADIENTS & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
83 See Note 2	MR1	MR7	9 (4.6)	753	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				754	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				755	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				756	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				757	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				758	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				759	1.64 (41.7)	Ring Terminals	Ring Terminals	
				760	1.64 (41.7)	Ring Terminals	Ring Terminals	
				761	1.64 (41.7)	Ring Terminals	Ring Terminals	
84 See Note 2	MR7	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
89	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
Note 1 Base system configuration used 8645 Gradient Cabinet (MR3) & cable group 27 Runs. 2 HiSpeed & EchoSpeed system configuraitons used 8645 Gradient Cabinet (MR3), GRAM Cabinet (MR7), and cable groups 82, 83, & 84 Runs. These Runs are removed onlu if GRAM Cabinet (MR7) is part of the existing system being upgraded.								

6-6-5 Signa Horizon 5.X Software Only & Signa Advantage 5.X System

Illustration 6-8 and Table 6-24 lists the existing 5.1 to 5.4 Signa Advantage and 5.5 software only system cables which are removed and discarded during upgrade to Horizon LX with ACGD configurations.



5.5 SOFTWARE ONLY SITE & 5.1-5.4 SIGNA ADVANTAGE UPGRADES REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-8

6-6-5 Signa Horizon 5.X Software Only & Signa Advantage 5.X System (Continued)

TABLE 6-24

5.5 SOFTWARE ONLY SITE & 5.1-5.4 SIGNA ADVANTAGE CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be placed to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be placed to meet Section 5, Power Requirements.
7	PD1	MR1	22 (6.70)	009	0.92 (23.4)	Ring Terminal	Ring Terminal	Power Cable, #8 AWG 5 wires
8	PD1	MR3	31 (9.45)	003	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
12	PD1	OC1	52 (15.8)	023	0.188 (4.78)	Ring Terminal	Ring Terminal	Ground wire, #10 AWG 1 wire
				050	1.02 (25.9)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
20	MR2	MR3	24 (7.31)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
21	MR2	CC1	11 (3.35)	064	0.78 (19.8)	2.00x3.75 (50.8x95.3)	2.00x3.75 (50.8x95.3)	Run 064 is power for CC1 from MR2.
				719	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
				787	0.48 (12.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
22	MR2	MR1	24 (7.32)	708	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 708 is flexible conduit containing fiber optic cables.
24	MR2	OC1	64 (19.51)	707	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				718	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
25	MR1	PP1	41 (12.5)	470	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
(Continued)								

6-6-5 Signa Horizon 5.X Software Only & Signa Advantage 5.X System (Continued)

TABLE 6-24 (Continued)
5.5 SOFTWARE ONLY SITE & 5.1-5.4 SIGNA ADVANTAGE CABLES REMOVED BY HORIZON UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
26	MR3	PP1	33 (10.0)	405	0.70 (17.8)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length of 100 ft (30.5 m) included both Equipment and Magnet Rooms. Refer to Group 45.
				406	0.70 (17.8)	Ring Terminals	Ring Terminals	
				407	0.70 (17.8)	Ring Terminals	Ring Terminals	
29	CC1	OC1	74 (22.56)	800	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				801	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				802	0.43 (10.9)	1.60x1.90 (40.6x48.3)	1.60x1.90 (40.6x48.3)	
				803	0.242 (6.15)	0.57x1.07 (14.5x27.2)	0.57x1.07 (14.5x27.2)	
				804	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
34	OC1	PP1	56 (17.1)	704	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
45	PP1	MG3	40 (12.2)	405	0.70 (17.8)	Ring Terminals	Ring Terminals	Runs 405, 406, and 407 were cut and connected to PP1 at site: total length of 100 ft (30.5 m) included both Equipment and Magnet Rooms. Refer to Group 26.
				406	0.70 (17.8)	Ring Terminals	Ring Terminals	
				407	0.70 (17.8)	Ring Terminals	Ring Terminals	
				412	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				474	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				475	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				476	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				477	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				478	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				711	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711 is flexible conduit containing fiber optic cables.
				712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 712 is flexible conduit containing fiber optic cables.
				(Continued)				

6-6-5 Signa Horizon 5.X Software Only & Signa Advantage 5.X System (Continued)

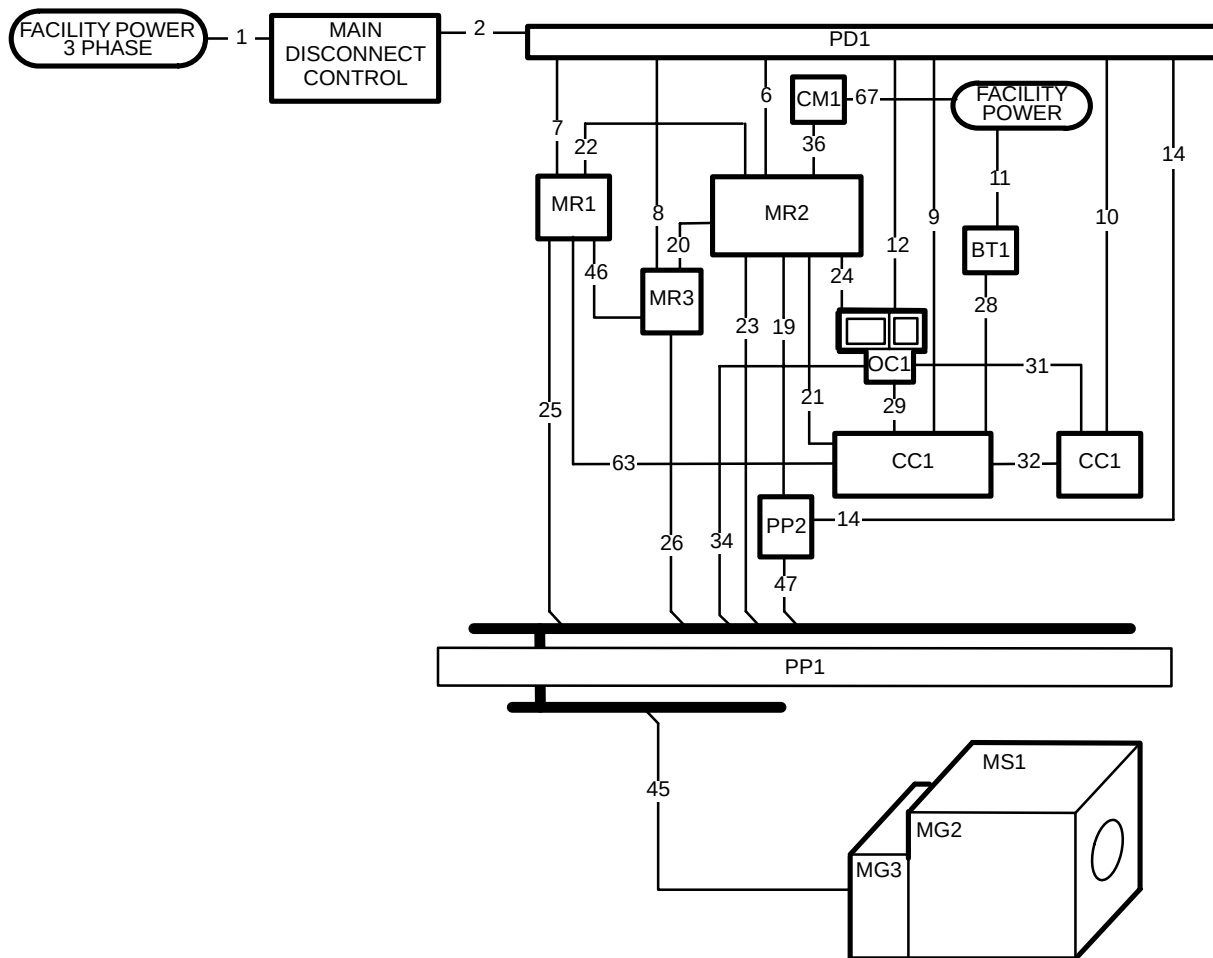
TABLE 6-24 (Continued)

5.5 SOFTWARE ONLY SITE & 5.1-5.4 SIGNA ADVANTAGE CABLES REMOVED BY HORIZON LX UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
46	MR1	MR3	24 (7.32)	709	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 709 is flexible conduit containing fiber optic cables.
47	PP1	PP2	21 (6.4)	384	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				386	0.53 (13.3)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				388	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				389	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				413	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	

6-6-6 Signa Advantage 4.X System

Illustration 6-9 and Table 6-25 lists the existing Signa Advantage 4.X system cables which are removed and discarded during upgrade to Horizon LX with ACGD configurations. Note, all Signa Advantage 4.X system configurations are covered, including both forward product and runs provided by various 4.X system upgrades.



SIGNA ADVANTAGE 4.X UPGRADES REMOVED INTERCONNECTS DIAGRAM

ILLUSTRATION 6-9

6-6-6 Signa Advantage 4.X System (Continued)

TABLE 6-25
4.X SIGNA ADVANTAGE CABLES REMOVED BY HORIZON UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be placed to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be placed to meet Section 5, Power Requirements.
6	PD1	MR2	32 (9.75)	002	0.635 (16.1)	Ring Terminals	2.00x4.30 (50.8x109)	Power cable, #12 AWG 3 wires
7	PD1	MR1	22 (6.70)	009	0.92 (23.4)	Ring Terminal	Ring Terminal	Power Cable, #8 AWG 5 wires
8	PD1	MR3	31 (9.45)	003	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
9	PD1	CC1	32 (9.75)	014	0.635 (16.1)	Ring Terminals	2.00x4.30 (50.8x109)	Power cable, #12 AWG 3 wires
				015	0.92 (23.4)	Ring Terminals	2.25x3.90 (57.2x99.1)	Power cable, #8 AWG 5 wires
				021	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire #10 AWG 1 wire
10	PD1	CC2	32 (9.75)	016‡ or 035**†	0.92 (23.4)	Ring Terminals	2.25x3.90 (57.2x99.1)	Power cable, #8 AWG 5 wires
				017**†	0.635 (16.1)	Ring Terminals	2.00x4.30 (50.8x109)	Power cable, #12 AWG 3 wires
				022**†	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire #10 AWG 1 wire
11	Facility Power	BT1	—	—	—	—	—	Customer Furnished.
12	PD1	OC1	52 (15.8)	012	0.635 (16.1)	Ring Terminals	2.00x4.30 (50.8x109)	Power cable, #12 AWG 3 wires
				013	0.635 (16.1)	Ring Terminals	2.00x4.30 (50.8x109)	Power cable, #12 AWG 3 wires
				023	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire #10 AWG 1 wire
14	PD1	PP2	32 (9.75)	027	0.635 (16.1)	Ring Terminals	2.40x5.00 (61x127)	Power cable, #12 AWG 3 wires
				028	0.850 (21.6)	Ring Terminals	2.03x4.30 (51.6x109)	Power cable, #8 AWG 3 wires
Note ** These runs may be present in systems that were upgraded to 4.X Signa Advantage systems only, not present in forward production 4.X Signa Advantage systems. † These runs are applicable to 4.5 to 4.8 Signa Advantage systems (have RF Cabinet Model 46–306900) only. ‡ These runs are applicable to 4.0 to pre–4.5 Signa Advantage systems (have RF Cabinet Model 46–282900) only.								
(Continued)								

6-6-6 Signa Advantage 4.X System (Continued)

TABLE 6-25 (Continued)
4.X SIGNA ADVANTAGE CABLES REMOVED BY HORIZON UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
19	MR2	PP2	44 (13.4)	417	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
20	MR2	MR3	24 (7.32)	200	0.5 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				201	0.5 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				202	0.5 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				203	0.5 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
21	MR2	CC1	59 (17.98)	228 †	0.42 (10.7)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				427	0.42 (10.7)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				428 or 448**	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				430 or 449**	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				431 or 450**	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
22	MR2	MR1	24 (7.32)	226	0.35 (8.90)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
23	MR2	PP1	41 (12.5)	419	0.30 (7.62)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	Not present with Run 454.
				425	0.525 (13.3)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Not present with Run 425.
				454	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
24	MR2	OC1	64 (19.5)	287	0.35 (8.9)	1.30x2.00 (33.5x50.8)	2.80x2.00 (70.4x50.8)	
25	MR1	PP1	41 (12.5)	416 ‡	0.212 (5.38)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				470**	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
Note ** These runs may be present in systems that were upgraded to 4.X Signa Advantage systems only, not present in forward production 4.X Signa Advantage systems. † These runs are applicable to 4.5 to 4.8 Signa Advantage systems (have RF Cabinet Model 46–306900) only. ‡ These runs are applicable to 4.0 to pre–4.5 Signa Advantage systems (have RF Cabinet Model 46–282900) only.								
(Continued)								

6-6-6 Signa Advantage 4.X System (Continued)

TABLE 6-25 (Continued)
4.X SIGNA ADVANTAGE CABLES REMOVED BY HORIZON UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
26	MR3	PP1	33 (10.0)	405	0.70 (17.8)	Ring Terminals	Ring Terminals	Runs 405, 406, and 407 were cut and connected to PP1 at site: total length of 100 ft (30.5 m) included both Equipment and Magnet Rooms. Refer to Group 45.
				406	0.70 (17.8)	Ring Terminals	Ring Terminals	
				407	0.70 (17.8)	Ring Terminals	Ring Terminals	
28	CC1	BT1	44 (13.4)	244 † or 429	0.464 (11.8)	2.10x2.10 (53.3x53.3)	2.10x2.10 (53.3x53.3)	
29	CC1	OC1	74 (22.6)	238	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				239	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				433	0.41 (10.4)	0.60x1.90 (15.2x48.3)	0.60x1.90 (15.2x48.3)	
				434	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
31	CC2	OC1	74 (22.6)	248 †	0.41 (10.4)	0.60x1.90 (15.2x48.3)	0.60x1.90 (15.2x48.3)	
				249 †	0.5 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
32	CC2	CC1	9 (2.74)	287**	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				301**	0.5 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				302**	0.5 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				325**	1.0 (25.4)	2.70x2.40 (68.6x60.1)	2.70x2.40 (68.6x60.1)	
				327**	1.0 (25.4)	2.70x2.40 (68.6x60.1)	2.70x2.40 (68.6x60.1)	
34	OC1	PP1	56 (17.1)	256	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
Note ** These runs may be present in systems that were upgraded to 4.X Signa Advantage systems only, not present in forward production 4.X Signa Advantage systems. † These runs are applicable to 4.5 to 4.8 Signa Advantage systems (have RF Cabinet Model 46-306900) only. ‡ These runs are applicable to 4.0 to pre-4.5 Signa Advantage systems (have RF Cabinet Model 46-282900) only.								
(Continued)								

6-6-6 Signa Advantage 4.X System (Continued)

TABLE 6-25 (Continued)
4.X SIGNA ADVANTAGE CABLES REMOVED BY HORIZON UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
36	MR2	CM1	73 (22.3)	346**	0.25 (6.35)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				347**	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
45	PP1	MG3	53 (16.2)	261	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Not present with Run 453.
				405	0.70 (17.8)	Ring Terminals	Ring Terminals	Runs 405, 406, and 407 were cut and connected to PP1 at site: total length of 100 ft (30.5 m) included both Equipment and Magnet Rooms. Refer to Group 26.
				406	0.70 (17.8)	Ring Terminals	Ring Terminals	
				407	0.70 (17.8)	Ring Terminals	Ring Terminals	
				411	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				412	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Not present with Run 261.
				415 ‡ or 476**	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				420	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				424	0.18 (4.57)	0.3x1.0–X2 (7.62x25.4)	0.3x1.0–X2 (7.62x25.4)	
				446 ‡	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				447 ‡	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				453	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				474**	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				475**	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
Note ** These runs may be present in systems that were upgraded to 4.X Signa Advantage systems only, not present in forward production 4.X Signa Advantage systems. † These runs are applicable to 4.5 to 4.8 Signa Advantage systems (have RF Cabinet Model 46–306900) only. ‡ These runs are applicable to 4.0 to pre–4.5 Signa Advantage systems (have RF Cabinet Model 46–282900) only.								
(Continued)								

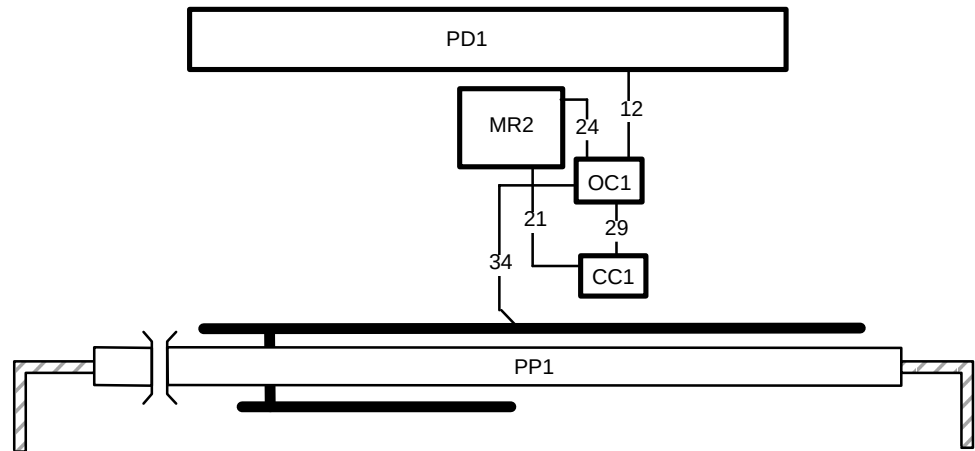
6-6-6 Signa Advantage 4.X System (Continued)

TABLE 6-25 (Continued)
4.X SIGNA ADVANTAGE CABLES REMOVED BY HORIZON UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
46	MR1	MR3	59 (17.98)	426	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
47	PP1	PP2	21 (6.4)	384	0.525 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				386	0.53 (13.3)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				388	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				389	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				410	0.525 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				413	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
63	MR1	CC1	59 (17.98)	224	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				274	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
67	Facility Power	CM1	—	—	—	—	—	Customer Furnished.
Note ** These runs may be present in systems that were upgraded to 4.X Signa Advantage systems only, not present in forward production 4.X Signa Advantage systems. † These runs are applicable to 4.5 to 4.8 Signa Advantage systems (have RF Cabinet Model 46-306900) only. ‡ These runs are applicable to 4.0 to pre-4.5 Signa Advantage systems (have RF Cabinet Model 46-282900) only.								

6-7 INTERCONNECTS REMOVED BY SIGNA HORIZON 5.X TO SIGNA LX COMPUTER ONLY UPGRADE

Illustration 6-10 and Table 6-26 lists the existing Signa Horizon 5.X system cables which are removed and discarded during upgrade to Signa Horizon LX Computer only configuration with no gradient subsystem change.



SIGNA HORIZON 5.X UPGRADE REMOVED INTERCONNECTS DIAGRAM

ILLUSTRATION 6-10

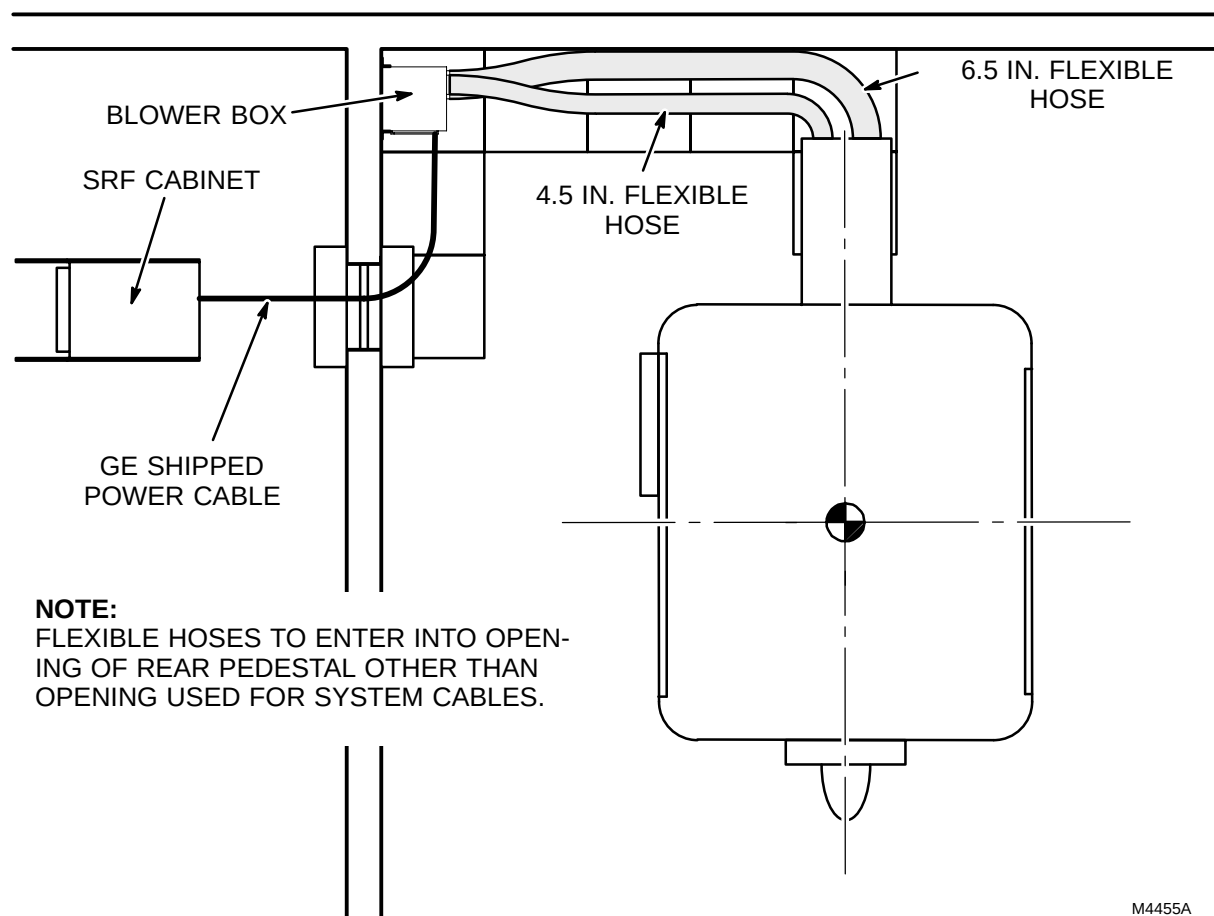
**6-7 INTERCONNECTS REMOVED BY SIGNA HORIZON 5.X TO SIGNA LX COMPUTER ONLY UPGRADE
(Continued)**

TABLE 6-26
SIGNA HORIZON 5.X CABLES REMOVED BY HORIZON LX COMPUTER ONLY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
12	PD1	OC1	52 (15.8)	023	0.188 (4.78)	Ring Terminal	Ring Terminal	Ground wire, #10 AWG 1 wire
				050	1.02 (25.9)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
21	MR2	CC1	11 (3.35)	064	0.78 (19.8)	2.00x3.75 (50.8x95.3)	2.00x3.75 (50.8x95.3)	Run 064 is power for CC1 from MR2.
				719	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
				787	0.48 (12.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
24	MR2	OC1	64 (19.51)	707	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				718	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
29	CC1	OC1	74 (22.56)	800	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				801	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				802	0.43 (10.9)	1.60x1.90 (40.6x48.3)	1.60x1.90 (40.6x48.3)	
				803	0.242 (6.15)	0.57x1.07 (14.5x27.2)	0.57x1.07 (14.5x27.2)	
				804	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
34	OC1	PP1	56 (17.1)	704	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-8 REAR PEDESTAL AND ENCLOSURE CABLE ACCESS

Signa Horizon has an increase of cabling and hoses routed through the Rear Pedestal cable access areas. These include the Blower Box, 2 flexible hoses, 3 new large Gradient cables, 2 water lines for Body Coil, and the remainder of the Magnet Enclosure interconnect cables. Rear Pedestal cable access and routing needs to be considered. Sites with cable access from under the Rear Pedestal should have enough cable access area. Sites using Rear Pedestal cable access on rear or one side will need to plan to utilize more than one access opening. Illustration 6-11 shows one example of Blower Box set-up with flexible hoses route separate from other interconnects.

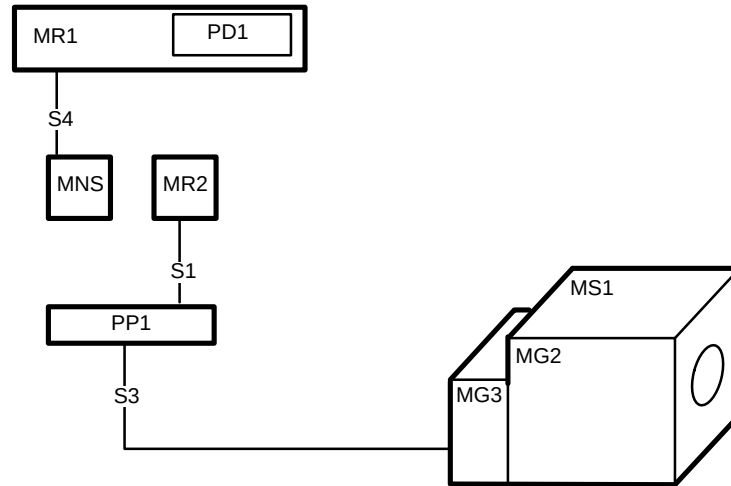


EXAMPLE OF SIGNA HORIZON BLOWER BOX SET-UP
ILLUSTRATION 6-11

6-9 SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH ACGD

6-9-1 M3090DA – 4.X-5.4 MNS Upgrade Kit For ACGD Site

Illustration 6-12 shows the 1.5T system group interconnects for the MNS Cabinet and Spectroscopy Modules added to the Penetration Panel and Magnet Enclosure by the spectroscopy upgrade options listed below. Table 6-27 contains the cable data for these interconnects.



CATALOG M3090DA, M3090DB, & M3090DC SPECTROSCOPY OPTION SYSTEM INTERCONNECTS
ILLUSTRATION 6-12

TABLE 6-27
CABLES PROVIDED BY CATALOG M1033MD, M3090DA, M3090DB, & M3090DC

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
S1	0.035 (22.8)	MR2	PP1	31 (9.45)	469	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnect to Table 6-3 Group 23.
S3	0.357 (231)	PP1	MG3	30 (9.15)	472	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnects to Table 6-3 Group 45.
S4	0.656 (425)	MR1	MNS	9 (2.74)	936	0.68 (17.3)	2.00x4.00 (50.8x102)	2.00x3.63 (50.8x92.2)	Cabinet Power Cable, #10 AWG 3 wires
					961	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					962	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					963	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					964	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					965	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	

6-9-2 M3090DB – 5.5 MNS Upgrade Kit For ACGD Site

Cables provided by catalog M3090DB are the same as catalog M3090DA. Refer to Illustration 6-12 and Table 6-27 cables details.

6-9-3 M3090DB – LX MNS Upgrade Kit For ACGD Site

Cables provided by catalog M3090DC are the same as catalog M3090DA. Refer to Illustration 6-12 and Table 6-27 cables details.

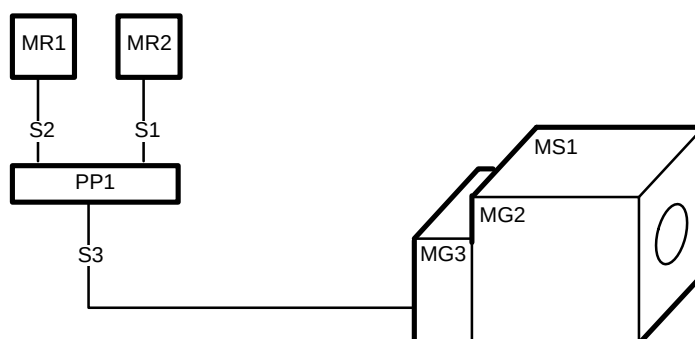
6-10 SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH SGD

Catalogs M1033MA and M1033MB utilize the already installed Spectroscopy option system interconnect cables.

Catalogs M1033MC and M1033MD supply additional system interconnect cables, refer to the following sub-sections for details.

6-10-1 M1033MC – MNS Kit For Signa LX (8.X) Systems That Do Not Have RF/PDU Cabinet

Illustration 6-13 shows the 1.5T system group interconnects for the Spectroscopy Modules added to the Penetration Panel and Magnet Enclosure by M1033MC. Table 6-28 contains the cable data for these interconnects.



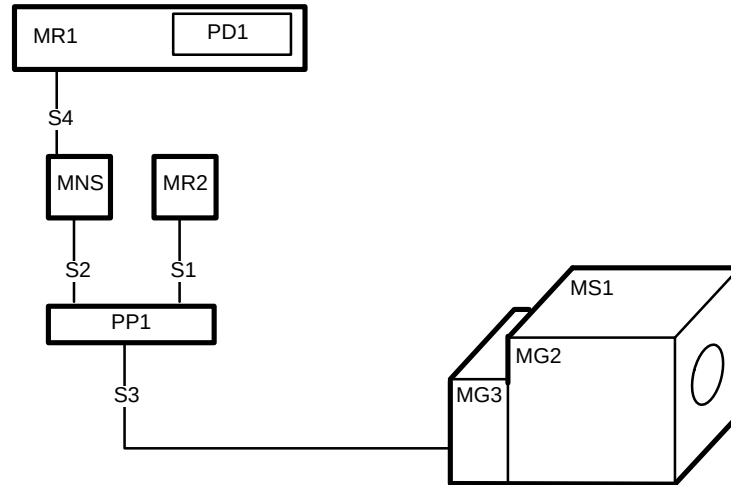
**CATALOG M1033MC SPECTROSCOPY OPTION SYSTEM INTERCONNECTS
ILLUSTRATION 6-13**

**TABLE 6-28
CABLES PROVIDED BY CATALOGS M1033MC**

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
S1	0.035 (22.8)	MR2	PP1	31 (9.45)	469	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnect to Table 6-3 Group 23.
S2	0.41 (267)	PP1	MR1	21 (6.40)	468	0.64 (16.3)	0.80x2.50 (20.3x63.5)	0.80x2.50 (20.3x63.5)	Additional interconnects to Table 6-3 Group 25.
S3	0.357 (231)	PP1	MG3	30 (9.15)	472	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnects to Table 6-3 Group 45.
					473	0.64 (16.3)	0.80x2.50 (20.3x63.5)	0.80x2.50 (20.3x63.5)	

6-10-2 M1033MD – MNS Kit For Signa LX (8.X) Systems That Have RF/PDU Cabinet

Illustration 6-14 shows the 1.5T system group interconnects for the MNS Cabinet and Spectroscopy Modules added to the Penetration Panel and Magnet Enclosure by the spectroscopy upgrade options listed below. Table 6-29 contains the cable data for these interconnects.



CATALOG M1033MD SPECTROSCOPY OPTION SYSTEM INTERCONNECTS

ILLUSTRATION 6-14

TABLE 6-29
CABLES PROVIDED BY CATALOG M1033MD

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
S1	0.035 (22.8)	MR2	PP1	31 (9.45)	469	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnect to Table 6-3 Group 23.
S2	0.41 (267)	PP1	MNS	21 (6.40)	468	0.64 (16.3)	0.80x2.50 (20.3x63.5)	0.80x2.50 (20.3x63.5)	Additional interconnects to Table 6-3 Group 25.
S3	0.357 (231)	PP1	MG3	30 (9.15)	472	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnects to Table 6-3 Group 45.
					473	0.64 (16.3)	0.80x2.50 (20.3x63.5)	0.80x2.50 (20.3x63.5)	
S4	0.656 (425)	MR1	MNS	9 (2.74)	936	0.68 (17.3)	2.00x4.00 (50.8x102)	2.00x3.63 (50.8x92.2)	Cabinet Power Cable, #10 AWG 3 wires
					961	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					962	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					963	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					964	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					965	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	

SECTION 7 – RF SHIELDED ROOM

TABLE OF CONTENTS

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
7-1	RF SHIELDED ROOM SPECIFICATION	7-3
7-2	VENTS	7-3
7-3	PLUMBING	7-3
7-4	ELECTRICAL	7-3
7-5	RF PENETRATION PANEL	7-3
7-6	ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM	7-5
	7-6-1 Magnet Mounting	7-5
	7-6-2 Blower Box	7-5
7-7	MAGNET ROOM EQUIPMENT MOUNTING	7-5
	7-7-1 Magnet Rundown Unit (MS4)	7-5
	7-7-2 Emergency Off Buttons	7-5
7-8	ROOM VENTILATION SWITCH	7-5

7-1 RF SHIELDED ROOM SPECIFICATION

The shielding effectiveness of the RF Shielded Room impacts the imaging ability of the system. Therefore, the RF Shielded Room needs to meet the attenuation specification before the upgrade. The RF Shielded Room RF attenuation specification with the GE system connected is 85 dB planewave (10MHz – 100MHz). If the RF Shield is not at this specification then RF vendor maintenance may be necessary.

7-2 VENTS

If an upgrade to an LCC magnet is being performed with the system electronics upgrade then refer to ***Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 7-2 VENTS*** for cryogenic & HVAC vents requirements and recommendations.

7-3 PLUMBING

The Signa LX system performance requires adherence to the specifications in ***Direction 2223170 Signa MR/i & CV/i Pre-Installation 7-3 PLUMBING*** for pipes, water, medical gasses, and sprinklers.

7-4 ELECTRICAL

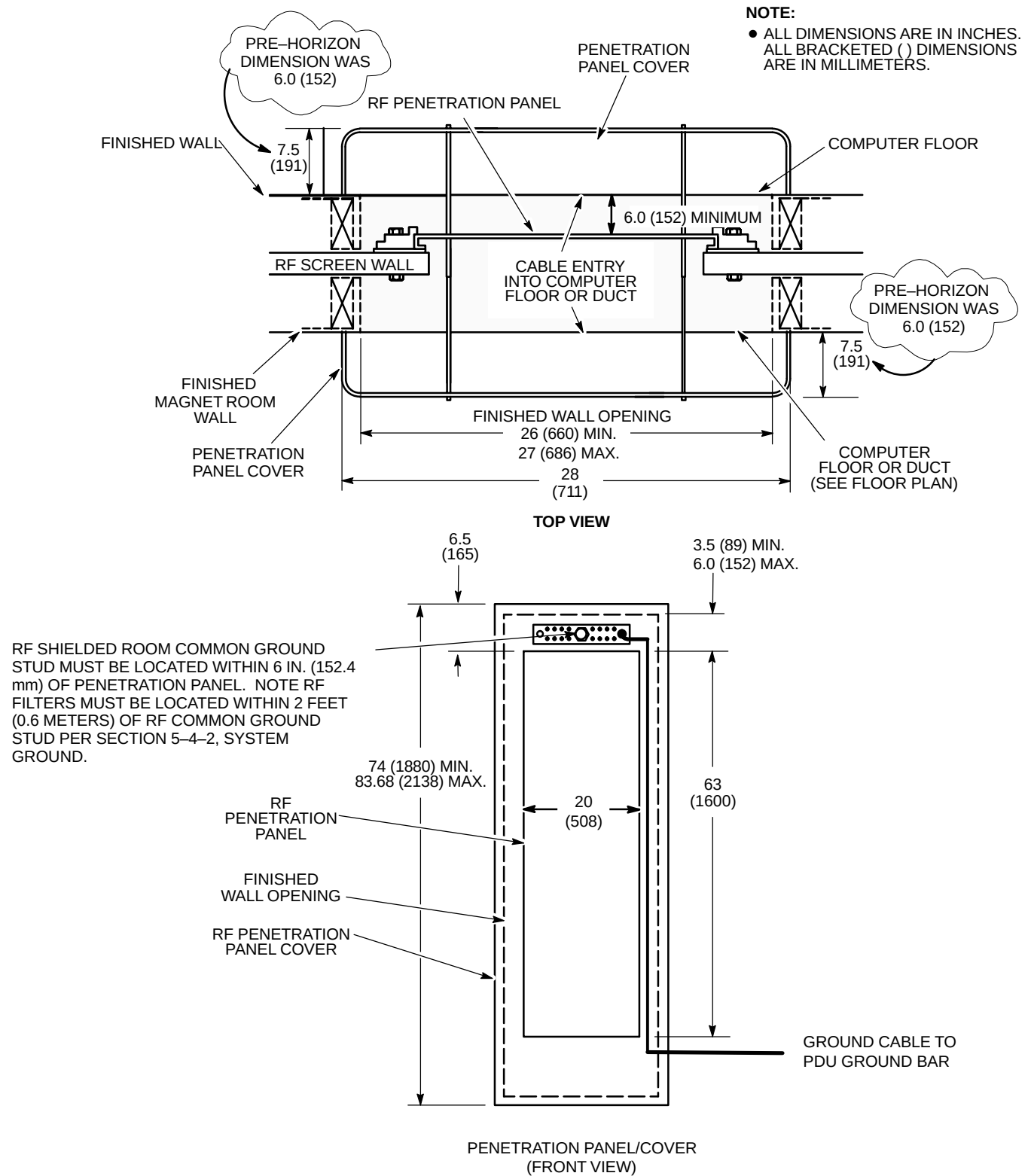
The Signa LX system performance requires adherence to the specifications in ***Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 7-4 ELECTRICAL*** for filter attenuation, power line grounding, lighting, and common ground stud proximity.

7-5 RF PENETRATION PANEL

The Penetration Panel is replaced during upgrade as shown in Section 1 – SYSTEM CONFIGURATION. Current panel mounting hardware will be used to mount new Horizon Penetration Panel. The Horizon/LX Penetration Panel is deeper therefore new deeper Penetration Panel Covers are provided. The mounting and clearance dimensions for the Penetration Panel covering are shown in Illustration 7-1. Refer to Section 2-8, COMPONENT DIMENSIONS, for specific upgrade catalogs provide new Penetration Panel and Covers.

Note

The Penetration Panel must be covered on both sides for safety. If GE supplied adjustable covers are not used, customer furnished covers or enclosures with key or tool are required for opening to limit access to the panel.



PENETRATION PANEL/COVERING MOUNTING REQUIREMENTS

ILLUSTRATION 7-1

7-6 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM

7-6-1 Magnet Mounting

If an upgrade to an LCC magnet is being performed with the Signa system upgrade then refer to ***Direction 2223170 Signa MR/i & CV/i Pre-Installation*** following sections for **Section 7-7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM** for magnet mounting requirements.

7-6-2 Blower Box

The Blower Box (MG6) provides cooling air for the Gradient Body Coil and the Patient Comfort Module in the Magnet Enclosure. The Blower Box will be mounted within the RF Shielded Room and connect to the Gradient Body Coil and the Patient Comfort Module by 6.5 inch (165.1 mm) OD and 4.5 inch (114.3 mm) OD flexible vinyl air ducting. **The Blower Box contains magnetic material and must be securely mounted to the floor of the Magnet Room or mounted to a raised support shelf on the Magnet Room wall or ceiling with support provided under the box. RF Shield integrity must be maintained for mounting the Blower Box within the Magnet Room.** The flexible vinyl air ducts route from the Blower Box through the Magnet Enclosure Rear Pedestal cable access and connect to the Gradient Coil. Refer to Illustrations 2-26 for Blower Box and mounting dimensions. Refer to ***Direction 2223170 Signa MR/i & CV/i Pre-Installation*** **Section 7-7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM** for Blower Box mounting RF Shield Integrity and electrical isolation requirements.

If an upgrade to an LCC magnet is being performed with the system electronics upgrade then refer to ***Direction 2223170 Signa MR/i & CV/i Pre-Installation*** **Section 7-7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM** for Magnet and Dock Assembly requirements and recommendations.

7-7 MAGNET ROOM EQUIPMENT MOUNTING

7-7-1 Magnet Rundown Unit (MS4)

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then the magnet rundown unit should be mounted 60 in. (1524 mm) above the Magnet Room floor near the front of the Magnet Room but outside the 200 gauss zone.

7-7-2 Emergency Off Buttons

Emergency off buttons to be located near the exit in the magnet and equipment rooms. These buttons must be clearly labeled, "Emergency Off". Refer to Section 5, POWER REQUIREMENTS, for emergency off power requirements.

7-8 ROOM VENTILATION SWITCH

Placement of the room ventilation switch should be near the Magnet Room door and is the responsibility of the architect and mechanical contractor.

SECTION 8 – SHIPPING AND DELIVERY DATA

TABLE OF CONTENTS

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
8-1	INTRODUCTION	8-3
8-2	UPGRADE CATALOGS SHIPPING INFORMATION	8-6

8-1 INTRODUCTION

Domestic transportation for the Signa LX upgrade will be via an air-ride moving van. Export transportation for overseas will be via air shipment in a pressurized cargo hold.

The tables in Section 8-2 UPGRADE CATALOGS SHIPPING INFORMATION (arranged by catalog numbers) provide estimates of shipping weights and dimensions for upgrade catalog major components. Refer to the Table 8-1 for page number location for a specific catalog.

Note

The weight and dimension information is estimate only, actual shipment packaging configurations for a catalog may vary.

TABLE 8-1
LIST OF CATALOG TABLES AND PAGE NUMBER LOCATION

TABLE	TITLE	PAGE
8-2	CATALOG M1000MW, TABLE FOR USE WITH ALL MONITORS	8-6
8-3	CATALOG M1000MZ, COLOR MONITOR	8-6
8-4	CATALOG M1000NZ, LCD COLOR MONITOR	8-6
8-5	CATALOG M1033MA, 8.X MNS Upgrade For Signa Advantage 4.X-5.4	8-6
8-6	CATALOG M1033MB, 8.X MNS Upgrade For Signa Horizon 5.5 And Later	8-7
8-7	CATALOG M1033MC, MNS Kit For Signa LX (8.X) That Do Not Have RF/PDU Cabinet	8-7
8-8	CATALOG M1033MD, MNS Kit For Signa LX (8.X) That Have RF/PDU Cabinet	8-8
8-9	CATALOG M1033MF, CERD TO UCERD UPGRADE KIT	8-8
8-10	CATALOG M1085BG, 1.5T BODY COIL FOR SIGNA HORIZON	8-8
8-11	CATALOG M1085KB, 50 HZ WATER-TO-AIR CHILLER FOR GRADIENT COIL COOLING	8-9
8-12	CATALOG M1085KC, 60 HZ WATER-TO-AIR CHILLER FOR GRADIENT COIL COOLING	8-9
8-13	CATALOG M1085KD, 1.5T 55 CM INTEGRATED BODY MODULE (CRM)	8-9
8-17	CATALOG M1090JM, OXFORD MAGNET ENCLOSURE UPGRADE FOR 5.5 FIXED SITE	8-10
8-18	CATALOG M1090JN, GE S-I MAGNET MAGNET ENCLOSURE UPGRADE FOR 5.5 FIXED SITE	8-11
8-19	CATALOG M1090JP, GE MAGNISHIELD MAGNET ENCLOSURE UPGRADE FOR 5.5 FIXED SITE	8-11
8-20	CATALOG M1090JR, ALL OTHER GE MAGNET ENCLOSURE UPGRADE FOR 5.5 FIXED SITE	8-12
8-21	CATALOG M1090JS, BRIDGE REPLACEMENT KIT FOR SYSTEM WITH OLD STANDARD BRIDGE	8-12
8-22	CATALOG M1090JT, BRIDGE FIBERGLASS KIT FOR SYSTEM WITH STANDARD NEW BRIDGE	8-12
8-23	CATALOG M1090JW, LASER TO HALOGEN ALIGNMENT LIGHTS UPGRADE	8-12
8-24	CATALOG M1090JY, FRONT RING ASSEMBLY FOR SYSTEM WITH HALOGEN ALIGNMENT LIGHTS	8-13
8-28	CATALOG M1090NY, FULL SIZE PDU (MODELS 46-224847G1/G2 OR 46-282368G1) UPGRADE FOR 5.5 TO LX	8-13
8-29	CATALOG M1090NZ, COMPACT PDU (MODELS 46-317099P1 TO P9) UPGRADE FOR 5.5 TO LX	8-13
8-30	CATALOG M1090RW, 128 MBYTE MEDCAM MEMORY BOARD	8-14
8-31	CATALOG M1090RZ, 4.0 TO PRE 4.5 BASE KIT ADDER FOR 1.5T SYSTEM	8-14
8-32	CATALOG M1090SC, BRIDGE FIBERGLASS KIT FOR SYSTEM WITH STANDARD NEW BRIDGE	8-14
8-33	CATALOG M1090SD, FRONT RING ASSEMBLY FOR MAGNISHIELD SYSTEM	8-14
(Continued)		

8-1 INTRODUCTION (Continued)

TABLE 8-1 (Continued)
LIST OF CATALOG TABLES AND PAGE NUMBER LOCATION

TABLE	TITLE	PAGE
8-34	CATALOG M1090SJ, 5.5 TO LX WITH SGD CABLE UPGRADE FOR FIXED SITE	8-14
8-35	CATALOG M1090TZ, VCR INTERFACE KIT FOR OCTANE COMPUTER	8-15
8-36	CATALOG M1090WG, HISPEED TO ECHOSPEED WITH SGD UPGRADE KIT FOR HORIZON LX WITH SGD 1.5T SYSTEMS	8-15
8-37	CATALOG M1090WJ, ECHOSPEED ADDER KIT FOR SITES WITHOUT SPECTROSCOPY ARE UPGRADING TO SGD ECHOSPEED	8-15
8-38	CATALOG M1090YT, BRIDGE FIBERGLASS KIT FOR SIGNA CVMR	8-15
8-39	CATALOG M1090ZA, FIBER OPTIC REPEATER PANEL KIT	8-15
8-40	CATALOG M1090ZJ, EXCITER KIT FOR 1.5T SYSTEM	8-15
8-41	CATALOG M1090ZK, EXCITER KIT FOR 1.0T SYSTEM	8-16
8-42	CATALOG M1091JC, S-II THROUGH S-V MAGNET ENCLOSURE UPGRADE FOR SIGNA CVMR FIXED SITE	8-16
8-43	CATALOG M1091JE, S-II THROUGH S-V MAGNET FIXED SITE BRM TO CRM UPGRADE	8-16
8-45	CATALOG M1191JE, GE S-I MAGNET ENCLOSURE UPGRADE FOR 5.5 TRUE MOBILES AND T/R	8-17
8-46	CATALOG M1191JF, ALL OTHER GE MAGNET ENCLOSURE UPGRADE FOR 5.5 TRUE MOBILES AND T/R	8-17
8-47	CATALOG M1191JG, ACTIVE SHIELD MAGNET ENCLOSURE ADDER KIT	8-18
8-48	CATALOG M1885BG, 1.0T BODY COIL FOR SIGNA HORIZON	8-18
8-49	CATALOG M3090AA, 1.5T 4.5 UPGRADE TO HISPEED LX WITH ACGD HISPEED PLUS PHASED ARRAY	8-18
8-50	CATALOG M3090AB, 1.5T 4.5 UPGRADE TO ECHOSPEED LX WITH 4 CHANNEL PHASED ARRAY	8-19
8-51	CATALOG M3090AD, 1.5T 5.1-5.4 U/G TO HORIZON LX WITH ACGD HISPEED PLUS PHASED ARRAY	8-19
8-52	CATALOG M3090AE, 1.5T 5.1-5.4 U/G TO HORIZON LX WITH ACGD ECHOSPEED PLUS PHASED ARRAY	8-20
8-53	CATALOG M3090AF, 1.5T 5.5 UPGRADE TO CV2 AND/OR NV1 CRM WITH ACGD	8-20
8-54	CATALOG M3090AG, 1.5T 5.5 TO HISPEED LX WITH ACGD, OCTANE COMPUTER & 4 CHANNEL PHASED ARRAY	8-20
8-55	CATALOG M3090AH, 1.5T 5.5 TO ECHOSPEED LX WITH ACGD, OCTANE COMPUTER & 4 CHANNEL PHASED ARRAY	8-21
8-56	CATALOG M3090AJ, 1.5T 5.X HORIZON UPGRADE TO CV AND/OR NV CRM WITH ACGD	8-21
8-57	CATALOG M3090BA, 1.0T ACGD UPGRADE FOR SYSTEM WITH SGD GRADIENT CABINET & RF/PENETRATION CABINET	8-21
8-58	CATALOG M3090BB, ACGD UPGRADE FOR SYSTEM WITH SGD GRADIENT CABINET & RF/PDU CABINET	8-22
8-59	CATALOG M3090BC, 1.5T ACGD UPGRADE FOR SYSTEM WITH SGD GRADIENT CABINET & RF/PENETRATION CABINET	8-22
8-60	CATALOG M3090BD, HISPEED PLUS PERFORMANCE OPTION	8-22
8-61	CATALOG M3090BE, ECHOSPEED PLUS PERFORMANCE OPTION	8-22
8-62	CATALOG M3090CA, 4.X TO LX WITH ACGD CABLE UPGRADE FOR FIXED SITE	8-23
8-63	CATALOG M3090CB, 5.1-5.4 U/G TO LX WITH ACGD CABLE UPGRADE FOR FIXED SITE	8-23
(Continued)		

8-1 INTRODUCTION (Continued)

TABLE 8-1 (Continued)
LIST OF CATALOG TABLES AND PAGE NUMBER LOCATION

TABLE	TITLE	PAGE
8-64	CATALOG M3090CC, 5.5 SOFTWARE ONLY SITE TO LX WITH ACGD CABLE UPGRADE FOR FIXED SITE	8-23
8-65	CATALOG M3090CD, 5.5 TO LX WITH ACGD CABLE UPGRADE FOR FIXED SITE	8-24
8-66	CATALOG M3090CH, 3.X-5.4 TO LX WITH ACGD CABLE UPGRADE FOR TRANSPORTABLE/RELOCATABLE (T/R) WITH REAR MOUNTED PENETRATION PANEL	8-24
8-67	CATALOG M3090CJ, 5.5 SOFTWARE SITE TO LX WITH ACGD CABLE UPGRADE FOR T/R WITH REAR MOUNTED PEN PANEL	8-24
8-68	CATALOG M3090CN, 5.5 TO LX WITH ACGD CABLE UPGRADE FOR T/R WITH REAR MOUNTED PENETRATION PANEL	8-25
8-69	CATALOG M3890AB, 1.0T 5.1-5.4 U/G TO HORIZON LX WITH ACGD HISPEED PLUS PHASED ARRAY	8-25
8-70	CATALOG M3890AC, 1.0T 5.5 TO BASE LX WITH ACGD, OCTANE COMPUTER WITHOUT CHANNEL PHASED ARRAY	8-25
8-71	CATALOG M3890AD, 1.0T 5.5 TO BASE LX OR HISPEED LX WITH ACGD, OCTANE COMPUTER & 4 CHANNEL PHASED ARRAY	8-26

8-2 UPGRADE CATALOGS SHIPPING INFORMATION

TABLE 8-2
CATALOG M1000MW, TABLE FOR USE WITH ALL MONITORS

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Operator Workspace Table for Color Monitor	45 x 54 x 37 (1143 x 1372 x 940)	180 (82)	box

TABLE 8-3
CATALOG M1000MZ, COLOR MONITOR

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Operator Workspace Color Monitor	27 x 33 x 27 (686 x 838 x 686)	125 (57)	skid

TABLE 8-4
CATALOG M1000NZ, LCD COLOR MONITOR

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Operator Workspace Color Monitor	27 x 33 x 27 (686 x 838 x 686)	125 (57)	skid

TABLE 8-5
CATALOG M1033MA, 8.X MNS UPGRADE FOR SIGNA ADVANTAGE 4.X-5.4

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Multi Nuclear Spectro MOD Option Key for 8X/LX	12 x 12 x 4 (305 x 305 x 102)	3 (1.5)	box
1.5T RF/Pen Spectro Internal Cable Kit	24 x 24 x 18 (610 x 610 x 457)	30 (14)	box
1.5T MNS Phosphorus Flex Coil Kit	18 x 24 x 6 (457 x 610 x 152)	5 (2.5)	box
LPCC MNS Magnet Enclosure Kit	18 x 18 x 6 (457 x 457 x 152)	8 (4)	box
MRS Phantom	16 x 16 x 16 (406 x 406 x 406)	10 (5)	box
UCERD to TNS and I/F Panel Cable	12 x 12 x 5 (305 x 305 x 127)	4 (2)	box

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-6
CATALOG M1033MB, 8.X MNS UPGRADE FOR SIGNA HORIZON 5.5 AND LATER

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Multi Nuclear Spectro MOD Option Key for 8X/LX	12 x 12 x 4 (305 x 305 x 102)	3 (1.5)	box
1.5T RF/Pen Style Cabinet MNSpectro Upgrade Kit	12 x 12 x 6 (305 x 305 x 152)	5 (2.5)	box
1.5T MNS Phosphorus Flex Coil Kit	18 x 24 x 6 (457 x 610 x 152)	5 (2.5)	box
LPCC MNS Magnet Enclosure Upgrade Kit	18 x 18 x 6 (457 x 457 x 152)	8 (4)	box
MRS Phantom	16 x 16 x 16 (406 x 406 x 406)	10 (5)	box
UCERD to TNS and I/F Panel Cable	12 x 12 x 5 (305 x 305 x 127)	4 (2)	box

TABLE 8-7
CATALOG M1033MC, MNS KIT FOR SIGNA LX (8.X) THAT DO NOT HAVE RF/PDU CABINET

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Multi Nuclear Spectro MOD Option Key for 8X/LX	12 x 12 x 4 (305 x 305 x 102)	3 (1.5)	box
1.5T RF/Pen Style Cabinet MNSpectro Upgrade Kit	24 x 24 x 18 (610 x 610 x 457)	30 (14)	box
1.5T MNS Phosphorus Flex Coil Kit	18 x 24 x 6 (457 x 610 x 152)	5 (2.5)	box
MRS Phantom	16 x 16 x 16 (406 x 406 x 406)	10 (5)	box
LX MNSpectro Magnet Enclosure Kit for HI/LO Carriage Covers	24 x 24 x 24 (610 x 610 x 610)	20 (9)	box
UCERD to TNS and I/F Panel Cable	16 x 16 x 6 (406 x 406 x 152)	8 (4)	box
Heliac Cable	36 x 36 x 4 (914 x 914 x 102)	20 (9)	box
Heliac Cable	36 x 36 x 4 (914 x 914 x 102)	20 (9)	box
Analogic RF Amplifier	24 x 24 x 48 (610 x 610 x 1219)	150 (68)	on pallet

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-8
CATALOG M1033MD, MNS KIT FOR SIGNA LX (8.X) THAT HAVE RF/PDU CABINET

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Multi Nuclear Spectro MOD Option Key for 8X/LX	12 x 12 x 4 (305 x 305 x 102)	3 (1.5)	box
1.5T MNS Phosphorus Flex Coil Kit	18 x 24 x 6 (457 x 610 x 152)	5 (2.5)	box
MRS Phantom	16 x 16 x 16 (406 x 406 x 406)	10 (5)	box
LX MNSpectro Magnet Enclosure Kit for HI/LO Carriage Covers	24 x 24 x 24 (610 x 610 x 610)	20 (9)	box
UCERD to TNS and I/F Panel Cable	16 x 16 x 6 (406 x 406 x 152)	8 (4)	box
Heliac Cable	36 x 36 x 4 (914 x 914 x 102)	20 (9)	box
Heliac Cable	36 x 36 x 4 (914 x 914 x 102)	20 (9)	box
MNS Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	0	on cabinet casters, wrapped with plastic
MNS Interconnect Cables	16 x 16 x 6 (406 x 406 x 152)	8 (4)	box

TABLE 8-9
CATALOG M1033MF, CERD TO UCERD UPGRADE KIT

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
UCERD	24 x 24 x 6 (610 x 610 x 152)	10 (5)	box
UCERD PSE DC power cord	16 x 16 x 6 (406 x 406 x 152)	8 (4)	box

TABLE 8-10
CATALOG M1085BG, 1.5T BODY COIL FOR SIGNA HORIZON

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Body Coil Assembly	35 x 94.5 x 55.88 (889 x 2400 x 1420)	3280 (1491)	Coil on cradle mated to installation cart (See Note 1)
RF Cable Kit	16 x 16 x 2 (406 x 406 x 51)	2 (1)	box
Body Coil Antifreeze mix (6 gallons)	16 x 16 x 14 (406 x 406 x 356)	60 (27)	box
Body Coil Antifreeze mix (5 gallons)	16 x 16 x 14 (406 x 406 x 356)	50 (23)	box
Note: 1 USA sites ONLY upgrade shipments will receive the new RF/Gradient coil on an aluminium (re-useable) cradle mated to coil cart with accessories. International sites upgrade shipments will have the RF/Gradient Coil shipped on a wooden (disposable) cradle with dimensions 35 x 84 x 39.5 (889 x 2134 x 1004) and weight 2425 lbs (1102). A forklift or crane/hoist rated for 4000 lbs (1818 kg) is required to position the coil/cradle assembly onto the cart for installing the coil into the magnet. GE Service needs to provide the coil cart and accessories for International Sites, refer to Section 10-5 INSTALLATION EQUIPMENT.			

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-11
CATALOG M1085KB, 50 HZ WATER-TO-AIR CHILLER FOR GRADIENT COIL COOLING

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Water Chiller	29 x 42 x 33 (737 x 1067 x 838)	320 (154) dry weight	crate

TABLE 8-12
CATALOG M1085KC, 60 HZ WATER-TO-AIR CHILLER FOR GRADIENT COIL COOLING

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Water Chiller	29 x 42 x 33 (737 x 1067 x 838)	320 (154) dry weight	crate

TABLE 8-13
CATALOG M1085KD, 1.5T 55 CM INTEGRATED BODY MODULE (CRM)

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
CRM Body Coil Assembly	35 x 84 x 39.5 (889 x 2134 x 1004)	2625 (1193)	cradle

TABLE 8-14
CATALOG M1090CP, 5.5 STEALTH CPU WITH 64 MBYTE MEMORY FOR SYSTEM WITH SUN 68020 CPU

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
5.5 Stealth CPU Upgrade	24 x 24 x 24 (610 x 610 x 610)	5 (2)	box

TABLE 8-15
CATALOG M1090FE, 128 MBYTE BAM EXPANSION

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Memory Board	25 x 25 x 5 (635 x 635 x 127)	5 (2)	box

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-16
CATALOG M1090JL, 5.X TO 5.5 CABLE UPGRADE FOR FIXED SITE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Technical Publications	12 x 20 x 12 (305 x 508 x 305)	40 (18)	box
Fixed Site Interconnect Cables Kit	18.5 x 18 x 17 (470 x 457 x 432)	60 (27)	box
Power Cables Kit	17 x 26 x 17 (432 x 660 x 432)	100 (45)	box
Penetration Panel Cover Kit	32.5 x 71 x 8.5 (826 x 1803 x 216)	65 (30)	box
Fiber Optic Cables Kit	27.5 x 31 x 33 (699 x 787 x 838)	50 (23)	box
Heat Exchanger Kit	22.5 x 22.5 x 17 (572 x 572 x 432)	60 (27)	box
Hose Kit	18.5 x 18 x 12.5 (470 x 457 x 318)	50 (23)	box

TABLE 8-17
CATALOG M1090JM, OXFORD MAGNET ENCLOSURE UPGRADE FOR 5.5 FIXED SITE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Front Cover Assembly	106 x 36 x 80 (2692 x 914 x 2032)	175 (80)	on cart
Rear Cover Assembly	106 x 36 x 80 (2692 x 914 x 2032)	175 (80)	on cart
Pedestal Covers Kit	12 x 52 x 30 (305 x 1321 x 762)	40 (18)	box
Rear Pedestal Assembly	23 x 49 x 32 (584 x 1245 x 813)	40 (18)	box
Side Cover	12 x 60 x 48 (305 x 1524 x 1219)	80 (36)	box
Standard Rear Ring Assembly	68 x 56 x 4 (1727 x 1422 x 102)	30 (14)	box
Front Endbell	47 x 48 x 22 (1194 x 1219 x 559)	75 (34)	box on skid
Rear Endbell	47 x 48 x 17 (1194 x 1219 x 432)	75 (34)	box on skid
Miscellaneous Hardware	14.5 x 30 x 10.5 (368 x 762 x 267)	10 (5)	box
Blower Hose Kit	40 x 40 x 46 (1016 x 1016 x 1168)	150 (68)	box on skid
Blower Assembly	30.5 x 38 x 26 (775 x 965 x 660)	70 (32)	box on skid
Patient Comfort Module	24.5 x 24 x 25 (622 x 610 x 635)	20 (9)	box on skid
PAC II Assembly	17 x 17 x 17 (432 x 432 x 432)	10 (5)	box on skid

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-18
CATALOG M1090JN, GE S-I MAGNET MAGNET ENCLOSURE UPGRADE FOR 5.5 FIXED SITE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Front Cover Assembly	106 x 36 x 80 (2692 x 914 x 2032)	175 (80)	on cart
Rear Cover Assembly	106 x 36 x 80 (2692 x 914 x 2032)	175 (80)	on cart
Pedestal Covers Kit	12 x 52 x 30 (305 x 1321 x 762)	40 (18)	box
Rear Pedestal Assembly	23 x 49 x 32 (584 x 1245 x 813)	40 (18)	box
Side Cover	14 x 80 x 48 (356 x 2032 x 1219)	80 (36)	box
Front Endbell, Standard	47 x 48 x 22 (1194 x 1219 x 559)	75 (34)	box on skid
Rear Endbell	47 x 48 x 17 (1194 x 1219 x 432)	75 (34)	box on skid
Miscellaneous Hardware	17.5 x 44.5 x 10 (445 x 1130 x 254)	100 (46)	box
Blower Hose Kit	40 x 40 x 46 (1016 x 1016 x 1168)	150 (68)	box on skid

TABLE 8-19
CATALOG M1090JP, GE MAGNISHIELD MAGNET ENCLOSURE UPGRADE FOR 5.5 FIXED SITE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Front Cover Assembly	120 x 36 x 80 (3048 x 914 x 2032)	200 (91)	on cart
Rear Cover Assembly	120 x 36 x 80 (3048 x 914 x 2032)	200 (91)	on cart
Pedestal Covers Kit	12 x 52 x 30 (305 x 1321 x 762)	40 (18)	box
Rear Pedestal Assembly	23 x 49 x 32 (584 x 1245 x 813)	40 (18)	box
Magnishield Front End Ring & Standard Rear Ring Assembly	46 x 67 x 6 (1168 x 1702 x 152)	40 (18)	box
Magnishield Front Endbell	48 x 48 x 15 (1219 x 1219 x 381)	50 (23)	box on skid
Rear Endbell	47 x 48 x 17 (1194 x 1219 x 432)	75 (34)	box on skid
Patient Comfort Module	24.5 x 24 x 25 (622 x 610 x 635)	20 (9)	box on skid
Blower Hose Kit	40 x 40 x 46 (1016 x 1016 x 1168)	150 (68)	box on skid
Miscellaneous Hardware	17.5 x 44.5 x 10 (445 x 1130 x 254)	100 (46)	box
Blower Assembly	30.5 x 38 x 26 (775 x 965 x 660)	70 (32)	box on skid
PAC II Assembly	17 x 17 x 17 (432 x 432 x 432)	10 (5)	box on skid

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-20
CATALOG M1090JR, ALL OTHER GE MAGNET ENCLOSURE UPGRADE FOR 5.5 FIXED SITE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Front Cover Assembly	106 x 36 x 80 (2692 x 914 x 2032)	175 (80)	on cart
Rear Cover Assembly	106 x 36 x 80 (2692 x 914 x 2032)	175 (80)	on cart
Pedestal Covers Kit	12 x 52 x 30 (305 x 1321 x 762)	40 (18)	box
Rear Pedestal Assembly	23 x 49 x 32 (584 x 1245 x 813)	40 (18)	box
Standard Rear Ring Assembly	46 x 67 x 6 (1168 x 1702 x 152)	20 (9)	box
Standard Front Endbell	47 x 48 x 22 (1194 x 1219 x 559)	75 (34)	box on skid
Rear Endbell	47 x 48 x 17 (1194 x 1219 x 432)	75 (34)	box on skid
Patient Comfort Module	24.5 x 24 x 25 (622 x 610 x 635)	30 (14)	box on skid
Blower Hose Kit	40 x 40 x 46 (1016 x 1016 x 1168)	150 (68)	box on skid
Miscellaneous Hardware	17.5 x 44.5 x 10 (445 x 1130 x 254)	100 (46)	box
Blower Assembly	30.5 x 38 x 26 (775 x 965 x 660)	70 (32)	box on skid
PAC II Assembly	17 x 17 x 17 (432 x 432 x 432)	10 (5)	box on skid

TABLE 8-21
CATALOG M1090JS, BRIDGE REPLACEMENT KIT FOR SYSTEM WITH OLD STANDARD BRIDGE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Bridge	24 x 161 x 10 (610 x 4090 x 254)	75 (34)	box
Pedestal Covers Kit	12 x 52 x 30 (305 x 1321 x 762)	40 (18)	box
Miscellaneous Installation Parts	37.5 x 30 x 25 (953 x 762 x 635)	50 (23)	box

TABLE 8-22
CATALOG M1090JT, BRIDGE FIBERGLASS KIT FOR SYSTEM WITH STANDARD NEW BRIDGE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Bridge	24 x 161 x 10 (610 x 4090 x 254)	75 (34)	box

TABLE 8-23
CATALOG M1090JW, LASER TO HALOGEN ALIGNMENT LIGHTS UPGRADE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Light Conversion Kit	46 x 67 x 6 (1168 x 1702 x 152)	30 (14)	box

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-24

CATALOG M1090JY, FRONT RING ASSEMBLY FOR SYSTEM WITH HALOGEN ALIGNMENT LIGHTS

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Standard Front Ring Assembly Kit	68 x 56 x 4 (1727 x 1422 x 102)	20 (9)	box

TABLE 8-25

CATALOG M1090MK, 32 MBYTE EXPANSION MEMORY FOR STEALTH CPU

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
5.5 32 MB Expansion for Stealth CPU	20 x 20 x 12 (508 x 508 x 305)	5 (2)	box

TABLE 8-26

CATALOG M1090NR, FULL SIZE PDU (MODELS 46-224847G1/G2 OR 46-282368G1) UPGRADE TO 5.5

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Contact Kit	18 x 18 x 18 (457 x 457 x 457)	25 (11)	box
Panel Cover & Circuit Breaker Kit	18 x 45 x 9 (457 x 1143 x 229)	40 (18)	box

TABLE 8-27

CATALOG M1090NS, COMPACT PDU (MODELS 46-317099P1 TO P9) UPGRADE TO 5.5

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
PDU Upgrade Kit	20 x 20 12 (508 x 508 x 305)	35 (16)	box

TABLE 8-28

CATALOG M1090NY, FULL SIZE PDU (MODELS 46-224847G1/G2 OR 46-282368G1) UPGRADE FOR 5.5 TO LX

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Panel Cover, Contactor, & Circuit Breaker Kit	18 x 45 x 10 (457 x 1143 x 254)	40 (18)	box

TABLE 8-29

CATALOG M1090NZ, COMPACT PDU (MODELS 46-317099P1 TO P9) UPGRADE FOR 5.5 TO LX

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
PDU Upgrade Kit	20 x 20 12 (508 x 508 x 305)	35 (16)	box

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-30
CATALOG M1090RW, 128 MBYTE MEDCAM MEMORY BOARD

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Memory Board	16 x 16 x 5 (406 x 406 x 127)	5 (2)	box

TABLE 8-31
CATALOG M1090RZ, 4.0 TO PRE 4.5 BASE KIT ADDER FOR 1.5T SYSTEM

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Carriage Cover Upgrade	24 x 24 x 12 (610 x 610 x 305)	40 (18)	box

TABLE 8-32
CATALOG M1090SC, BRIDGE FIBERGLASS KIT FOR SYSTEM WITH STANDARD NEW BRIDGE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Bridge	24 x 161 x 10 (610 x 4090 x 254)	75 (34)	box

TABLE 8-33
CATALOG M1090SD, FRONT RING ASSEMBLY FOR MAGNISHIELD SYSTEM

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Magnishield Front End Ring	46 x 67 x 6 (1168 x 1702 x 152)	20 (9)	box

TABLE 8-34
CATALOG M1090SJ, 5.5 TO LX WITH SGD CABLE UPGRADE FOR FIXED SITE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Technical Publications	12 x 20 x 12 (305 x 508 x 305)	40 (18)	box
Fixed Site Interconnect Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	80 (36)	box on skid
Power Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	100 (45)	box on skid

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-35
CATALOG M1090TZ, VCR INTERFACE KIT FOR OCTANE COMPUTER

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
VCR Video Board	16 x 16 x 3 (406 x 406 x 76)	6 (3)	box

TABLE 8-36
CATALOG M1090WG, HISPEED TO ECHOSPEED WITH SGD UPGRADE KIT FOR HORIZON LX WITH SGD 1.5T SYSTEMS

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
SGD Hi-Slew Gradient Cabinet without Fan Module installed	24 x 37 x 71 (610 x 940 x 1803)	1170 (532)	on cabinet casters, wrapped with plastic
Fan Module for SGD Hi-Slew Gradient Cabinet	28 x 38 x 15 (711 x 965 x 381)	160 (73)	on pallet
Fast Receiver Kit	26 x 26 x 12 (660 x 660 x 305)	35 (16)	box

TABLE 8-37
CATALOG M1090WJ, ECHOSPEED ADDER KIT FOR SITES WITHOUT SPECTROSCOPY ARE UPGRADING TO SGD ECHOSPEED

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Receiver 0 Kit	26 x 26 x 4 (660 x 660 x 102)	25 (11)	box

TABLE 8-38
CATALOG M1090YT, BRIDGE FIBERGLASS KIT FOR SIGNA CVMR

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Bridge	24 x 161 x 10 (610 x 4090 x 254)	75 (34)	box

TABLE 8-39
CATALOG M1090ZA, FIBER OPTIC REPEATER PANEL KIT

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Repeater Assembly Kit	16 x 24 x 16 (406 x 610 x 406)	20 (9)	box

TABLE 8-40
CATALOG M1090ZJ, EXCITER KIT FOR 1.5T SYSTEM

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Repeater Assembly Kit	18 x 22 x 3 (457 x 559 x 76)	5 (2)	box

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-41
CATALOG M1090ZK, EXCITER KIT FOR 1.0T SYSTEM

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Repeater Assembly Kit	18 x 22 x 3 (457 x 559 x 76)	5 (2)	box

TABLE 8-42
CATALOG M1091JC, S-II THROUGH S-V MAGNET ENCLOSURE UPGRADE FOR SIGNA CVMR FIXED SITE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Front Cover Assembly	106 x 36 x 80 (2692 x 914 x 2032)	175 (80)	on cart
Rear Cover Assembly	106 x 36 x 80 (2692 x 914 x 2032)	175 (80)	on cart
Pedestal Covers Kit	12 x 52 x 30 (305 x 1321 x 762)	40 (18)	box
Rear Pedestal Assembly	23 x 49 x 32 (584 x 1245 x 813)	40 (18)	box
Standard Rear Ring Assembly	46 x 67 x 6 (1168 x 1702 x 152)	20 (9)	box
Standard Front Endbell	47 x 48 x 22 (1194 x 1219 x 559)	75 (34)	box on skid
Rear Endbell	47 x 48 x 17 (1194 x 1219 x 432)	75 (34)	box on skid
Patient Comfort Module	24.5 x 24 x 25 (622 x 610 x 635)	30 (14)	box on skid
Blower Hose Kit	40 x 40 x 46 (1016 x 1016 x 1168)	150 (68)	box on skid
Miscellaneous Hardware	17.5 x 44.5 x 10 (445 x 1130 x 254)	100 (46)	box
Blower Assembly	30.5 x 38 x 26 (775 x 965 x 660)	70 (32)	box on skid
PAC II Assembly	17 x 17 x 17 (432 x 432 x 432)	10 (5)	box on skid

TABLE 8-43
CATALOG M1091JE, S-II THROUGH S-V MAGNET FIXED SITE BRM TO CRM UPGRADE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
CRM Quad Head Coil Assembly	24 x 24 x 24 (610 x 610 x 610)	30 (14)	box
RF Cable Kit	16 x 16 x 2 (406 x 406 x 51)	2 (1)	box
Body Coil Antifreeze mix	16 x 16 x 14 (406 x 406 x 356)	60 (27)	box
Fiber Optic Bundle	16 x 16 x 14 (406 x 406 x 356)	25 (11)	box
Bridge	24 x 161 x 10 (610 x 4090 x 254)	125 (57)	box
Endbell Kit	40 x 48 x 32 (1016 x 1219 x 813)	150 (68)	box on skid

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-44
CATALOG M1191JB, 5.5 CABLE UPGRADE FOR T/R WITH REAR MOUNTED PENETRATION PANEL

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Technical Publications	12 x 20 x 12 (305 x 508 x 305)	40 (18)	box
Mobile Site Interconnect Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	100 (45)	box on skid
Fiber Optic Cables Kit	35 x 35 x 33 (889 x 889 x 838)	50 (23)	box
Heat Exchanger	22.5 x 22.5 x 17 (572 x 572 x 432)	60 (27)	box
Hose Kit	18.5 x 18 x 12.5 (470 x 457 x 318)	50 (23)	box
Fiber Optic Cables Kit	27.5 x 31 x 33 (699 x 787 x 838)	50 (23)	box

TABLE 8-45
CATALOG M1191JE, GE S-I MAGNET ENCLOSURE UPGRADE FOR 5.5 TRUE MOBILES AND T/R

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Front Cover Assembly	106 x 36 x 80 (2692 x 914 x 2032)	175 (80)	on cart
Rear Pedestal Assembly	23 x 49 x 32 (584 x 1245 x 813)	40 (18)	box
Front Endbell, Standard	47 x 48 x 22 (1194 x 1219 x 559)	75 (34)	box on skid
Rear Endbell	47 x 48 x 17 (1194 x 1219 x 432)	75 (34)	box on skid
Miscellaneous Hardware	17.5 x 44.5 x 10 (445 x 1130 x 254)	100 (46)	box
Blower Hose Kit	40 x 40 x 46 (1016 x 1016 x 1168)	150 (68)	box on skid
Blower Assembly	30.5 x 38 x 26 (775 x 965 x 660)	70 (32)	box on skid
Rear Cover – RF Tube Extension	30 x 30 x 28 (762 x 762 x 711)	35 (16)	box

TABLE 8-46
CATALOG M1191JF, ALL OTHER GE MAGNET ENCLOSURE UPGRADE FOR 5.5 TRUE MOBILES AND T/R

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Front Cover Assembly	106 x 36 x 80 (2692 x 914 x 2032)	175 (80)	on cart
Rear Pedestal Assembly	23 x 49 x 32 (584 x 1245 x 813)	40 (18)	box
Front Endbell, Standard	47 x 48 x 22 (1194 x 1219 x 559)	75 (34)	box on skid
Rear Endbell	47 x 48 x 17 (1194 x 1219 x 432)	75 (34)	box on skid
Standard Rear Ring Assembly	68 x 56 x 4 (1727 x 1422 x 102)	30 (14)	box
Miscellaneous Hardware	17.5 x 44.5 x 10 (445 x 1130 x 254)	100 (46)	box
Blower Hose Kit	40 x 40 x 46 (1016 x 1016 x 1168)	150 (68)	box on skid
Blower Assembly	30.5 x 38 x 26 (775 x 965 x 660)	70 (32)	box on skid
Rear Cover – RF Tube Extension	30 x 30 x 28 (762 x 762 x 711)	35 (16)	box

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-47
CATALOG M1191JG, ACTIVE SHIELD MAGNET ENCLOSURE ADDER KIT

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Weld Mount Brackets Kit	45 x 18 x 9 (1143 x 457 x 229)	50 (23)	box

TABLE 8-48
CATALOG M1885BG 1.0T BODY COIL FOR SIGNA HORIZON

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Body Coil Assembly	35 x 85 x 63 (889 x 2159 x 3759)	3250 (1477)	Coil on skid on shipping cart
RF Cable Kit	16 x 16 x 2 (406 x 406 x 51)	2 (1)	box
Body Coil Antifreeze mix (6 gallons)	16 x 16 x 14 (406 x 406 x 356)	60 (27)	box
Body Coil Antifreeze mix (5 gallons)	16 x 16 x 14 (406 x 406 x 356)	50 (23)	box

TABLE 8-49
CATALOG M3090AA, 1.5T 4.5 UPGRADE TO HISPEED LX WITH ACGD HISPEED PLUS PHASED ARRAY

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
System Control Cabinet	24 x 33 x 78 (610 x 838 x 1981)	494 (225)	on cabinet casters, wrapped with plastic
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic
Penetration Panel	22 x 40 x 25 (559 x 1016 x 635)	70 (32)	box
Head SNR Phantom Set	16 x 16 x 16 (406 x 406 x 406)	20 (9)	box
Body SNR Phantom Set	14 x 33 x 20 (356 x 838 x 508)	65 (30)	box
4.5 Enclosure Upgrade Kit	24 x 35 x 23 (610 x 889 x 584)	100 (46)	box
Operator Workspace Equipment	24 x 36 x 23 (610 x 914 x 584)	60 (27)	box
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 x 15 (711 x 965 x 381)	160 (73)	on pallet

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-50
CATALOG M3090AB, 1.5T 4.5 UPGRADE TO ECHOSPEED LX WITH 4 CHANNEL PHASED ARRAY

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
System Control Cabinet	24 x 33 x 78 (610 x 838 x 1981)	494 (225)	on cabinet casters, wrapped with plastic
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic
Penetration Panel	22 x 40 x 25 (559 x 1016 x 635)	70 (32)	box
Head SNR Phantom Set	16 x 16 x 16 (406 x 406 x 406)	20 (9)	box
Body SNR Phantom Set	14 x 33 x 20 (356 x 838 x 508)	65 (30)	box
4.5 Enclosure Upgrade Kit	24 x 35 x 23 (610 x 889 x 584)	100 (46)	box
Operator Workspace Equipment	24 x 36 x 23 (610 x 914 x 584)	60 (27)	box
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet

TABLE 8-51
CATALOG M3090AD, 1.5T 5.1-5.4 U/G TO HORIZON LX WITH ACGD HISPEED PLUS PHASED ARRAY

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
System Control Cabinet	24 x 33 x 78 (610 x 838 x 1981)	494 (225)	on cabinet casters, wrapped with plastic
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic
Penetration Panel	22 x 40 x 25 (559 x 1016 x 635)	70 (32)	box
Operator Workspace Equipment	24 x 36 x 23 (610 x 914 x 584)	60 (27)	box
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-52
CATALOG M3090AE, 1.5T 5.1-5.4 U/G TO HORIZON LX WITH ACGD ECHOSPEED PLUS PHASED ARRAY

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
System Control Cabinet	24 x 33 x 78 (610 x 838 x 1981)	494 (225)	on cabinet casters, wrapped with plastic
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic
Penetration Panel	22 x 40 x 25 (559 x 1016 x 635)	70 (32)	box
Operator Workspace Equipment	24 x 36 x 23 (610 x 914 x 584)	60 (27)	box
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet

TABLE 8-53
CATALOG M3090AF, 1.5T 5.5 UPGRADE TO CV2 AND/OR NV1 CRM WITH ACGD

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
System Control Cabinet	24 x 33 x 78 (610 x 838 x 1981)	494 (225)	on cabinet casters, wrapped with plastic
Penetration Panel	22 x 40 x 25 (559 x 1016 x 635)	70 (32)	box
Operator Workspace Equipment	24 x 36 x 23 (610 x 914 x 584)	60 (27)	box
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
Octane Computer	22 x 30 x 24 (559 x 762 x 610)	50 (23)	box on skid
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic

TABLE 8-54
CATALOG M3090AG, 1.5T 5.5 TO HISPEED LX WITH ACGD, OCTANE COMPUTER & 4 CHANNEL PHASED ARRAY

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
System Control Cabinet	24 x 33 x 78 (610 x 838 x 1981)	494 (225)	on cabinet casters, wrapped with plastic
Operator Workspace Equipment	24 x 36 x 23 (610 x 914 x 584)	60 (27)	box
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-55

CATALOG M3090AH, 1.5T 5.5 TO ECHOSPEED LX WITH ACGD, OCTANE COMPUTER & 4 CHANNEL PHASED ARRAY

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
System Control Cabinet	24 x 33 x 78 (610 x 838 x 1981)	494 (225)	on cabinet casters, wrapped with plastic
Operator Workspace Equipment	24 x 36 x 23 (610 x 914 x 584)	60 (27)	box
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic

TABLE 8-56

CATALOG M3090AJ, 1.5T 5.X HORIZON UPGRADE TO CV AND/OR NV CRM WITH ACGD

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
System Control Cabinet	24 x 33 x 78 (610 x 838 x 1981)	494 (225)	on cabinet casters, wrapped with plastic
Operator Workspace Equipment	24 x 36 x 23 (610 x 914 x 584)	60 (27)	box
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
Octane Computer	22 x 30 x 24 (559 x 762 x 610)	50 (23)	box on skid
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic

TABLE 8-57

CATALOG M3090BA, 1.0T ACGD UPGRADE FOR SYSTEM WITH SGD GRADIENT CABINET & RF/PENETRATION CABINET

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Power Cables Kit	24 x 24 x 18 (610 x 610 x 457)	50 (23)	box
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-58
CATALOG M3090BB, ACGD UPGRADE FOR SYSTEM WITH SGD GRADIENT CABINET & RF/PDU CABINET

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Power Cables Kit	24 x 24 x 18 (610 x 610 x 457)	50 (23)	box
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic

TABLE 8-59
CATALOG M3090BC, 1.5T ACGD UPGRADE FOR SYSTEM WITH SGD GRADIENT CABINET & RF/PENETRATION CABINET

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Power Cables Kit	24 x 24 x 18 (610 x 610 x 457)	50 (23)	box
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic

TABLE 8-60
CATALOG M3090BD, HISPEED PLUS PERFORMANCE OPTION

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Software & Install document	9 x 12 x 1 (229 x 305 x 25)	2 (1)	envelope

TABLE 8-61
CATALOG M3090BE, ECHOSPEED PLUS PERFORMANCE OPTION

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Software & Install document	9 x 12 x 1 (229 x 305 x 25)	2 (1)	envelope

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-62
CATALOG M3090CA, 4.X TO LX WITH ACGD CABLE UPGRADE FOR FIXED SITE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Technical Publications	12 x 20 x 12 (305 x 508 x 305)	40 (18)	box
Fixed Site Interconnect Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	80 (36)	box on skid
Penetration Panel Cover Kit	32.5 x 71 x 10 (826 x 1803 x 254)	65 (30)	box
Power Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	100 (45)	box on skid
Fiber Optic Cables Kit	27.5 x 31 x 33 (699 x 787 x 838)	50 (23)	box
Heat Exchanger Hose Kit	18.5 x 18 x 12.5 (470 x 457 x 318)	50 (23)	box
SPT Phantom Set	34 x 32.5 x 60 (864 x 826 x 1524)	350 (159)	on cart casters with box cover
Gradient To Coil Power Cables	40 x 40 x 18 (1016 x 1016 x 457)	620 (282)	box on skid

TABLE 8-63
CATALOG M3090CB, 5.1-5.4 U/G TO LX WITH ACGD CABLE UPGRADE FOR FIXED SITE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Technical Publications	12 x 20 x 12 (305 x 508 x 305)	40 (18)	box
Fixed Site Interconnect Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	80 (36)	box on skid
Penetration Panel Cover Kit	32.5 x 71 x 10 (826 x 1803 x 254)	65 (30)	box
Power Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	100 (45)	box on skid
Fiber Optic Cables Kit	27.5 x 31 x 33 (699 x 787 x 838)	50 (23)	box
Heat Exchanger Hose Kit	18.5 x 18 x 12.5 (470 x 457 x 318)	50 (23)	box
SPT Phantom Set	34 x 32.5 x 60 (864 x 826 x 1524)	350 (159)	on cart casters with box cover
Gradient To Coil Power Cables	40 x 40 x 18 (1016 x 1016 x 457)	620 (282)	box on skid

TABLE 8-64
CATALOG M3090CC, 5.5 SOFTWARE ONLY SITE TO LX WITH ACGD CABLE UPGRADE FOR FIXED SITE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Technical Publications	12 x 20 x 12 (305 x 508 x 305)	40 (18)	box
Fixed Site Interconnect Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	80 (36)	box on skid
Penetration Panel Cover Kit	32.5 x 71 x 10 (826 x 1803 x 254)	65 (30)	box
Power Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	100 (45)	box on skid
Fiber Optic Cables Kit	27.5 x 31 x 33 (699 x 787 x 838)	50 (23)	box
Heat Exchanger Hose Kit	18.5 x 18 x 12.5 (470 x 457 x 318)	50 (23)	box
SPT Phantom Set	34 x 32.5 x 60 (864 x 826 x 1524)	350 (159)	on cart casters with box cover
Gradient To Coil Power Cables	40 x 40 x 18 (1016 x 1016 x 457)	620 (282)	box on skid

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-65
CATALOG M309CD, 5.5 TO LX WITH ACGD CABLE UPGRADE FOR FIXED SITE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Technical Publications	12 x 20 x 12 (305 x 508 x 305)	40 (18)	box
Fixed Site Interconnect Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	80 (36)	box on skid
Power Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	100 (45)	box on skid
Fiber Optic Cables Kit	27.5 x 31 x 33 (699 x 787 x 838)	50 (23)	box

TABLE 8-66
CATALOG M3090CH, 3.X-5.4 TO LX WITH ACGD CABLE UPGRADE FOR TRANSPORTABLE/RELOCATABLE (T/R) WITH
REAR MOUNTED PENETRATION PANEL

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Technical Publications	12 x 20 x 12 (305 x 508 x 305)	40 (18)	box
Cryo Monitor Support Bracket Kit	30 x 30 x 5 (762 x 762 x 127)	20 (9)	box
Power Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	100 (45)	box on skid
Fiber Optic Cables Kit	27.5 x 31 x 33 (699 x 787 x 838)	50 (23)	box
Fixed Site Interconnect Cables Kit (4 boxes)	24 x 24 x 30 (610 x 610 x 762)	25 (11)	4 boxes
Heat Exchanger Hose Kit	18.5 x 18 x 12.5 (470 x 457 x 318)	50 (23)	box
SPT Phantom Set	34 x 32.5 x 60 (864 x 826 x 1524)	350 (159)	on cart casters with box cover
Gradient To Coil Power Cables	40 x 40 x 18 (1016 x 1016 x 457)	620 (282)	box on skid

TABLE 8-67
CATALOG M3090CJ, 5.5 SOFTWARE SITE TO LX WITH ACGD CABLE UPGRADE FOR T/R WITH REAR MOUNTED PEN PANEL

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Technical Publications	12 x 20 x 12 (305 x 508 x 305)	40 (18)	box
Cryo Monitor Support Bracket Kit	30 x 30 x 5 (762 x 762 x 127)	20 (9)	box
Power Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	100 (45)	box on skid
True Mobile Site Interconnect Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	100 (45)	4 boxes on one skid
Fiber Optic Cables Kit	35 x 35 x 33 (889 x 889 x 838)	50 (23)	box
Heat Exchanger Hose Kit	18.5 x 18 x 12.5 (470 x 457 x 318)	50 (23)	box
Gradient To Coil Power Cables	40 x 40 x 39 (1016 x 1016 x 991)	620 (282)	box on skid
SPT Phantom Set	34 x 32.5 x 60 (864 x 826 x 1524)	350 (159)	on cart casters with box cover

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-68
CATALOG M3090CN, 5.5 TO LX WITH ACGD CABLE UPGRADE FOR T/R WITH REAR MOUNTED PENETRATION PANEL

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Technical Publications	12 x 20 x 12 (305 x 508 x 305)	40 (18)	box
True Mobile Site Interconnect Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	100 (45)	box on skid
Power Cables Kit	40 x 40 x 23 (1016 x 1016 x 584)	100 (45)	box on skid

TABLE 8-69
CATALOG M3890AB, 1.0T 5.1-5.4 U/G TO HORIZON LX WITH ACGD HISPEED PLUS PHASED ARRAY

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
System Control Cabinet	24 x 33 x 78 (610 x 838 x 1981)	494 (225)	on cabinet casters, wrapped with plastic
Penetration Panel	22 x 40 x 25 (559 x 1016 x 635)	70 (32)	box
Operator Workspace Equipment	24 x 36 x 23 (610 x 914 x 584)	60 (27)	box
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
Octane Computer	22 x 30 x 24 (559 x 762 x 610)	50 (23)	box on skid
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic

TABLE 8-70
CATALOG M3890AC, 1.0T 5.5 TO BASE LX WITH ACGD, OCTANE COMPUTER WITHOUT CHANNEL PHASED ARRAY

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
System Control Cabinet	24 x 33 x 78 (610 x 838 x 1981)	494 (225)	on cabinet casters, wrapped with plastic
Operator Workspace Equipment	24 x 36 x 23 (610 x 914 x 584)	60 (27)	box
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
Octane Computer	22 x 30 x 24 (559 x 762 x 610)	50 (23)	box on skid
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-71
CATALOG M3890AD, 1.0T 5.5 TO BASE LX OR HISPEED LX WITH ACGD, OCTANE COMPUTER & 4 CHANNEL PHASED ARRAY

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
System Control Cabinet	24 x 33 x 78 (610 x 838 x 1981)	494 (225)	on cabinet casters, wrapped with plastic
Operator Workspace Equipment	24 x 36 x 23 (610 x 914 x 584)	60 (27)	box
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
Octane Computer	22 x 30 x 24 (559 x 762 x 610)	50 (23)	box on skid
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic

SECTION 9 – PREINSTALLATION CHECKLIST

TABLE OF CONTENTS

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
9-1	INTRODUCTION	9-3
9-2	PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE DELIVERY	9-3
9-3	GENERAL PREINSTALLATION REMINDERS	9-5
9-4	SAFETY	9-5
9-5	PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE LCC MAGNET DELIVERY AND MOVING INTO MAGNET ROOM	9-6
9-6	PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE LCC MAGNET CRYOGEN TRANSFER/MAGNET FILL	9-7
9-7	PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE LCC MAGNET RAMP-UP	9-7

9-1 INTRODUCTION

“Preinstallation” refers to work necessary to plan and prepare a site for delivery and installation of equipment. Delay, confusion, and waste of manpower can be avoided by completing preinstallation work.

The purpose of this section is to outline key areas of concern in the preparation of a customer's site for Signa Horizon upgrade. It is intended as a guide for GE's Installation Specialists and/or Installation Leaders when making on-site inspections during the installation planning phase of the upgrade. Note site inspections by GE are intended to aid the overall site preparation process. They do not relieve the customer and project architect from responsibility for the design, engineering, and coordination efforts necessary to ensure a successful MR project.

During the course of the site preparation process, GE's Installation Specialists and/or Installation Leaders may observe that the Requested Arrival Date (RAD) for the upgrade equipment is not realistic. If this is the case, appropriate actions must be taken by the GE Field Sales/Service Team to adjust the RAD accordingly.

All work must be in compliance with national and local safety codes.

9-2 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE DELIVERY

The following items must be completed by the installation Specialist or Site Field Engineer prior to upgrade equipment delivery. An on-site inspection by a GE Representative to confirm site readiness must be completed at least 2 weeks prior to the Scheduled Arrival Date (SAD) of the upgrade equipment.

- ☐ 1. Does the system to be upgraded include an S-IV magnet?
If yes, an additional 2 days of customer downtime may be needed immediately prior to the start of present system de-install process to allow for magnet mechanical plotting. Contact Magnet Support at 1-800-324-7937 to determine if magnet requires mechanical plotting and for the details of efficient mechanical plotting.
- ☐ 2. Are there concerns about properly shimming the existing magnet after the Horizon upgrade components have been installed?
If there are concerns, consult with the appropriate MAC (Magnet And Cryogenics) Team member for the site. Additional customer downtime may have to be planned into the upgrade installation schedule to allow for special magnet shimming requirements.
- ☐ 3. Has appropriate upgrade ordering checklist been completed and FAXed to Americas Customer Center, ATTENTION: Bob Healy (FAX number 262-827-3248) at least 4 weeks prior to SAD?
Checklists documents are:
 - *Direction 2246012 Pre-Sale/Site Survey For Upgrades To CXK4 Magnet With WideOpen Enclosure*
 - *Direction 2221391 Signa CVMR Upgrade Ordering Checklist*
 - *Direction 2171144 Signa Horizon LX Upgrade Ordering Checklist.*
 If no, the upgrade SAD will be rescheduled to a later date and will increase the risk of incomplete order shipment.
- ☐ 4. Does system being upgraded have a Full-size PDU and have arrangements been made for old PDU removal? Refer to **Section 2-5-1 Power Distribution Unit (PDU)**.
- ☐ 5. Has a site equipment plan been completed, reviewed with the customer, and finalized?
- ☐ 6. Has a cable routing plan been completed? Are all cable lengths sufficient? If lengths are not sufficient, a Special Order Inquiry (SOI) will be required. The GE Installation Specialist or Site Field Engineer must submit a SOI Request Form at least 8 weeks prior to SAD to allow for proper processing of the request.
The SOI Request Form is available on the Web at:
<http://medquickplace02.ge.com/QuickPlace/mr/PageLibrary862569F30075E524.nsf?OpenDatabase>

9-2 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE DELIVERY (Continued)

- ☐ 7. Does incoming power available at PDU meet the requirements in **Section 5 – POWER REQUIREMENTS** of this document? This includes correctly sized facility transformer, feeder wire size, Main Disconnect Control wire sizes and circuit breaker rating, and system ground requirements.
- ☐ 8. Have local and national codes been check to determine if an electrician will be required to provide and/or install new feeder wires to the PDU? If required then have arrangements been made for an electrician?
- ☐ 9. Have the Blower Box installation and mounting details been planned for with the Site Field Engineer?

Note

Blower Box mounting requires customer supplied hardware (ie. lag bolts, screws, etc.) appropriate for the surface on which the box will be mounted. The Blower Box is shipped with only the screws to mount the brackets to the box.

- ☐ 10. Does the site's HVAC (heating, ventilation, and air conditioning) system meet requirements in **Section 4-2 TEMPERATURE AND HUMIDITY SPECIFICATIONS** and **Section 4-3 COOLING REQUIREMENTS** of this document? If these specifications are not strictly adhered to, failure of the SGD Gradient Cabinet modules may occur.
- ☐ 11. Has the site's vibration environment been assessed including system stability tests performed as defined in Section 4-11, VIBRATION?
- ☐ 12. Has delivery route been defined for equipment especially enclosure covers (refer to **Section 2-3, MINIMUM DOOR SIZES**) including plans for protecting flooring for heavy equipment & carts (ie. RF/Gradient Body Coil & installation cart)?
- ☐ 13. Have all upgrade exceptions, such as special cabling and options unique to the site including customer equipment, been identified and clearly tagged **DO NOT REMOVE** ?
- ☐ 14. For system with Spectroscopy option: Has the site Field Engineer removed the Spectral Module from the Exciter Board in the Systems Cabinet and is the module available at site for re-installation during upgrade.
- ☐ 15. Has a holding area for old equipment and installation debris been agreed upon with the customer? Making provision for a holding area will allow for efficient unloading and reloading of the delivery vehicle.
- ☐ 16. Has a holding area for old equipment and installation debris been agreed upon with the customer? Making provision for a holding area will allow for efficient unloading and reloading of the delivery vehicle.
- ☐ 17. Has RF Shielded Room been checked to meet specifications in **Section 7 – RF SHIELDED ROOM** ?
- ☐ 18. If an upgrade to an LCC magnet is being performed with the Signa system electronics upgrade and the RF Shield is opened, modified, upgraded, etc. then are there plans to recertify (test) the RF Shielded Room integrity after magnet installation to meet or exceed the new system RF Shielded Room specifications. Refer to refer to **Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 7-1 RF SHIELDED ROOM SPECIFICATION**.
- ☐ 19. If an upgrade to an LCC Magnet is being performed then the cryogenic venting must comply with the specifications in **Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 4-8 CRYOGENIC VENTING**. The customer is to supply a copy of the Vent Inspection Report to GE Medical Systems for the customer's record file stating compliance to this section.

9-3 GENERAL PREINSTALLATION REMINDERS

- ☐ 1. Has the customer's architect/contractor fully reviewed final site installation drawings using the guidelines provided by MR Siting and Shielding (MRSS)?
- ☐ 2. Have vehicle parking arrangements been made for installation personnel?
- ☐ 3. Is temporary storage space available for use during installation?
- ☐ 4. What is hospital smoking policy?
- ☐ 5. Are a first aid kit and non-ferrous fire extinguishers available at site?
- ☐ 6. Have facility arrangements been made for refuse disposal during installation?

9-4 SAFETY**WARNING!**

THE FOLLOWING SAFETY ITEMS MUST BE ADHERED TO PRIOR TO MOVING THE MAGNET FROM THE TRUCK INTO THE ROOM. THE MAGNET WILL BE EXHAUSTING HELIUM GAS DURING THIS TIME. GASEOUS HELIUM IS AN INVISIBLE, ODORLESS GAS THAT CAN CAUSE ASPHYXIATION WHEN DEPLETING THE AMBIENT OXYGEN SUPPLY. HELIUM GAS IS LIGHTER THAN AIR AND WILL RISE TO THE CEILING.

- ☐ 1. Have all of the required room ventilation items been installed and tested to make sure sufficient ventilation is available? Is exhaust fan and fan control installed and functional? The customer is to supply a copy of the Exhaust Fan Test Report to GE Medical Systems for the customers record file. Refer to **Section 4-7 ROOM VENTILATION** for specific requirements.
- ☐ 2. Are a first aid kit and non-ferrous fire extinguishers available at site?
- ☐ 3. Are removable chain or strap available for securing cryogen gas cylinders in an upright position to prevent the cylinders from falling which may cause injury or damage?
- ☐ 4. Are functioning telephones as defined in **Section 2-5-7 System Monitoring & Support Connectivity Requirements** available at site for duration of installation? Phone lines are needed to dial out in case of an emergency.
- ☐ 5. Has the cryogenic vent been installed complete from the Magnet Room RF shield penetration to the final exit and been inspected to final exit outside of the building? The customer is to supply a copy of the Vent Inspection Report to GE Medical Systems for the customers record file. Refer to **Section 4-8, CRYOGENIC VENTING**.
- ☐ 6. Have plans been made to connect the cryogenic vent to the magnet after magnet installation in the room? Refer to **Section 4-8, CRYOGENIC VENTING**.
- ☐ 7. Is there adequate ventilation for helium dewar storage area. Refer to **Section 4-7, ROOM VENTILATION**.

9-5 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE LCC MAGNET DELIVERY AND MOVING INTO MAGNET ROOM

The following items must be completed prior to magnet delivery. A site inspection by GE Service Representative must be completed prior to magnet delivery to ensure site readiness.

- ☐ 1. Has Main Disconnect Panel been installed, electrician wiring complete, MDP operational with power available? Refer to **SECTION 5-2 CRITICAL POWER REQUIREMENTS**.
- ☐ 2. **Is 24 hour/day, 7 days/week power and water cooling available Shield/Cryo Cooler Compressor cabinet and Magnet Monitor equipment?**
- ☐ 3. Are all walls and ceiling in magnet room essentially complete except for removable section?
- ☐ 4. Has a clear route to the magnet room been defined for magnet installation (refer to **Section 2-4, MINIMUM DOOR SIZES**)?
- ☐ 5. Is a secure space available to store equipment on site?
- ☐ 6. Have arrangements been made for the use of special rigging equipment for moving the magnet into the magnet room?
- ☐ 7. Has the raised floor been installed in the equipment/computer room in preparation for the superconducting and resistive power supplies?
- ☐ 8. Has work in the magnet room been completed or suspended and the magnet room closed off to provide a dust-free, closed environment?
- ☐ 9. Has all equipment been removed from the magnet room to allow space for the magnet with rigging equipment and cryogen dewars?
- ☐ 10. Are equipment anchors installed in floor as required for magnet room equipment? Refer to **Section 7-6 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM**.
- ☐ 11. For all magnet room anchors, has pull test been performed and installation torque value defined? Test results recorded by RF Shield Room Vendor and information forwarded to GE Medical Systems for the customers record file? Refer to **Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 7-7-1 Clamping Force And Pull Test**.
- ☐ 12. For all magnet room anchors, has ground impedance test been performed? Test results recorded by RF Shield Room Vendor and documentation given to GE Medical Systems for the customers record file? Refer to **Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 7-7-3 Electrical Isolation**.

9-6 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE LCC MAGNET CRYOGEN TRANSFER/MAGNET FILL

- ☐ 1. Have arrangements been made for cryogen delivery and storage?
- ☐ 2. Does a clear path exist to magnet room for delivery of cryogens?
- ☐ 3. Is the magnet room adequately vented to allow exhaust of cryogen boil-off?
- ☐ 4. Is the magnet room clear to provide room for cryogen dewars during filling?

9-7 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE LCC MAGNET RAMP-UP

- ☐ 1. Is the cryogenic vent connected to the magnet?
- ☐ 2. Has the magnet room been completely closed (removable section closed up and sealed)?
- ☐ 3. Has the magnet room been tested to ensure that the RF shielding meets the electrical isolation and the 100dB requirement for electric/plane waves and the 90dB requirement for magnetic waves refer to **Section 7-1 RF SHIELDED ROOM INTRODUCTION**? The customer is to supply a copy of RF shielded room vendor test reports.
- ☐ 4. Have all ferrous metal objects been removed from the magnet room?
- ☐ 3. Have adequate signs (Safety and Exclusion Zones) been posted to warn personnel about dangers of magnetic field?
- ☐ 4. Have hospital personnel been informed of magnet safety precautions and procedures?
- ☐ 7. Has power been connected from the Main Disconnect Panel to Power Distribution Unit?
- ☐ 8. Has the penetration panel been installed?
- ☐ 9. Is all contractor construction work completed?
- ☐ 10. Has all contractor equipment that could affect shimming been removed from within the three gauss zone?
- ☐ 11. Have precautions been taken to prevent movement of large metal objects within the three gauss zone?
- ☐ 12. Has outside emergency personnel (e.g. fire department and other outside ambulance or emergency personnel) been informed of unique characteristics (e.g. strong magnetic field, cryogenics, etc.) of magnet and correct precautions to take in event of emergencies?

SECTION 10 – TOOLS AND TEST EQUIPMENT

TABLE OF CONTENTS

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
10-1	INTRODUCTION	10-3
10-2	MOVING METAL MEASUREMENT EQUIPMENT	10-3
10-3	RIGGER/CUSTOMER SUPPLIED EQUIPMENT	10-4
10-4	CRYOGENIC EQUIPMENT	10-5
10-5	INSTALLATION EQUIPMENT	10-6
10-7	TEST EQUIPMENT	10-8
10-8	CALIBRATION TOOLS/FIXTURES	10-9
10-9	TOOL KIT	10-10

10-1 INTRODUCTION

The following tables lists the tools and test equipment needed to install a Signa LX Upgrade and calibrate the system.

10-2 MOVING METAL MEASUREMENT EQUIPMENT

TABLE 10-1
MOVING METAL MEASUREMENT EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	—	Magnetometer sensitive to 50 μ gauss Flux gate or equivalent Example: Barrington Instruments Ltd. Model Mag-03MC 500 μ T
2	—	Spectrum analyzer capable of moving metal measurements include the GenRad 2515, B&K 2032 or HP 3560A Dynamic Signal Analyzer.
3	—	Accelerometer capable of measuring from 0.2 Hz beyond 250 Hz.

10-3 RIGGER/CUSTOMER SUPPLIED EQUIPMENT

Note

Table 10-2 contains an overview of equipment for initial planning purposes, refer to *Direction 2226318, 1.5T & 1.0T LCC Magnet Signa MR/i Delivery* and Installation Manual for details of equipment and requirements.

TABLE 10-2
RIGGER/CUSTOMER SUPPLIED EQUIPMENT

SUPPLIED BY	ITEM	DESCRIPTION
Rigger	1	Crane for removing magnet crate from delivery truck: 12 ton (10,909 kg)
	2	Motorized tow vehicle for pushing/pulling magnet when it is on dollies (e.g. an electric fork lift).
	3	Steel floor plates (8) to cover floors while transporting magnet, 36 in. x 12 in. x 0.25 in. (914 mm x 305 mm x 6.4 mm).
	4	Wood blocks, assorted sizes.
	5	Jacks and accessories for lifting magnet: 20,000 lbs (9091 kg) total capacity
	6	Spreader Beam
	7	For LCC magnet: Lifting straps which meet specifications in <i>Direction 2226318, 1.5T & 1.0T LCC Magnet Signa MR/i Delivery/Installation Manual</i>
	8	For BRM Installation: The following equipment to meet specifications in Section 2-3-1 Gradient Coil Assembly, Cradle, & Cart Requirements - Forklift or crane/hoist rated for 4,000 lbs (1814 kg) - spreader beam - lifting straps
Customer	9	Equipment for off loading electronics and other miscellaneous components. (e.g. fork lift, hand trucks, straps, etc.)
	10	Panel lifters for computer flooring: Smooth floor Standard carpeted floor Level loop carpet floor Indicon Industries PL2DC; Indicon Industries PL30P; Indicon Industries SCLV1.
Customer/ Contractor	11	Resistance meter: - Megger Insulation Tester – PREFERRED TEST UNIT - Analog d'Arsonval Meter (meter must have test source >9 VDC per specification in <i>Direction 2223170 Signa MR/i & CV/i Pre-Installation Appendix B-6 ENCLOSURE POWER REFERENCE INSOLATION</i>)

10-4 CRYOGENIC EQUIPMENT

TABLE 10-3
CRYOGENIC EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	46-306734G1	Low Pressure Regulator Kit (Non-magnetic gas cylinder regulator / hose assembly consisting of regulator, hose, and case) (See Note 2)
2	—	Liquid helium in non-magnetic dewars is needed for refilling the magnet (See Note 2). Refer to appropriate magnet manual for helium volume.
3	46-306717G1	Non-magnetic gaseous helium cylinder cart (See Note 2)
4	46-294705G1	Universal Fill Line Kit (See Note 2)
5	46-294511P1	250 Liter Dewar Stinger (See Note 2)
6	46-294511P2	500 Liter Dewar Stinger (See Note 2)
7	46-294512P1 (12 ft) 46-294512P2 (8 ft)	Transfer Line (See Note 2)
8	46-282336P2	Dewar stinger assembly (High Efficiency) (See Note 2)
9	46-306812G1	Dewar tube "Thumper" tool (See Note 2)
10	2253802	Burst Disk and Gasket Assembly for LCC Magnet Helium Vessel (See Note 1)
11	46-271136G1	Dewar Adapting Kit including O-ring Kit for Dewar Adapting kit (46-271135P9) (See Note 2)
12	46-271137G1	Cryogen Safety Shield Kit (Includes gloves, face shield, and safety glasses.) (See Note 2)
13	46-294804G1	Non-magnetic Aeroquip wrench Set (See Note 2) includes the following: • 46-294800G1 case for bronze wrench kit • 46-294805P1 bronze 1 ⁵/₈ in. open end wrench • 46-294805P2 bronze 1 ³/₈ in. open end wrench • 46-294805P3 bronze 1 ³/₁₆ in. open end wrench • 46-294805P4 bronze 1 ¹/₈ in. open end wrench • 46-294805P5 bronze 1 in. open end wrench
14	46-281088G3	Shield Cooler Installation / Maintenance Kit (See Note 2)
15	46-265273G1	Liquid Helium Level Meter Kit (See Note 2)
16	46-306781G1	Helium Mechanical Gas Flowmeter (See Note 2)
17	46-301477G1	Cryogenic Thermometer, Lakeshore 208 Cryogenic Thermometer Kit or alternate Diode Temperature current source 46-317543G1 (See Note 2)
18	46-252210P1	3 Inch Valve Operator (vacuum break tool) (See Note 1)
19	46-294872G2	SAV-CON / Instrumentation Lead Service Kit (See Note 2)
20	46-318784G2	Shield Cooler Test Kit (See Note 2)
21	46-318696G1	Water Tee for Shield Cooler Water Samples access (See Note 2)
22	46-294052G1	Water Flow Meter Kit (for checking flow of water to Shield Cooler Compressor) (See Note 2)
23	2171219	RUO Temperature Monitor for LCC 1.5T & 1.0T Magnet (See Note 2)
Note		
1	Supplied as part of Signa.	
2	Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.	

10-5 INSTALLATION EQUIPMENT

TABLE 10-4
INSTALLATION EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	—	Ramp for removing cabinets from pallets for International shipments (See Note 1)
2	—	Wrecking bar
3	—	Claw hammer, 3/4 lb
4	—	Forklift or crane/hoist rated for 4000 lbs (1818 kg) FOR INTERNATIONAL SITES ONLY International will receive body coil on metal cradle. Forklift or crane/hoist needed for setting body coil on cradle onto Coil Cart for installation.
5	2164744-4	Gradient Coil Insertion Tool includes Coil Lift Kit 2326753
6	2144093	Standard Long Room coil cart with cradle for use in 20.44 ft (6.23 m) or longer Magnet Room
7	2144093-2	Short Room coil cart with cradle for use in <20.44 to 19.6 ft (<6.23 to 5.97 m) long Magnet Room
8	46-271138G1	Restricted Access Control Kit. Contains two plastic warning signs for posting at site during installation and service activity.
9	2230707	For LCC Magnet: Aluminum platform ladder, 47.5 inches (1206.5mm) (See Note 1)
10	2134776	Gradient Cable Crimper/Stripper Kit (Note 2) consisting of: • 2134586 Cable stripping tool • 2134586-2 Stripping tool replacement blade • 2134587 Cable slicer • 46-282853P1 Ratcheting crimper • 2135839 1/2 inch terminals • 2135839-2 3/8 inch terminals
11	46-306763G1	Subminiature-D Connector Removal/Re-termination Repair Kit for Robinson-Nugent Sub-D connectors (See Note 2)
12	46-320362G1	Heliac Cable Cutter / Stripper Kit (See Note 2)
13	46-301450G1	Fiber optic connector repair kit (See Note 2)
14	46-320273G3 or G4	Non-Magnetic Tool Kit – Universal (See Note 2) Both metric and inch Non-Magnetic Tool Kits needed. May substitute both of the following kits: • 46-320273G1 Non-Magnetic Tool Kit – Metric • 46-320273G2 Non-Magnetic Tool Kit – Inch
Note		
	1	Supplied as part of Signa.
	2	Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.

10-6 RF SURVEY APPARATUS

TABLE 10-5
RF SURVEY APPARATUS

ITEM	GE PART NUMBER	DESCRIPTION
15	46-265169P1	Tuned dipole antenna (two required)
16	46-265169P2	Tripod for dipole antenna (two required)
17	—	0.305 m diameter loop antenna
18	46-255835P2	Spectrum analyzer
19	46-265171P1	RF Preamplifier
20	46-265170P1	RF Linear Power Amplifier

10-7 TEST EQUIPMENT

TABLE 10-6
TEST EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	2284754	Textronic TDS3012 digital scope
2	2284763	Fluke FLK196M scopemeter for medical applications
1	46-194427P226	Dual trace oscilloscope, 100 MHz bandwidth, 2 channel, digital storage, Textronic 2232
2	46-194427P222	Dual trace oscilloscope, 350 MHz bandwidth, 4 channel, Textronic 2465
3	46-194427P284	Battery operated digital multi-meter, Fluke 87, 4.5 digits with frequency counter and capacitance
4	46-208572P9	Clamp-on Ammeter, 1-200A, AC/DC
5	46-255838P1	Attenuator for testing and troubleshooting RF subsystem, 10 dB
6	46-255838P2	Attenuator for testing and troubleshooting RF subsystem, 70 dB
7	46-317160G1	Vector impedance meter kit includes: • meter • case • connector/adaptor kit
8	46-317724P14	Attenuator/ load, 200 Watt, 30 dB attenuator
9	46-317724G1	RF Power Measurement Kit without Attenuator/ load, 200 Watt, 30 dB attenuator
10	46-317724G2	RF Power Measurement Kit with Attenuator/ load, 200 Watt, 30 dB attenuator 46-317724P14
11	2218826	RPM 1650 Power Analyzer Kit
12	46-328143G3	Dranetz 626 analyzer kit with 1 and 3 phase modules
13	Catalog E6320DA or 46-194427P144	Densitometer
14	46-306801G1	Bell Gauss Meter Kit
15	46-194427P248	Microguard or ECOS leakage tester
16	46-306797G1	Fogg (ECG) Simulator and Memory Module
17	46-306668G2	RS 232/422 Breakout Kit
18	46-320433P1	Infrared Scanner 0 – 100° C
19	46-317830G1	Fiber Optic Light Meter Kit

10-8 CALIBRATION TOOLS/FIXTURES

TABLE 10-7
CALIBRATION TOOLS/FIXTURES

ITEM	GE PART NUMBER	DESCRIPTION
1	46-307500G1	Longitudinal Drive Force Gauge
2	2133388-6	System Performance Test (SPT) Phantom Set for Systems with Cx (Active Shield) systems (See Note 1) consisting of the following: • 2125247-10 Nesting Plate • 2125245 Large Volume Shim Phantom Assembly • 2125244 Short Loader Assembly • 2131027-2 Daily Quality Assurance (DQA III) Phantom • 2134213 SPT Shipping/Storage Cart (Not used for Mobile Systems) • 2170481 EPI Foam Positioner • 2141454 Quick Shim Positioner
3	2133388	System Performance Test (SPT) Phantom Set for Systems with Non-Cx Magnets (See Note 1) consisting of the following: • 2125247 Nesting Plate • 2125245 Large Volume Shim Phantom Assembly • 2125244 Short Loader Assembly • 2131027-2 Daily Quality Assurance (DQA III) Phantom • 2134213 SPT Shipping/Storage Cart (Not used for Mobile Systems) • 2170481 EPI Foam Positioner • 2141454 Quick Shim Positioner
4	46-287900G3	Head TLT Sphere, one 100 mm Nickel Chloride sphere, and Loader (See Note 1)
5	46-328021G1	Enmet Oxygen Monitor Calibration consisting of the following: • 1800 PSI cylinder of 20.9% Oxygen in Nitrogen • 1800 PSI cylinder of 17% Oxygen in Nitrogen • Regulators with calibration adapter For use with portable or permanent Oxygen Monitor.
6	2106236	Portable Oxygen Monitor (Connecticut Analytical)
7	2107184	Permanent Oxygen Monitor (Enmet)
8	46-255816G1	RF Test Cables Kit (See Note 2)
9	46-307164G4	1.5T / 1.0T GRAFIDY II Kit (See Note 2)
10	46-307164G6	1.0T GRAFIDY II Kit (See Note 2)
11	46-301549G1	TPS RF Test Cable/Adapter Kit (See Note 2)
12	46-301927G1	TPS RF Service Interface Kit (See Note 2)
13	46-306712G1	Torque driver kit (See Note 2)
14	2101360	Power Supply Calibration Kit (See Note 2)
15	46-265826G3	Head Signal-to-noise (SNR) sphere (See Note 3)
16	46-265635G4	Body Signal-to-noise (SNR) sphere (See Note 3)
17	46-265635G2	Body T2 sphere (See Note 3)
18	46-328480G1	100 mm Nickel Chloride sphere (46-317586G1) and Universal Phantom Holder (46-328383P1) See Note 2
Note 1 Supplied as part of Signa. 2 Supplied by GE until turnover of system to customer, then available as part of a GE Service Contract. 3 Customer may purchase these items.		

10-9 TOOL KIT

TABLE 10-8
TOOL KIT

ITEM	GE PART NUMBER	DESCRIPTION
1	—	Extension cords, with ground conductor
2	—	Power strip, grounded type, with minimum of five outlets
3	—	Soldering iron, pencil type with solder
4	—	Solder sucker
5	—	Assortment of Brady Quick labels
6	—	Micro clip leads
7	—	14 pin and 16 pin DIP clips
8	46-258218P3	Vinyl electrical tape
9	46-258218P4	Copper tape, 3 in. wide
10	46-258218P5	Copper tape, 2 in. wide
11	46-258218P6	Copper tape, 1 in. wide
12	—	Alcohol cleaning solution
13	—	Plastic or aluminum flashlight
14	—	Plastic or aluminum pen light
15	AMP No. 458994-1	Pin extractor, Universal Mate'n'Lock
16	AMP No. 305183-R	Pin extractor, M-series
17	46-237072P1	Pin extractor, Sub-D
18	46-307307G1	Crimping tool for coax cable and BNC connectors Inserts for RG8, 9, 11, 214 46-255841P103 RG58, 223 46-255841P100 RG59 46-255841P101 RG174 46-255841P102 Die Removal Tool 46-255841P201
19	—	Assorted crimp tools.
20	—	Non-magnetic level
21	—	Non-magnetic tape rule, 12 ft
22	—	Assorted drill bits
23	—	Inspection mirror
24	—	Tap set, standard, and tap handle, T-type
25	—	6 in. rule, standard and metric markings
26	—	Alignment tool (tweaker)
27	—	Hex/alignment tool
28	—	Hemostat, 5 in., curved
29	46-198094P1	Wrist grounding strap
(Continued)		

10-9 TOOL KIT (Continued)TABLE 10-7 (Continued)
TOOL KIT

ITEM	GE PART NUMBER	DESCRIPTION
30	Xcelite 110CG	Diagonal Cutting Pliers, 4-1/2 in.
31	—	Screw Starter, aluminum or plastic shaft
32	—	Hobby and utility knives
33	—	Spring scales, 0-10 lbs and 0-50 lbs
34	46-313413P1	Extractor for 20 – 100 pin PLCC Chips

APPENDIX A –MR SITE VIBRATION TEST GUIDELINES

A-1 TEST MEASUREMENTS

- Vibration measurements are in the range of 10^{-6} g. Test equipment must have the required sensitivity to these levels.
- All analyses are to be narrowband Fast Fourier Transforms (FFT's) over the frequency bands listed in Table A-1.

TABLE A-1
FREQUENCY BANDS FOR FFT'S

FREQUENCY BAND	FREQUENCY RESOLUTION
0.2 to 50 Hz	$\Delta f = 0.125$ Hz

- Time histories of the vibration must be recorded (i.e. acceleration levels vs. time). The resolution of the time history must be adjusted to clearly capture the transient events. The analyzer set-up will be site dependent.

A-2 EQUIPMENT (SPECTRAL ANALYZER) SET-UP

- Frequency average a minimum of 20 linear averages (do not use peak hold or 1/3 octave analysis)
- Hanning window must be applied to the entire spectra

Spectrum analyzers capable of these measurements include the GenRad 2515, B&K 2032 or HP 3560A Dynamic Signal Analyzer. Accelerometers must have the capability to measure from 0.2 Hz beyond 50 Hz. Time histories can be recorded using any of the analyzers mentioned above. Good quality strip chart recorders may also be sufficient. Please note that the equipment mentioned are for example only. It is the responsibility of the Engineering test firm to provide equipment that will follow the test requirements.

A-3 DATA COLLECTION

A-3-1 Ambient Baseline Condition:

All of the measurements defined in Section A-1 and A-2 must be made in a 'quiet' environment. That is, in areas where excessive traffic, subway trains, etc. exists, a vibration measurement must also be made during periods without traffic or during periods of light traffic. Measurements must define the lowest levels of vibration possible at the site.

The source of any steady state vibration whose levels exceed the specifications in the appropriate following sections must be identified as to the source of the vibration disturbance:

Section 4-11-3 Specifications For 1.5T Oxford, S-I, S-II, & S-III (Unshielded) Magnets and 1.5T S-III Magnet With Magnishield

Section 4-11-4 Specifications For 1.5T S-IV, S-V, & CX And 1.0T S-X & CX Magnets

Section 4-11-5 Specifications For 1.5T And 1.0T LCC Magnets Without Gradient Isolation

Section 4-11-6 Specifications For 1.5T And 1.0T LCC Magnets With Gradient Isolation

A second measurement should be made with all of the identified contributors powered down if possible. In situations where it is not possible to power down equipment, vibration data must be collected to identify specific source of the vibration concern. The majority of steady state vibration problems can be negated by isolating the vibration source.

A-3-2 Normal (Operating) Condition

All of the measurements listed above must be repeated during periods of 'normal' environmental conditions including the FFT's and time histories. The transient measurements must be provided to define the dynamic disturbances the MR system might be exposed to. This transient disturbance is required for a true assessment of the site.

Transient vibration is difficult to assess and to eliminate. Environments exceeding the 500 micro-g, zero to peak trigger level, requires analysis of the building's dynamic response to the transient event. The building response must remain below the vibration specifications in the appropriate following sections:

Section 4-11-3 Specifications For 1.5T Oxford, S-I, S-II, & S-III (Unshielded) Magnets and 1.5T S-III Magnet With Magnishield

Section 4-11-4 Specifications For 1.5T S-IV, S-V, & CX And 1.0T S-X & CX Magnets

Section 4-11-5 Specifications For 1.5T And 1.0T LCC Magnets Without Gradient Isolation

Section 4-11-6 Specifications For 1.5T And 1.0T LCC Magnets With Gradient Isolation

The dynamic response following the transient event must exceed the specifications for sustained periods of time, i.e. for random dynamic response of duration less than a few seconds will have less impact on system performance than response of longer duration.

Building structures excited by transient events exceeding the specifications in the appropriate following sections must either provide recommendations to reduce levels to meet specifications or an alternative location should be identified:

Section 4-11-3 Specifications For 1.5T Oxford, S-I, S-II, & S-III (Unshielded) Magnets and 1.5T S-III Magnet With Magnishield

Section 4-11-4 Specifications For 1.5T S-IV, S-V, & CX And 1.0T S-X & CX Magnets

Section 4-11-4 Specifications For 1.5T And 1.0T Active Shield Magnets

Section 4-11-5 Specifications For 1.5T And 1.0T LCC Magnets Without Gradient Isolation

Section 4-11-6 Specifications For 1.5T And 1.0T LCC Magnets With Gradient Isolation

A second vibration measurement should be made to help identify a more stable location.

A-4 PRESENTATION/INTERPRETATION OF RESULTS

The recommended format for site vibration data collection, presentation, and analysis is illustrated in the examples shown in Illustrations A-1 through A-2. Presentation of the data in any other format (linear units only) may result in an incorrect interpretation and diagnosis of the site. Additional data collection or presentation methods is at the option of the vibration testing service.

It is the responsibility of the customer's vibration testing service to interpret the results and determine if that site meets GE's specifications. Illustrations A-1 and A-2 are examples provided to assist a test consultant in the use of GE Steady State specifications (vibration specifications above ambient baseline). If the vibration levels are too high, additional data acquisition may be necessary to:

- determine the source of the vibration
- propose a solution to the problem
- find an alternate site location.

Illustrations A-3 and A-4 are examples provided to assist a test consultant in the use of GE Transient specifications. The 500 micro-g, zero to peak trigger level identifies data collection to begin assessment of the site vibration analysis. The response of the transient must be assessed relative to the Steady State vibration specifications in the appropriate following sections:

Section 4-11-3 Specifications For 1.5T Oxford, S-I, S-II, & S-III (Unshielded) Magnets and 1.5T S-III Magnet With Magnishield

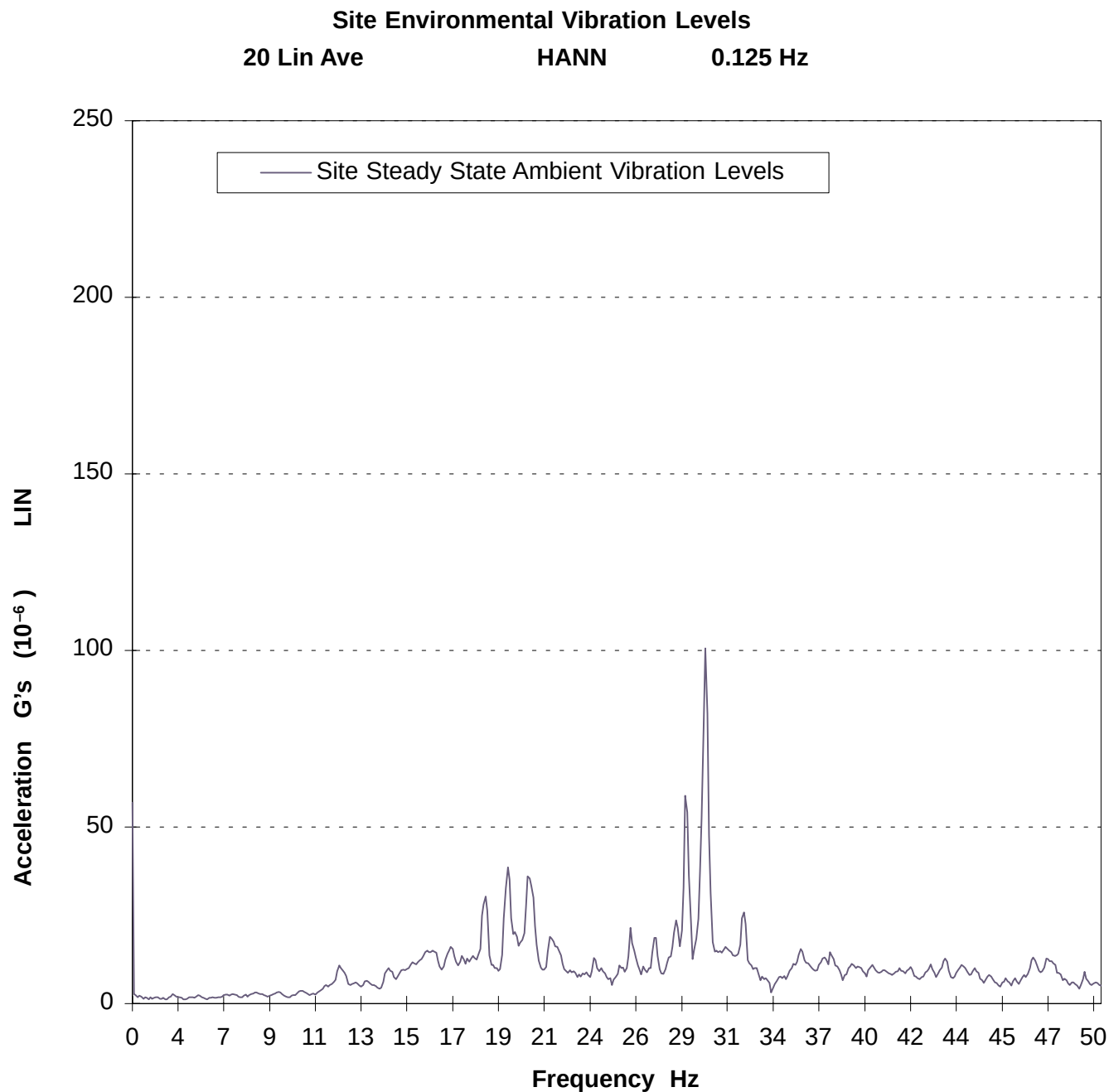
Section 4-11-4 Specifications For 1.5T S-IV, S-V, & CX And 1.0T S-X & CX Magnets

Section 4-11-5 Specifications For 1.5T And 1.0T LCC Magnets Without Gradient Isolation

Section 4-11-6 Specifications For 1.5T And 1.0T LCC Magnets With Gradient Isolation

Any questions regarding test equipment requirements, test parameters, or general questions should be discussed with your GE Installation Specialist.

A-4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)



EXAMPLE SITE ENVIRONMENTAL VIBRATION ILLUSTRATION A-1

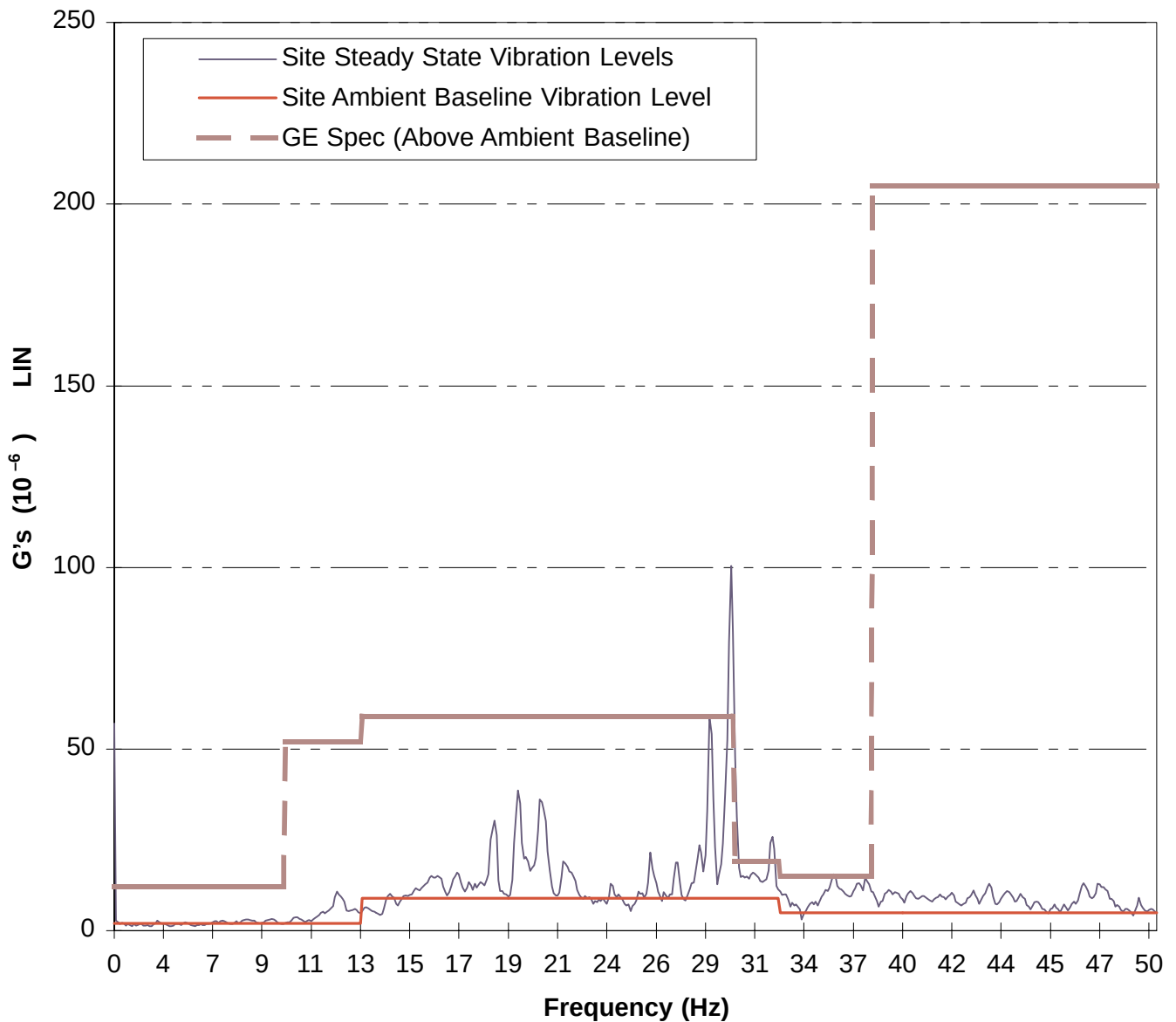
A-4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)

EXAMPLE: Site Environmental Vibration vs. GE Spec. for 1.5T K4 Magnet

20 Lin Ave

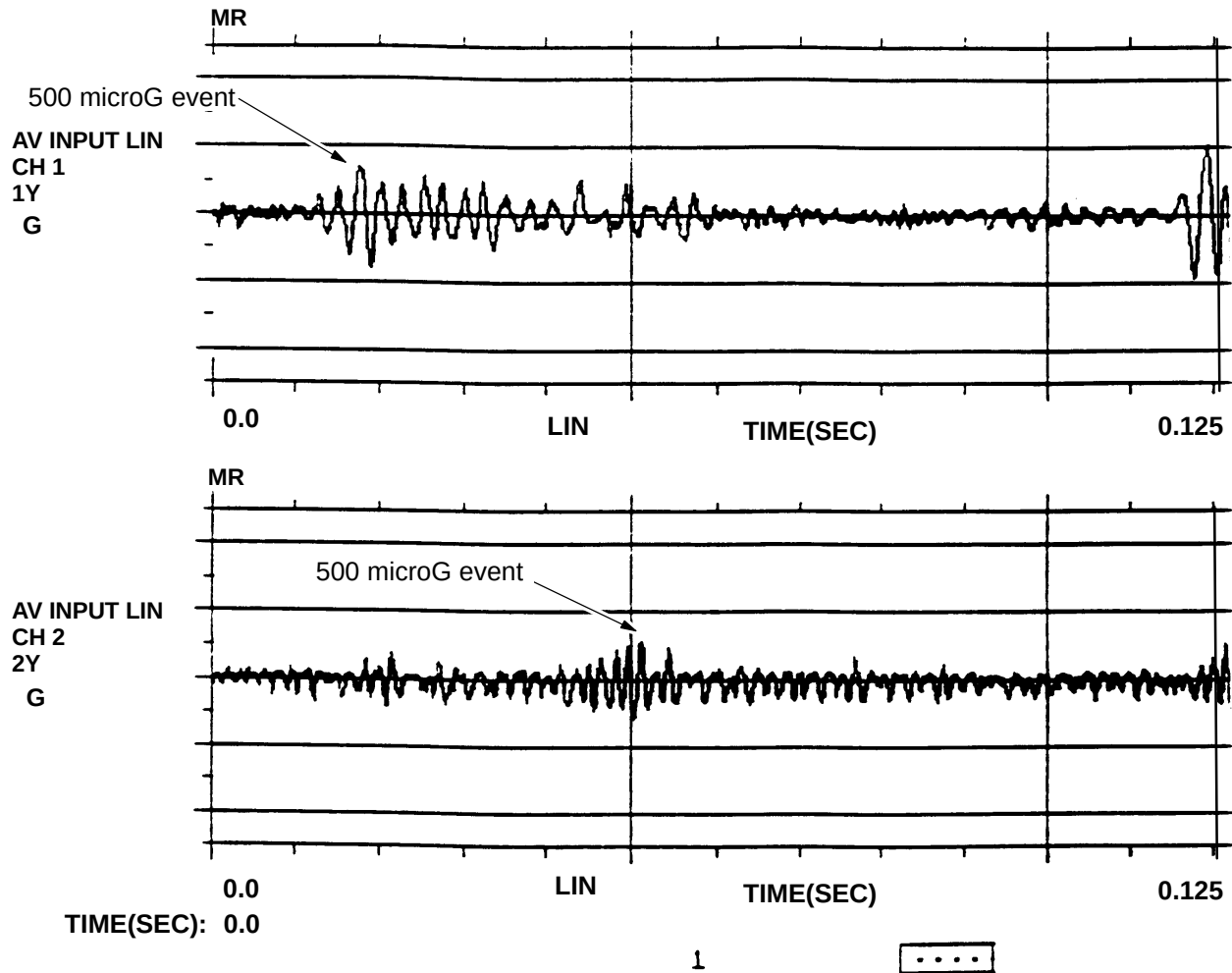
Hann

0.125Hz



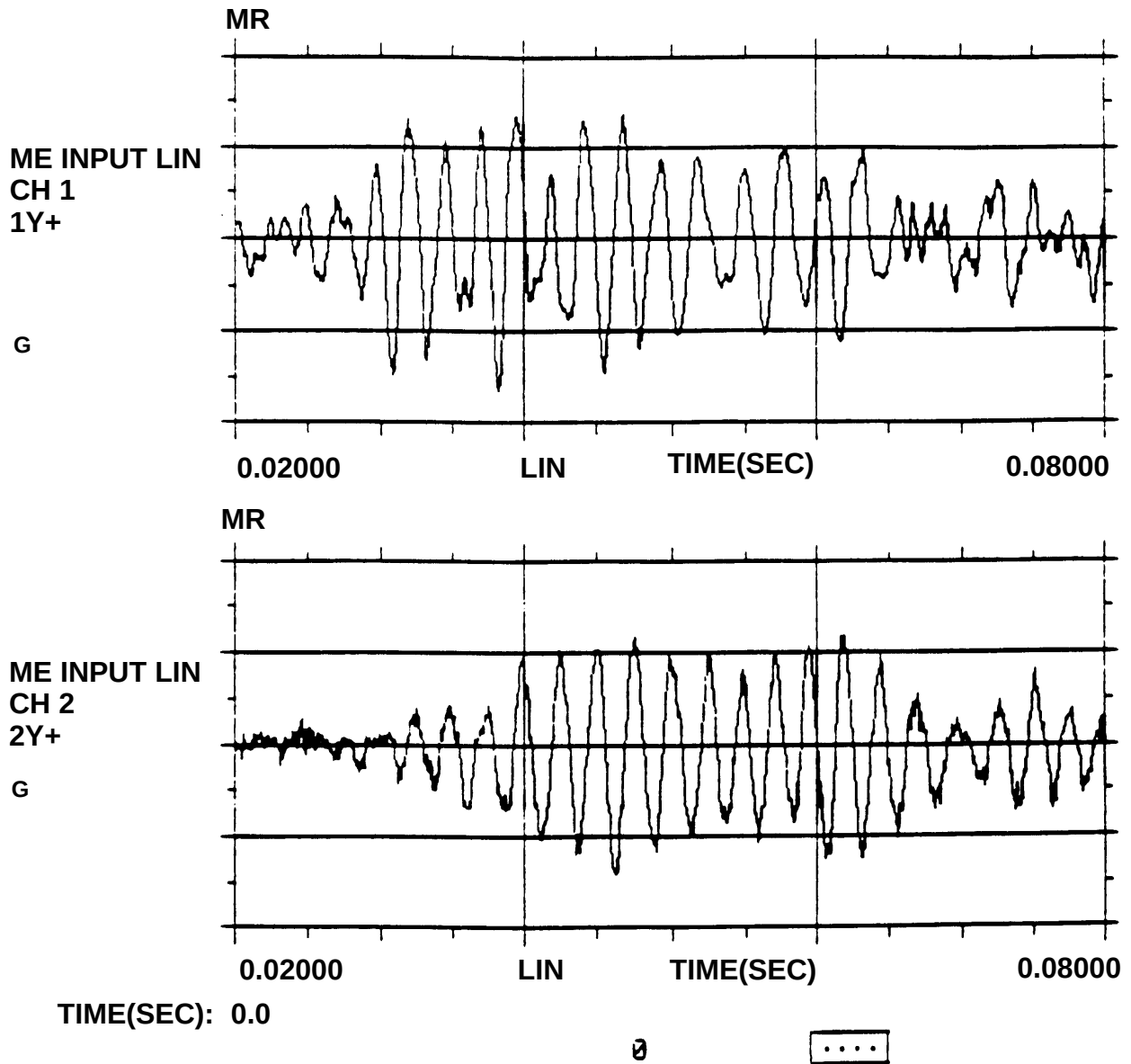
EXAMPLE SITE ENVIRONMENTAL VIBRATION VS. GE SPECIFICATION FOR 1.5T LCC MAGNET WITHOUT GRADIENT ISOLATION
 ILLUSTRATION A-2

A-4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)



ACCELERATION TIME HISTORY
ILLUSTRATION A-3

A-4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)



ACCELERATION TIME HISTORY (ZOOM IN ON 500 microG TRANSIENT EVENT)
ILLUSTRATION A-4